

Antennal Movements and Mechanoreception: Neurobiology of Active Tactile Sensors

Erich M. Staudacher^{a*}, Michael Gebhardt^{b†} and Volker Dürr^c

^a*Center for Computational Biology, Montana State University, 1 Lewis Hall, Bozeman, MT 59717, USA*

^b*Lehrstuhl für Zoologie, TU München, Lichtenbergstr. 4, D-85748 Garching, Germany*

^c*Abteilung Biologische Kybernetik und Theoretische Biologie, Fakultät für Biologie, Universität Bielefeld, Postfach 10 01 31, D-33501 Bielefeld, Germany*

1	Introduction	50
2	The antennal motor system	51
2.1	Morphological types and model systems	51
2.2	Biomechanical and kinematic considerations	59
2.3	Efferent innervation of the antenna	65
2.4	Physiology of antennal motoneurons and muscles in insects	70
3	Sensory physiology of antennal mechanoreceptors	74
3.1	Sensory structures and transduction	74
3.2	Distribution of mechanoreceptors	84
3.3	Physiology of antennal mechanosensory neurons	100
4	Neuroanatomy of antennal mechanosensory and motor pathways	103
4.1	Mechanosensory neuropils	103
4.2	Connections to other parts of the CNS	109
4.3	Immunocytochemistry	113
4.4	Antennal motoneurons in insects	118
5	Central neurophysiology of antennal mechanoreception	122
5.1	Processing of antennal mechanosensory information by local brain interneurons	122
5.2	Descending antennal mechanosensory interneurons	126
6	Behavioural physiology of the antennal tactile sense	129
6.1	Passive sensing	130
6.2	Active sensing	143
7	Biomimetics and ‘antennal engineering’	168
7.1	Steering insects and robots	169
7.2	Engineering of active tactile sensors	170
8	Conclusions	172
8.1	The antennal tactile sense of insects and Crustacea	172

*Present address: Lehrstuhl für Zoologie, TU München, Lichtenbergstr. 4, D-85748 Garching, Germany.

†Present address: Institut für Zoologie, Abteilung für vergleichende Neurobiologie, Universität Bonn, Poppelsdorfer Schloß, D-53115 Bonn, Germany.

8.2	Active and passive mechanical sensing	174
8.3	Levels of behavioural complexity	176
	Acronyms and abbreviations	178
	Acknowledgements	179
	Note added in proof	180
	References	180

1 Introduction

The tactile sense provides animals with vital information about their immediate environment, comprising cues on object location and surface properties. In insects and other arthropods, the antennae are probably the most important part of this tactile sense.

The paired insect antenna is regarded as a true, segmented limb arising from the cranium, although there is still some debate about the precise number of segments in the head of the Mandibulata (Crustacea, Myriapoda and Insecta; Schmidt-Ott and Technau, 1992; Schmidt-Ott *et al.*, 1994; Damen *et al.*, 1998; Abzhinov and Kaufman, 1999; Boyan and Williams, 2002). Hennig (1986) uses the name *Antennata* as a synonym for the Mandibulata, expressing the fact that this taxon comprises all living arthropod species with antennae. The most ancient arthropods, the extinct trilobites, also carried antennae, which is why the 1st pair of antennae is typically considered to be the first pair of limbs, i.e. articulated appendages, in the ground plan of all arthropods. Accordingly, the Chelicerata are thought to have secondarily lost their antennae. Within the Mandibulata, the Crustacea carry two pairs of antennae: the antennules (first pair) are appendages of the deutocerebral segment and are considered to be homologous to the insect antenna, whereas the second pair of antennae are the appendages of the so-called tritocerebral segment (Hennig, 1986; Brusca and Brusca, 1990; Telford and Thomas, 1998; Boyan *et al.*, 2002). Insects have lost the second pair of antennae during evolution.

Antennae are equipped with various types of mechanoreceptors, including large numbers of exteroceptive sensory hairs, but also proprioceptors such as internal chordotonal organs, external hair plates, and campaniform sensilla.

Being moveable, the antennae can actively sample the space surrounding the animal, thus aiding the localisation of obstacles, recognition of conspecifics or predators, active tracking of objects, or probing of surface structures. Because both passive and active movements can occur during tactile sensing, the tactile sensory system requires integration of proprioceptive and exteroceptive information in order to locate the stimulus source. Moreover, self-induced sensory stimulation may have to be predicted from motor commands, and compared with the actual sensory input. Therefore, understanding the neurobiology of the tactile sense of insects and crustaceans requires an integrative view of the motor physiology and biomechanics on the one hand, and of the

neurophysiology of the participating sensory and interneuronal structures on the other. In the light of this knowledge, the functional significance of the various system components can be interpreted for a range of natural behaviours.

The last integrative view on insect antennae that included all known sensory modalities was given by Schneider (1964). Since then, much progress has been made, not only in the field of insect olfaction (e.g. Hansson, 1999) and the sensory physiology of mechanoreceptors (e.g. McIver, 1975, 1985; Zacharuk, 1985), but also in elucidating key components in tactile behaviours such as the localisation of obstacles or physical interactions in intraspecific communication. To illustrate this body of knowledge, and to extend and contrast the mentioned reports on mechanoreception, it is the aim of the present review to combine aspects of antennal biomechanics, mechanoreception, processing of tactile information, and the behavioural repertoire involving the antennal tactile sense. Each of these aspects will be treated in separate sections, concentrating on prominent animal model systems, such as the cockroach (mainly *Periplaneta americana* L. and *Blaberus* sp.), stick insect (*Carausius morosus* Br.), cricket (*Gryllus* sp., *Teleogryllus* sp. and *Acheta domesticus* L.), locust (*Locusta migratoria* L. and *Schistocerca gregaria* Forskål) and the honeybee (*Apis mellifera* L.). For comparison with a related animal group that employs very similar, though not homologous, organs in tactile sensing, we will expand our view on decapod crustaceans, notably crayfish and lobsters. Finally, we discuss the potential of active tactile sensing from a technical perspective, to point at a helpful mutual exchange between engineering and animal physiology.

2 The antennal motor system

To set the framework for discussing the anatomical and physiological results, the following sections present an overview of antennal morphology (Section 2.1); the basic physical properties of antennae with respect to tactile sensing (Section 2.2); the components of the contractile machinery (Section 2.3) and their physiological properties (Section 2.4).

2.1 MORPHOLOGICAL TYPES AND MODEL SYSTEMS

By far the most physiological studies have dealt with a small number of animal model systems, whose anatomical and physiological properties are considered to be representative for many related species. The following sections introduce the representative groups of insect model systems (Section 2.1.1). For comparison with an analogous tactile organ, Section 2.1.2 will then introduce the second antenna of decapod Crustacea.

2.1.1 *Insect model systems*

In principle, there are two types of antennae in insects: the segmented and annulated type, that differ primarily in articulation and musculature of the distal segments. In both types, the first two segments are called scape and pedicel. These two are followed by a varying number of further, distal sections (Matsuda, 1965; Weber and Weidner, 1974; Zacharuk, 1985; Brusca and Brusca, 1990; Boyan *et al.*, 2002).

2.1.1.1 *The segmented antennae of the Entognatha.* Segmented antennae are found in Entognatha. They consist of scape, pedicel and a number of more or less uniform articulated segments. Antennal muscles arise from all except the last segment, thus all articulations are true joints. For example, detailed descriptions of the antennal musculature of the dipluran genera *Japyx* and *Campodea*, and of the collembolan genus *Orchesella* are given by Imms (1939). The Collembola have a tendency to reduce the number of antennal segments. The pedicel of segmented antennae lacks Johnston's organ (Matsuda, 1965; Zacharuk, 1985), a chordotonal organ that is characteristic of the antennae of higher insects (see Section 3.2.3.3).

Owing to the serial arrangement of segmental muscles in the flagellum of the Entognatha, repetitive sets of motoneurons presumably exist that innervate these muscles. To our knowledge, there is no information available concerning antennal motoneurons of these insects.

2.1.1.2 *The annulated antennae of higher insects.* Annulated antennae are found in the Ectognatha (bristletails, silverfish and winged insects). They consist of three functional segments (Fig. 1), the scape, pedicel, and the flagellum or clavola. Paralleled by the functional importance of a strong head-scape (HS) musculature, the Ectognatha have evolved an increasingly complex internal head skeleton for its proximal insertion: the tentorium. It consists of the posterior tentorium that is common to all Ectognatha, the anterior tentorium of the Dicondylia and the corpo-tentorium of the Pterygota (Koch, 2000). Apart from antennal muscles, some muscles of the mouthparts also insert on the tentorium.

Currently, it is still debated whether scape and pedicel are homologous to coxa and trochanter, respectively, of an ancestral appendage (Haas *et al.*, 2001; Boyan *et al.*, 2002). The flagellum has a varying number of annuli or meristal segments, the morphology of which is not necessarily uniform (Weber and Weidner, 1974; Brusca and Brusca, 1990). The annuli are not true segments because they are not articulated by true joints, i.e. they do not contain muscles (Imms, 1939). Since the flagellum in its entirety essentially functions as a multi-sensory probe, it can be considered as a third segment in functional terms (e.g. Lee and Strausfeld, 1990; Kloppenburg *et al.*, 1997; Dürre *et al.*, 2001), if not in morphological terms. As the pedicel does not contain muscles and, thus, the flagellum cannot be actively moved relative to the pedicel, the connection between pedicel and flagellum will be called pedicel-flagellum (PF) junction, to

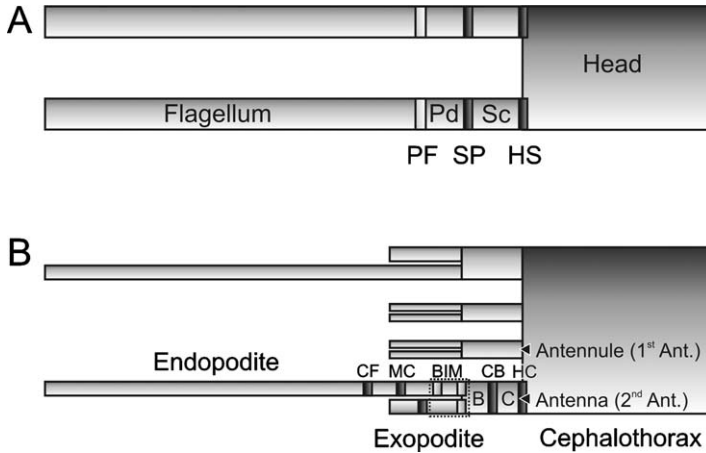


FIG. 1 Antennal joints and segments: (A) Schematic top view of an insect head. The antennae consist of three functional segments: the scape (Sc), pedicel (Pd) and an annulated flagellum. In all model systems discussed in this review, the annuli of the flagellum are approximately cylindrical and fairly uniform. The articulations head-scape (HS) and scape-pedicel (SP) are true joints (dark bars), and can be moved by musculature. The junction between the pedicel and the flagellum (PF, light bar) allows only passive deflection. (B) Crayfish and lobsters (Astacidea) and spiny lobsters (Palinurida) carry two pairs of antennae: The 1st pair are called antennules, the 2nd pair is called antennae. The basal segments of the antenna, coxopodite (C) and basipodite (B), connect to the multi-segmented endopodite that includes the long flagellum (distal to CF). In the Astacidea, the basipodite connects to a second, smaller branch, the exopodite, or scaphopodite (dotted). The latter is reduced in Palinurida. There are five functional joints: The two basal articulations 'head'-coxopodite (HC) and coxopodite-basipodite (CB), as well as the distal articulations meropodite-carpopodite (MC) and carpopodite-flagellum (CF) are simple joints with one joint axis (dark bars). The BIM-joint of the Astacidea is a complex articulation involving the basipodite-ischiopodite and ischiopodite-meropodite joints of the endopodite and the basal joint of the exopodite (dotted rectangle, containing three light bars). In the Palinurida, the first three segments are fused. As a consequence, the CB-joint is absent and the BIM-joint is only formed by the ischiopodite and meropodite.

emphasise the difference to a true joint. Variations of the morphology and number of the flagellar annuli account for the different types described for annulated antennae (Table 1; e.g. filiform, setaceous, lamellate; for illustrations cf. Weber and Weidner, 1974; Zacharuk, 1985; Brusca and Brusca, 1990).

The first, i.e. the most proximal, joint of the annulated antenna is the articulation of the scape with the head cuticle (HS-joint). The second joint connects the scape to the pedicel (scape-pedicel SP-joint, Fig. 1A). The HS joint is constructed differently in different insect groups. A hinge-joint occurs in locusts, crickets and stick insects (Gewecke, 1972b; Honegger *et al.*, 1990a; Dürre *et al.*, 2001), whereas a ball-and-socket joint is found in other insects. The axis of the hinge-joint is determined by two points of articulation of the scape with the head, thus restricting the movements of the scape to one plane.

TABLE 1 The morphology of the antennae of the insect model organisms

Species	Type of antenna	Number of annuli	Length mm	Reference(s)
<i>Periplaneta americana</i> L., female	Filiform	139 ± 26	42	Schafer and Sanchez (1973)
<i>P. americana</i> L., male	Filiform	135 ± 33	42	Schafer and Sanchez (1973)
<i>Acheta domesticus</i> L., 13th instar	Filiform	~150	50	Toh (1977)
		~255	30	Fudalewicz-Niemczyk and Rosciszewska (1973)
<i>Locusta migratoria</i> L.	Filiform	24	14	Gewecke (1972)
<i>Schistocerca gregaria</i> Forskål	Filiform	24	14	Ochieng <i>et al.</i> (1998)
<i>Carausius morosus</i> Br.	Filiform	40–46	~39	Slifer (1966)
		40–46	35	Monteforti <i>et al.</i> (2002)
<i>Apis mellifera</i> L., female	Geniculate	10	Not reported	Snodgrass (1956)
<i>A. mellifera</i> L., male	Geniculate	11		Snodgrass (1956)
<i>A. mellifera</i> L., worker	Geniculate	10	3.9	Esslen and Kaissling (1976)
				Slifer and Sekhon (1961)
			2.4	Esslen and Kaissling (1976)

In locusts, a protrusion of the head cuticle, the antennifer (Snodgrass, 1935), contributes to form the lateral point of articulation which is more rigid than the median articulation point (Gewecke, 1972b). This configuration requires some initial contraction of the extrinsic antennal muscles, i.e. the ones moving the HS-joint, to tense the membrane of the median articulation before it can function as the second pivot of the HS-joint. In *Locusta*, this joint allows for 60° levation and 50° depression. The scape of *Locusta* is rotated laterally by 40° with respect to the sagittal plane such that, at rest, the scape points antero-laterally (Gewecke, 1972b). In crickets too, this plane of movement allows levation and depression of the scape, but is arranged parallel to the sagittal plane (Honegger, 1981; Honegger *et al.*, 1990a). In stick insects, the axis of the HS-joint is rotated dorsally by 149° relative to the horizontal plane, and medially by 10° relative to the frontal plane. The angular range of the HS-joint in walking stick insects is about 100°, as calculated from behavioural measurements by Dürr *et al.* (2001). The HS-joints of the cockroach (*Periplaneta americana*), the honeybee, ants, and the moth *Manduca* are different from the HS-joints of locusts, crickets and stick insects in that they work like a ball-and-socket-joint, allowing movements in more than one plane (Snodgrass, 1956; Kloppenburg, 1995; Ehmer and Gronenberg, 1997a; Kloppenburg *et al.*, 1997). In *Periplaneta*,

the HS-joint is described as a “ball-joint-like structure” (Okada and Toh, 2001) that allows movement of the antenna in any direction (Okada and Toh, 2000, 2001). According to Ellerkmann (1994), movements of this joint cover about one-fourth of a semi-sphere in bees, when moving dorsally, ventrally or laterally. Medial movements do not cross the midline. The scape of *Manduca* is free to pivot into all directions (Kloppenburger *et al.*, 1997).

In all insect species where the SP-joint has been investigated, it is a hinge-joint which moves the pedicel, together with the flagellum, within a single movement plane (locust: Snodgrass, 1956; Gewecke, 1972b; cricket: Honegger, 1981; Honegger *et al.*, 1990a; stick insect: Dürri *et al.*, 2001; honeybee: Kloppenburger, 1995; ant: Ehmer and Gronenberg, 1997a; *Manduca*: Kloppenburger *et al.*, 1997; see also Fig. 2).

In the locust, one dorsal and one ventral point of articulation form the pivots of the hinge, allowing for 45° abduction and 45° adduction (Gewecke, 1972b). In crickets, the SP-joint is also composed of a dorsal and a ventral pivot, and allows horizontal movement of the pedicel/flagellum. Here, the pedicel locks against the scape during abductions at 90° lateral relative to the body long axis. Movements further laterally require the scape to be lifted. Abductions beyond 80°–85° occur only rarely during natural antennal movements (Honegger, 1981). Therefore, in locusts and crickets, the two antennal joint axes are approximately orthogonal to each other.

This is different in stick insects, where the axis of the SP-joint axis is rotated dorsally by 28° with respect to the horizontal plane. Therefore, the joint axes of the stick insect antenna are not orthogonal to each other. With the HS-joint axis being rotated by 149°, the smaller angle between the HS-joint and the SP-joint axes is 59° (Dürri *et al.*, 2001; see also Section 2.2.1). In the honeybee and in ants, the SP-joint resembles an ‘elbow-like’, dicondylic hinge joint that allows flexion and extension in a single plane (Snodgrass, 1956; Ehmer and Gronenberg, 1997b). Kloppenburger *et al.* (1997) describe the movement of the SP-joint of *Manduca* as “more limited [with respect to the HS-joint], hinge-like”.

2.1.2 Crustacean model systems

It is the 2nd pair of antennae that is typically referred to when speaking of the ‘crustacean antenna’. This convention will be kept throughout this review. Being the larger and more robust pair of antennae, it is this pair of appendages that is particularly relevant to the tactile sense of decapod crustaceans. Accordingly, this review will nearly exclusively cover the second pair of antennae in Crustacea.

Most of the available information on crustacean antennal movements concerns clawed species of crayfish and true lobsters on the one hand (Astacidea), and the clawless spiny lobsters on the other (Palinurida). Within the Astacidea, all of the presented information has been collected on the marine lobster genus *Homarus* and on species of three families of crayfish, belonging to the European Astacidae (*Astacus*), the American Cambaridae (*Cambarus*,

Procambarus and *Orconectes*) and the Australian Parastacidae (*Cherax* and *Euastacus*). Within the Palinurida, all studies concern the genera *Palinurus* and *Panulirus*.

The decapod antenna consists of two basal segments, the coxopodite and basipodite, followed by two branches of segments: the lateral exopodite and the medial endopodite (Fig. 1B). The endopodite consists of four segments: the ischiopodite, meropodite, carpopodite and flagellum. With one exception, each pair of adjacent antennal segments forms a single joint between them. The exception is a complex joint formed by articulations of the basipodite, ischiopodite and meropodite (see below). The latter can be viewed as one functional joint which will subsequently be called the BIM-joint. Thus, the proximal to distal sequence of joints connecting the 'head' (cephalothorax) to the basal segments and the segments of the endopodite will be called HC-, CB-, BIM-, MC- and CF-joints (Figs. 1B and 2B). The exopodite of the Eumalacostraca, a taxon that contains the Decapoda, is the scale-shaped scaphopodite. The secondary reduction of this scaphopodite is a synapomorphy of the Palinurida (Hennig, 1986). Associated with this reduction is a different arrangement of the BIM-joint (see below). In the Astacidea, the proximal segment of the scaphopodite acts as the antagonist of the BIM-joint and, thus, is instrumental to rotational movements of the endopodite.

The most detailed description of the anatomy and kinematics of the decapod antenna was given by Habig and Taylor (1982a) for crayfish of the genera *Procambarus* and *Cambarus* (Figs. 1B and 2B). The most basal HC-joint rotates the coxopodite in a vertical plane relative to the head. It is moved by two muscles that act as a depressor and a levator. The CB-joint rotates the basipodite in a horizontal plane relative to the coxopodite. It is moved medially by four adductors and laterally by one abductor.¹ Two adductors of the CB-joint are bi-articulate muscles.

continued

rotated around its long axis is uncertain. The honeybee (*Apis*) has a geniculate antenna with a three-degree-of-freedom HS-joint. The pedicel is flexed/extended using the SP-joint. Note that segment lengths are not drawn to scale: the relative length of the scape is much exaggerated, and species differ greatly in flagellum length. Compared to the insect model systems, crayfish and lobsters have much more complex antennal kinematics. The HC- and CB-joints have limited action range. They abduct/adduct and levate/depress the endopodite, respectively. The BIM-joint rotates the endopodite around its long axis (bold arrow), thus slanting the movement plane of the flagellum. The two distal MC- and CF-joints have nearly parallel joint axes, and act synergistically to flex/extend the flagellum.

¹Note that Habig and Taylor (1982a) use a different nomenclature than is used here, to account for the serial homology with leg musculature: depressor and levator are synonymous to Habig and Taylor's remotor and promotor, respectively. Adductor and abductor are synonymous to Habig and Taylor's depressor and levator, respectively.

The basipodite connects to two distal segments: the first segments of the endopodite and exopodite, respectively. Both articulations control a roll movement of the endopodite in an antagonistic manner. The remotor muscle of the basipodite rotates the exopodite medially (BE-joint) and presses it against the lateral surface of the meropodite. This causes the meropodite to rotate outward, i.e. the lateral side moves to a ventral position. The two ischioceritis muscles and the two meroceritis muscles of the BIM-joint synergistically cause an inward rotation of the entire endopodite. One meroceritis muscle is a bi-articulate muscle, spanning the entire BIM-joint.

In essence, the complicated arrangement of the BE- and the BIM-joints makes them antagonistic components of one functional rotator joint, allowing the endopodite roll around its long axis by some 90° . Together with the two basal joints, this allows all three degrees of freedom of rotation of the antenna, although the working ranges of the HC- and CB-joints are relatively small. The distal MC- and CF-joints have approximately parallel joint axes, allowing them to act as synergists with an expanded common flexion/extension range of the flagellum of almost 180° . Both joints are moved by a pair of flexor/extensor muscles. Upon full extension, the flagellum points straightforward, parallel to the body long axis. Full flexion rotates the flagellum to point rearward, where the arc described by the tip is a semi-circle above or lateral to the animal, depending on the rotation angle of the BIM-joint. Thus, the function of the three basal joints of the crayfish antenna is to orient the extension/flexion plane of the flagellum between a horizontal plane and a vertical plane.

The musculature of the distal antennal segments in the crayfish *Euastacus armatus* is similar to that of cambarid species, with the addition of a small accessory extensor muscle spanning the MC- and CF-joint (Sandeman and Wilkens, 1983). For the distal segments of the lobster antenna (*Homarus americanus*), Sigvardt (1977) gives the joint angular ranges as 104° for the MC-joint and 95° for the CF-joint. Other than in crayfish, the MC- and CF-joint axes are not parallel, but form an angle of 17° . In its resting posture, the meropodite of the *Homarus* antenna is rotated 90° medially by the BIM-joint, so that the flagellum normally moves in a nearly vertical plane.

In the more distantly related palinurid species, the major difference to the crayfish antenna is probably related to the reduction of the scaphopodite. In spiny lobsters the first three segments are fused (Balss, 1940; cf. Sigvardt, 1977), simplifying the complex BIM-joint to a single joint between the ischiopodite and meropodite. In the rock lobster *Palinurus vulgaris*, angular ranges of the three distal antennal joints are 80° for the inward/outward rotation about the oblique IM-joint axis (Clarac *et al.*, 1976), and 60° each for flexion/extension about the parallel MC- and CF-joint axes (Clarac and Vedel, 1975). The IM-joint is moved by three internal rotator muscles and two external rotator muscles (Clarac *et al.*, 1976). Both distal joints are moved by a pair of flexor/extensor muscles, and an additional accessory extensor muscle spans both joints (similar to *Euastacus*).

2.2 BIOMECHANICAL AND KINEMATIC CONSIDERATIONS

To allow a combined functional interpretation of anatomical, physiological and behavioural findings on antennal movements and mechanoreception, it is helpful to discuss some theoretical aspects of the various antennal ‘designs’ that have evolved (Section 2.2.1), but also the physical framework that the nervous system has to deal with when employing a flagellum as a tactile sensor (Section 2.2.2).

2.2.1 *Kinematics, workspace and tactile acuity*

Physiological limitations for tactile localisation of a contact point are set by three aspects. First, the spacing of sensilla on the flagellum (see Section 3.2), and the angular resolution of the proprioceptive coding set the upper bound for the sensory resolution of surface structures and external coordinates of a contact point. Second, there is a limit to positioning accuracy set by the antennal musculature and its motoneurons, affecting the reliability of trajectories or sampling positions in repetitive contacts. Third, the properties of the kinematic chain, i.e. the sequence of joint axes and segments, determine the workspace of the flagellum, but also affects positioning accuracy within the workspace.

Concerning the accuracy of proprioceptive coding on the one hand (Sections 3 and 5), and positioning accuracy on the other (see Section 2.4), there is virtually no information as to how precise the nervous system can control these aspects.

Concerning the significance of kinematic parameters on the sampling strategy, Dürre and Krause (2002) have offered a preliminary framework to assess the physical limitations set by the ‘construction’ of a given antenna (Fig. 2A). Following a suggestion on the functional significance of slanted joint axes in the stick insect *Carausius morosus* (Dürre *et al.*, 2001), they show how the angle between the HS- and SP-joint, the inter-joint angle, influences the shape of the antennal workspace. In case of orthogonal joint axes, the workspace has a torus-like shape, where the length of the scape, determines the deviation from a sphere (the deviation being larger with increasing length of the scape). As the inter-joint angle becomes larger, the holes of the torus-shaped workspace become larger. Thus, the volume out of reach of the flagellum increases (Dürre and Krause, 2002; Krause and Dürre, 2004). On the other hand, provided that there is a limit of angular accuracy in each antennal joint (both sensory and motor), and noise in both joints is independent of each other, the impact of this inaccuracy on positioning decreases with the inter-joint angle (Dürre *et al.*, 2001). In conclusion, slanting the joint axes to narrow the inter-joint angle gives rise to an out-of-reach zone, but increases tactile accuracy in the within-reach zone.

Slanting the axis of the flagellum on the SP-joint axis introduces an asymmetry into the torus-shaped workspace. The result of this asymmetry is that the out-of-reach zone narrows on the side to which the flagellum points, and widens on the other. Thus, assuming that the flagella of an insect were slanted laterally, i.e. diverge with respect to the midline, the widened out-of-reach zone would be located at the postero-median side of the workspace (facing towards the head). Assuming bilaterally symmetrical antennae, the medially widened out-of-reach zone could be compensated by the presence of the contralateral antenna. It follows that the kinematic chain that describes the mobility of an antenna, also describes some physical limits that are likely to be of behavioural significance (Krause and Dürre, 2004).

The kinematic chains of the model systems discussed in this review are illustrated in Fig. 2B. Among the three model systems with two hinge joints (locust, cricket and stick insect) the major differences lie in the length of the flagellum (shortest in the locust) and in the inter-joint angle (non-orthogonal in the stick insect). It is reasonable to assume that the axes of the stick insect are slanted for an evolutionary reason. However, whether or not stick insects benefit from an improved tactile acuity in a dorso-ventral direction, as the joint axis orientations would suggest, remains to be shown. Also, environmental constraints may bias the benefits of one construction over another.

The ball-and-socket construction of the HS-joint in cockroaches and bees does not allow the full three degrees of freedom of rotation that a genuine ball joint, like the human shoulder joint, would allow. Rather, the degree of rotation around the axis of the scape is limited, due to the mechanical constraints of the joint membrane and by the small set of four HS muscles. In effect, the action range of these HS-joints may be more similar to that of a Cardan joint rather than a ball-and-socket joint, mainly allowing for levation/depression and abduction/adduction. The introduced redundancy does not add much mobility compared to the hinge-joint construction. One implication of the ability to rotate the flagellum around the long axis of the scape would be that a given point in the workspace could be approached from any angle, and that the contact surface could be adjusted. In a hinge-joint or Cardan construction, the contact surface is determined by the direction of movement. Although the range of rotation around the scape is not addressed explicitly by any of the available studies, Heran (1959) describes postural adjustments in the flying honeybee which imply such rotations. Thus, it is likely, that the honeybee can actively rotate the scape to touch an object with a particular part of the flagellar surface (see Section 3.2 for distribution of sensilla and Section 6.2.3 for tactile sensing in honeybees).

Crayfish and lobsters can actively rotate the flagellum around its long axis by means of the BIM-joint. It acts to orient a movement plane, within which the MC- and CF-joints adjust the orientation of the flagellum. Owing to the small action ranges of the joints proximal to the BIM-joint, crayfish and lobsters have little freedom to adjust the movement plane other than by rotating it.

Therefore, the preferred contact surfaces of the flagellum in decapod crustaceans are the median and lateral surface (with respect to a resting posture with a horizontal movement plane). Even if a lobster antenna hits an opponent from above, i.e. by rapid extension of the MC- and CF-joints, it will hit him with the medial surface that is rotated by the BIM-joint to face downward.

2.2.2 *Biomechanics of the flagellum*

Antennal mechanoreception occurs in two ‘mechanical contexts’. In the ‘non-contact context’, the antenna is deflected or vibrated by flow or pressure oscillations of the surrounding medium. In the ‘contact context’, the antenna touches or is touched by a solid object, and is either bent by the applied force or oscillates upon release of the imposed deflection. Although both contexts have been studied, there is more detailed information available on non-contact deflections. All of the available information concerns the deflections of the flagellum and the corresponding torque or vibration transmitted to the PF-junction. This is because the flagellum has both the largest surface and the largest lever arm of all antennal segments. Moreover, it is the PF-junction that has been shown repeatedly to be the crucial site for mechanoreception in this context (Heran, 1959; Gewecke, 1970; Saager and Gewecke, 1989; Dreller and Kirchner, 1993a; see Section 6.2.3). In the ‘contact context’, under natural conditions, the site of object contact typically lies on the flagellum. Even in the honeybee, where the length of the scape relative to the total length of the antenna is much longer compared to the other model systems, the geniculate posture of the antenna makes contacts by the scape unlikely. Thus, it is important to understand the mechanical properties of the flagellum, and how they modify the mechanical stimulus received by the sensory structures.

2.2.2.1 *Static properties.* To our knowledge, there is no information about specific properties of the antennal cuticle. For instance, variations of cuticle thickness along a flagellum have not been measured. All of the studies available concern deflection properties of the flagellum.

Antennal deflection induced by air flow is of behavioural relevance to all flying insects. In response to steady frontal air flow, the flagellum of the locust *Locusta migratoria* behaves similar to a glass cylinder of equal proportions (i.e. 13.9 mm length, 0.25 mm diameter and 3.17 mm² exposed surface area). The torque increases with antennal angle, α , by less than $\sin(\alpha)$, but by more than $\sin^2(\alpha)$, irrespective of air speed (Gewecke and Heinzel, 1980). However, the dependency of torque, T , on air speed, V , does not follow Newton’s law of aerodynamic drag (the relation is $T \sim V^{1.44}$ rather than $T \sim V^2$), indicating that frictional forces acting on the flagellum are not negligible. Rather, friction gains importance with increasing air speed. Moreover, because the estimated drag coefficient of the flagellum is always larger than that of an ideal, infinitely long cylinder, Gewecke and Heinzel (1980) argue that the force acting on the

flagellar tip is significant. As the locust decreases the antennal angle with increasing air-speed (see Section 6.1.3.1), it actively decreases the torque acting on the PF-junction and, thus, maintains the angular deflection at the base of the flagellum, δ , within an almost linear range ($\delta \sim V^{1.2}$). In other words, the locust actively adjusts the antennal posture so that the restoring torque of the PF-junction approximately maintains the properties of a linear spring.

The stiffness of the locust flagellum is not uniform in all directions (Gewecke and Heinzel, 1980). Pulling the tip of the flagellum into different directions by the same force causes forward deflections that are three times larger than the corresponding rearward deflections. Thus, the locust flagellum is least compliant to forces applied in the direction of air flow during flight. This direction also matches the locust's preferred contact direction during active obstacle contacts: locusts often touch an object repeatedly with the medial surface of the flagellum (Saager and Gewecke, 1989).

The above observations also apply to the flagellum of the honeybee, where stiffness is smallest in the ventral direction and largest in the dorsal direction (Heran, 1959). During flight, the honeybee holds its flagellum slightly elevated above the horizontal plane, a posture that exposes the ventral surface to dorsal deflections by the air current. Much like the locust, the honeybee actively decreases the angle between flagellum and air current with increasing flight speed (Heran, 1959). In a contact context, non-uniform compliance is advantageous, because an external force opposing extension of the flagellum may cause the flagellum to bend, whereas the same force would be resisted during flexion. This is likely to be of functional significance in the honeybee's active tactile scanning behaviour. When presented with an object within the range of the antennae, honeybees scan the object with a movement sequence that is characterised by fast repetitive flexion of the SP-joint (Erber and Pribbenow, 2000). Owing to its non-uniform compliance, the flagellum remains stiff during object contacts, which is a necessary prerequisite to allow a sensory estimate of the contact point without taking the curvature of the flagellum into account. On the other hand, the lower compliance in the opposite direction reduces the chance of the antenna getting caught by surfaces or edges during extension.

The stiffness of the flagellum of crayfish also is non-uniform. The flagellum of *Cherax destructor* is tapered, consisting of 220–250 annuli that increase in length and decrease in diameter towards the tip (Sandeman, 1989). The flagellum is oval in cross-section and exerts the largest mechanical resistance against medial deflection and the lowest mechanical resistance against dorsal deflection. Medial mechanical resistance is $0.0014 \text{ N m rad}^{-1}$ at the base of the flagellum and decreases by almost 1.5 orders of magnitude towards the tip. When bent passively, the curvature is confined to a small portion of the flagellum. With increasing force applied, the apex of the bend moves towards the base (Sandeman, 1989). In *Procambarus spiculifer*, the mechanical impedance (stiffness $\times$ mass) of the tapering flagellum decreases logarithmically with length (Taylor, 1975). If the tip of the flagellum is pressed against a flat surface,

the lateral deflection of any point between the most proximal contact point and the base of the flagellum is well described by an exponential function.

An important mechanical aspect, which has been largely neglected by biologists, concerns the impact of the antennal shape on the bending characteristics of the flagellum. The importance of this is shown by an engineering study that investigates the difference of contact-induced bending as opposed to drag-induced bending of a tapered probe (Barnes *et al.*, 2001). Transferred to the biological situation of a crayfish antenna, sensing the curvature at several locations along the flagellum is likely to be sufficient for a crayfish to tell whether the bend is caused by flow of the surrounding medium, or rather by physical contact with an obstacle. In an environment with still water, sensing a drag-induced bend must be due to active motion of the antenna, thus allowing the animal to distinguish between active and passive bending.

A biological example of the importance of drag forces caused by the antennae is the speed-dependent adduction of antennae in queues of migrating spiny lobsters (Bill and Herrnkind, 1976). During long migration in water, reduction of drag forces can save a lot of energy. Indeed, a queue of five spiny lobsters walking at 0.35 m s^{-1} , reduce the summed drag of individuals by some 1.5 N. This is equivalent to more than 60% of drag reduction. Spiny lobsters also adapt their antennal posture to the walking speed of the queue. Compared to a mean abduction angle of 46° , the queue of five adducts their antennae to 20° , resulting in further drag reduction by 0.3 N. The speed-dependent adduction is reminiscent of antennal adduction of flying locusts and bees (see Section 6.1.3).

2.2.2.2 Dynamic properties. Three experimental paradigms have been applied to study the dynamic properties of the flagellum. The ‘contact context’ has been studied by the application of oscillations by means of a mechanical probe, or by vibration of the head with the flagellum being free to oscillate in air. The ‘non-contact context’ has been studied by exposing the flagellum to an air current of oscillating speed. Generally, studies have measured the passive properties of the flagellum of dead animals. Thus, it is important to note that the properties measured may be subject to active modification by the living animal, as has been shown to be the case in mosquitoes (Göpfert and Robert, 2001a).

Imposing oscillatory movement upon either the tip or the base of an excised flagellum of *Procambarus spiculifer* causes travelling waves that take a characteristic form, depending on the contact site (Taylor, 1975). Deceleration of the travelling speed from base to tip sometimes shows profound non-linearities, which Taylor (1975) attributes, at least in part, to the unequal decrease of mass and stiffness from base to tip. Imposing oscillations on a locust’s head causes the flagellum to resonate at frequencies near 74 Hz (female) or 104 Hz (male). Apart from the sex difference, the resonance frequency also depends on age (being lower in young animals) and the direction of the oscillation (being largest in the horizontal plane; Heinzel and Gewecke, 1987). When superimposing the passive

oscillations with an air-current directed towards the head of the animal, the peak resonance frequency changes only little, but the amplitude of flagellar oscillations are strongly dependent on the antennal angle with respect to the current (oscillations being larger at large angles). Release of a passive static deflection at the PF-junction by angles between 0.1° and 1° (the sensitive range of the campaniform sensilla on the distal pedicel) causes the flagellum to return to its resting position with a time constant of 10 ms (Heinzel and Gewecke, 1987). If the PF-junction is deflected by a contact point at the middle of the flagellum, the return is aperiodic. If, however, the PF-junction is deflected by flagellar contact at the tip, the return time course is superimposed by damped oscillations at frequencies slightly below the resonance frequency. Thus, free oscillations after obstacle contact contain information about the contact location, a fact that has also been suggested to be exploited in technical applications of tactile sensors (Ueno and Kaneko, 1995; Ueno *et al.*, 1998; see Section 7.2). In the honeybee, passive vibration of the flagellum via oscillation of the head produces peak amplitudes at frequencies between 240 and 300 Hz, which is close to the wing beat frequency of the bee (275 Hz; Heran, 1959).

When exposing the flagellum of the locust to a sinusoidally modulated air current (Heinzel and Gewecke, 1987), the shape of the frequency tuning curve of the flagellum depends on whether air-particle displacement or air-particle velocity is kept constant. In the first case, the curve is of similar shape as when the head is vibrated, but the resonance frequency is shifted to higher frequencies by some 50 Hz. If, on the other hand, air current velocity is kept constant, the lower frequency range is less attenuated in comparison with passive vibration of the head. A natural example, where a gregarious locust is likely to encounter an air current of oscillating speed is during flight in a swarm. Indeed, the wing beat of a locust ahead can deflect the flagellum at amplitudes that are sufficient to stimulate the campaniform sensilla on the pedicel (Heinzel and Gewecke, 1987).

In the honeybee, mechanical properties of the flagellum are suitable for passive transmission of airborne vibrations with high air-particle velocities (Kirchner, 1994). When exposed to vibration stimuli similar to those that occur near the abdomen of a waggle-dancing honeybee (Section 6.2.4.2), the flagellum follows the air particle movements like a stiff rod held by the pedicel. The amplitude of flagellar vibration is the same in all radial directions and increases linearly with air-particle displacement. The 10 Hz frequency component caused by the wagging movement of the dancing bee, as well as the 265 Hz frequency component caused by vibratory wing movements of the dancer, are transmitted with similar gain. The sound of the waggle dance is transmitted to the pedicel, where it is transduced by Johnston's organ.

As in the honeybee, the antenna of the fruit fly *Drosophila melanogaster* also acts to transmit airborne sound to Johnston's organ inside the pedicel. The specialised antenna of the brachyceran flies has a bulbous flagellum with a feathered lever arm, the arista. Behavioural experiments show that female fruit

flies perceive the air current and sound produced during the courtship display of a male (Petit, 1958). The arista is an obligatory component for sound perception during courtship, while an intact pedicel, including Johnston's organ, is not sufficient (Manning, 1967). Fixation of the PF-junction, without affecting the insertion of the arista, diminishes sound perception, showing that the whole of the flagellum acts in concert to transmit airborne sound in *Drosophila* (Manning, 1967). Recently, Göpfert and Robert (2001b) showed that the arista turns the flagellum around its long axis, thus rotating a chitinous hook inside the pedicel, much like a key's bit inside a lock. The exerted movement inside the pedicel can be sensed. Interestingly, loading of the arista does not affect courtship success (Manning, 1967), suggesting that the resonance frequency can be altered with no impact on female's ability to perceive male courtship signals, despite the fact that hearing, in a strict sense, should be affected.

2.3 EFFERENT INNERVATION OF THE ANTENNA

2.3.1 *Muscle innervation in insects*

An overview of the number of antennal muscles and their innervating neurons in *Gryllus*, *Locusta*, *Carausius*, *Manduca* and *Apis* is given in Table 2. The innervation of antennal muscles of the insect model systems, introduced in Section 2.1, reflects the basic innervation scheme commonly found in insect musculature. One skeletal muscle is typically innervated by two, sometimes more, excitatory motoneurons and one common inhibitory motoneuron (Hoyle, 1978; for review see Burrows, 1996). The insect antennal motor system differs from that of Crustacea (Section 2.3.2) in that insects lack common excitatory motoneurons. Also, insects have a higher total number of motoneurons innervating the antennal musculature. In addition to motoneurons, efferent modulatory dorsal unpaired median (DUM) neurons ramify on most insect skeletal muscles (for review, see Bräunig and Pflüger, 2001).

The extrinsic, or tentorio-scapal, muscles of the HS-joint are located in the head capsule and have their proximal insertion points on the tentorium in the head. Distally they are attached to the basal cuticle of the scape via apodemes. The intrinsic, or scapo-pedicellar, muscles of the SP-joint originate on the cuticle of the scape and they are attached to the pedicellar base via apodemes (e.g. Imms, 1939).

According to Gewecke (1972b), locusts have two extrinsic and two intrinsic antennal muscles. Imms (1939), on the contrary, depicts bipartite scapal levator and depressor muscles and an additional intrinsic muscle. In the locust antenna, the extrinsic muscles are innervated by the nerve *Nervus musculus tentorio scapalis* which originates separately from the main antennal nerve, *N. antennalis*, from the deutocerebrum (Gewecke, 1972b). The *N. musculus tentorio scapalis* splits into three branches which supply the two scapal muscles and a hair plate between the compound eye and the antennal base. Two nerves,

TABLE 2 Antennal muscles and muscle innervation of the insect model organisms

Species	Antennal joint	Movement	Muscles	Number of excitatory motoneurons		Number of inhibitory motoneurons	Total number of motoneurons	Number of DUM neurons
				Fast	Slow			
<i>Gryllus</i> ^{a,b}	HS	Levation	3	5		1 (except M5b)	18	2
		Depression	2	5				
	SP	Adduction	1	1	2			
		Adduction	1	2/3	2/1			
<i>Locusta</i> ^{b,c,d}	HS	Levation	1	9		1	19	2
		Depression	1					
	SP	Adduction	1	10				
		Adduction	1					
<i>Carausius</i> ^c	HS	Levation	2	4		1	13–16	
		Depression	1	4–5				
	SP	Adduction	1	3				
		Adduction	1	2–4				
<i>Manduca</i> ^{f,g}	HS		5	≥ 7			≥ 12	
	SP	Adduction	2	≥ 5				
		Adduction	2					
<i>Apis</i> ^{h,i,j}	HS		4	9		Probably none	15	
	SP	Flexion	1–2	1	2			
		Extension	1	3				

^aHonegger *et al.* (1990a).

^bBräunig *et al.* (1990).

^cBauer and Gewecke (1991).

^dGewecke (1972).

^eDürr *et al.* (2001).

^fFleming (1968).

^gKloppenburg *et al.* (1997).

^hKloppenburg (1995).

ⁱErber *et al.* (2000).

^jSchäfer and Bicker (1986).

branching off the *N. antennalis* at variable positions, supply the intrinsic antennal muscles with motor innervation: one branch, the *N. scapalis medianus*, innervates the median adductor muscle, whereas the *N. scapalis lateralis* innervates the lateral abductor muscle (Gewecke, 1972b).

The innervation of the antennal musculature is very similar in *Carausius* (Dürr *et al.*, 2001) and *Gryllus* (Honegger *et al.*, 1990a). For a schematic overview, see Fig. 3. The depressor muscles are innervated by the pure motor nerve N4 (cricket), which leaves the deutocerebrum via a root separate from the root of the main antennal nerve (N1). One of the cricket depressor muscles (M5a) receives additional innervation from the nerve N3, which branches off N1. The levator muscles are supplied by N3, which is a mixed sensory-motor nerve in crickets (Honegger *et al.*, 1990a). The intrinsic scapo-pedicellar muscles receive their motor innervation by branch N4B and by N2, which is also a side branch of N1. N2 and N4 form a conspicuous anastomosis in the scape with single motoneurons projecting across it to innervate their specific target muscle. The motor axons either leave the brain via N4 and project through the anastomosis to the adductor muscle M6 or they leave the brain via N1 and N2 to innervate the abductor muscle M7. The motor branch of N2 contains also sensory axons in crickets and stick insects (Honegger *et al.*, 1990a; Dürr *et al.*,

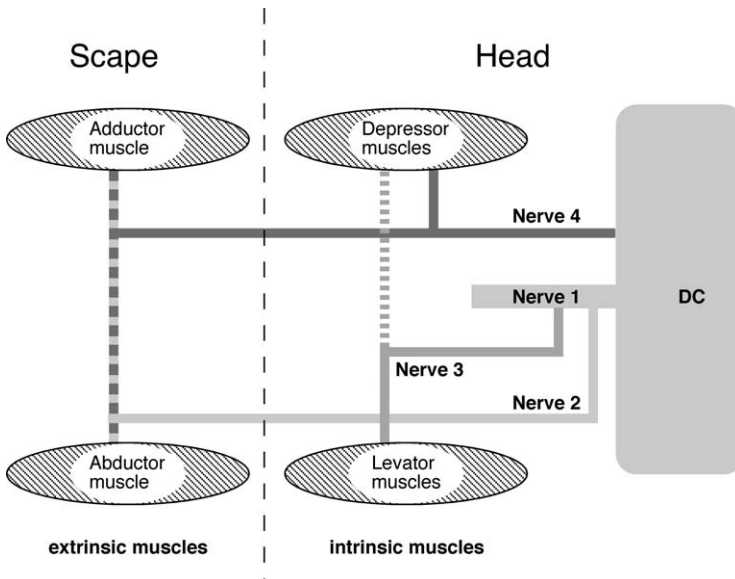


FIG. 3 Diagrammatic innervation pattern of antennal muscles in stick insects and crickets. The anastomosis in the nerve running between the abductor muscle and the adductor muscle (dotted line) contains the axons of motoneurons leaving the brain via N2 (light grey) and N4 (dark grey). The extension (broken line) of N3 to the depressor muscle (medium grey) is found in crickets only. DC, deutocerebrum.

2001). The branching pattern of these antennal nerves varies between different specimens in both species.

In the honeybee, the extrinsic muscles are innervated by a motor nerve which branches off the main antennal nerve N1 in the scape. The intrinsic muscles are supplied by nerves that branch off N1 in the scape (Snodgrass, 1956). Similarly, the intrinsic antennal muscles of *Manduca* are innervated by small side branches of the main antennal nerve within the scape. However, the extrinsic muscles of *Manduca* do not have a distinct motor nerve, but are innervated by side branches of N1 (Kloppenburger *et al.*, 1997).

In locusts, the four antennal muscles are innervated by 19 motoneurons (Bauer and Gewecke, 1991) plus two DUM neurones with their somata in the suboesophageal ganglion (SOG) (Bräunig, 1990, Bräunig *et al.*, 1990). Bauer (1991) performed immunohistochemical staining of antennal muscles and locust brains. She found proctolin-immunoreactive profiles on all antennal muscles, and up to 12 somata in regions of the deutocerebrum where antennal motoneurons reside. This suggests that proctolin is a modulator in the locust antennal motor system.

Extra- and intracellular recordings have revealed that the extrinsic muscles of the locust, which consist of approximately 65 fibres each, receive slow, intermediate, fast and inhibitory innervation. Recordings from the median depressor muscle of the scape predominantly reveal slow motoneurons, whereas fast and slow excitatory junction potentials (EJPs) can be recorded in fibres of the lateral levator muscle. Bauer (1991) and Bauer and Gewecke (1991) interpret these findings in the light of possible functional specialisations: the depressor muscle must overcome the headwind drag during flight by a sustained force production, whereas the lateral levator muscle is phasically active during most resistance reflexes and, therefore, requires fast fibres. The inhibitory innervation of both muscles appears to be also biased, because IJPs were never observed in levator fibres.

In crickets, the seven antennal (five extrinsic, two intrinsic) muscles are innervated by a total of 18 motoneurons. Backfills of single antennal nerves, immunohistochemical stainings and electrophysiological experiments show that the adductor muscle (M6) receives innervation by one fast, two slow and one inhibitory motoneuron. The abductor muscle (M7) is innervated, alternatively, either by two fast, two slow and one inhibitory motoneurons, or by three fast, one slow and one inhibitory motoneurons (Honegger *et al.*, 1990a,b). Again, this indicates an asymmetric innervation of two antagonistic muscles of one joint. The inhibitory motoneuron is GABAergic (GABA: γ -aminobutyric acid; Honegger *et al.*, 1990b), the excitatory motoneurons are glutamatergic (Honegger *et al.*, 1995). Fourteen of the 17 excitatory motoneurons can be labelled with an antiserum against the neuropeptide proctolin (Bartos *et al.*, 1994), which modulates the contraction properties of antennal muscles (Allgäuer and Honegger, 1993, see Section 2.4). The two intrinsic muscles have been stained for their myofibrillar ATPase under different pH

conditions. Both, the adductor and the abductor muscles are subdivided into two subunits each, containing fast fibres in the periphery of the muscle and intermediate as well as slow fibres in the centre (Honegger *et al.*, 1995). In addition, two suboesophageal DUM neurons project into all major antennal nerves and ramify on antennal muscles (Honegger *et al.*, 1990b).

In bees, the four extrinsic muscles that move the HS-joint, are controlled by nine excitatory motoneurons, the two intrinsic muscles in the scape are controlled by six excitatory motoneurons (Kloppenburger, 1995). An inhibitory neuron seems to be missing in the bee antennal motor system, based on immunocytochemical evidence (Schäfer and Bicker, 1986a). Work on the antennal muscles in bees has focussed on the two intrinsic muscles, since they control fast extension/flexion movements of the flagellum (reviewed by Erber and Pribbenow, 2000; see also Section 6.2.3). The extensor muscle contains 23 individual muscle fibres, which are innervated by three motoneurons (Erber *et al.*, 2000). The flexor muscle consists of two clearly separated fibre groups, a dorsal group of 23 fibres and a ventral group of 20 fibres. A combination of backfills of antennal nerves and extra- and intracellular recordings have identified the ventral fibre group as a fast fascicle with probably one fast excitatory motoneuron. The ventral fibres are also of larger diameter than the dorsal fibres. A single motoneuron controls these fast flexor fibres (Erber *et al.*, 2000).

The extrinsic muscles of the antenna are innervated by a minimum of five to six motoneurons in bees (Ehmer and Gronenberg, 1997a). Additionally, two ventral unpaired median (VUM) neurons can be labelled by retrograde muscle stainings, which is similar to the situation in locusts and crickets (Bräunig *et al.*, 1990). These motoneurons apparently project via a separate nerve to the tentorio-scapal muscles.

2.3.2 *Muscle innervation in crustacea*

The musculature of the crayfish antenna is innervated by the main antennal nerve and a number of smaller side nerves, all of which leave the tritocerebrum (*Procambarus*: Habig and Taylor, 1982a; *Cherax*: Sandeman and Wilkens, 1983).

Extracellular recordings from antennal motor nerves in the rock lobster *P. vulgaris* reveal four distinct units innervating the CF-extensor and three distinct units innervating the CF-flexor (Clarac and Vedel, 1975). Intracellular muscle fibre recordings reveal both excitatory and IJPs. Each one of the two extensor and flexor muscles is innervated by a 'dedicated' tonic motoneuron that terminates at only one muscle, and by the common inhibitor neuron, that innervates all muscles (Vedel, 1980). The CF-extensor receives additional innervation from a phasic excitatory motoneuron. Furthermore, the two extensor muscles share a common extensor motoneuron and the two flexors share a common flexor motoneuron. These common motoneurons reflect the synergistic action of the parallel MC- and CF-joints. The accessory extensor muscle is

innervated by a single motoneuron. From nerve recordings, Neil *et al.* (1982) conclude that at least 11 motoneurons innervate the musculature of the IM-joint in *Palinurus*.

In the lobster *Homarus americanus*, each muscle of the distal antennal segments (meropodite and carpopodite) is innervated by a dedicated motoneuron (Sigvardt, 1977). The extensors are innervated by two common extensor motoneurons, while the flexors share only one common motoneuron.

In the crayfish *E. armatus*, muscle innervation is almost the same as in the lobster *Homarus* (Sandeman and Wilkens, 1983). Other than in *Homarus*, where Sigvardt (1977) found no evidence for a common inhibitor neuron, IJPs can be observed in flexor muscles of *Euastacus* (Sandeman and Wilkens, 1983). Also, attenuation of EJPs in extensor muscles indicates the presence of inhibitory innervation. In spite of this difference, resistance reflexes in *Euastacus* and *Homarus* are very similar.

2.4 PHYSIOLOGY OF ANTENNAL MOTONEURONS AND MUSCLES IN INSECTS

As will be discussed in Section 6, the antennae of insects perform highly variable and adaptable movements in a variety of behavioural contexts. Several studies demonstrate ways in which the performance of the antennal musculature can be modified to obtain more flexibility in the power output and time course of muscular contractions. The functional role of antennal motoneurons and their relationship to contraction properties of their target muscles will be described first for 'regular' excitatory motoneurons (Section 2.4.1), followed by inhibitory innervation and other sources of efferent modulation (Section 2.4.2).

2.4.1 *Excitatory innervation*

Kloppenborg *et al.* (1999) have for the first time investigated the properties of ion channels of antennal motoneurons. Using *in vitro* and *in vivo* patch-clamp methods in a honeybee preparation, they identified four different voltage-activated currents including two K^+ -currents and two inward currents (Na^+ , Ca^{2+}). These currents resemble those found in other insect neurons. The ratio between the currents, however, differed between different motoneurons. These differences provide the basis for future work that relates the different physiological properties of different motoneurons to their biophysical membrane properties.

Studies on the physiology of antennal muscles are available for ants (Ehmer and Gronenberg, 1997b), locusts (Saager and Gewecke, 1989; Bauer, 1991; Bauer and Gewecke, 1991) and crickets (Honegger *et al.*, 1990a; Allgäuer and Honegger, 1993; Bartos *et al.*, 1994; Bartos and Honegger, 1997; Gebhardt, 2004).

The work of Ehmer and Gronenberg (1997b) focuses on functionally relevant muscle properties, i.e. muscle fibre diameter, sarcomere length and

myofibrillar ATPase activity, to help explain the production of fast antennal movements. Their major finding is that the lengths of the sarcomeres negatively correlate with maximum speeds of an antennal retraction reflex in eight ant species belonging to three subfamilies. Highest speeds are observed in trap-jaw ants of the genera *Odontomachus* and *Anochetus*, which produce angular velocities of up to 12 200 and 13 200° s⁻¹, respectively. Ants of these species have sarcomeres as short as 1.1 and 1.3 µm, respectively. The shortest sarcomeres are localised in the thickest and longest fibres in the periphery of the muscles. This fibre design with many, serially arranged, short sarcomeres should facilitate the generation of high contraction velocities. The peripheral fibres stain strongly against a pH-labile isoform of mATPase, suggesting that a fast mATPase is located in the peripheral fibres. Moreover, the antennal retractor muscle, which withdraws the antenna during the retraction movement, occupies almost 70% of the total volume of the antennal musculature in *Odontomachus*. This muscle is, therefore, pre-disposed for a high-power output. All these muscle properties show considerable variation between different muscles within a species and between species. Taking their data together, Ehmer and Gronenberg (1997b) conclude that the musculature of ant antennae is tuned to produce a large range of contraction velocities including the very fast movements produced by the large retractor muscle in trap-jaw ants.

Saager and Gewecke (1989) report on different antennal reflexes of locusts, including tactile avoidance reflexes and resistance reflexes of the antennae during air-current-induced flight (see Section 6.1.2.1). They find that extracellular myogram recordings of the lateral and median scapo-pedicellar muscles reveal the activity of two to three and, occasionally, three to four units. The muscle potentials are different in size and shape, and their occurrence correlates with antennal movements. For example, the largest amplitude units occurring in the lateral muscle cause large single twitch contractions without complete tetanisation. Bauer and Gewecke (1991) report fast and slow EJPs, together with IJPs in scape muscle fibres. The latter have resting potentials around -40 mV. Fast and inhibitory innervations are different between both intrinsic muscles. In the lateral scape muscle, fast EJPs occur frequently, but IJPs are lacking, whereas in the median intrinsic muscle, slow, intermediate and, rarely, fast EJPs occur together with IJPs.

Incremental open-loop electrical stimulation of the antennal nerve, together with registrations of muscular tension, reveal up to five steps of motoneuron recruitment in the force output. This suggests the presence of at least five motoneurons innervating a single muscle. In the median intrinsic muscle (adductor), evoked muscle twitches are facilitated above 10 Hz stimulation frequency; summation occurs beyond 40 Hz; and tetanic contractions are found in response to stimulation frequencies above 100 Hz. Tetanic contractions are followed by post-contractions of up to 5 s duration. The resulting forces are 10⁻⁶ N in magnitude. In the lateral intrinsic muscle (abductor),

facilitation, summation and tetanus occur at 4, 20 and 50 Hz, respectively. Forces measured from the lateral muscle are 10^{-5} – 10^{-4} N in magnitude.

The different properties of the lateral and medial intrinsic muscles may match behavioural requirements in activation speed, in that the median muscle, with its slow innervation, maintains the prolonged antennal flight posture, whereas the fast lateral muscle is involved in fast avoidance reflexes.

In locusts, intracellular recordings from scapal muscle fibres, combined with bath applications of proctolin, show that proctolin alone does not alter the resting membrane potential of the muscle fibres, nor does it change the amplitudes of spontaneous EJPs (Bauer, 1990). In contrast, proctolin modulates contractions evoked by excitatory motoneuron activity: both the amplitude and the duration of single contractions are enhanced by bath-applied proctolin. Additionally, proctolin induces a slow tension component underlying single contractions elicited by trains of electrical stimuli. Bauer (1991) also finds that electrically quiescent extrinsic muscles exposed to proctolin for hours respond with an increase of their basal tension in a concentration-dependent manner. Bauer (1991) concludes that such an increase in basal tension is useful whenever muscular long-term contractions are required, particularly during the fixed antennal flight posture.

In crickets, the presence of fast and slow motor units was demonstrated by Honegger *et al.* (1990a). Extracellular stimulation of an adductor motoneuron (Ad1/2) leads to fast, twitch-like contractions of the antennal adductor muscle, whereas stimulation of another Ad1/2 motoneuron at stimulus frequencies above 40 Hz leads to slow adduction. Similarly, intracellular recordings from an abductor motoneuron of the group Ab2–4 reveal close correlation of single motoneuron action potentials with short antennal twitches, yet bursts of action potentials in a different Ab2–4 motoneuron are followed by slow antennal movements. Stimulation of the slow adductor motoneuron Ad3 causes slow force production above a firing frequency threshold of 30 Hz. In comparison, maximum tetanic forces of 3×10^{-5} N occur above 120 Hz stimulation frequency (Allgäuer and Honegger, 1993). Stimulation of fast motoneurons (e.g. Ad1) yielded single twitches with forces of $2\text{--}5 \times 10^{-6}$ N, with maximum tetanic force at 11.5×10^{-5} N. Summation occurs in both muscles at stimulus frequencies above 40 Hz.

Although there is some comparative detail available on the contraction forces produced by antennal muscles, the lack of knowledge about lever arms makes it difficult to relate forces to the kinematics of antennal movements, e.g. to predict the resulting torques. Hence, more information about the biomechanics of antennal joints would be desirable (see Section 2.2).

2.4.2 *Inhibitory and modulatory innervation*

Various modulators influence the action of antennal muscles (Allgäuer and Honegger, 1993). In extracellular recordings from a cricket antennal

neuromuscular preparation, the common inhibitor neuron of the antennal muscles is spontaneously active at an average frequency of 13.2 Hz while the antenna is at rest. During active antennal movements, the CI increases its firing rate before the onset of the activity of excitatory motoneurons. The average firing rate during antennal movements is 182 Hz with a maximum of 353 Hz. Evoked common inhibitor activity modulates the properties of slow contractions only: the peak force, contraction rate and resting tension decrease, while relaxation rate increases. For instance, CI activity starts lowering Ad3-evoked peak force of adductor muscle M6 at firing rates above 20 Hz. The higher the Ad3 activity, the higher the CI activity has to be to achieve a maximal force reduction (Allgäuer and Honegger, 1993). The CI effects can be blocked reversibly by application of picrotoxin, an antagonist of GABA. This supports the view that the CI is GABAergic (Honegger *et al.*, 1990b).

In contrast, DUM neuron activity has an effect on both slow and fast contractions. DUM neurones innervating antennal muscles are either silent or have low rates of spontaneous resting activity. DUM neuron activity does not generally precede bouts of antennal movements (Allgäuer and Honegger, 1993). In general, the effect of DUM neuron activity on force generation is weak and can only be elicited in 10% of the crickets tested. During DUM neuron stimulation at 10 Hz, a slow tetanic contraction, evoked by Ad3, is attenuated by an average of 25% of the peak force, and contraction speed decreases. However, forces of fast contractions are increased by an average of 11.9%, accompanied by an increase in contraction rate and a decrease in relaxation rate. In conclusion, DUM neurons have contrary effects on slow and fast contractions, and could, therefore, act to support fast force production by damping the effects of slow motoneuron activity. The transmitter of antennal DUM neurons is likely to be octopamine (Bräunig, 1991; Spörhase-Eichmann *et al.*, 1992), and application of octopamine at concentrations of 10^{-7} – 10^{-6} M onto the antennal musculature parallels the effect of DUM neuron activity (Bartos *et al.*, 1994).

In crickets, treatment of antennal muscles with proctolin results in the increase of forces of slow and fast contractions, accompanied by a decrease of relaxation rate (Bartos *et al.*, 1994). Application of 10^{-8} M proctolin results in a force gain of up to 330%. An increase of the basal muscle tonus after proctolin exposure, as observed in locusts (Bauer, 1991), is absent in crickets. The behavioural significance of proctolin is further substantiated by Bartos and Honegger (1997). Using densitometric immunocytochemistry against proctolin, they show that proctolin stores in two antennal motoneurons, the slow adductor motoneuron Ad3 and the depressor motoneuron D5, are reversibly depleted during prolonged episodes of flight. During flight, crickets adopt a posture that includes an active and sustained forward position of both antennae, similar to what is known in locusts (Gewecke, 1972a). This suggests that proctolin is released onto antennal muscles during prolonged muscle contractions. Since motoneuron firing rates decrease in the course of a flight episode

and this decrease is accompanied by only small changes in antennal position, proctolin might play a decisive role in the maintenance of the antennal flight posture. Gebhardt (2004) tests this hypothesis by injections of anti-proctolin antiserum into the scape of flying crickets. The injections are followed by a reversible loss of the antennal flight posture, suggesting that the antiserum inactivated proctolin released onto the muscles. In conclusion, proctolin is indeed a component of the antennal muscular system that is necessary for the adoption of a sustained antennal posture as it occurs during flight.

Though more studies are required to understand the basis of the differential intrinsic properties of motoneurons, it is apparent that the antennal musculature and its modulatory innervation can account for substantial plasticity of antennal movements. A variation in muscular structure alone contributes to the generation of variable movements. Modulatory substances, the release of which is controlled by efferent neurons, increase the dynamic range of muscle contractions beyond the limits set by the fast, slow and inhibitory innervation. The mechanisms used are efficient since a single modulator (e.g. octopamine) has differential effects across a muscle fibre population.

3 Sensory physiology of antennal mechanoreceptors

3.1 SENSORY STRUCTURES AND TRANSDUCTION

In the following section, the term sensillum is understood as the whole assembly of cuticular structures, which may or may not be external, sensory cell(s), and other structures required to obtain a functional unit, which responds to mechanical and/or other stimuli.

The insect antenna is a compound sensory organ carrying a large variety of receptors, including olfactory (Keil, 1982, 1989; Lee and Strausfeld, 1990; Steinbrecht, 1997; Shields and Hildebrand, 1999a,b), gustatory or contact-chemosensory (Rüth, 1976; Schaller, 1978; Tichy and Loftus, 1983; Chapman, 2003), hygro- (Loftus, 1976; Tichy, 1979; Tichy and Loftus, 1990, 1996), thermosensory (Loftus, 1966, 1968, 1969; Altner, 1977b; Altner *et al.*, 1977, 1981, 1983; Altner and Prillinger, 1980; Tominaga and Yokohari, 1981; Yokohari, 1981; Tichy and Loftus, 1987) and mechanosensory (Markl, 1962; Gewecke, 1972b; Schmidt, 1973; Altner and Prillinger, 1980; McIver, 1985; Zacharuk, 1985). The axons from all these receptor cells project directly to the deutocerebrum of the brain (cf. Zacharuk, 1985; Rospars, 1988). There are four basic types of mechanosensory structures: (1) hairs, (2) campaniform sensilla, (3) chordotonal organs and (4) strand organs or stretch receptors. All of these are found on and in various segments of insect antennae, and, can, thus, in principle function as proprio- and exteroceptors. They can be found as single receptors or arranged in groups, e.g. to form hair plates (Markl, 1962; Gewecke, 1972b; Ehmer and Gronenberg, 1997b), Johnston's organ (vande Berg, 1971) or

rings around a segment (Knyazeva, 1974; Toh, 1981), and their distribution on the antennae varies with sensillum type and insect species.

3.1.1 *Mechanosensory hairs*

A wide variety of mechanosensory hairs can be found on the insect antenna (Schneider, 1964; Altner and Prillinger, 1980; Zacharuk, 1985) and their morphology allows a crude classification. But only detailed studies of their fine structures led to descriptions that are adequate to infer their function (Altner, 1977a). Names like sensilla chaetica or sensilla trichoidea are used in different ways by different authors. Therefore, they do not allow a proper classification of the various hair types (cf. Table 1 of Altner and Prillinger, 1980). The two hair types with mechanosensory function are hairs without pores, but with flexible sockets (Section 3.1.1.1), and hairs with a terminal pore and a flexible socket (Section 3.1.1.2; Altner and Prillinger, 1980).

3.1.1.1 *Tactile hairs.* Hairs without pores, but with flexible sockets (Fig. 4) are often called sensilla chaetica (Toh, 1977, 1981), or sometimes sensilla trichoidea (Esslen and Kaissling, 1976). Their shank is of variable form, length and wall thickness, and sometimes they have longitudinal grooves reaching up to the tip. Their sockets are often asymmetric, allowing movement of the hair in only one direction, which is part of the mechanism underlying the directional responses of the sensory neurons. Usually, these hairs are innervated by one bipolar sensory cell. The inner dendritic segment of this neuron contains many mitochondria, as is the case in other sensory cells, a ciliary body and a rather short ciliary root. The outer dendritic segment contains arrays of parallel microtubules, which are connected by an amorphous substance at the distal end. Such a distal end is called a 'tubular body' (Thurm, 1964). In some cases, the tubular body is flattened, which provides another basis for directional sensitivity (Thurm *et al.*, 1975). The proximal part of the outer segment lies within an extracellular space filled with receptor lymph. The outer dendritic segment, including the tubular body, is enclosed by cuticle, which is called the dendritic sheath. The distal part of the outer segment is attached to the base of the hair and surrounded by a cap, which probably consists of resilin (Thurm, 1964).

The attachment site is on the side where the movement of the hair is most restricted. When the hair is moved in the preferred direction, the inner structures of the joint distort the tubular body of the sensory cell. This mechanical change in the outer dendritic segment is the basis for the mechano-electric transduction in insect sensory hairs (Thurm, 1964, 1965, 1983; Gaffal *et al.*, 1975; French and Sanders, 1979, 1981). It is still debated, however, how significant the microtubules and the tubular body are for the transduction (Moran and Varela, 1971; Moran *et al.*, 1977; Schafer and Reagan, 1981; Kuster *et al.*, 1983; Zacharuk, 1985). We will not deal with the molecular basis

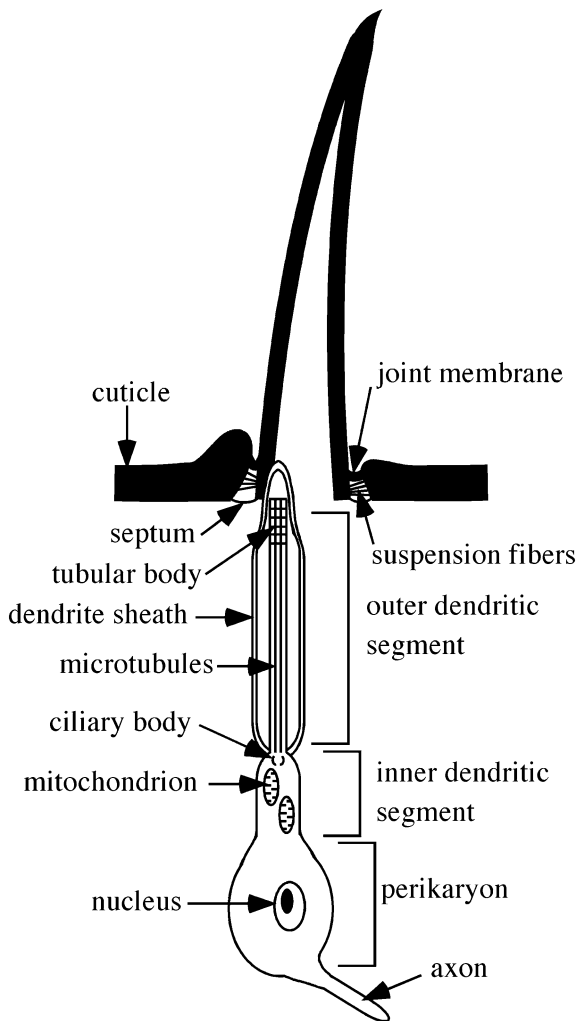


FIG. 4 Schematic view of the different elements comprising a mechanosensory hair, i.e. a mobile hair without a pore. In this and the following schemes, accessory cells are omitted for clarity.

of the mechanosensory transduction, but refer the reader to a recent review (Gillespie and Walker, 2001).

3.1.1.2 Contact-chemosensory hairs. Hairs with flexible sockets and terminal pores vary with regard to their length, and external structure (Fig. 5). They have been described under various names, such as sensilla chaetica, trichoidea, basiconica, styloconica (cf. Altner and Prillinger, 1980). Functionally, this hair type is considered to be mechanosensory and gustatory, i.e. contact-chemosensory. A

variable number of sensory cells, usually three to five, which send their dendrites into the hair shank, are considered to be chemoreceptive. As in the other hair type (see Section 3.1.1.1), only a single cell attaches to the base of the hair, and this neuron is considered to be mechanosensory. This cell is attached in the same manner to the hair base as in hairs without pores. Therefore, the mechanism for mechanosensory transduction is considered to be very similar to tactile hairs, if not the same.

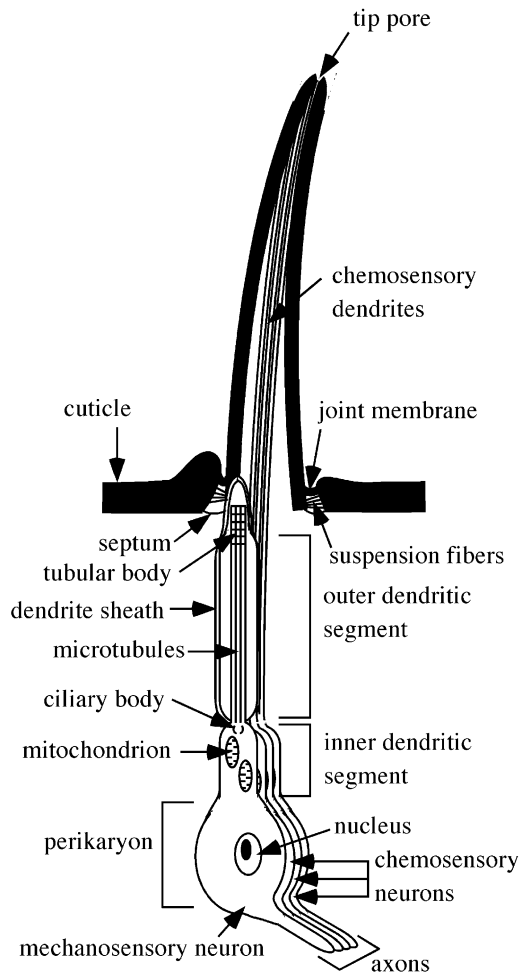


FIG. 5 Schematic view of a contact-chemosensory or gustatory hair, with an emphasis on the mechanosensory neuron. Note, that this hair type has a single pore at the tip.

3.1.2 *Campaniform sensilla*

Campaniform sensilla (Fig. 6; Hicks, 1857) are externally recognisable sensory structures, which, like mechano- and chemosensory hairs, chordotonal organs and strand receptors, have been found and described in many locations on the bodies of many insect species (e.g. fly haltere: Pringle, 1948; Smith, 1969; cicada: Klein *et al.*, 1988; Coleoptera: Kim and Yamasaki, 1996; moth: Schneider and Kaissling, 1957; cockroach maxillary palp: Pringle, 1938a,b). Some authors refer to single campaniform sensilla as Hick's organ (Snodgrass, 1935), while others call a prominent group of campaniform sensilla on the distal pedicel Hick's organ (Gewecke, 1972b). To avoid confusion, we will not use this term but simply refer to these structures as campaniform sensilla. Typically, an elliptic or round, dome-like structure, which sometimes has a small central pit, lies in the middle of a slight cuticular indentation. The elliptic shape of the cuticular structures provides the basis for the directional sensitivity (Heinzel and Gewecke, 1979; see Section 3.3) with the axis of greatest sensitivity aligned along the short axis of its ellipsoid shape (Thurm *et al.*, 1975). Round campaniform sensilla are considered omni-directional. This sensory structure is usually innervated by a single bipolar sensory cell with a tubular body, which attaches to the central part of the cuticular dome. The outer segment is located within a receptor lymph space, and the tip, which

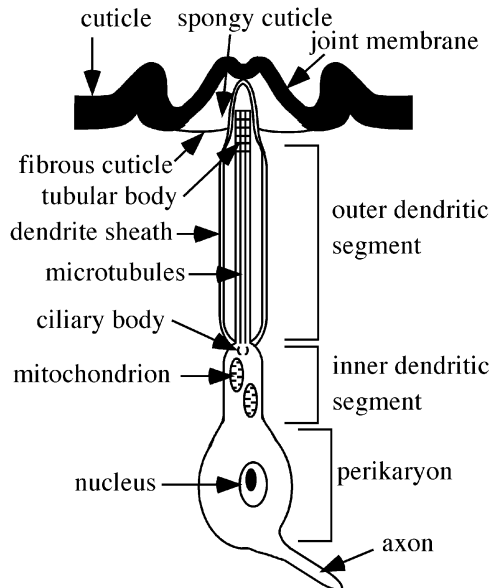


FIG. 6 Schematised section through a campaniform sensillum.

contains the tubular body, is enclosed by a cap, which is reminiscent of the situation in mechanosensory hairs.

The so-called marginal sensilla have been described for the antennae of *Periplaneta americana* L. by Schafer and Sanchez (1973) and Toh (1977) and of *Blattella* (Campbell, 1972). According to Toh (1977), it seems as if these sensory structures, which are innervated by a single bipolar neuron with a tubular body, are just a morphological variation of campaniform sensilla.

Pringle (1938a) was the first to report the responses of campaniform sensilla to pressure on the cuticle and to bending of a joint. Consequently, he called them "stress receptors". As in sensory hairs, the shape of the outer dendritic element is considered to be important for the directional sensitivity of the sensillum (Thurm *et al.*, 1975). Given the many structural similarities of campaniform and hair sensilla, it is not astonishing that their transduction mechanism was proposed to be the same (Thurm, 1964; Thurm *et al.*, 1975, but see Chapman *et al.*, 1973).

3.1.3 *Chordotonal organs*

Chordotonal organs consist of arrangements of a variable number of scolopidia, a functional unit comprised of sensory and accessory cells (Fig. 7; for review see Field and Matheson, 1998). Sometimes, they are subdivided into distinct groups of scolopidia, which then are referred to collectively as scoloparia. Chordotonal mechanosensory neurons are viewed as completely internalised and modified homologues of the receptors associated with cuticular structures, i.e. hairs and campaniform sensilla (Schmidt, 1973). The receptor cells show some similarity with the mechanosensory cells described above, in that their bipolar neurons also have an inner and an outer dendritic element. The inner segment, however, is usually much longer than in the other sensory cells, and sometimes has a bulbous widening. Furthermore, the ciliary root extends from near the tip throughout the entire inner segment into the soma. The outer segment contains microtubules, like in other mechanosensory neurons, and often ends with a small dilatation (McFarlane, 1953; Howse, 1965; Howse and Claridge, 1970; vande Berg, 1971; Schmidt, 1973). In holometabolous insects, the distal end of the microtubular structure is dense and, thus, may be considered a tubular body (Schmidt, 1973). One or more sensory cells are usually associated with two accessory cells, the scolopale and the attachment cell, and only in a few cases a separate cap cell is described (Schmidt, 1973). Unlike the other mechanosensory neurons, chordotonal sensory cells are not directly attached to the cuticle, but indirectly through the attachment cell. Another specialisation of chordotonal organs is the scolopale cell, which surrounds a good part of the inner dendritic segment and the outer dendritic segment, nearly up to the distal end of the neuron. The invaginations of this cell form an extracellular space around the outer dendritic segment. The scolopale cell also contains the scolopale rods, tubular structures filled with

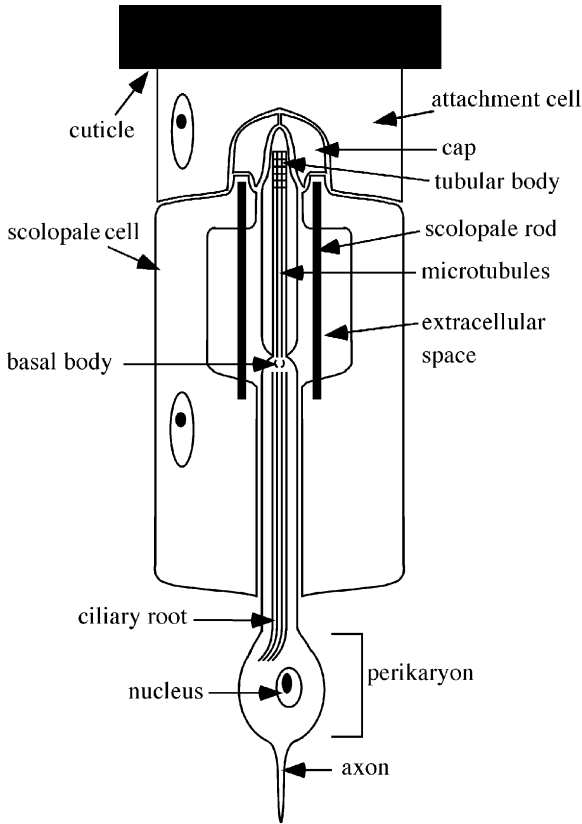


FIG. 7 Scheme of a scolopidial sensillum, as found in ordinary chordotonal organs and Johnston's organ.

electron-dense material, which surround the dendrite of the neuron from the distal end of the inner segment to shortly proximal of the distal tip. The distal end of the sensory cell reaches into an electron-dense cap, which is surrounded by the attachment cell. The latter also encloses the distal end of the scolopale cell.

Together, the scolopale rods, the cap and the ciliary roots modify the transduction mechanism of chordotonal sensory cells. The details of the precise role of the different structural elements are still a matter of debate (cf. Field and Matheson, 1998). Chordotonal sensory neurons are stretch- or relaxation sensitive. It has been suggested that transduction occurs due to either a resulting pinching (Thurm, 1965), just by stretching (Young, 1970) or by bending (Moran *et al.*, 1977) of the distal end of the sensory cell. It is assumed that mechanically activated channels play a major role in these and the other insect mechanosensory cells. Although they have been described for vertebrate sensory cells some time ago (Guharay and Sachs, 1984; Kros *et al.*, 1992),

mechanically activated channels have only recently been reported in insects (Torkkeli and French, 1999a,b). Cultured antennal mechanosensory cells from *Manduca sexta*, which presumably stem from Johnston's organ, respond to positive and negative pressure perpendicular to the long axis of the dendrite (Torkkeli and French, 1999a,b). This is in accordance with the hypothesis that compression or expansion of their tips leads to transduction.

A structural comparison of the mechanosensory and gustatory hairs with campaniform sensilla and chordotonal organs indicates that they may be homologues. The major similarities are of structural and developmental, rather than functional nature. Each of these sensilla has one or more sensory cells, all of which have ciliary structures. Furthermore, besides the sensory neuron, there are usually three accessory cells per sensillum. The cuticular structures ensheathing the outer dendritic segment of the sensory cells are secreted from the same cell type. During moulting, the structures of these sensilla undergo very similar sequences of events. These arguments, made by Schmidt (1973), lead to the general acceptance of the homology of these sensory structures. However, recently reported results from developmental and molecular genetics indicate that homologising these sensory structures may not be as simple. Some genes, that are important for the development of the peripheral nervous system determine which type of mechanosensory sensillum or olfactory organ will develop (Ato olfactory: Gupta and Rodrigues, 1997; Jhaveri *et al.*, 2000; Ato mechanosensory: Bodmer *et al.*, 1987; Brewster *et al.*, 2001; Amos olfactory: Goulding *et al.*, 2000; Jhaveri *et al.* 2000; Amos mechanosensory: Brewster and Bodmer, 1995; Brewster *et al.*, 2001). However, there are two major differences between mechanosensory and olfactory organ development. First, olfactory sensilla develop from one to three secondary precursor cells, which are only associated with the olfactory progenitor cell (Sen *et al.*, 2003). Second, the genes, which lead to the development of the different cell types are different from the genes activated during mechanosensory organ development (olfactory: Pros, Elev, Svp, cf. Sen *et al.*, 2003). Thus, the different mechanosensory sensilla, including gustatory hairs, are more closely related to one another than to olfactory sensilla.

In contrast to olfactory sensilla, all the mechanosensory organs develop directly from the sensory organ precursor (e.g. Brewster *et al.*, 2001). The three key genes are Cut, Ato and Amos, and their activation leads to different types of mechanosensory structures. If Cut is expressed in the sensory organ precursor, external sensory organs, i.e. campaniform sensilla, mechanosensory or chemosensory hairs, or stretch receptors develop (Brewster *et al.*, 2001). The activation of BarH1/BarH2 leads to campaniform sensilla, while hairs develop in their absence (Higashijima *et al.*, 1992). A chemosensory hair is induced if Poxn is active, while a mechanosensory hair develops if it is absent (Dambly-Chaudiere *et al.*, 1992; Nottebohm *et al.*, 1992; Ghysen and Dambly-Chaudiere, 1993). This is in contrast to chordotonal organs and some stretch receptors, which are Ato dependent, and only develop in the absence of Cut (Blochliger *et al.*, 1991;

Bodmer *et al.*, 1997; Brewster *et al.*, 2001). Another group of stretch receptors is Amos dependent (Brewster and Bodmer, 1995; Brewster *et al.*, 2001). From all this follows, that (1) the different mechanoreceptors are closer related to each other than to the various olfactory sensors, and that (2) campaniform sensilla, mechanoreceptive and gustatory hairs are closer related with each other than with chordotonal organs. The only indication of homology remaining is that in external and chordotonal organs, the neuron and one supporting cell, thecogen or scolopale cell, arise from the same secondary precursor lineage (Merritt, 1997; Younossi-Hartenstein and Hartenstein, 1997).

3.1.4 *Strand receptors*

Strand or stretch receptors are typically found in the joints of insect appendages (e.g. locust wing hinge; leg). In contrast to the other sensors (cf. Sections 3.1.1–3.1.3), they consist of multipolar neurons, some of which have central somata (leg: Bräunig and Hustert, 1980). Their dendrites are usually embedded in a sheath of glial cells and connective tissue, but they are not associated with such obvious structures as scolopale cells. Unlike other mechanoreceptors, their dendrites do not contain specialised structures, like tubular bodies or long ciliary roots that are visible in the electron microscope. Stretching of the free dendritic endings is supposed to induce transduction (Kuster *et al.*, 1983).

So far, an antennal strand receptor has been described only in the locust (Bräunig, 1985; see also Section 3.3). The locust strand organ spans from the tentorium to the base of the scape and runs parallel to the levator muscle of the scape (Bräunig, 1985). The length of the strand decreases with antennal elevation. In contrast to any other known antennal mechanoreceptor, the somata of the four to five receptor neurons are not located in the periphery. Rather, they are located in the medial protocerebrum, anterior to the central complex. These neurons arborise in the anterior dorsal deutocerebrum, i.e. close to, if not in, the dorsal lobe (DL). The finding of four somata in a similar location in the protocerebrum of a cricket indicates that a strand receptor may be present in the head of this species, too (Staudacher and Schildberger, 1999).

3.1.5 *Mechanoreceptors of the crustacean antenna*

Lobster and crayfish antennae bear many sensory structures, some of which may seem to be very similar to sensilla on insect antennae. Among these receptors are short smooth hairs, long smooth upright hairs, and short round peg hairs (*Cherax destructor*; Sandeman, 1989). Spine-like structures, smooth hairs, campaniform sensilla, and hydrodynamic receptors occur on the antennae of the rock lobster, *Palinurus vulgaris* (Vedel, 1985).

However, there also are specialised hydrodynamic receptors and feathered hairs in crayfish (Tautz *et al.*, 1981; Vedel, 1985), which are not innervated, but

aid in support movement detection by transmitting forces to nearby smooth hairs (Bender *et al.*, 1984). Internal mechanoreceptors, i.e. chordotonal organs are found in all appendages of crustacea (Mill, 1976), including the antennae (e.g. Rossi-Durand and Vedel, 1981). Even though many of these receptors seem superficially similar to insect structures, they may not be if details are compared. For example, the mechanoreceptive hairs of insects and crustacea differ in the ultrastructure of the outer dendritic segment (Crouau, 1997). For chordotonal organs, details of fine structural differences to insect chordotonal organs have been reported (Howse, 1968). Because of the reported differences in ultrastructure, it is conceivable that the transduction mechanisms may also be different. It is, however, striking, that these receptor types appear to have similar functions, despite structural differences. Another example of the difficulty of ascribing the same function to superficially similar structures, are the crustacean campaniform sensilla. Here, the ultrastructure of what had been proposed to be campaniform sensilla led to a reclassification as bimodal sensilla and dermal glands (Gnatzy, 1984; Gnatzy *et al.*, 1984; Schmidt and Gnatzy, 1984). The structures most likely resembling insect campaniform sensilla contain two scolopidial sensory cells, instead of one receptor cell with a tubular body (Schmidt, 1990). In both, crustacea and insecta the number of annuli of their flagella varies greatly, and is species dependent. Crayfish antennae have about 100 annuli, while rock lobsters have up to 400 (Tautz *et al.*, 1981; Vedel, 1985). The bee has only 10, while a cockroach may have about 140 annuli (cf. Table 1). The situation is similar, if the numbers of mechanosensory structures on the flagella are compared. Rocklobsters have about 6000 such structures, while crayfish may only have about 2000 (Tautz *et al.*, 1981; Vedel, 1985). About 1200 external mechanosensory structures can be counted on the flagellum of the bee, while there may be more than 7000 in a cockroach (cf. Table 8). Thus, a comparison of the mere numbers of external structures does not indicate a clear-cut trend in the difference in the numbers of sensilla on crustacean and insect antennae. A comparison of the number of sensilla per annulus leads to a different picture. Crayfish have 17 sensilla/annulus, rocklobsters 15 sensilla/annulus, while a cockroach has 50 sensilla/annulus and a bee 120 sensilla/annulus. This indicates that at least some insects have a higher density of sensory structures than some crustacea. However, a comparison based on external structures should be interpreted with caution, because the number of sensory cells associated with each sensillum may be very different. This is, for example, the case in crustacean olfactory sensilla, which can have up to 200 sensory cells, while insect olfactory hairs usually are associated with only 3–7 sensory neurons (Hallberg and Hansson, 1999). The situation seems to be similar for spider mechanosensory sensilla. Most spider mechanosensory hairs are innervated by three sensory cells, instead of a single neuron as in insects (Foelix, 1996; Barth, 2002). The so-called slit sensilla are elongated structures on the body of spiders. In contrast to campaniform sensilla, there are two dendrites running towards the exocuticle, only one of which enters a

canal and attaches to the membrane within the slit (Foelix, 1996). This sensory cell, however, has a tubular body, which is typical for arthropod mechanosensory cells (Barth, 2002). Functionally, campaniform and slit sensilla are very similar, and it would be interesting to know if they were homologous or analogous structures.

3.2 DISTRIBUTION OF MECHANORECEPTORS

The mechanosensory neurons of the antenna need to represent two different aspects of the world, (1) antennal movement induced by the animal itself, and (2) external stimulation of the antenna. The various sensory structures responding to mechanical stimulation have different characteristics, based on their morphological and anatomical properties, their different ways of responding to mechanical stimulation and their different response dynamics. Furthermore, animal species differ in their morphology, life styles and behaviour. These factors, together with differing receptor characteristics are likely to be reflected in the species-specific distribution of the various sensor types. Two variations of a general theme are conceivable: first, certain sensory structures will be distributed in very similar ways in different species, because they are especially suited to detect certain stimulus features, which are of behavioural significance to these species. An example could be the campaniform sensilla on the distal edge of the pedicel that are suitable to sense torque exerted at the PF-junction (cockroach: Schafer and Sanchez, 1973; cricket: Knyazeva *et al.*, 1975, 1978; locust: Gewecke, 1972b; Knyazeva, 1974; stick insect: Rosciszewska and Petryszak, 1986). Second, other receptors will show larger differences in their distribution, because of species-specific tasks, e.g. sensory hairs on the tip of the bee antenna (Martin and Lindauer, 1966).

3.2.1 *Hair plates*

Hair plates, fields or rows are distinct arrangements of similar hairs in close proximity. On the antenna, they are located near the basal joints, close to the proximal margin of the distal segment. Due to their specific features, hair plates are discussed separately from other sensory hairs (see also Section 3.3). In Lepidoptera, the hair plates on the basal antennal segments were described early on by Böhm (1911). Therefore, the hairs in these plates are sometimes called 'Böhm bristles' (cf. Kloppenburg *et al.*, 1997). We will, however, not use this term, but use the more general term 'hair plates' instead.

Each hair plate is distinct from the surrounding tactile hairs and consists of a variable number of hairs (Table 3), which in all cases reported so far, are mechanoreceptive hairs, innervated by a single sensory cell (Schafer and Sanchez, 1973; Okada and Toh, 2001). The precise location of the hair fields reflects the species-specific arrangement of the joint axes, and indicates how

TABLE 3 Hair plates and joint axes in the insect model organisms

	Periplaneta	Cricket	Locust	Carausius	Bee
Head-scape joint axes	Ball joint: Okada and Toh (2000)	Hinge joint: vertical plane: up/down; Honegger et al. (1990)	Hinge joint: 60° up and 50° down Gewecke (1972a)	Hinge joint: vertical plane: up/down; Dürr et al. (2001)	Ball joint; Snodgrass (1956)
Scapal hair plates	Ventral, dorsal, lateral (24, 25, 10 hairs; Petryszak, 1975) Dorsal, lateral, median (120, 60, 30 hairs; Okada and Toh, 2000)	Dorsal, ventrolateral, ventromedian (60, 20, 20 hairs; Knyazeva et al. (1975))	Dorsal, ventral, median, ventromedian; Gewecke (1972b)	Medio-dorsal, dorso-latero-ventral, medio-ventral (16–20, 10–16, 16–18 hairs; Urvoy et al., 1984) Medio-ventral, ventro-latero-dorsal, medio-dorsal (Dürr, unpublished observation)	One plate extending laterally and medially (90–120 hairs; Markl, 1962)
Scape-pedicel joint axes	Hinge joint: vertical plane: up/down Okada and Toh (2000)	Hinge joint: horizontal plane: lateral/median; Honegger et al. (1990)	Hinge joint: 45° lateral and median; Gewecke (1972a)	Hinge joint: horizontal plane: lateral/median; Dürr et al. (2001)	Hinge joint: pedicel movement: up/down; Snodgrass (1956)
Pedicellar hair plates	Dorsal, ventral, dorsolateral; Schaffer and Sanchez (1973) , Roszczewska and Petryszak (1986) , Okada and Toh (2000)	<i>Gryllus</i> : two hair plates; Kammerer and Honegger (1988) <i>Acheta</i> : lateral, median (15, 15 hairs; Knyazeva et al., 1975 ; Roszczewska and Petryszak, 1986)	Lateral, median; Gewecke (1972b)	Ventro-dorso-external, lateral-internal, centro-ventral (14–22, 10, 7 hairs; Urvoy et al., 1984 ; Roszczewska and Petryszak, 1986)	Lateral, medio-dorsal, medio-ventral (25–30, 15, 25 hairs; Markl, 1962)

antennal movements may be modulated by the activation of one or more hair plates (Kammerer and Honegger, 1988; Okada and Toh, 2000, 2001).

The scape of *Periplaneta* has at least three hair plates (Petryszak, 1975; Okada and Toh, 2000), yet the descriptions of their location and the number of hairs in each plate differs greatly. Schafer and Sanchez (1973) report four hair plates, but no hair count. Petryszak (1975) reports ventral, dorsal and lateral hair plates with 24, 25 and 10 hairs, respectively. According to Okada and Toh (2000), there are dorsal, lateral and medial hair plates with 120, 60 and 30 hairs, respectively. These hairs are reported to have no pores (Schafer and Sanchez, 1973; Okada and Toh, 2001). The cricket scape has three hair plates (Roscziszewska and Petryszak, 1986). Knyazeva *et al.* (1975) report a dorsal scapal hair plate with some 60 hairs, plus one ventro-lateral and one ventro-medial scapal hair plate with about 20 hairs. Four hair plates are reported for the locust scape. They are located dorsally, ventrally, ventro-medially and laterally (Gewecke, 1972b). According to reports by Urvoy *et al.* (1984) and Roscziszewska and Petryszak (1986), the scape of the stick insect *Carausius morosus* has three hair plates, labelled as medio-dorsal, dorso-latero-ventral and 'medio-ventral' with 16–20, 10–16 and 16–18 hairs, respectively.²

The scape of the honeybee inserts on the head with a ball joint (see Fig. 2). There is only one hair plate reported, which contains about 90–120 hairs (Markl, 1962). This plate, however, seems to extend laterally and medially around the base of the scape. Functionally, it may not matter, if one or more hair plates can be discerned. It is most likely that the movement of this joint is represented by the relative number and positions of the bent hairs.

The pedicel is usually smaller than the scape and the movement axis of the SP joint is often nearly perpendicular to the axis of the HS joint (see Sections 2.1.1.2 and 2.3.1). It also bears hair plates near its proximal margin, adjacent to the SP-joint membrane. It is very likely, that they are comprised of the same type of mechanosensory hairs as the hair plates on the scape, i.e. they have no pores.

The pedicel of *Periplaneta* bears three hair plates: dorsal, ventral and dorso-lateral (Schafer and Sanchez, 1973; Roscziszewska and Petryszak, 1986; Okada and Toh, 2000). In *Gryllus*, two pedicellar hair plates have been described (Kammerer and Honegger, 1988). *Acheta* also has two hair plates: lateral and medial, with about 15 hairs in each (Knyazeva *et al.*, 1975; Roscziszewska and Petryszak, 1986). However, the type of hair is not clear (Knyazeva *et al.*, 1975, 1978). A very similar arrangement of pedicellar hair plates is reported for a locust. Gewecke (1972b) described lateral and medial hair plates on the pedicel. This may be due to a similar joint arrangement in these species (Section 2.1.1.2).

²Comparison with another laboratory colony of *Carausius morosus* reveals a remarkably similar description of antennal hair plates, yet with reversed orientation of dorsal and ventral sides. Probably, the labelling and corresponding text description of the dorsal and ventral sides in Fig. 1 of Urvoy *et al.* (1984) were accidentally swapped (Dürr, unpublished observation).

The three hair plates of *Carausius* are ‘ventro-dorso-external’, ‘latero-internal’ and ‘centro-ventral’ and have 14–22, about 10 and seven sensilla, respectively (Urvoy *et al.*, 1984; Rosciszewska and Petryszak, 1986). For *Apis*, Markl (1962) describes three hair plates on the base of the pedicel. Like on the scape, these hairs appear to be mechanosensory (Schneider and Kaissling, 1957). The lateral hair plate with 25–30 hairs probably responds to an extension of the antenna, while the medio-dorsal and medio-ventral hair plates, with 15 and 25 hairs, respectively, respond to bending of the antenna (Markl, 1962).

3.2.2 Campaniform sensilla

Campaniform sensilla can be found all over the insect antennae in variable numbers and arrangements (Table 4). Their total number on the antenna may well be species-specific (cockroach: 140–400; Campbell, 1972; Schafer and Sanchez, 1973; Petryszak, 1975; cricket: *Acheta*: 238–285, Knyazeva *et al.*, 1975, 1978; *Gryllus*: 13–16; Yildiz and Gebhardt, 2003; stick insect: 0–50; Slifer, 1966; Weide, 1960; honeybee: 44–48; Dietz and Humphreys, 1971; Esslen and Kaissling, 1976).

Campaniform sensilla are most obvious on the distal edge of the pedicel, where they are usually arranged evenly in one or two rings around the rim. The number of rings, and sensilla per ring varies between species, but nothing is

TABLE 4 Campaniform sensilla in the insect model organisms

Species	Named	Number of mechanosensory cells	Location	Reference(s)
<i>Periplaneta americana</i> L.	Sensilla campaniformia (sensory domes)	Not reported	Scape, pedicel (distal margin), flagellum (single, dorsal)	Schafer and Sanchez (1973)
<i>P. americana</i> L.	Campaniform sensillum	One with a tubular body	Not reported	Toh (1977)
<i>P. americana</i> L.	Campaniform sensillum	One with a tubular body	1 Row of about 28 around distal part of pedicel	Toh (1981)
<i>Acheta domesticus</i> L. (<i>Gryllus domesticus</i> L.)	Sensilla campaniformia	One bipolar neuron	Distal part of pedicel	Knyazeva <i>et al.</i> (1978)
<i>Locusta migratoria</i> L.	Campaniform sensilla	70 Sensilla	Around distal pedicel	Gewecke (1972b)
		Not reported	Two rows at distal pedicellus	Knyazeva (1974)
<i>L. migratoria</i> L.	Campaniform sensilla	Not reported	Soft cuticle of scape-pedicel joint	Gewecke (1972b)
<i>Carausius morosus</i> Br.	Campaniform sensilla	N/a	None identified with certainty	Slifer (1966)
<i>Apis mellifica</i> L., worker	Sensilla campaniforme	One (?)	Flagellar segment 1 and 3–10	Esslen and Kaissling (1976)
<i>A. mellifica</i> L., drone	Sensilla campaniforme	One (?)	Flagellar segment 3–10	Esslen and Kaissling (1976)

known about the functional relevance of these differences, e.g. their impact on angular resolution of deflection sensitivity at the PF-junction. *Periplaneta* has one ring with about 27 sensilla (Petryszak, 1975). In crickets, the number and arrangement of the campaniform sensilla are different in the two species. While *Acheta* has two rings with 60 sensilla, *Gryllus* has only one ring with 13–16 sensilla at the distal rim of the pedicellus (Knyazeva *et al.*, 1975, 1978; Rosciszewska and Petryszak, 1986; *Gryllus*: Yildiz and Gebhardt, 2003). Two parallel rings are found in locusts (Knyazeva, 1974), in which Gewecke (1972b) counted about 70 campaniform sensilla. For *Carausius*, details are not yet published. In *Apis*, McIndoo (1914) mapped campaniform sensilla on the whole body, but not on the antenna. Thus, together with other sensory structures of the pedicel, the campaniform sensilla of the distal edge of the pedicel can probably play an important role for flight steering and maintenance (cockroach: Yagodin, 1980; locust: Gewecke, 1972a; see Section 6.1.3.1).

Except for the sensilla on the distal pedicel, only a few campaniform sensilla are located on the first two antennal segments, and they are usually scattered over their surfaces (cockroach: Petryszak, 1975; Schafer and Sanchez, 1973; cricket: Knyazeva *et al.*, 1975, 1978; stick insect: Weide, 1960). The only exception is the basal scape of *Periplaneta*, which contains a single row of campaniform sensilla (Rosciszewska and Petryszak, 1986).

It is common across the five model species that the annuli of the flagellum bear only a small number of campaniform sensilla. In the cockroach the campaniform sensilla on the flagellum have been called ‘marginal sensilla’ (Campbell, 1972). There are usually one or two marginal sensilla on the dorsal surface of each annulus (Schafer and Sanchez, 1973; Petryszak, 1975). In *Periplaneta*, *Acheta* and *Carausius*, the situation is very similar (Knyazeva *et al.*, 1975, 1978; Rosciszewska and Petryszak, 1986). They are found close to the distal edges of the annuli. Therefore, marginal sensilla may be well suited to respond to antennal bending, i.e. to deflections of one or more annuli relative to their neighbours. So far, their descriptions do not allow inferences about their directional sensitivity, and one may wonder, how well the small number of sensilla can cover the stimulus range for flagellar bending. It is conceivable, however, that the whole population of marginal sensilla is quite efficient in representing the direction and strength of antennal bending.

A more complicated arrangement is reported for *Apis*. Esslen and Kaissling (1976) report eight sensilla on the first, four on the sixth, and 18 on the 10th annulus, while Dietz and Humphreys (1971) count none on the first, eight on the seventh, and 10 on the last annulus. All other annuli bear between one and four campaniform sensilla. Despite the differences in the number of sensilla, it seems clear that, first, the honeybee flagellum bears more campaniform sensilla per annulus than reported in other species, and second, that there is an obvious concentration of these sensilla at the tip. These observations probably have some significance, given the many ways bees use their antennae, e.g. in comb building (Martin and Lindauer, 1966).

3.2.3 Chordotonal organs

Chordotonal organs are common in insect antennae (Table 5). The most prominent one, Johnston's organ, is a rather large and specialised example. Chordotonal organs associated with antennal movements are found in the head and the two basal antennal segments, but they have not been reported for the flagellum.

In the honeybee, a chordotonal organ inside the head (Janet's organ) connects the tentorium with the membrane of the HS joint (Janet, 1911). It is attached to the dorsal side of the joint membrane, suggesting that it may respond to upward movements of the scape. Janet's organ has also been described in ants (Ehmer and Gronenberg, 1997b), but, so far, not outside the hymenoptera.

TABLE 5 Chordotonal organs in the insect model organisms

Species	Named	Number of mechanosensory cells	Location	Reference(s)
<i>P. americana</i> L.	Johnston's organ	Not reported About 150	Pedicel Completely surrounding the distal pedicel	Schafer and Sanchez (1973) Toh (1981)
<i>P. americana</i> L.	Connective chordotonal organ	About 50	Ventral hypodermis of distal pedicel	Toh (1981)
<i>A. domesticus</i> L. (<i>G. domesticus</i> L.)	Chordotonal organ	One scoloparium ea., medial with 10, lateral with five scolopidia	Medial and lateral in scape	Fudalewicz-Niemczyk and Rosciszewska (1973)
<i>A. domesticus</i> L. (<i>G. domesticus</i> L.)	Chordotonal organ	Five scolopidia	Upper two-thirds in pedicel	Fudalewicz-Niemczyk and Rosciszewska (1973)
<i>A. domesticus</i> L. (<i>G. domesticus</i> L.)	Johnston's organ	12 scoloparia with four to seven scolopidia ea.	Pedicel	Fudalewicz-Niemczyk and Rosciszewska (1973)
<i>A. domesticus</i> L. (<i>G. domesticus</i> L.)	Johnston's organ	Not reported	Antennae	Knyazeva <i>et al.</i> (1978)
<i>L. migratoria</i> L.	Chordotonal organ	Not reported	Proximal scape to proximal pedicel; lateral and medial	Gewecke (1972b)
<i>L. migratoria</i> L.	Chordotonal organ	Not reported	Proximal to distal pedicel; medial	Gewecke (1972b)
<i>L. migratoria</i> L.	Johnston's organ	Six–seven scoloparia	Proximal to distal pedicel; medial	Gewecke (1972b)

3.2.3.1 *Chordotonal organs of the scape.* For some species, chordotonal organs are described in the scape. Distally, they are attached to structures of the SP joint. Depending on the location of the attachment site(s), they respond to antennal movement in this joint. In *Periplaneta*, one organ with four scoloparia with a total of 20 scolopidia reaches from the medio-dorsal scape into the proximal pedicel (Petryszak, 1975). Thus, it probably responds to downward flexion of the pedicel. The 13th instar of *Acheta* possesses a medial and a lateral scoloparium with 10 and five scolopidia, respectively (Fudalewicz-Niemczyk and Rosciszewska, 1973). A lateral and a median chordotonal organ reach from the proximal edge of the scape to the ventral edge of the pedicel. Thus, they are suitable to respond to medial and lateral rotation of the pedicel (Gewecke, 1972b).

3.2.3.2 *Ordinary chordotonal organs of the pedicel.* Movements in the PF-junction can be registered by small chordotonal organ(s), the number of which may be species-specific. In contrast to Johnston's organ (Section 3.2.3.3), the smaller, 'ordinary' chordotonal organs can only respond to certain movements, because of their restricted attachment sites.

The pedicel of *Periplaneta* houses two chordotonal organs (Petryszak, 1975; Toh, 1981). The medial organ consists of three scoloparia with 6–10 scolopidia in each, while the more distally situated organ has 5 scolopidia. Toh (1981) mentions a single connective chordotonal organ with 50 scolopidia and two neurons in each of these. However, he does not describe the location of this organ in the pedicel. One chordotonal organ with five scolopidia in one scoloparium is reported for 13th instar *Acheta* (Fudalewicz-Niemczyk and Rosciszewska, 1973), and possibly two chordotonal organs exist in the pedicel of adult *Gryllus* (Yildiz and Gebhardt, 2003).

The locust pedicellar chordotonal organ reaches from the medial wall of the pedicel to the joint membrane between pedicel and flagellum (Gewecke, 1972b), and, thus, may be suited to respond to lateral movement.

3.2.3.3 *Johnston's organ.* Johnston's organ varies in size from species to species. Because it reaches around the whole PF-junction in a ring-like fashion, it can respond to flagellar movements in all directions.

In *Periplaneta*, Johnston's organ encircles the inner surface of the PF-junction (Petryszak, 1975; Toh, 1981). According to Petryszak (1975), this organ consists of 12–14 scoloparia with 4–5 scolopidia in each, while Toh (1981) counted 150 scolopidia with three receptor neurons in each.

In the 13th instar of *Acheta*, Johnston's organ consists of 12 scoloparia with 4–7 scolopidia each (Fudalewicz-Niemczyk and Rosciszewska, 1973; Knyazeva *et al.*, 1975). Unfortunately, details are missing for adult crickets (Knyazeva *et al.*, 1975), and it is not clear if sensory cells are being added during post-embryonic development to antennal chordotonal organs. For the more complicated tibial chordotonal organs of the *Teleogryllus* front leg, Ball and Young (1974) report that sensory cells are added during postembryonic development,

while Klose (1996) reports for the same species that these chordotonal organs are complete at the end of the embryonic development. For a pedicellar chordotonal organ in a cockroach, the number of neurons increases by about 10 during postembryonic development (Bloechl and Selzer, 1988).

The scolopidia of the locust Johnston's organ can be subdivided into a lateral and a medial portion, each of which contains six to seven scoloparia, each one containing a few scolopidia. They attach at the soft membrane of the PF joint, just distal to the ring of campaniform sensilla (Gewecke, 1972b). In the honeybee, the scolopidia of Johnston's organ run towards the joint membrane of the PF-junction (McIndoo, 1922).

It is not clear, whether the differences in the number of chordotonal organs reported for the different species reflects the difficulties in obtaining reliable counts of these delicate structures or, whether they are real, and of functional significance. Unlike studies on sex-specific differences, comparative studies with regard to this functional question are missing. The sex-specific difference in the number of sensory cells of Johnston's organ in mosquitoes is of functional significance, because males use this organ to hear the wing beat frequency of the females (e.g. Risler, 1977; Göpfert *et al.*, 1999), but a comparative study on sex-differences in other species is lacking.

3.2.3.4 Chordotonal organs in crustacea. Chordotonal organs of the crustacean antennae have received much attention (e.g. Clarac and Vedel, 1971, 1975; Sigvardt, 1977; Rossi and Vedel, 1980; Vedel, 1980; Rossi-Durand and Vedel, 1981; Vedel and Monnier, 1983; Sandeman, 1985).

In the crayfish *C. destructor*, two chordotonal organs sense movement of the distal three segments (Sandeman, 1985). The proximal chordotonal organ spans the IM-joint. The distal chordotonal organ spans the MC-joint with an additional side branch spanning the CF-joint. This is virtually identical in *Homarus americanus* (Sigvardt, 1977) and in *P. vulgaris* (Clarac and Vedel, 1975). The proximal chordotonal organ of *Cherax* (Sandeman, 1985) is also in the same location as the myochordotonal organ in *Palinurus* (Vedel and Monnier, 1983), but lacks the accessory muscle of the latter. Sensory units of the chordotonal organs are directionally selective and discharge in a tonic or phasic-tonic manner, coding a mix of position, velocity and acceleration, which is similar for insects (cf. Section 3.3). Yet, Sandeman (1985) also documents a unit the average spike rate of which codes the logarithm of the angular velocity of the IM-joint. The modulation of tonic discharge rate with joint angle may be as small as a 1 Hz increment per 10°. However, given the range fractionation properties of chordotonal organs (Matheson, 1992), it is impossible to use the response properties of single afferents to assess the overall angular resolution of the proprioceptive information that is available to the animal.

In the rock lobster *P. vulgaris*, the sensory neurons of the MCF chordotonal organ (that spans the MC- and the CF-joint) can be categorised according to their static or dynamic response characteristic, according to their location in

the organ, and according to their sensitivity to extension and/or flexion of the MC- and CF-joint (Rossi and Vedel, 1980). The ratio of static and dynamic units is 7:3. Among the static units, some selectively code angular position of the MC-joint or the CF-joint, others respond most strongly if the two joints are moved into the same direction (homo-directional units) or in opposite direction (heterodirectional units). The joint-selective units can be assigned to a proximal and a distal cell group in the organ. The homo- and heterodirectional units are assigned to a central cell group. An analog case cannot be made for antennal chordotonal organs in insects, because (1) no known insect chordotonal organ spans two joints (cf. Field and Matheson, 1998), and (2) the physiology of the antennal chordotonal organs is not known well enough. However, it is known that a chordotonal organ in the locust hindleg contains five populations of receptor cells, each of which is sensitive to different parameters of the stimulus or combinations of stimulus parameters (Kondoh *et al.*, 1995).

3.2.4 *Sensory hairs*

Purely mechanosensory (Table 6) and bimodal mechano-/chemosensory hairs (Table 7) account for a notable number of sensilla on insect antennae. As mentioned earlier, these hairs can have a variety of shapes and lengths. Their numbers and specific arrangement on the antennae, especially on the flagellum are quite variable in different species. It is also important to note obvious sexual dimorphisms in the number of sensilla (e.g. *Periplaneta*: Lambin, 1973; Schafer and Sanchez, 1973, 1976). In the case of *Periplaneta*, the numbers reported by different authors vary greatly, and, thus, should be used and interpreted with caution.

Environmental factors and population density are reported to have an impact on the number of chemosensory sensilla of locust antennae and mouthparts (Greenwood and Chapman, 1984; Chapman and Lee, 1991; Rogers and Simpson, 1997). Recently, it has been shown that the numbers of purely mechanoreceptive hairs on the femora of locust hindlegs are also affected by such factors (Rogers *et al.*, 2003). Some of these changes in locusts (Greenwood and Chapman, 1984; Chapman and Lee, 1991; Rogers and Simpson, 1997) are supposed to be due to changes in the chemical, rather than due to changes in mechanosensory input (Rogers and Simpson, 1997). In some cases, the number of contact-chemosensory hairs changes, and, as a consequence, the number of mechanoreceptive neurons does too (Rogers and Simpson, 1997). It is, however, not known if, and how, such changes affect the sensory neuropils in the deutocerebrum.

Forming additional sensory structures on any part of the insect body requires, among other prerequisites, changes in the cuticle. Therefore, new sensilla can be acquired only when insects moult (e.g. Shepherd and Murphey, 1986; Kämper, 1992). Thus, changes in external structures may only be expected in hemimetabolous insects. Consequently, the number of sensilla is most

TABLE 6 Mechanosensory hairs in the insect model organisms

Species	Named	Number of mechanosensory cells per sensillum	Location	Reference(s)
<i>Periplaneta americana</i> L.	Sensilla chaetica A (Hair plate sensilla)	One	Head-scape and scape-pedicel joints	Schafer and Sanchez (1973)
<i>P. americana</i> L.	Sensilla chaetica B	One with a tubular body	Scape, pedicel	Toh (1981)
<i>Acheta domesticus</i> L. (<i>Gryllus domesticus</i> L.)	Stout hairs	Not reported	Scape	Fudalewicz-Niemczyk and Rosciszewska (1973)
<i>Acheta domesticus</i> L. (<i>G. domesticus</i> L.)	Middle sense hairs	Not reported	All segments	Fudalewicz-Niemczyk and Rosciszewska (1973)
<i>A. domesticus</i> L. (<i>G. domesticus</i> L.)	Smallest hairs	One	All segments	Fudalewicz-Niemczyk and Rosciszewska (1973)
<i>Locusta migratoria</i> L.	Borsten (= bristles)	Not reported	Scape, pedicel	Gewecke (1972b)
<i>Carausius morosus</i> Br.	Tactile hairs	One with a tubular body	Flagellum	Slifer (1966)
<i>C. morosus</i> Br.	NP sensilla	One with a tubular body	Last annulus	Monteforti <i>et al.</i> (2002)
<i>Apis mellifica</i> L.	Sensillum trichoideum B1	Not reported	Flagellum	Esslen and Kaissling (1976)
<i>A. mellifica</i> L.	Sensillum trichoideum B2	Not reported	Flagellum	Esslen and Kaissling (1976)

likely fixed in adults of holometabolous species, such as bees. In holometabolous insects, environmental changes may have effects, but these will probably only affect structures within the central nervous system (CNS).

3.2.4.1 *Distribution along the flagellum.* Usually, two to five long and a variable number of intermediate hairs are found near the distal edges of the scape and the annuli of the antenna of *Periplaneta* (Table 8; Petryszak, 1975). Intermediate hairs are also evenly scattered on the pedicel and form rings around the annuli of the flagellum. The number of rings increases with distance from the base of the antenna (Petryszak, 1975). It remains to be shown if the hairs closest to the distal edge of an annulus respond to relative deflection of adjacent annuli.

Furthermore, the number of contact-chemoreceptors increases from antennal base to the middle of the flagellum, i.e. the 50th–75th annulus (male: 130; female: 100 sensilla), and then drops towards the tip (male 70; female: 30 sensilla; Lambin, 1973). A similar distribution was reported for another cockroach

TABLE 7 Gustatory hairs in the insect model organisms

Species	Named	Number of mechanosensory cells per sensillum	Location	Reference(s)
<i>Periplaneta americana</i> L.	Sensilla chaetica B (Thick-walled chemoreceptors)	One	All segments, especially flagellum	Schafer and Sanchez (1973)
<i>P. americana</i> L.	Sensilla chaetica (B)	One with a tubular body	Flagellum, scarce on scape and pedicel	Toh (1977, 1981)
<i>P. americana</i> L.	Sensillum chaeticum	One with a tubular body	Flagellum	Rüth (1976)
<i>P. americana</i> L.	Tp (terminal-pore)-sensilla	One with a tubular body	Flagellum	Schaller (1978)
<i>Schistocerca gregaria</i> Forskål	Sensillum chaeticum	One with a tubular body	Flagellum	Ochieng <i>et al.</i> (1998)
<i>Carausius morosus</i> Br.	TP sensilla	One with a tubular body	Last annulus	Monteforti <i>et al.</i> (2002)
<i>Apis mellifica</i> L.	Hakenborste (= hooked bristle)	One with a tubular body	Flagellum	Martin and Lindauer (1966)
<i>A. mellifica</i> L.	Sensillum trichoideum D	Not reported	Flagellum	Esslen and Kaissling (1976)

species (Schaller, 1978). This distribution is in contrast with some, but not all of the other model organisms, where the number of sensilla often increases towards the tip of the flagellum (Weide, 1960; *Apis*: Martin and Lindauer, 1966; Esslen and Kaissling, 1976).

For *Acheta*, two large hairs are reported to be located on the distal scape (Fudalewicz-Niemczyk and Rosciszewska, 1973), while medium and small hairs are scattered on the surfaces of scape and pedicel (Fudalewicz-Niemczyk and Rosciszewska, 1973). In addition, medium length hairs form circles around the first few annuli, but seem to be irregularly placed on more distal annuli (Fudalewicz-Niemczyk and Rosciszewska, 1973).

Mechanosensory hairs are reported to form regular rings around the annuli near their distal edges (Rosciszewska and Petryszak, 1986).

In the cricket *Gryllus bimaculatus*, the number of sensory hairs increases with distance from the antennal basis, and this seems to be similar in females and males (Fig. 8; Staudacher, unpublished observation). This seems to be similar in another cricket species, *Teleogryllus commodus*, where no structural differences between male and female flagella are reported (Rence and Loher, 1977). This arrangement is somewhat reminiscent of the situation in the cockroach, where each annulus carries a ring of contact-chemosensory hairs. The fact that different types of hairs (mechanosensory in a cricket and gustatory in a cockroach), are arranged in a similar way along the flagellum, could reflect a

TABLE 8 Number and distribution of sensory structures in the insect model organisms (*continued*)

Species	Campaniform sensilla	Mechano- sensory hairs	Gustatory hairs	Chordotonal organs	Johnston's organ	Strand receptors	Olfactory sensilla	Reference(s)
<i>A. domesticus</i> L., adult		> 110						Knyazeva <i>et al.</i> (1975) Rosciszevska and Petryszak (1986)
<i>Gryllus bimaculatus</i> DeGeer						May be four		Staudacher and Schildberger (1998)
<i>Locusta migratoria</i> , L.	> 70				6–7 Scoloparia, each with a few scolopidia			Gewecke (1972b)
<i>Schistocerca gregaria</i> Forskål			295 gregarious 315 solitary					Ochieng <i>et al.</i> (1998) Bräunig (1985)
<i>Carausius morosus</i> Br.	40–50	4000–4500	600–700			4–5 sensory cells		Weide (1960) Slifer (1966) Monteforti <i>et al.</i> (2002)

					ANTENNAL MOVEMENTS AND MECHANORECEPTION
		> 46–54			Rosciszewska and Petryszak (1986) Urvoy <i>et al.</i> (1984)
<i>Apis mellifera</i> L., male	42	388	130	18971	Snodgrass (1956) Esslen and Kaissling (1976)
<i>A. mellifera</i> L., worker: <i>bee 1</i>	44	995	100	4781	Slifer and Sekhon (1961) Esslen and Kaissling (1976)
<i>bee 2</i>	48	1271	132	4870	Dietz and Humphreys (1971)
<i>A. mellifera</i> L., unspecified		> 155–190			Markl (1962)

difference in the relative importance of tactile and contact-chemoreceptive senses for these animals. This possibility may not be too surprising, e.g. given the sexual dimorphism of the number of olfactory structures in cockroach and bee (Schafer and Sanchez, 1973; Esslen and Kaissling, 1976).

In contrast to the mechanosensory hairs on the first two segments (Gewecke, 1972b), only little is known about mechanosensory hairs on the flagellum of the locust antennae. Slightly fewer contact-chemosensory hairs are found on the flagellum of gregarious than solitary animals (Ochieng *et al.*, 1998). The number of contact-chemosensory hairs is the highest on the first annulus, with a second maximum at the tenth annulus. Distal to this second peak, the number declines steadily towards the tip of the antenna (Ochieng *et al.*, 1998).

Weide (1960) counts 4000–4500 sensilla chaetica and trichoidea on the flagellum of *Carausius*, amounting to about 85% of the total number of sensory structures he reported. It is not clear, however, how many of these sensilla are mechanosensory or gustatory. Slifer (1966) describes two types of mechanosensory hairs with different distributions along the flagellum: long, heavy hairs

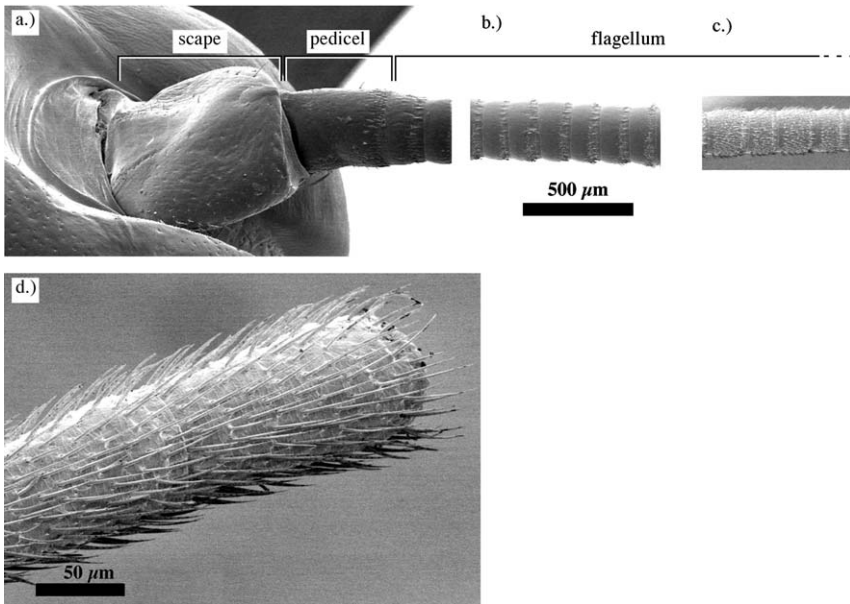


FIG. 8 SEM pictures of different sections along the antenna of a cricket (*G. bimaculatus* De Geer) exemplify the distribution of sensory structures along the antenna. With increasing distance from the base of the antenna, more sensors are located on each annulus. (a) Scape (Sc), pedicel (Pd) and first two annuli (A1, A2). (b) Six annuli from the first quarter of the antenna. (c) Four annuli from the second quarter of the antenna. (d) Tip of the flagellum. Figs. 5a–c: same scale. Fig. 5d has a different scale.

with blunt tips are found on the proximal third of the antenna, whereas short, delicate ones with sharp tips occur on the distal two-thirds of the antenna. In addition, there are thick-walled chemoreceptor pegs with a terminal pore and a flexible socket, which are inclined towards the tip of the antenna (Slifer, 1966). Their distribution along the antenna has not been reported, but Weide (1960) notes that the number of sensilla in general increases along the flagellum and reaches a maximum at the tip. It would be interesting to know if these two classes of mechanoreceptive hairs respond differently to mechanical stimulation and, if so, in which way. This could help shed more light on the way the antenna is used for obstacle avoidance by this animal (see Section 6.2.1).

The different sensilla of bee antennae are, in general, most abundant on the tip (Esslen and Kaissling, 1976). Like in the stick insect, two types of mechanosensory hairs are reported for the honeybee flagellum. Furthermore, these two classes are also distributed in very distinct ways along the antenna. The first class, sensillum trichoideum B1, is absent on the first two annuli, increases in number until the fifth or sixth annulus, and reaches the highest number at the tip (Esslen and Kaissling, 1976). Sensors of the other class, sensillum trichoideum B2 (the 'hooked hair' or "Hakenborste" of Martin and Lindauer, 1966), only occur on the tip (Esslen and Kaissling, 1976). Contact-chemoreceptive hairs, sensillum trichoideum D (or trichoideum D II; "spear-like hair" or "Spiessborste" of Martin and Lindauer, 1966), occur in high numbers on the first and around the sixth annulus, but are most abundant at the tip (Dostal, 1958; Esslen and Kaissling, 1976). Drones and workers are similar regarding the distribution of the different mechanoreceptive structures, but the numbers of sensilla may be slightly different. This difference, however, is not as large as for the olfactory sensilla, which are reported to be 19 000 in drones compared to only about 4800 in workers (Esslen and Kaissling, 1976). Besides the gradients of sensilla along the flagellum, the form of the tip of the honeybee antenna is also very peculiar. In both drone and worker, the last annulus has a flat, oval area that faces postero-laterally ("sensory plate" of Martin and Lindauer, 1966; Esslen and Kaissling, 1976). Furthermore, most mechanosensors are concentrated close to the tip on the antero-lateral and medial side (Esslen and Kaissling, 1976). Here, a special arrangement of the second class of mechanosensory 'hooked hairs', and contact-chemosensory 'spear-like' hairs is reported. A single gustatory hair is surrounded by a number of mechanosensory hairs, which are arranged in a circle; and there are three such rings at the tip (Martin and Lindauer, 1966). This special arrangement of sensory structures probably is very significant for the precision of combcell building by workers (Martin and Lindauer, 1966), and may be also for pattern recognition on flower petals (see Section 6.2.3). It is not clear if another specific arrangement of sensory hairs and/or the concentration of sensory structures at the tip of the antenna play a role for discerning microstructures on the surfaces of flowers (Kevan and Lane, 1985). Noteworthy, though not yet convincingly explained, is the fact that the ventral side of the honeybee flagellum, i.e. the side

facing the scape when flexed, is nearly devoid of sensory organs (Dietz and Humphreys, 1971).

3.2.4.2 Considerations about functional significance. In insects, sensory hair plates are present on the scape and pedicel, and most of the model insects have a ring of campaniform sensilla around the distal edge of the pedicel. In some species, the distribution of mechanosensory and gustatory hairs shows a maximum near the middle of the flagellum, but, most often, a second maximum of sensillum density is located at the tip of the flagellum. Furthermore, the highest concentration of sensory structures occurs on the very tip of the flagellum. Such a concentration, and in some cases, special pattern of sensillum arrangement is found across species. The number of chordotonal organs and the number and distribution of gustatory hairs as well as other sensory structures varies between species. In some cases, there is also sexual dimorphism.

Different receptors respond to different ways of mechanical stimulation, e.g. stress in the exoskeleton, bending of the flagellum in a certain joint, or bending of one or more hairs due to touch, and are distributed in different patterns over the antenna. Thus, the activity pattern of a specific group of antennal sensilla can represent only one aspect of the movement of the antenna or external stimulation of only one part of the antenna. Only the pattern of activity of all the different antennal sensors across the antenna can fully represent movement or stimulation of the antenna as a whole. However, in principle, all antennal mechanosensory structures can function as both, proprio- and exteroceptors, depending on the source of the transduced force. Moreover, because the antennae are actively moved, the representation of the external space by active antennal sensing is ambiguous. A somatotopic arrangement of afferent fibres in the CNS alone will not be sufficient to represent the complex stimulus pattern arising from self-motion-induced and external stimulation of the antennal sensors. Possible mechanosensory self-stimulation due to active movement cannot easily be discerned from an external stimulation, thus, implying the necessity for central modification or gating of the sensory input, e.g. the postulated “re-afference principle” of Holst and Mittelstaedt (1950; see also Section 5.25.2). Moreover, biomechanical properties of the antenna may also help to discriminate between exteroceptive and proprioceptive cues (cf. Sections 2.2.2.1 and 7.2).

3.3 PHYSIOLOGY OF ANTENNAL MECHANOSENSORY NEURONS

The mechanical stimuli detected by the antennae include tactile stimuli, proprioceptive signals and graviception (Horn, 1983; Horn and Bischof, 1983; Section 6.1.1), which involves a variety of internal and external mechanosensory neurons. The plethora of anatomical data on the structure of antennal sensilla (Section 3.1) or the knowledge about olfactory coding (see Shields and Hildebrand, 2001, or Stocker, 2001, for recent reviews) is contrasted by the

small number of electrophysiological studies of antennal mechanosensors. In this section, we will discuss studies that focus on aspects of proprioceptive physiology relevant for behaviour.

Bending of the locust flagellum is detected by a ring of some 70 campaniform sensilla around the distal end of the pedicel. The directional selectivity of each sensillum follows a cosine function of deflection angle, with excitation being strongest if the flagellum is bent away from it, i.e. by increasing the longitudinal strain across it (Heinzel and Gewecke, 1979). If the flagellum is deflected towards the sensillum, i.e. compressing the cuticle, the response ceases. The step response of each sensillum consists of a phasic-tonic receptor potential and spike train. Both the plateau of the receptor potential and the spike frequency increase proportionally with increasing deflection of the flagellum until a saturation level is reached at 1.3° .

An antennal strand receptor organ is characterised by Bräunig (1985). This receptor is present in locusts and, probably, in crickets (cf. Staudacher and Schildberger, 1999). The strand spans the HS-joint and is innervated by four to five sensory neurons with central cell bodies (see Section 3.1.4). Depression of the HS-joint leads to an elongation of the strand which is linear between 0° (HS-joint completely elevated) and 90° ; further depressions yield little or no additional increase in length (Fig. 9A). A full depression over 150° corresponds to a 1.5-fold increase of the resting (0°) length of the strand. The sensory neurons of the strand receptor organ are tonically active at rest. Elongation of the strand leads to an increase in the firing frequency of at least three discernible units. Accordingly, relaxation of the strand leads to a decrease in firing frequency (Fig. 9B).

Allgäuer (1989) describes the responses of sensory neurons of the scapal chordotonal organ (sCO) of crickets (and of unidentified pedicellar sensory neurons) to forced deflections of the antennal base, as revealed by intracellular recordings. As the y-shaped sCO spans the SP-joint, it is well suited to monitor position and horizontal movement of the pedicel. The sensory neurons encountered by Allgäuer (1989) are position-, velocity-, and acceleration-sensitive to various extent and, therefore, resemble the sensory neurons of other chordotonal organs (e.g. Hofmann *et al.*, 1985; Matheson, 1990). Accordingly, ablation of the sCO leads to a decrease of the precision of horizontal antennal tracking movements (Honegger, 1981) and to a decrease of the overall angular range of the movements (Kammerer and Honegger, 1988).

The scapal hair plate of cockroaches (*Periplaneta americana*) plays a major role for detecting objects in the horizontal plane (Okada and Toh, 2000; Okada *et al.*, 2002; see Section 6.2.2). Each hair of the hair plate sits in a moveable socket with a V-shaped ridge restricting the proximal movements of the hairs and thus imposing directional selectivity. Tungsten electrode recording from the base of individually deflected hairs reveals no resting activity (Okada and Toh, 2001). Upon increasing deflection, hairs start to fire in a phasic-tonic manner beyond a deflection threshold of 6° . The hairs can be kept activated for as long

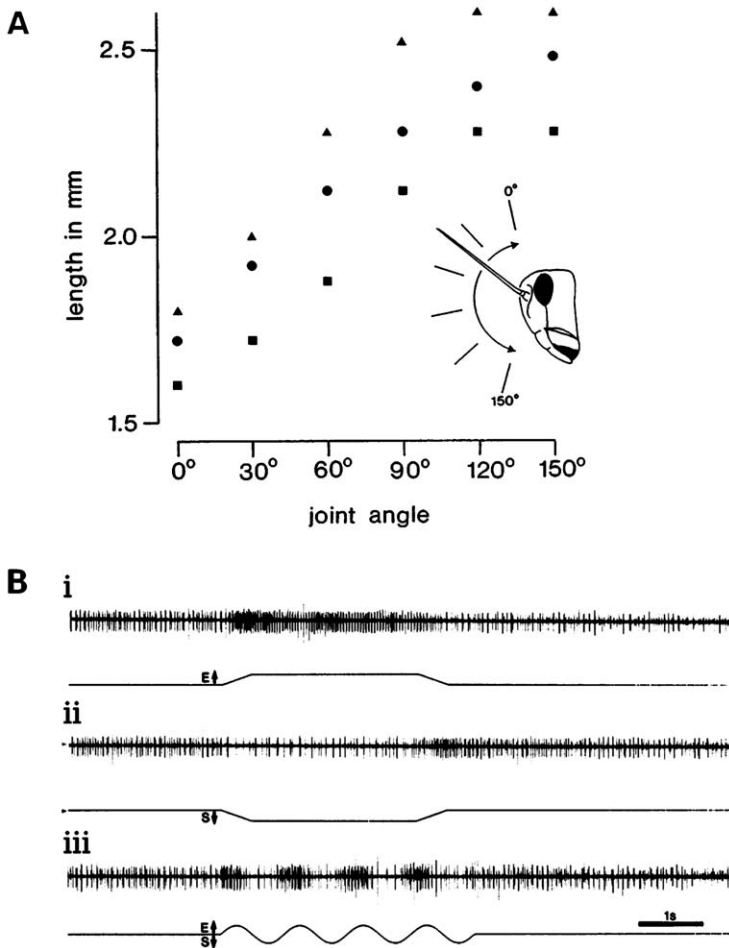


FIG. 9 (A) The length of the connective tissue strand of the antennal strand receptor depends on the angle of the HS-joint. Inset explains the definition of the angle. The length of the strand increases almost linearly with increasing depression of the scape in the range between 0° and 90° in three locusts (different symbols). (B) Extracellular recording from strand receptor neurons (upper traces, recordings; lower traces, stimulus monitors in Bi, ii, iii). (Bi) The firing frequency increases upon elongation of the receptor strand (E) and decreases upon shortening (S in Bii). The firing rate of the neurons reflects the sinusoidal stimulation of the receptor strand (Biii). Adapted from Bräunig (1985), with permission from Wiley and Sons, Inc.

as 10–20 min. Both the phasic and the tonic components of the response are dependent on the amplitude of deflection: the phasic, but not the tonic component increases with increasing deflection velocity. Phasic spike rates exceed 200 Hz upon large and fast deflections. The hair plate is subdivided into medial, dorsal and lateral portions, which consist of 30, 120 and 60 hairs, respectively

(Okada and Toh, 2001). Most importantly, the hairs are covered and thus deflected by the cutaneous membrane of the HS-joint in a way that depends on the angular position of the HS-joint. Depending on their relative location, different afferents will be bent by a different degree. The pattern of activated hairs and the strength of their activation is, thus, suitable to code for horizontal and vertical HS-joint angles for sustained periods of time. This is the case in the cockroach HS-joint investigated by Okada and Toh (2001), but also in other body parts, e.g. the cervical bristle fields of the bee (Thurm, 1963). The number of hairs, their respective angle of bending and the distribution of the bent hairs across the hair plates accounts for an ‘across-fibre-pattern’ (Pfaffmann, 1955), representing a population code of antennal posture.

The studies discussed above cover campaniform sensilla, strand receptors, chordotonal organs and hair plates. The behavioural relevance of other antennal mechanoreceptive neurons, such as the mechanosensory or gustatory hairs, remains so far to be studied. The picture emerging from these studies implies that the physiology of antennal mechanoreceptive neurons does not fundamentally differ from that of mechanoreceptive neurons of other body parts, e.g. the legs. The ring-like arrangement of the campaniform sensilla of the distal pedicel can be considered to be an antennal specialisation reflecting the connection between the pedicel and the flagellum which can be deflected into all directions. Other specialisations, which can be expected, such as a tuning of chordotonal neurons to high movement velocities (see Section 6.2.1), have still to be investigated.

4 Neuroanatomy of antennal mechanosensory and motor pathways

4.1 MECHANONSENSORY NEUROPILS

4.1.1 *Insects*

In contrast to the antennal olfactory pathways (Boeckh and Tolbert, 1993; Hildebrand, 1996; Laurent, 1996; Menzel and Müller, 1996), antennal non-olfactory afferent projections have not been adequately elucidated, despite the behavioural importance of the arthropod tactile sense (cf. Gewecke, 1975; Pelletier and McLeod, 1994; Dürre and Krause, 2001; see Section 6). Thus, the part of the deutocerebrum outside the antennal lobe (AL) can only be described as a diffuse neuropil, anatomically and metaphorically. Some regions of the diffuse neuropil are often referred to as ‘antennal mechanosensory and motor center’ (AMMC, Rospars, 1988; Homberg *et al.*, 1989), which does not take into account that there are a number of different mechanosensory structures at different locations along the antenna. So far, only a few studies provide data on antennal mechanosensory afferents and their arborisations in the DL of the deutocerebrum of a locust, and these form a basis for ascribing

functional significance to these areas (Gewecke, 1979; Bräunig *et al.*, 1983; Bräunig, 1985). A single study in *Gryllus* includes afferent fibres from the flagellum and shows three distinct areas of termination of antennal receptors: the AL, the DL and the ventral area of flagellar afferents (VFA; Staudacher and Schildberger, 1999).

Antennal sensory afferents, regardless of their modality, project towards the deutocerebrum via side branches of the two main branches of the antennal nerve N1 (see Fig. 10; Gewecke, 1972b; Bräunig, 1985). Receptors from the flagellum enter the brain exclusively via the antennal nerve (Nervus antennalis; N1). In the pedicel, two additional side branches contain the axons of Johnston's organ, the pedicellar hair plates, and other sensory structures. The axons of the sensory structures of the scape, scapal hair plates and chordotonal organs, enter N1 via separate side branches in the scape. The Nervus antennalis lateralis and Nervus antennalis medianus (N1l and N1m, respectively) branches fuse to a joint N1, either within the scape or shortly after entering the head capsule. The antennal stretch receptor enters N1 via a separate nerve within the head. Variations of these branching patterns may be expected in other species, because this picture is based on the situation in the locust.

In all insects, the antennal nerve N1 enters the deutocerebrum from ventral and more or less lateral, with respect to the embryonic neuroaxis (cf. Boyan *et al.*, 1993). At the point of entry into the deutocerebrum, the antennal nerve contains the axons of all the antennal sensory cells and some motoneurons. Immediately upon entering the brain, however, the axons divide into several bundles, or tracts, projecting to different areas in the deutocerebrum, where they terminate in distinct neuropils. To date, there is not a single systematic study about the branching patterns of the axons from different parts of the antenna and/or different modalities in any single insect. Thus, a general picture about the gross neuroanatomy of the sensory neuropils and the origin of the axon end terminals in the different neuropils can only be assembled from studies in different species (bee: Pareto, 1972; Suzuki, 1975; Arnold *et al.*, 1985; locust: Gewecke, 1979; Bräunig *et al.*, 1983; Bräunig, 1985; cricket: Staudacher and Schildberger, 1999; cockroach: Staudacher, unpublished). However, because it seems that reported differences in the numbers of tracts are likely to be due, at least in part, to the experimental approaches (for example, in the honeybee, four, six, and seven tracts have been discerned by Pareto, 1972; Suzuki, 1975 and Arnold *et al.*, 1985, respectively), the emerging picture should be regarded with some caution.

In all insects studied so far, a large portion of the fibres entering the deutocerebrum, namely those of tracts one to four (T1–4) terminate in the glomeruli of the (olfactory) AL (Fig. 11; e.g. moth: Rospars and Hildebrand, 1992; butterfly: Kim *et al.*, 1990; honeybee: Pareto, 1972; Suzuki, 1975; Arnold *et al.*, 1985; cricket: Staudacher and Schildberger, 1999; locust: Hildebrand, 1996; Laurent, 1996; cockroach: Distler and Boeckh, 1996; fly: Strausfeld, 1976). Most of the afferents branching in the AL are considered to be olfactory. Tracts one to four

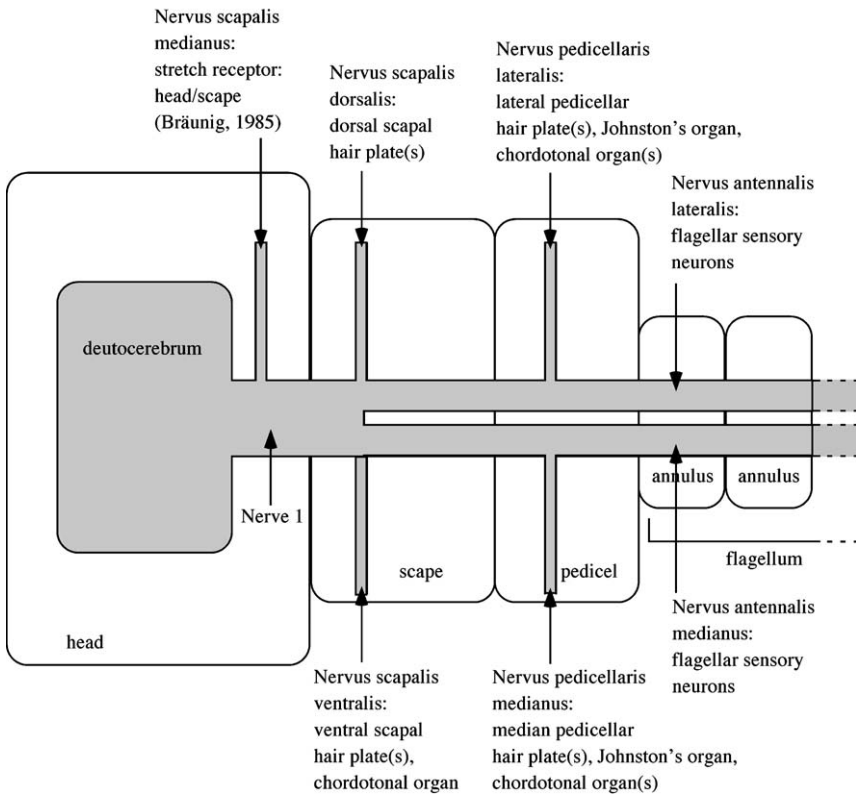


FIG. 10 Schematic view of the antennal nerve and the side branches, which run towards the various sensory structures inside and on the surface of the locust antenna. The nerves and branches are named according to Gewecke (1972b) and Bräunig (1985).

separate from the other afferents early on, i.e. ventrally. Fibres from tracts one and two run towards the core of the AL. From there, they enter the glomeruli of the AL, where they arborise (*Apis mellifera* L.: Pareto, 1972; Suzuki, 1975; *Gryllus bimaculatus*: Staudacher and Schildberger, 1999). Tract three fibres run along the posterior margin of the AL to enter some glomeruli from there (bee: Pareto, 1972; Suzuki, 1975). The fourth tract also runs along the posterior margin of the AL, and some of its fibres also enter the glomeruli, while others form a small, not very dense neuropil between antennal and dorsal lobes (bee: Suzuki, 1975). In the cricket *Gryllus bimaculatus* (Staudacher and Schildberger, 1999), and the cockroach *Gromphadorhina portentosa* (Staudacher, unpublished observation) the situation seems to be very similar, indicating that these tracts may be homologous.

Upon entry of the antennal nerve into the deutocerebrum, tracts five and six separate from the others and bypass the AL. They are reported to run posterior to the AL in the honeybee (Suzuki, 1975), but dorsal to the AL in the cricket

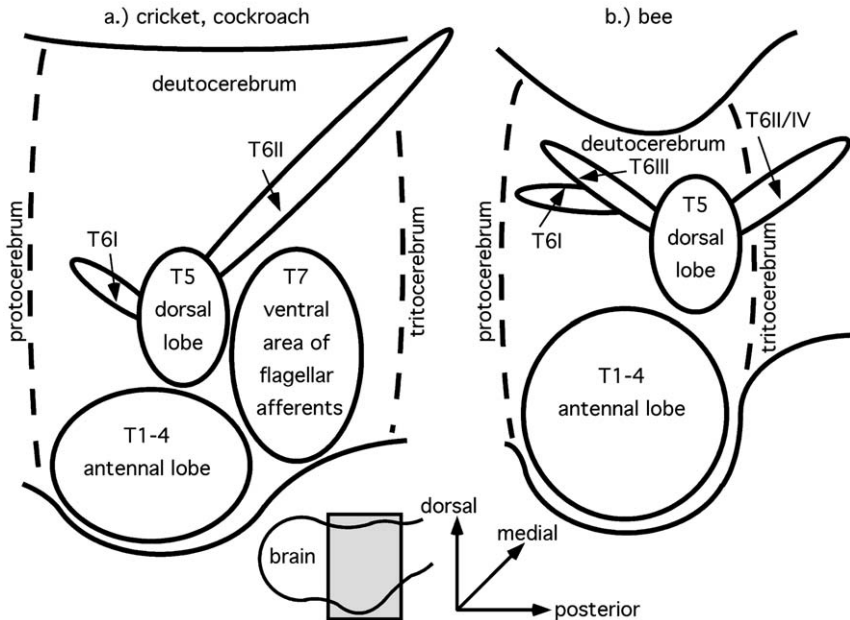


FIG. 11 Comparison of the antennal neuropils in the deutocerebrum of (a) cricket and cockroach (Staudacher and Schildberger, 1999; Staudacher, unpublished) and (b) the honeybee (Pareto, 1972; Suzuki, 1975; Arnold *et al.*, 1985; Maronde, 1991). The similarities in the layout of the deutocerebrum are obvious. However, there are also differences. Tracts T6III and T6IV have not been described for a cricket or cockroach, while the T7 (vfa area) has not been described in the bee. Tracts running from the antennal nerve to the different neuropils are named T1—T7. Broken lines indicate the borders of the deutocerebrum. Note, that the embryonic neuroaxis, which is tilted by about 90° relative to the head axis, is used in both schemes.

(Staudacher and Schildberger, 1999). Tract five contains sensory cells, which terminate and arborise in an area specifically called the DL. Tract six neurons have arborisations in the DL, but also send collaterals into the posterior protocerebrum (bee: T6I and T6III of Suzuki, 1975; Arnold *et al.*, 1985) or into the anterior deutocerebrum (cricket, cockroach: T6I). Furthermore, the same cells have collaterals that project into the SOG (T6II). In the bee, the T6II fibres have some bilateral arborisations in the SOG (Pareto, 1972; Maronde, 1991), and some ipsilateral collaterals reach into the more posterior SOG (T6IV of Maronde, 1991). In the cricket and a cockroach, these collaterals stay ipsilaterally in the dorsal to medial SOG (Staudacher and Schildberger, 1999; Staudacher, unpublished). The branching patterns of the fibres in tract six are similar to what has been reported for afferents from the scapal and pedicellar hair plates, the campaniform sensilla of the distal pedicel and Johnston's organ in locusts (Gewecke, 1972b; Bräunig *et al.*, 1983). It remains to be shown, however, that these receptor groups have the same arborisation patterns in

cricket, bee, and in other insects, too. Recent work in progress on *Periplaneta* attempts to do this (Baba and Comer, 2003).

So far, there is only a single description of an additional tract seven, found in the cricket *G. bimaculatus* (Staudacher and Schildberger, 1999). Tract seven fibres separate from the main bundle ventrally to run in a posterior–medial direction. Numerous small, closely packed fibres terminate and form a neuropil of dense appearance with conspicuous parallel bands in the ventral posterior deutocerebrum. This neuropil is, thus, postero-lateral to the AL, and postero-ventral to the DL. In the cricket, and the cockroach *Gromphadorhina portentosa* (Staudacher, unpublished observation), this is the third antennal neuropil in the deutocerebrum. Because the afferents terminating in this area are presumed to arise from the flagellum, this neuropil is called VFA (Staudacher and Schildberger, 1999). The striped appearance of this neuropil may be an indication for a somatotopic arrangement of the receptor terminals. This would not be surprising, since many insect first-order neuropils have been shown to receive orderly afferent projections with regard to location or functional properties (e.g. auditory afferent fibres; Römer, 1983; Römer *et al.*, 1988), leg hair afferents in *Drosophila* (Murphey *et al.*, 1989), in the locust middle leg (Mücke and Lakes–Harlan, 1995; Newland *et al.*, 2000), locust hind leg (Newland, 1990; Burrows and Newland, 1993) and the cricket cercal system (Jacobs and Theunissen, 1996; Paydar *et al.*, 1999). Under the assumption of a somatotopic arrangement, one might speculate on the behavioural role of the VFA neuropil (cf. Camhi and Johnson, 1999). However, only a detailed analysis of the VFA anatomy and physiology will elucidate, if this area has a functional significance for active tactile sensing.

The emerging picture seems to be that the insects investigated so far are very similar with regard to tracts T1–4, which mainly contain axons terminating in the AL (Fig. 12). There seem to be some slight differences with regard to tracts T5 and T6, which could reflect species-specific differences. Another point is that tract T7 and the associated VFA area have not been described in the honeybee. Could this reflect the difference in length between honeybee and cricket or cockroach antennae, or be linked to the different ways in which these insects use their antennae? Some hints may also come from a comparison with crustacea.

4.1.2 Crustacea

The insect antenna is homologous to the crustacean antennule, which seems to be less important as an active tactile organ than the crustacean antenna (which lacks a homologue in insects). Therefore, it is interesting to see if the functions of the two crustacean head appendages and their different usage have an impact on brain anatomy. One question would be: ‘do structures related to the antenna show analogies to the insect deutocerebrum?’

Afferents from the first appendage, the antennule, project to the deutocerebrum. All olfactory afferents run towards the olfactory lobe, while the other

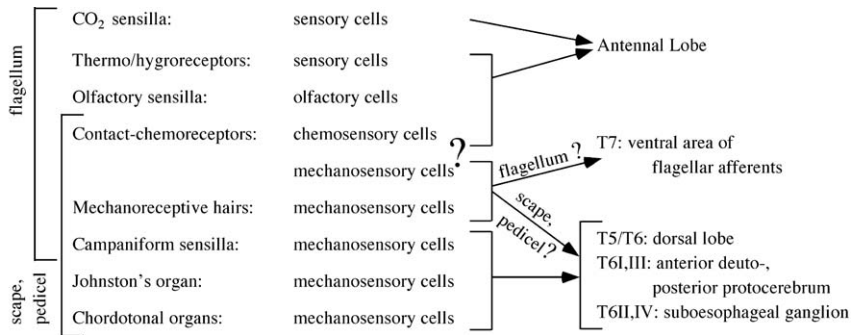


FIG. 12 Schematic representation of how different types of sensilla from various locations along the antenna project via various tracts to the different neuropils in the CNS, especially the deutocerebrum. Arrows indicate projections of the sensory cells, while the labels above the arrows indicate the receptor cell locations along the antenna. Hypothesized projections are marked with question marks (?).

afferents project into the lateral and median antenna I neuropil (Sandeman *et al.*, 1992; Schmidt and Ache, 1992). Non-olfactory afferents from the antennule enter the deutocerebrum via the nerve of antenna I, and project into the bi-lobed lateral antennular neuropile (LAN; Schmidt *et al.*, 1992). Thin and medium-sized axons are usually oriented along the long axis of this neuropil, and many send perpendicular branches into the neuropil, which stay within the same lobe, and are regularly spaced (50–70 μm ; Schmidt *et al.*, 1992). Motoneurons project only into the ventral areas of the LAN (Schmidt and Ache, 1993). In the median LAN, they branch in a conspicuous pattern with regularly spaced side branches (50–70 μm), while the arborisations are not as regular in the lateral LAN (Schmidt and Ache, 1993). Physiological results confirm what was suggested by the anatomical results, which is that the LAN is an area of integration of mechanosensory and olfactory information, and is a 'lower' motor centre for the antennule (Schmidt and Ache, 1996).

The antenna II neuropil is located in the tritocerebrum, i.e. the primary projection area of the antennal afferents; has an elongated tubular appearance. It receives input from antennal sensory neurons and projections from motoneurons and various interneurons (Sandeman *et al.*, 1992). In *Procambarus clarkii*, about 2000 sensory hairs from the flagellum project to the brain (Tautz *et al.*, 1981). Afferents run in four tracts along the margin of the AL, but with regularly spaced (25–35 μm), perpendicular branches reaching into the lobe. Ascending and descending interneurons, and some local interneurons show similar arborisation patterns in the AL (Tautz and Müller-Tautz, 1983). This could be interpreted as a somatotopic arrangement in this neuropil. Motoneurons, which supply the muscles of the second antenna, enter the tritocerebrum through a variety of nerve branches. They send regularly spaced branches into the AL, and collaterals into the diffuse areas of the tritocerebrum

(Habig and Taylor, 1982a,b). Sandeman *et al.* (1993) report a striking observation. They compared the structures of the antennal neuropils across different Decapod species. In the crab, which has the shortest antenna, the neuropil is least well structured. The neuropil is highly structured in species with the longest antennae, both of which are shrimps. An intermediate situation is found in crayfish, the antennae of which are shorter than in shrimps and longer than in crabs (Schram, 1986; Sandeman *et al.*, 1993; Ruppert and Barnes, 1994). It may prove useful to quantify such correlations with regard to body vs. antenna length, distribution of mechanosensory structures, etc. in both crustacea and insects. Knowledge about the different life styles, and use of the antennae may also prove to be useful in this regard. The compilation of such diverse information may lead to a better understanding of how brain structures are modified through evolution.

The afferents from both crustacean antennae project to highly structured brain areas, which may be regarded as integration centres for antennal motor control. But their structure also indicates somatotopic arrangements of afferents and the branches of other neurons (Habig and Taylor, 1982a,b; Tautz and Mueller-Tautz, 1983; Schmidt and Ache, 1993), which may be well suited for roles in active tactile sensing. Two of the projection areas of antennule afferents in the deutocerebrum are laminated and receive both mechanosensory and chemosensory inputs (Schmidt *et al.*, 1992; Schmidt and Ache, 1996; Strausfeld, 1998). Insects like firebrats (*Thermobia*) and silverfish (*Lepisma*) have uniform flagellar segments and laminated neuropils in the deutocerebrum, which seem to be structurally very similar to the LAN and medial antennular neuropile (MAN), but these insects do not have an olfactory lobe (Strausfeld, 1998; Strausfeld *et al.*, 1998). Is there a link between these areas and the VFA, not only structurally, but also functionally? Another question arises from the correlation of antennal length and the degree of order in the tritocerebral neuropils of Crustacea (Sandeman *et al.*, 1993). Could the difference with regard to the VFA between cricket and cockroach, on the one hand, and the honeybee on the other, indicate a similarity of these cases with respect to the length of the flagellum?

4.2 CONNECTIONS TO OTHER PARTS OF THE CNS

Mechanosensory stimulation of the antenna and the corresponding activity of mechanosensory afferents are important in a wide variety of behavioural contexts. Thus, it is not surprising that input of mechanosensory afferents is integrated with input from other sensory modalities through a variety of connections within the CNS. Anatomically, this is manifested by the overlap of afferents and many different types of neurons like motoneurons, local interneurons, some of which project to other brain neuropils, and ascending and descending interneurons.

4.2.1 *Connections with motoneurons*

The finding that mechanosensory afferents overlap with antennal motoneurons is not surprising. In the cricket, the locust and the stick insect, antennal motoneurons project to the dorsal deutocerebrum, where they may overlap with some, probably mechanosensory, antennal afferents (Honegger *et al.*, 1990a; Bauer and Gewecke, 1991). In *Manduca* and *Apis*, afferents and motoneurons project to the same area in the deutocerebrum (Kloppenburg, 1995; Kloppenburg *et al.*, 1997). Double labelling experiments have further substantiated this overlap in the bee (Kloppenburg, 1995). As yet, it is not clear if this overlap means direct connections between those classes of cells.

4.2.2 *Connections with local brain neurons*

A variety of local brain neurons have been described, which connect the antennal mechanosensory areas with the AL (Section 4.2.2.1) and various other brain regions (Section 4.2.2.2). Moreover, there also are connections to other parts of the CNS (Section 4.2.3).

4.2.2.1 *Antennal lobe (AL).* Some neurons recorded in the AL respond to mechanical stimulation. Because nothing is known about direct mechanosensory input to the AL, this is interpreted as evidence for indirect connections between olfactory and mechanosensory neuropils in the deutocerebrum (Waldow, 1975, 1977; Boeckh *et al.*, 1983, 1984; Kanzaki *et al.*, 1989; Zeiner and Tichy, 1998; Hill *et al.*, 2002). For some of these neurons in cockroach, direct connections to other neuropils can be ruled out (Boeckh *et al.*, 1984). A number of projection neurons of the male moth AL respond to clean air stimulation, which should only affect the mechanosensory afferents, but none of these neurons has deutocerebral branches outside the AL (Kanzaki *et al.*, 1989). Similarly, bee AL interneurons respond also to stimulation with air alone, even though they may not have branches outside the olfactory neuropil (Flanagan and Mercer, 1989). However, only little is known about neurons which actually have projections in both neuropils. A distinct class of bee AL interneurons (“D(M)Ho”) with branches in the AL, but also in other areas of the deutocerebrum, responds to olfactory and mechanosensory stimulation (Flanagan and Mercer, 1989). Within this group, the morphology is known for only one neuron, which has arborisations in the glomeruli of the dorsal AL and part of the mechanosensory areas in the cockroach (Malun *et al.*, 1993). This neuron sends a collateral into the outer antenno-cerebral tract, and arborises *en route* (Malun *et al.*, 1993). It is obvious, that all these cells take part in the integration of olfactory and mechanosensory input, but their precise role and impact on the behaviour of the animal is, as yet, elusive.

4.2.2.2 *Other brain regions.* Maronde (1991) reports three areas of overlap between antennal afferents and visual interneurons from the lobula, namely (1)

T6I, (2) T6III and (3) the DL (= T5). These three areas get input from sensory structures of the scape and pedicel, and from visual neurons of the lobula. These interneurons have one arborisation area in the ipsilateral lobula and blebby branches in either the ipsilateral, contralateral or both DLs. They show a variety of responses to visual stimulation (Hertel and Maronde, 1987). Connections like these may have functional roles in behavioural responses like the direction-sensitive antennal response, which can be elicited by stimulation of the ipsilateral compound eye (Kloppenburg and Erber, 1995).

4.2.3 *Connections to ascending and descending neurons*

The antennal mechanosensory areas are connected to other parts of the CNS via cells ascending from, and neurons descending to the ventral nerve cord. Some of these interneurons have also been labelled in immunohistochemical studies, and will, therefore, be mentioned later (see Section 4.3).

Some DUM neurons from the SOG project to the antennal mechanosensory and motor system, and may have significant arborisations in the deutocerebrum. Some of the cells, which were described in the locust (SA 2, SA 4, SA 5 and SAD 1), have rather obvious arborisations in the dorsal deutocerebrum (Bräunig, 1991). One of these cells, SA 5, has branches in both the olfactory and mechanosensory areas of the deutocerebrum. They may be octopaminergic and have input regions in the SOG and output areas in the brain. This may indicate, that these neurons are involved in neuromodulation of activity in their target areas of the brain (Bräunig, 1991), but direct physiological evidence is missing. Arborisations of two other, non-DUM, ascending neurons, one suboesophageal neuron and another cell with unknown soma location, have been shown in the cricket AMMC (Horseman *et al.*, 1997). The ascending input of these, and other cells is reported to have a rather strong impact on the function of the antenna (Horseman *et al.*, 1997). Even more striking is the finding of release sites of a suboesophageal DUM neuron onto antennal muscles that have been reported for locust and cricket (Bräunig *et al.*, 1990). These indicate a direct modulation of muscle activity by ascending input.

A large number of ipsi- and contralateral descending brain neurons (DBNs) have been described (e.g. Strausfeld *et al.*, 1984; Hörner and Gras, 1985; Brodfuehrer and Hoy, 1990; Gronenberg and Strausfeld, 1990; Kanzaki *et al.*, 1991; Böhm and Schildberger, 1992; Hensler, 1992). Some of these cells have branches in the mechanosensory areas of the brain, and probably receive input from antennal mechanosensory neurons, but may also have output regions in this area (Okada *et al.*, 2003). The antennal mechanosensory input is integrated with other modalities (Hörner and Gras, 1985; Staudacher, 1998a,b, 2001; Staudacher and Schildberger, 1998, 1999) and relayed to the thoracic and abdominal ganglia of the ventral nerve cord, where descending neurons usually branch in the dorsal to medial neuropil areas (Hörner, 1992; Staudacher and Schildberger, 1998; Staudacher, 2001).

A variety of descending neurons in the locust get ocellar input, and input in the dorsal protocerebrum and deutocerebrum. Three of these cells (DNI, DNM, DNC; Griss and Rowell, 1986) are of major importance for locust flight stabilisation (Reichert and Rowell, 1986; Reichert, 1993). Another large descending neuron with branches in the non-olfactory areas of the deutocerebrum is TCG (Bacon and Möhl, 1983), which has a homologue in *Periplaneta* (Gundel and Penzlin, 1995). This neuron is wind sensitive, which may in part be due to input from hairs on the head of the locust, but it also responds to antennal movement (Bacon and Möhl, 1983). It would be interesting to know, if these cells receive input from Johnston's organ and the campaniform sensilla on the distal pedicel, and how important these inputs are for their functional role.

Most descending neurons have at least some branches in the mechanosensory deutocerebrum, but most striking are those that have rather large arborisations in these areas. In *Periplaneta*, a large contralateral descending neuron (DM1a-1) has a number of branches in the medial protocerebrum, but major arborisations in an area that may resemble the VFA in crickets (Burdoan and Comer, 1990, 1996; Ye and Comer, 1996). This neuron is reported to play a major role in a cockroach escape reaction, which is triggered by tactile and wind stimulation of the flagellum (Comer *et al.*, 1994; Stierle *et al.*, 1994). A possible homologue of this cell (DBNc2-2) has been described anatomically, but never tested physiologically (Staudacher, 1998b). Six contralateral and two ipsilateral descending neurons with major branches in the mechanosensory areas of the deutocerebrum were described in the cricket (Staudacher, 1998a,b; Staudacher and Schildberger, 1999; Gebhardt and Honegger, 2001). All these cells have branches in other brain areas and, thus, can be considered multimodal, but their main arborisations are in the non-glomerular areas of the deutocerebrum. A detailed description of the branching areas of two ipsilateral (DBNi1-2 and DBNi2-1) and one contralateral (DBNc2-3) cell demonstrates a correlation with the arborisations of simultaneously labelled antennal afferents (Staudacher and Schildberger, 1999). The three neurons show differential overlap with the afferents. DBNi2-1 branches overlap exclusively with afferents in the DL (T5 and T6). The branches of the other two cells overlap only with afferents in the VFA. Here, the contralateral descending neuron arborises in the entire neuropil, while the ipsilateral descending cell overlaps with the afferents at median depth in the neuropil (Staudacher and Schildberger, 1999). These results illustrate the need for more detailed studies on afferent projection areas, because to date it is uncertain if and how the afferent inputs are spatially separated in the deutocerebrum. Thus, one can only speculate about the functional significance of the differences in the branching patterns of the interneurons.

Like other inputs to the CNS, the mechanosensory afferents are connected to a variety of areas in the CNS. There are local deutocerebral connections to the AL and other parts of the deutocerebrum. Via brain interneurons, the

mechanosensory neuropils are also connected to visual neuropils, the lobula and the medulla. A large number of descending neurons has a few branches in this area of the deutocerebrum. Eight descending cells have major arborisations there, which seem to be restricted to very specific sub-regions of the mechanosensory neuropils. A number of ascending cells from the SOG and the ventral nerve cord project into this region. One has to be very cautious in assigning directions of information flow to these connections, because it has been shown in a couple of cases, that input and output sites may be close to each other, that information can travel in two directions (Hughes and Wiersma, 1960; Watson and Burrows, 1983; Watson and Pflüger, 1989; Malun, 1991; Homberg, 1994). Nevertheless, the variety of connections show, that mechanosensory input from the antenna is integrated on a variety of levels with other input to, and activity in the CNS. It is an exciting task now, to study how these anatomically defined connections affect the control of the antennal movements, and how significant they are for the integration of antennal mechanosensory responses with ongoing behaviour or changes in behaviour.

4.3 IMMUNOCYTOCHEMISTRY

So far, only relatively basic studies of the neuroanatomy of antennal mechanosensory afferents are available, and the corresponding antennal mechanosensory circuits in the deutocerebrum (DC) are not known in detail. Immunohistochemistry, if reported in sufficient detail, and combined with knowledge of the underlying circuits may permit the formulation of hypotheses about the functional properties of these pathways. Some immunohistochemistry has been done on insect antennae (Section 4.3.1), on motoneurons (Section 4.3.2), and other neurons with arborisations in the brains of some of the model organisms (Sections 4.3.3).

4.3.1 *Antennal mechanoreceptors*

4.3.1.1 *Acetylcholine.* The most abundant transmitter in the insect nervous system (Breer, 1981; Breer and Sattelle, 1987), acetylcholine (ACH), is also present in antennal sensory cells. Indirect evidence for cholinergic cells has been provided by immunohistochemistry of the enzyme acetylcholine esterase (ACHE) in the bee brain. In this study, the glomeruli of the AL were labelled, but a much stronger stain was reported for the DL, including the posterior DC (= dorsal DC), fibres projecting into the tritocerebrum (TC), and the SOG (Kreissl and Bicker, 1989). The latter projections may be T6I and T6II, which were described by Suzuki (1975; Section 4.1.1). A similar result is reported for a locust, where the antennal nerve, AL and AMMC stain strongly with an ACHE antibody (Rind and Leitinger, 2000). A recent study using *in situ* hybridisation of an ACH receptor subunit in the bee brain provides more indirect evidence that ACH is the transmitter of mechanosensory afferents. Here, the

DL is weakly stained, a hint towards the presence of neurons, which are postsynaptic to cholinergic cells (Thany *et al.*, 2003). Further evidence comes from an autoradiography study that shows that α -Bungarotoxin, which specifically blocks nicotinic acetylcholine receptors, strongly stains the antennal and the DL of the bee (Scheidler *et al.*, 1990). Most important, however, these studies are in accord with results reported for moth antenna. Here, a number of mechanoreceptor neurons on the antennae of *Manduca* contain ACHE, which is widely accepted as an indicator for cholinergic neurons (cf. Stengl *et al.*, 1990). These comprise the mechanosensory hairs on the scape and the pedicel, Johnston's organ in the pedicel, mechanosensory hairs (sensilla chaetica), on the annuli of the flgellum, and possibly campaniform sensilla on the tip of the antenna of male moths (Stengl *et al.*, 1990). Thus, even though not all of these results stem from the same species, and even though many other species have not been studied, it may be stated that acetylcholine probably is the predominant transmitter of the mechanosensory afferents.

4.3.1.2 Serotonin. In *Periplaneta*, a few serotonergic axons are described at the base of the antennal nerve, and these can be traced into the DL (Salecker and Distler, 1990). They might be the axons of neurons from internal mechanosensory organs, if the result of a study on locust leg sensilla can be transferred: both, external and internal mechanosensory neurons of the locust leg can be stained with an ACHE antibody, suggesting ACH as a transmitter (Lutz and Tyrer, 1988). Moreover, internal and external receptors can be discerned by their immunoreactivity with antibodies against serotonin, which only stain internal receptors (Lutz and Tyrer, 1988). Such a method could, in principle, be used to identify and discern the branching areas of internal and external antennal mechanoreceptors. This could be an important tool to gain more detailed insight into the functional organisation of the mechanosensory areas in the deutocerebrum.

4.3.1.3 Nitric oxide. Sensory afferents from hairs, chordotonal organs and campaniform sensilla in locust leg show immunoreactivity against soluble guanylyl cyclase- α , which indicates that these neurons might be targets of interneurons using nitric oxide (NO) as a modulator (Ott *et al.*, 2000). Nitric oxide synthase (NOS), which is found in NO-releasing cells, generates NO and can indirectly be stained by NADPH diaphorase. In cockroach and locust there is weak staining in the non-glomerular DC, and there are no individually discernible fibres (Ott and Elphick, 2002). The latter result could mean that antennal mechanosensory fibres are not the target of NO releasing interneuron, provided it could be shown, that they are not stained with antibodies against soluble guanylyl cyclase- α .

4.3.1.4 Taurine. So far, no purely taurinergeric neurons (Bicker, 1992) have been described. Taurine is colocalised with octopamin, GABA, and therefore considered a neuromodulator, rather than a transmitter (Stevenson, 1999).

During bee larval development, all the different types of receptor neurons on their antenna react with an antibody against the amino acid taurine (Eichmüller and Schäfer, 1995). In the last larval stage, cells in the scape and pedicel, and their axons, which can be traced into the deutocerebrum, can be stained with taurine antibodies. Taurine-like immunoreactivity (taurine-like IR) then decreases during the pupal stage. At pupal ecdysis, taurine-IR is reported for cells in the flagellum: both types of mechanoreceptive hairs (trichoid B1 and B2) and gustatory hairs (trichoid D: 5–7 cells) are stained at this stage (Eichmüller and Schäfer, 1995). The finding that all the receptor types show taurine-like IR during development contrasts with the results of a study on adult bees (Schäfer *et al.*, 1988). In adult bees, the glomeruli of the AL and some unidentified neurons show taurine-like IR. However, only weak taurine-like IR and no discrimination of single branches or cells is reported for the DL, while no presumed mechanosensory axons are stained (Schäfer *et al.*, 1988).

4.3.2 *Antennal motoneurons and muscles*

The antennal muscles are innervated by excitatory and inhibitory motoneurons, the numbers of which are species specific (cf. Table 2; Sections 2.3 and 2.4). Immunohistochemical studies reveal similarities regarding excitatory and inhibitory transmitters, and modulatory substances despite the differences in the innervation patterns of antennal musculature.

4.3.2.1 *Glutamate.* Glutamate, the putative excitatory transmitter of insect motoneurons, and a precursor of GABA, shows a staining pattern similar to GABA in the bee brain. But the intensity of the labelling is significantly weaker in GABAergic neurons, and, thus, Glutamatergic and GABAergic neurons can be discerned (Bicker *et al.*, 1988). In the bee, some large axons from the DL, which show glutamate-like IR, project into the antennal nerve, and are presumed to be motoneurons. Ten of these axons are stained in antennal motor nerve cross-sections and three distinct profiles in the sensory nerve (Bicker *et al.*, 1988). The number of profiles counted in the immunohistochemical study (13; Bicker *et al.*, 1988) is not expected, because 15 motoneurons are described (Kloppenburger, 1995), and there is no indication for inhibitory, GABAergic motoneurons in the bee (cf. below; Schäfer and Bicker, 1986b). The transmitter of the two remaining motoneurons is unknown. Seventeen excitatory motoneurons are described for the cricket (Bräunig *et al.*, 1990; Honegger *et al.*, 1990a). Two of these motoneurons are glutamatergic (Bartos *et al.*, 1995), but according to a recent study, this number could be higher (Schürmann *et al.*, 2000). An ongoing study is reported to show, that locust antennal motoneurons are glutamatergic (cf. Homberg, 2002). All these publications strongly indicate that glutamate is the excitatory transmitter of many insect antennal motoneurons.

4.3.2.2 *Gamma-aminobutyric acid.* GABA is an important inhibitory transmitter in insect CNS (Bicker *et al.*, 1985; Schäfer and Bicker, 1986b; Honegger *et al.*, 1990b). In *Manduca*, some axons were stained in the antennal mechanosensory and motor tract, and indicate the presence of an unknown number of GABAergic, inhibitory motoneurons (Hoskins *et al.*, 1986). Thus, it remains to be shown, how many of the at least 12 motoneurons (Kloppenburger *et al.*, 1997) in *Manduca* are inhibitory. The bee DLs are homogeneously stained, and a number of perikarya in the rind nearby are marked with GABA antibodies (Schäfer and Bicker, 1986b). However, no GABAergic stain is reported in the antennal nerve of the bee (Schäfer and Bicker, 1986a). This could mean that, unlike cricket (Honegger *et al.*, 1990a), locust (Bauer and Gewecke, 1991), and stick insect (Dürr *et al.*, 2001), each of which has a single inhibitory motoneuron, the bee has no common inhibitor. In the cricket, the antennal common inhibitor neuron was filled retrogradely, and then double labelled with an anti-GABA (Honegger *et al.*, 1990a), and in a double labelling experiment shown to contain Proctolin, like 13 other cricket antennal motoneurons (Bartos *et al.*, 1995; Schürmann *et al.*, 2000).

4.3.2.3 *Proctolin.* Proctolin seems to play an important role in the modulation of the insect antennal motor system by prolonging muscle contraction (Section 2.4.1; Bartos and Honegger, 1997). In crickets, proctolin has been shown to be a co-transmitter in 14 of the 18 antennal motoneurons (Bartos *et al.*, 1994, 1995; Bartos and Honegger, 1997). In the locust, where it has similar effects as in the cricket, immunoreactive terminals are found on the antennal muscles, and some somata are stained in the rind of the deutocerebrum (Bauer, 1991).

Thus, one may conclude that Glutamate is the main excitatory, and GABA the inhibitory transmitter of antennal motoneurons. So far, proctolin is the only modulator shown to be present in, and important for the antennal motor system. In some cases, precise numbers of motoneuron axons stained in immunohistochemical experiments are missing or do not seem to account for all of the neurons expected to be found. Therefore, it is possible that other transmitters are present. Further studies are needed to elucidate this issue. In an immunohistochemical study on ACH, large profile neurons, which project into, and through the antennal nerve, and, thus, may resemble motoneurons, were not stained (Kreissl and Bicker, 1989). Thus, at least in the bee, ACH can be excluded from being a transmitter of the antennal motoneurons.

4.3.3 *Other immunohistochemical results*

The other immunohistochemical studies focus on describing the distribution of a given antigen within the brain. In most cases, they do not contain sufficiently detailed information to make them valuable for a detailed understanding of active tactile sensing. However, they contain hints with regard to transmitters

and modulators, which could be valuable for future experiments. Thus, they will be mentioned without lengthy discussions.

Local GABAergic (*Manduca*: Homberg *et al.*, 1987) and serotonergic cells (locust: Tyrer *et al.*, 1984), octopaminergic (for review: Stevenson and Spörhase-Eichmann, 1995; bee: Mercer *et al.*, 1983; Kreissl *et al.*, 1994; locust: Konings *et al.*, 1988; Stern *et al.*, 1995; Stern, 1999) are described. In the bee, serotonin decreases and octopamine increases visually induced movements of a direction-specific antennal reflex (Erber and Kloppenburg, 1995). Several dopaminergic neurons have been identified (locust: Wendt and Homberg, 1992; bee: Schäfer and Rehder, 1989; physiology, bee: Flanagan and Mercer, 1989; cockroach: Waldow, 1975).

A number of DBNs have been stained with immunohistochemical methods. Some are reported to be glutamatergic (bee: Goodman, 1981; Bicker *et al.*, 1988), others to be GABAergic (tritocerebral dwarf, locust: Tyrer *et al.*, 1988), some serotonergic (bee: Rehder *et al.*, 1987; PI(2), locust: Williams, 1975; Tyrer *et al.*, 1984; Hensler, 1988; PM3, *Manduca*: Homberg and Hildebrand, 1989), and some octopaminergic (bee: Kreissl *et al.*, 1994; locust: Williams, 1975; Bacon and Tyrer, 1978; Bacon and Möhl, 1983; Konings *et al.*, 1988).

Other studies describe branching patterns of ascending, and yet unidentified neurons. GABAergic (cricket: Schoch *et al.*, 2001, 2002), and histaminergic fibres (cricket, bee: Bornhauser and Meyer, 1997), ascending branches of vasopressin-like immunoreactive neurons (locust: Thompson and Bacon, 1991; Thompson *et al.*, 1991), leucokinin IR cells (cockroach: Naessel *et al.*, 1992) are described. Allatostatin immunoreactivity is widespread (Stay *et al.*, 1994; cricket, cockroach: Schildberger and Agricola, 1992; cricket: Neuhauser *et al.*, 1994; locust: Vitzthum *et al.*, 1995, 1996). FMRFamide (cockroach: Verhaert *et al.*, 1985; locust: Myers and Evans, 1987; Remy *et al.*, 1988; *Manduca*: Homberg *et al.*, 1990) and pigment-dispersing hormone IR (stick insect: Homberg *et al.*, 1991) are reported. Gastrin-cholecystokinin does not stain branches in the DC (cockroach, locust: Tamarelle *et al.*, 1988, 1990).

4.3.4 Immunohistochemistry summary

The results of all the neuroanatomical studies show a growing body of knowledge about the antennal mechanosensory system. However, to understand this system on a functional level, and to gain insight about the impact of the system on the behaviour of the animal, more details are needed. It seems to be of major importance to systematically compare the projection patterns of single receptor types from different locations along the antenna, and to do this across species. On the basis of this knowledge, descriptions of connections to other areas of the brain and the ventral nerve cord can, if reported in sufficient detail, reveal the overlap between interneurons and very specific areas of the afferent layer. Only, then will it be possible to understand the functional significance of the connections between mechanosensory afferents and different areas of the

CNS. Information gained from immunohistochemical studies needs to be more detailed with regard to mapping the precise areas of the stain, and more attempts are needed to identify single neurons. Such detailed results could provide testable hypotheses for neuroethological experiments.

4.4 ANTENNAL MOTONEURONS IN INSECTS

The peripheral innervation pattern of insect antennal motoneurons is outlined in Section 2.3.1 with an emphasis on the comparative layout of antennal nerves. Here, we concentrate on the aspects of the anatomy of antennal motoneurons within the deutocerebrum (or AMMC). Data are available for crickets (Honegger *et al.*, 1990a), locusts (Bauer and Gewecke, 1991), honeybees (Kloppenburg, 1995), ants (Ehmer and Gronenberg, 1997b), the tobacco hornworm (Kloppenburg *et al.*, 1997) and stick insects (Dürr *et al.*, 2001). Moreover, Baba (2000) shows the reconstruction of one antennal motoneuron of the cockroach *Periplaneta americana*. Several important characteristics of the central motoneuron anatomy, sometimes shared by two or more species, emerge from these studies, as discussed in the following section. To enable inter-species comparisons, the embryonal neuroaxes will be used as an anatomical frame of reference (Boyan *et al.*, 1993).

4.4.1 General organisation

All antennal motoneurons investigated so far, are restricted to the deutocerebral hemisphere ipsilateral to the muscles innervated, though dendrites may be located in the tritocerebrum in the cricket (Honegger *et al.*, 1990a) or may almost reach the SOG of the fused brain in the honeybee and in *Manduca* (Kloppenburg, 1995; Kloppenburg *et al.*, 1997). This resembles the unilateral layout of thoracic leg motoneurons, e.g. in the locust (for review, see Burrows, 1996). As a consequence of this unilateral layout, bilaterally co-ordinated antennal movements must be controlled by bilateral interneurons (cf. Section 5.2). Since antennal movements are sometimes co-ordinated with posterior body appendages like the pro-thoracic legs (Horseman *et al.*, 1997; Dürr *et al.*, 2001), again, interneurons have to connect between antennal and leg motor centres (see Sections 4.2 and 5.2).

4.4.2 Are antennal motoneurons “identified neurons”?

Most of the antennal motoneurons studied so far have been characterised on the population level. In the honeybee, for instance, antennal motoneurons are difficult to discriminate because of their similar central branching patterns, but groups of motoneurons are discernible on the basis of their soma clusters (Kloppenburg, 1995). The situation is similar in *Manduca*, where the central ramifications of the motoneuronal dendrites overlap to a great extent. Only the

intrinsic muscle motoneuron somata tend to cluster in one region of the deutocerebrum (Kloppenburger *et al.*, 1997). Bauer and Gewecke (1991) found the variation in the soma locations to be too large to identify motoneurons by their soma position, though the HS-joint and the SP-joint motoneuron somata tend to be separate in the dorsal ('posterior' after Bauer and Gewecke, 1991) deutocerebrum. In crickets, the antenno-motor system contains five morphologically identifiable individual motoneurons and five groups of two to three morphologically very similar motoneurons. The two 'structural sets' (Honegger *et al.*, 1990a) of the intrinsic muscle motoneurons can further be discriminated on the basis of their physiological properties, but two of the three Ab2–4 abductor motoneurons are indistinguishable (Honegger *et al.*, 1990a). Dürr *et al.* (2001) also describe the antennal motoneurons as individuals, with two sets of morphological twin motoneurons. It is to be expected from other insect motor systems that all of the antennal motoneurons could probably be individually identified upon an increase of knowledge of their properties, e.g. physiology, co-transmitter content or target muscle-fibres.

4.4.3 *Soma positions*

The study of Kloppenburger (1995) clearly demonstrates that, in the honeybee, somata of antennal motoneurons cluster in six groups. These clusters are located anterior and posterior to the DL and contain 15 somata. The numbers of somata per cluster vary between individuals. The three clusters of the extrinsic muscle motoneurons for instance, contain a maximum of nine somata with two somata consistently located in the posterior dorsal cluster; the remaining two clusters may contain three and four or two and five somata, respectively. Ehmer and Gronenberg (1997b) depict two sets of extrinsic antennal motoneurons of two single scapal muscles. The somata of these motoneurons lie in close vicinity to each other, suggesting that ant antennal motoneurons also have clustered somata. In *Manduca*, clustering of antennal motoneuron somata depends on the joint innervated: the motoneuron somata of extrinsic muscles are scattered over a wide area in the dorso-anterior soma band between the deuto- and protocerebrum, whereas the motoneurons of intrinsic muscles are concentrated in the dorso-lateral region of the soma band (Kloppenburger *et al.*, 1997).

In stick insects, the motoneuron somata are grouped into three areas, located in the lateral, the ventro-anterior and the ventro-posterior deutocerebrum (lateral, posterior- and anterior-dorsal, respectively, relative to the body-long axis in Dürr *et al.*, 2001). The somata of functionally different motoneurons controlling different antennal joints and/or movement directions are not located in separate soma clusters (Dürr *et al.*, 2001).

Most of the cricket antennal motoneuron somata are located in the lateral deutocerebrum, in the vicinity of the root of the antennal nerve N1. Some

motoneuron somata lie in the ventral deutocerebrum (Honegger *et al.*, 1990a). As in locusts (Bauer and Gewecke, 1991), no information on the formation of soma clusters is available. Some attention has been paid to the question of how the arrangement of neuronal somata in clusters may reflect a relationship of the adjacent neurons, e.g. a grouping according to a functional affinity. The antennal motoneuron somata clusters are unlikely to mirror a functional relationship, as the composition of stick insect soma clusters demonstrates. On the other hand, extrinsic and intrinsic muscle motoneuron somata of the honeybee separate in different clusters, though the somata of both motoneuron types lie close to each other in one area (Kloppenburger, 1995). Although the presence of distinct clusters does not appear to be a necessary feature of insect antennal motoneurons, the organisation of some antenno-motor systems is reminiscent of the findings of Siegler and Pousman (1990b) on the arrangement of meta-thoracic excitatory motoneurons into stereotyped clusters delimited by a glial sheath, which reflect a developmental rather than a functional relationship (Siegler and Pousman, 1990a). Thus, developmental studies may clarify, for example, whether the lateral band of motoneuron somata in crickets represents a 'cluster' in the sense that these neurons are the progeny of one neuroblast.

4.4.4 *Dendritic arborisation pattern*

All antennal motoneurons investigated so far have their dendritic arborisations in a part of the deutocerebrum outside the AL, the 'AMMC' (antennal mechanosensory and motor centre: Rospars, 1988; Homberg *et al.*, 1989; or 'DL': Suzuki, 1975). Honegger *et al.* (1990a) extend this neuropil by adding the dorsal tritocerebrum which contains motoneuronal projections, but no antennal mechanosensory projections ('AMC').

In honeybees and in *Manduca*, the dendrites of the motoneurons form a well-defined lateral, semi-circular border of the DL, where highest fibre densities occur. Fibre densities are reduced medially towards the esophageal foramen. The motoneuronal branchings end dorsal to the AL, and some fibres extend posteriorly close to the SOG (Kloppenburger, 1995; Kloppenburger *et al.*, 1997). Taken together, the antennal motoneurons appear to constitute a common projection field with a similar projection pattern of individual motoneurons. Similarly, in ants the median portion of the DL receives no motoneuronal projections, though antennal motoneurons occupy two-thirds of the DL (Ehmer and Gronenberg, 1997b).

The antennal motoneurons of crickets and stick insects resemble each other in that some individuals can be identified on the basis of their morphology in both taxa. The motoneurons of both insects will, therefore, be discussed in a greater detail in the following paragraphs, although a direct comparison of the arborisation patterns of potentially homologous motoneurons is hindered by the different brain morphologies of crickets and stick insects (Dürr *et al.*, 2001).

The levator motoneuron branchings of *Carausius* are restricted to the latero-posterior ('antero-lateral' in Dürre *et al.*, 2001) and to the medio-anterior to latero-anterior ('postero-medial' to 'postero-lateral' in Dürre *et al.*, 2001) parts of the AMMC. A circular patch in the latero-anterior AMMC is devoid of levator motoneuron arborisations. Depressor motoneurons ramify in lateral parts of the AMMC (Dürre *et al.*, 2001). Branchings of adductor motoneurons extend from the latero-anterior AMMC to its centre, whereas abductor motoneurons arborise mainly laterally in an area similar to that occupied by the depressor motoneurons. The abductor motoneurons also show a circular patch devoid of stained profiles. Thus, despite sharing some common projection areas, functionally distinct motoneurons differ in arborisation patterns specific for a group, e.g. the levator motoneurons with unique arborisations in the central AMMC.

In crickets, the anatomy of the antennal motoneurons is detailed by Honegger *et al.* (1990a). The two levator motoneuron groups L1/2 and L3-5 have overlapping dendrites (branches 1) that extend medially into the AMMC. In the depressor motoneurons, D1 and D4 overlap with many fine branches in the lateral AMMC, and the secondary neurites of D2/3 overlap with those of D5. Branch 1 of D4 overlaps with branches 1 and, partly, branches 2 of D2/3. Depressor and levator motoneurons differ in the extent of their branchings along the dorso-ventral axis of the brain in that depressor motoneurons extend more ventrally than do levator motoneurons. The intrinsic muscle motoneurons (see Fig. 13) ramify slightly more anteriorly at the level of N1 with respect to the extrinsic muscle motoneurons which have their main arborisation at the level of N4. Within the adductor motoneurons, Ad1/2 and Ad3 have one branching area in common (branch 1 of Ad3). Interestingly, the antagonistic Ad1/2 and Ab1 resemble each other in their general morphology. Stick insects lack the adductor twin pair that is present in crickets (Dürre *et al.*, 2001).

The resulting overall picture suggests that functionally related motoneurons often share projection areas, and that the projection areas of functionally different motoneurons can be displaced along one axis in crickets. Individual motoneurons within a functional group, however, are nevertheless characterised by unique dendritic branches. Thus, crickets and stick insects on the one hand, and bees, *Manduca* and possibly ants on the other, seem to differ in the 'individuality' of their antennal motoneurons. Whether this is due to specific differences in the projection patterns of antennal mechanosensory afferents and interneurons is unknown. Further possible explanations may be provided by differences in the general structure of the dorsal deutocerebrum. Bees, for instance, have a very well-marked DL with a compact structure (Suzuki, 1975), that may concentrate dendritic ramifications to a larger extent than in crickets and, possibly, stick insects, which have a less well defined and more diffuse dorsal deutocerebral neuropil (Honegger *et al.*, 1990a).

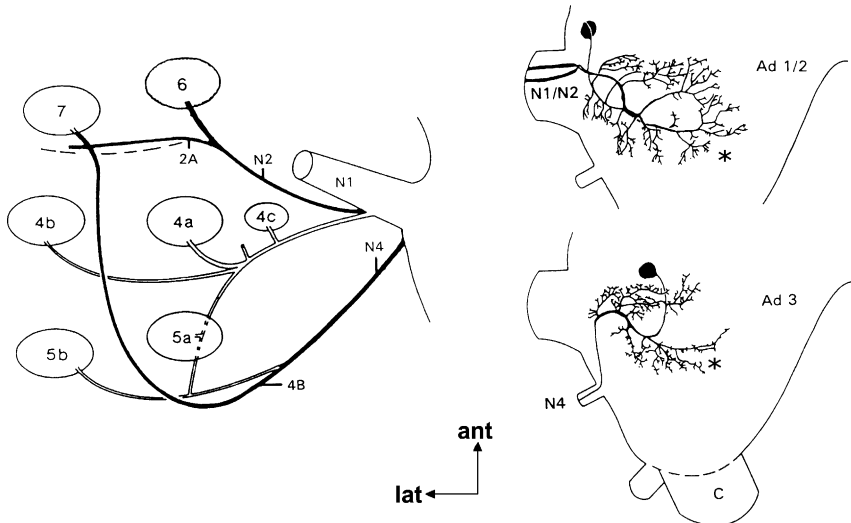


FIG. 13 Motoneurons innervating the adductor muscle M6 of the cricket (*Gryllus* sp.). Left: schematic layout of antennal muscles and nerves. 4a, b, c, levators; 5a, b, depressors; 6, adductor; 7, abductor; N1, antennal nerve; stippled line: scape-pedicle border. Right: the morphological twins Ad1/2 project to M6 via N2. Ad3 innervates M6 via N4B and N2A through the anastomosis between N2 and N4. c, circumoesophageal connective. Asterisks mark dendrites ramifying in the same brain area. After Honegger *et al.* (1995) with permission of the authors.

5 Central neurophysiology of antennal mechanoreception

The aim of this section is to provide an overview of the central neurophysiology of the processing of antennal mechanosensory information in insects. Section 5.1 will cover the local processing of antennal mechanosensory information; Section 5.2 concentrates on the physiology of descending brain interneurons, which convey antennal mechanosensory information to the ventral nerve cord.

5.1 PROCESSING OF ANTENNAL MECHANOSENSORY INFORMATION BY LOCAL BRAIN INTERNEURONS

It has to be expected from the organisation of the circuitry for generating and controlling leg movements in insects that local interneurons, intrinsic to the deutocerebrum, play a major role in the primary integration of sensory information from the body appendages. The emerging picture is that presynaptic inhibition of sensory neurons, interactions of excitation and inhibition in local networks, the involvement of non-spiking interneurons and reflex reversal shape the final motor output (e.g. Burrows, 1992; Bässler, 1993; Wolf and

Laurent, 1994; Wolf and Burrows, 1995; Bässler and Büschges, 1998; Büschges and El Manira, 1998). The knowledge of the antennal sensory-motor circuitry has not yet grown to an extent that allows us to draw a similarly detailed picture for the antennal control circuits.

Local brain interneurons have been characterised in crickets, which process antennal proprioceptive information and visual information from the compound eyes (Gebhardt, 1996). These local interneurons belong to at least seven morphological types, some of which are intrinsic to one deutocerebrum (Fig. 14B) and others of which extend into other brain neuropils, like the contralateral deutocerebrum (Fig. 14C), the tritocerebrum (Fig. 14D), the protocerebrum (Fig. 14E–G) or into the optical neuropils (Fig. 14H). The vast majority of them are unimodally driven by passive deflection of the SP-joint in an angular range between 0° and 100° (0° representing the body long axis). The responses of the interneurons depend on the angular velocities of deflections, their directions and on the angular position of the SP-joint. Some interneurons exhibit a high specificity of their responses to certain joint angles, e.g. the interneuron shown in Fig. 15, which responds best with phasic bursts of spikes to abductions between 0° and 13°, i.e. slightly lateral to the body long axis (Gebhardt, 1996).

Another interneuron that directly connects visual neuropils to the AMMC, as found in bees (Maronde, 1991), responds best to forced antennal adductions in the range between 0° and 60° and to wide-field “light on”-stimuli (Gebhardt, 1996). Local spiking and non-spiking interneurons, which are able to drive movements of the SP-joint upon current injection, reside in the dorsal-most area of the AMMC (Gebhardt, 1996). Thus, there is evidence that antennal proprioceptive information is processed diversely in the AMMC: on the one hand, some local interneurons simply convey information from antennal sensory neurons, as estimated from the coding properties of scapal chordotonal organ sensory neurons (Allgäuer, 1989). On the other hand, some local and intersegmental interneurons are precisely tuned to a very narrow range of movement parameters, thus indicating very selective proprioceptive input and/or extensive computations of proprioceptive information in the deutocerebrum.

Boeckh *et al.* (1984) summarise data on the convergence of olfactory, mechanosensory and thermal information from the antenna to deutocerebral, local and projection interneurons in the cockroach *Periplaneta* (O + M neurons). As displacement of the antennal base does not drive O + M neurons, the sources of the mechanosensory response are probably flagellar campaniform sensilla which are activated by relative movements between flagellar segments (Ernst and Boeckh, 1983). Zeiner and Tichy (1998) demonstrate that ~40% of local and projection interneurons sampled in the central AL, are responsive to deflection of the flagellum and to olfactory stimuli (lemon oil). The interneurons either respond to sinusoidal flagellar deflections in a cycle-to-cycle manner or display habituating phasic-tonic responses to a whole stimulus train. Interactions

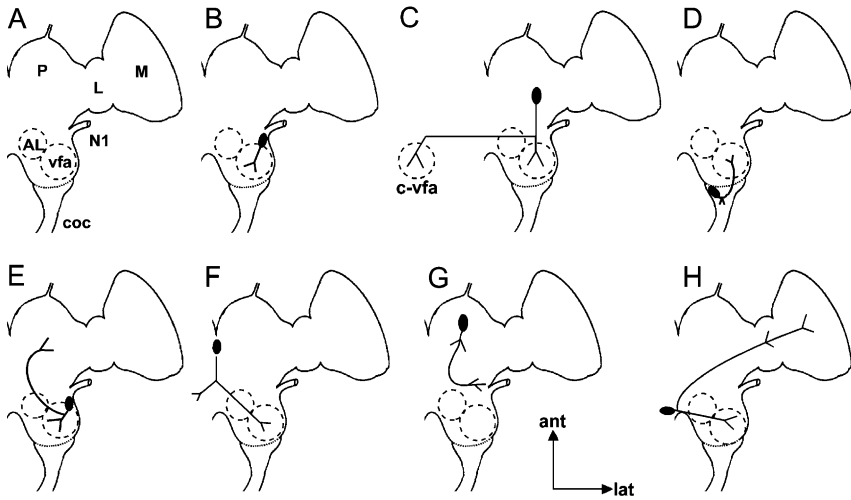


FIG. 14 Sketch of morphological types of antennal mechanosensory interneurons in the cricket brain. (A) Schematic drawing of one brain hemisphere: P, protocerebrum; L, lobula; M, medulla; AL, antennal lobe; vfa, ventral area of flagellar afferents; c-vfa: contralateral vfa; coc, circumoesophageal connective. Diagrammatic representation of the morphologies of different types of mechanosensory brain interneurons: local interneurons of the deutocerebrum (B); interneurons, which connect the deutocerebra of both hemispheres (C); tritocerebral interneurons with arborisations in the ipsilateral deutocerebrum (D); interneurons with somata and dendrites in the deutocerebrum and an axon ascending into the ipsilateral lateral protocerebrum (E); protocerebral interneurons with arborisations in one deutocerebrum (F); local interneurons of the protocerebrum (G); interneurons, which connect the deutocerebrum with the ipsilateral optical neuropils (H).

between mechanical and olfactory stimuli in the neuron population tested were observed frequently (61%). In the majority of neurons, the interaction consists of a response enhancement, in the others of a response depression.

Homborg (1984) and Schildberger (1984) report on extrinsic mushroom body interneurons in the honeybee and in the cricket *Acheta*, respectively. These local brain interneurons respond to combinations of olfactory, visual, acoustical, gustatory and antennal mechanosensory stimuli in a complex way. In the honeybee, one antenno-cerebral tract interneuron, which represents a commonly encountered response type, is inhibited by different scents, with a trailing pronounced rebound excitation, but is strongly excited by antennal mechanical stimulation (touching the antenna with a glass probe). Other antenno-cerebral interneurons are not responsive to mechanical stimulation of the antenna. At the level of extrinsic α -lobe interneurons, visual responses join in, and antennal mechanosensory responses (touch of the ipsilateral antenna) are preserved at least in one out of six interneurons tested (Homborg, 1984).

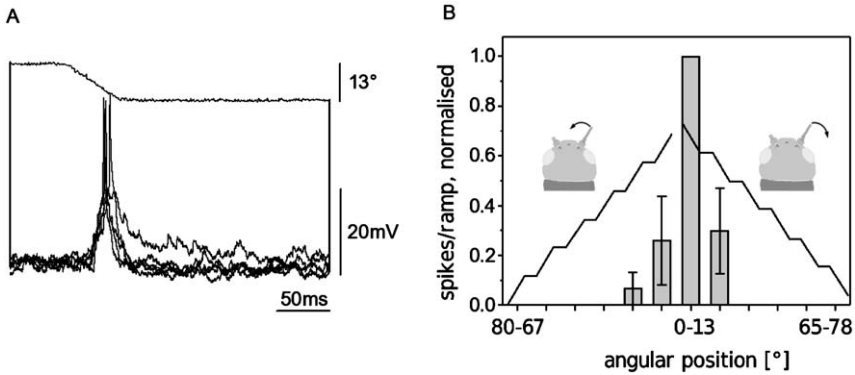


FIG. 15 (A) Intrinsic deutocerebral interneuron responds to forced deflections (upper trace, stimulus monitor) of the SP-joint with EPSPs and spikes (lower trace, five superimposed sweeps of the intracellular recording). (B) The maximum response (normalised spike count per ramp) occurs selectively for abductions slightly lateral to 0° (= body long axis). Note, there is a small offset of 4° at the turning points of the movement. Insets illustrate the direction of the forced antennal movements. (Data from Gebhardt, 1996.)

The response patterns encountered in crickets are also complex. Schildberger (1984) notes that interneurons may change their response pattern during an experiment. Nevertheless, antennal mechanosensory responses are dominant in local projection neurons and mushroom body lobe neurons, with approximately 86% and 68%, respectively, responding to odourless air puffs to the antenna.

Two bilateral, serotonin-immunoreactive interneurons, which ramify in various neuropils including the mushroom bodies, are assumed to provide feedback from the protocerebrum to the AL (*Periplaneta*: Salecker and Distler, 1990; *Manduca*: Sun *et al.*, 1993). These neurons were recorded by Hill *et al.* (2002) in the moth *Bombyx mori*. Air puffs to the antennae carrying odours and air puffs without odour elicit identical excitatory responses in these neurons, which led Hill *et al.* (2002) to the conclusion that the response is of mechanosensory nature.

In an attempt to sample the variety of modalities coded for in extrinsic output neurons of the mushroom bodies of *Periplaneta americana*, Okada *et al.* (1999) grouped recordings from neurons in the ventral pedunculus and β -lobe into sensory, movement-related and sensory-motor units. There is no clear difference in the distribution among these groups. The majority of mechanosensory extrinsic neurons in the pedunculus are multisegmental in that they received inputs from various parts of the body. For example, Okada *et al.* (1999) present recordings from a mechanosensory unit that is activated by tactile stimulation of the ipsilateral antenna and legs, but not of contralateral appendages. Multimodal sensory units were common. In general, the activity

of most mechanosensory units was related to tactile or air-current stimuli to the antennae. Movement-related units appeared to be specialised to various aspects of movement, such as onset or direction of movement, or the movement of the front legs. Often, these interneurons are not activated by proprioceptors of the respective limbs, but are thought to be related to the generation or control of a movement command. Okada *et al.* (1999) also describe a sensory-motor unit recorded at the pedunculus-lobes junction of the mushroom body that is activated by tactile stimuli to both antennae and by movement of the ipsilateral antenna, thus integrating exteroceptive and proprioceptive information.

These findings suggest that antennal mechanosensory information received in the deutocerebrum from antennal sensory neurons is not only unimodally processed in the AMMC, but also combined with other modalities (olfaction) in first-order interneurons (projection interneurons). The multimodal responses of mushroom body output interneurons indicate that antennal mechanosensory information is also preserved in higher brain centres. Its pronounced representation in interneurons of *Acheta* (Schildberger, 1984) emphasises its significance. Since local interneurons of the cricket AMMC never responded to olfactory stimuli (Gebhardt, 1996), and since tactile stimulation of the flagellum or air puffs to one antenna were used in many studies on insect olfaction, one can hypothesise that proprioceptive information from the antennal base is processed separately from tactile information from the flagellum. The latter appears to converge with olfactory information at an early stage of processing. Staudacher (1998a,b) and Staudacher and Schildberger (1999) suggest that flagellar sensory neurons project to the neuropil VFA separate from the dorsal deutocerebral neuropils in the cricket, which might support this hypothesis.

5.2 DESCENDING ANTENNAL MECHANOSENSORY INTERNEURONS

The mechanosensory information sampled by antennal mechanosensory neurons has to be distributed to motor centres in the ventral nerve cord to adaptively adjust the behaviour of the animal. Approximately 200–230 descending brain interneurons have been found in the cricket (Staudacher, 1998) and in the cockroach (Okada *et al.*, 2003). Some of these have been characterised with respect to their sensory physiology and their influence on locomotion (e.g. cockroach: Burdohan and Comer, 1990, 1996; cricket: Bentley, 1977; Boyan and Williams, 1981; Hörner and Gras, 1985; Richard *et al.*, 1985; Böhm and Schildberger, 1992; Gras and Hörner, 1992; Hedwig, 1996; Staudacher and Schildberger, 1998; Gebhardt and Honegger, 2001; Staudacher, 2001; locust and grasshopper: Rowell and O'Shea, 1976a,b; Bacon and Tyrer, 1978; Möhl and Bacon, 1983; Rowell and Reichert, 1986; Hensler and Rowell, 1990; Hensler, 1992; Hedwig, 1994; *Manduca*: Milde and Strausfeld, 1990; Kanzaki *et al.*, 1991). A recent review discusses the behavioural significance of

descending brain interneurons with a focus on the control of stridulation, walking and flight (Heinrich, 2002).

In the studies cited above and in related work, much attention has been paid to the analysis of the visual sense and its behavioural impact. However, antennal mechanosensory inputs to a descending brain interneuron have been described in the context of flight control. In locusts, the tritocerebral commissure giant (TCG) interneuron receives inputs from five bilateral wind-sensitive hair fields, from the antenna and the compound eye (Bacon and Tyrer, 1978). Bacon (1980) describes how the relative weight of these inputs is shifted in three different insects. The TCG of locusts receives strongest inputs from the numerous head hairs, but weak antennal inputs. In *Gryllus campestris*, which has many fewer head hairs, but long antennae, no head hair inputs to the TCG are present and strongest inputs arise from the antenna (bending the flagellum), similar to the mantid *Sphodromantis lineola*, the TCG of which responds weakly to flagellar stimulation. Interestingly, the TCG of all three species displays a conserved directionality with best responses for downward deflections of the antenna. The output connections of the TCG and, hence, its flight control function remains unaffected by the shifting sensory inputs.

An antennal mechanosensory pathway mediates escape responses in cockroaches (Comer *et al.*, 1988; Burdohan and Comer, 1990; Keegan and Comer, 1993; see Section 6.1.3.3). Extracellular recordings from one neck connective reveal a correlation between the increased spiking activity in the connective contralateral to the stimulated antenna and the direction of the escape turn. Using intracellular recordings and stainings, two descending interneurons have been identified, which contribute to this increased spiking activity (Burdohan and Comer, 1990, 1996; Ye and Comer, 1996). One of them, DM1a-1, is a contralaterally descending brain interneuron, which extends at least as far as the first abdominal ganglion. DM1a-1 responds to mechanical stimulation of the flagellum ipsilateral to the cell body. The response consists of a phasic burst of action potential with a latency of 6–8 ms. The action potential travel down the axon of DM1a-1 with a conduction velocity of 4.6 ms^{-1} . An analogous descending interneuron has recently been described in the cricket (Mathenia *et al.*, 2003). A second large interneuron, DM1b-1, has its cell body in the SOG and also extends at least to the first abdominal ganglion. Mechanical stimuli to either antenna give rise to brief phasic responses with latencies of ~ 8 ms (antenna ipsilateral to the soma) and 10 ms (ipsilateral antenna) in DM1b-1. Its conduction velocity is 4.7 ms^{-1} . The receptors feeding into these descending interneurons appear to be located in the antennal base, as immobilisation of the antennal base abolishes spiking, whereas ablation of the flagellum has no effects (Comer *et al.*, 2003). The sensitivity of both neurons to flagellar displacement is high: deflection by 0.05 and 0.5 mm reliably triggers spikes in DM1a-1 and DM1b-1, respectively. Their sensitivity for wind stimuli, however, is rather weak compared to the sensitivity of the cercal system. DM1a-1 displays a relatively weak directionality which is slightly emphasised for lateral

stimuli, whereas DMib-1 has a much more pronounced directionality for stimuli from the front for both antennae. Thus, the two DMI neurons constitute a rostral escape circuit which detects tactile stimuli and, therefore, complements the cercal system (see Section 6.1.3.3). To address the question, which of the parameters of the activity of the descending interneurons correlates with the turning direction and the turning angle, Ye and Comer (1996) performed extracellular neck connective recordings in tethered cockroaches walking in an open-loop paradigm. They found that the turning direction can be predicted from the latency difference for the first spike, and less reliably, from the spike count difference between both connectives. Cockroaches turn away from the touched antenna, towards the side of the first DMI spike, and towards the side of higher neuronal activity. The amplitude of the turn, however, co-varies with the spike count difference alone. Accordingly, when electrically stimulating one neck connective, turns are directed towards the side stimulated and the turning angle could be increased by either increasing the stimulus voltage or by increasing the pulse frequency in trains of stimulus pulses. Hence, a thoracic comparator seems to analyse the bilateral pattern of neuronal activity, evaluating latency and spike count differences differentially for the control of turning direction and turning amplitude.

Several studies deal with descending brain interneurons in crickets, but their focus is on sensory modalities other than antennal mechanoreception: they concern auditory (Boyan and Williams, 1981), visual (Richard *et al.*, 1985), wind-evoked escape (Hörner, 1992) or the effect of various sensory stimuli on walking control (Hörner and Gras, 1985).

Hörner and Gras (1985) have recorded from local and descending brain interneurons of *Acheta*, while presenting a variety of sensory stimuli to the crickets. The interneurons recorded belong to different morphological types with branchings in various brain neuropils except the mushroom bodies. Responses to mechanical stimulation of the antenna (wind streams with 0.8 ms^{-1}) are found in 25% of the interneurons sampled. Other effective modalities which elicit responses in single interneurons, are mechanical stimulation of the cerci (41%), visual stimuli (19%) and acoustical stimuli (15%). Antennal responses appear to be widely present in a descending multimodal interneuron set, indicating the behavioural relevance of antennal mechanosensory information.

Böhm and Schildberger (1992) report on different descending brain interneurons and their responses to visual, acoustical, antennal and cercal (air puffs) stimuli and their involvement in the control of walking. They depict two descending interneurons, which apparently have dendrites in or, at least close to, the deutocerebrum. One of them, an ipsilaterally descending interneuron has a dendrite reaching into the lateral protocerebrum and another projecting towards the deutocerebrum. Despite this pattern of branching, this interneuron responds, only to omit acoustic stimuli. Another interneuron with a dendrite close to the deutocerebrum and a contralaterally descending axon also

responds unimodally to directional visual stimuli. Whether the absence of antennal responses in both interneurons is due to their dendrites located outside the mechanosensory portion of the deutocerebrum, remains unknown.

Two ipsilaterally and six contralaterally descending brain interneurons with a dendrite in the mechanosensory deutocerebrum have been characterised by Gebhardt (1996), Staudacher (1998a,b) and Staudacher and Schildberger (1999), five of which have been recorded intracellularly by Gebhardt and Honegger (2001). The ipsilaterally descending interneurons project at least to the first abdominal ganglion. All of these interneurons respond phasically to forced deflections of the SP-joint; one (DBNi1–2, nomenclature according to Staudacher, 1998b) is also driven by visual inputs from the ipsilateral compound eye. The antennal mechanosensory responses of all these interneurons are directionally selective, e.g. with DBNi1–2 responding best to deflections between 30° and 60° lateral (0° being the body long axis), whereas other interneurons respond best close to lateral or medial extreme positions of deflections. Thus, it appears that these interneurons represent the total angular range of SP-joint movements in a fractionated manner. Owing to only fast deflection, but never static SP-joint posture, elicit spikes, these interneurons might serve the detection and localisation of external objects ahead of the cricket. For such a task, a mechanism must exist to discriminate between excitation resulting from external stimuli or from self-induced stimulation, i.e. active antennal movements. Such mechanisms were proposed by Holst and Mittelstaedt (1950; the ‘re-afference principle’) and by Sperry (1950; the ‘corollary discharge’). Poulet and Hedwig (2002, 2003) have recently shown that a corollary discharge mechanism modulates the auditory Ω -interneuron in the prothoracic ganglion of crickets. Similarly, the synaptic inputs to DBNi1-2 are suppressed in a gradual manner depending on the strength of an antennal motor command. This suppression is of a central nervous origin, since it persists in a preparation in which one antenna was rendered motionless and, thus, peripheral feedback loops were not active. It can be hypothesised that the antennal mechanosensory information conveyed to the posterior ventral cord is filtered to remove the self-induced response component. The resulting, so far unknown, behaviour modulated by the descending interneuron would then be influenced by external stimuli only.

6 Behavioural physiology of the antennal tactile sense

Having reviewed the neurophysiology of antennal mechanoreceptors and corresponding neuropils in the insect brain, the following section gives an overview of the various behavioural functions of antennal mechanoreception.

It is hard to draw a line between mechanical stimuli in general and tactile stimuli in particular. One reason for this is that, in spite of the neuroanatomical description of different neural tracts (see Section 4.1), it is still difficult to

determine the extent to which proprioceptive and exteroceptive information is being processed separately in the deutocerebrum. Another reason is that tactile contact information is typically accompanied by information about active or passive movement of the antenna. As a result, mechanoreceptor signals other than those associated with the actual object contact, will qualify the information in terms of the behavioural context in which antennal contact has been made. Thus, from a behavioural perspective, proprioceptive and exteroceptive cues should not be separated.

The following sections will deal with a wide range of behavioural situations and consider many different submodalities of mechanoreception. Some of the examples, e.g. the role of the antennae in flight or hearing, will appear to be not of primary concern to tactile sensing, either because the behavioural situation excludes tactile contacts (as in flight), or because no contacts are necessary (as in hearing). However, because antennae are involved in several aspects of steering in locomotion, some of which involve tactile contacts and some of which do not, a comparison of these behavioural contexts can shed light on general control mechanisms. Similarly, hearing in honeybees involves mechanoreceptors that are most likely stimulated in behavioural contexts where tactile information is relevant. Moreover, honeybee hearing is known to be of particular importance in dance communication, a behavioural situation in which the animal is also likely to make tactile contact. Finally, tracking and orienting movements of antennae will be treated in cases that are likely to increase the likelihood of tactile contact, although they must not necessarily lead to tactile contact.

In the following sections, behavioural functions of the antennae will be grouped according to whether the sensory act is typically characterised by passive displacement (Section 6.1), or rather by active antennal movements (Section 6.2).

6.1 PASSIVE SENSING

Behavioural contexts in which antennal mechanoreception is characterised by passive sensing concern the role of the antennae in the sense of gravity (Section 6.1.1), reflex actions that remain restricted to the antennal joints (Section 6.1.2), and primarily passive sensory functions in locomotion, as in steering and escape reactions (Section 6.1.3).

6.1.1 *Graviception*

Unlike decapod crustaceans, which possess a statocyst in the basal segment of the antennule (e.g. Hennig, 1986), insects have no dedicated gravity sensor. Rather they infer their body orientation relative to gravity from proprioceptors of the limbs or of the body segments. The contribution of antennal mechanoreception in sensing the gravity vector has been demonstrated for

behavioural contexts of gaze stabilisation in crickets, and negative geotaxis in stick insects.

The cricket *Gryllus bimaculatus* uses sensory cues from antennae and cerci to sense the roll angle around the body long axis relative to an upright posture. This can be investigated in stationarily walking crickets that are held by a support, and rotated together with the substrate on which they walk. If the substrate (e.g. a ball) is held by a bearing, only the appendages can signal a deviation from gravity. In this situation, passive roll by $\pm 90^\circ$ causes the walking cricket to counter-roll its head by some $\pm 10^\circ$ in males, and by $\pm 15^\circ$ in females (Horn and Bischof, 1983). Walking is a necessary condition for this response, indicating context-dependent gating of the neural pathway.

Antennectomy at the level of the mid-scape reduces the compensatory head roll to $\pm 5^\circ$, with no sex difference remaining. Additional removal of the cerci abolishes the response completely. Ablation of a single antenna or of a single cercus is sufficient to reduce compensatory head roll, and various combinations of cut antennae and cut cerci reveal that the impact of the cerci is larger than that of the antennae. Also, the impact of any combination of removed appendages is smaller than the linear sum of the individual effects (Horn and Föller, 1985).

If crickets are rotated while the substrate is not held by a bearing, there is additional stimulation of leg proprioceptors. In this situation, the head counter-roll angle increases 2.5- to 3-fold compared to the ball-on-bearing situation. As this response is stronger than that elicited by the leg proprioceptors alone, antennal and cercal graviceptive information appear to facilitate the effect of leg proprioceptors (Horn and Föller, 1985).

The source of antennal proprioceptive information that elicits the head counter-roll response is unknown. A first potential source could be that of hair plates at the SP-joint signalling passive horizontal deflection of the pedicel. In this case, antennectomy at the scape would preclude the ability to detect the deviation from the horizontal plane. Indeed, unilateral antennectomy has little effect on crickets that walk on a horizontal surface (Horn and Föller, 1985), a result that has been used to show that sensory cues from the antennae and cerci cause no tonic head counter-roll. However, if the SP-joint was involved in graviception, a putative tonic effect would be zero when walking on a horizontal plane. A second potential source of information could be proprioceptors of the HS-joint that detect passive vertical deflection of the scape, i.e. a deviation from the vertical plane. In this case, antennectomy causes a reduction of antennal mass and of the corresponding static downward force acting on the HS-joint.

The stick insect *Carausius morosus* shows negative geotaxis, an orientation behaviour that mainly depends on leg proprioceptors (Bässler, 1971) but involves supplementary information of the antennae. When walking on a vertical plane in the dark, stick insects normally tend to walk upward. If the leg proprioceptors are intact, antennectomy has a negligible effect on body

orientation. However, if the animal's own weight is compensated for by an upward-pulling force, intact animals perform equally well as before, but animals without antennae no longer show geotactic behaviour (Wendler, 1965). Using a similar approach, but recording the preferred body orientation at rest, Bässler (1971) demonstrated that the main sensors of gravity in the stick insect are the hair plates of the thorax-coxa joints of the legs. Again, antennectomy of otherwise intact animals does not affect the magnitude of the orientation angle relative to gravity, but increases the number of angles with reversed sign, i.e. animals facing downwards. In animals with shaved coxal hair plates, the antennae are sufficient to signal the correct sign, so animals preferably face upward. Applying an upward-pulling force to reverse the gravitational force vector sensed by leg proprioceptors without affecting the orientation sensed by the antennae, the number of positive geotactic responses increases in intact animals. It is only after antennectomy that negative geotaxis becomes apparent again, i.e. with animals now facing downward. Bässler (1971) concludes that the hair plates of the thorax-coxa joint determine the magnitude of the orientation angle relative to gravity, while the antennae only indicate the sign of gravity. Furthermore, the impact of the antennae on the sign must be larger than that of the leg proprioceptors.

As in the head counter-roll response of crickets, the source of antennal sensory information in stick insect geotaxis is still unknown. What is interesting about the mentioned studies is that both crickets and stick insects typically move their antennae during walking (see Section 6.2.1). To what extent active antennal movement interferes with or is important for the antennal role in graviception remains to be investigated.

6.1.2 *Avoidance, assistance and resistance reflexes*

Imposing passive movement on antennal joints, touching the flagellum at rest, or resisting antennal movements with a stop causes reflex activity in antennal muscles. Depending on the effect of the motor pattern in relation to the stimulus, these reflexes act to avoid contact, assist or resist imposed movement. Reflexes are integral parts of all behaviours that involve antennal movements, and it is clear that they are not due to unmodifiable, hardwired feedback circuits, but rather involve both central and sensory modulation. In general, antennal reflexes bear many similarities to leg reflexes, including state-dependent reflex reversal (e.g. see Bässler and Büschges, 1998).

6.1.2.1 *Antennal reflexes in insects.*

Locusts (*Schistocerca gregaria*) respond to passive or active contact between their flagellum and an object either by moving the antenna away from the object, or by a sequence of abduction and adduction movements that can last for several seconds (Saager and Gewecke, 1989). The latter sequence has been called a reflex chain, but because the trigger for the transition from the avoidance reflex to the subsequent resistance reflex

is unknown, it may well be an active exploratory movement sequence that requires more central control by the brain than would be expected in case of a reflex chain. In fact similar spontaneous oscillations, which the authors observed in “very active” animals, suggest that it may be a centrally patterned movement that does not rely on sensory information. Finally, because the movements reported by Saager and Gewecke (1989) were restrained to the SP-joint, possible interactions with the HS-joint that could be involved in exploratory movements were prevented.

The avoidance reflex can be elicited by flagellar deflections as small as 0.05° . In a standing animal, the movement of one antenna in *Schistocerca* is not neurally coupled to that of the other. This is not so during flight, when the antennae assume a symmetrical posture, pointing medio-laterally. During flight, the response to passive deflection of an antenna turns into a resistance reflex of that antenna. Moreover, “mediad deflection of one antenna with a needle causes lateral movement of the other” (Saager and Gewecke, 1989, p. 525). The reflex reversal can be elicited by a frontal air-current that triggers abduction of the flagellum, much like the tactile avoidance response, which turns into an active adduction as soon as the flight muscles are activated. During flight, co-activation of adductor and abductor muscles maintains a stable posture of the antenna (within 1° deviation). Sinusoidal deflection of the flagellum in this behavioural context always elicits a resistance reflex that acts to stabilise the posture by increasing the force as expected from a negative feedback loop. After switching off the air current, the animal stops flying, whereupon the antennal reflex reverses back to an avoidance reflex. Saager and Gewecke (1989) further demonstrate that fixing the SP-joint, thus eliminating the sensory input from the hair plates at this joint, abolishes the tonic component of the scape muscle electromyogram (EMG). In this case, only phasic avoidance reflex activity remains. On the other hand, by immobilisation of the PF-junction, i.e. reducing the stimulation of pedicellar campaniform sensilla, the tonic EMG component is increased. This can be attributed to the resistance reflex.

Cockroaches of the genera *Periplaneta* and *Blattella* show a tactile antennal avoidance reflex if the antenna is touched with a dry glass needle at more than 1 cm away from the head. At more proximal contact points, the insects move the maxillary palps towards the source of stimulation (Frings and Frings, 1949; cf. Cornwell, 1968, p. 114). Since Comer *et al.* (1994) use very similar stimuli to elicit escape behaviour in cockroaches, it has to be noted that such manual stimulus presentations can vary considerably, even if the same tool is used for stimulation. On the basis of observations on the cockroach *Periplaneta americana*, Okada *et al.* (2002) suggest that the HS-joint hair plates subserve a direction-selective negative feedback loop that pulls the HS-joint towards a median position around 40° . This interpretation is supported by two findings following ablation of the HS-joint hair plates: first, the effective workspace of the flagellum widens (-40 to 130° vs. -50 to 150°) and abduction–adduction

cycles lengthen. Second, far lateral contacts during adduction of the antennae are more often followed by abduction than before. Object contacts during abduction are not affected, however. According to the hypothesis of Okada *et al.* (2002), the tendency to adduct increases with increasing abduction angle, and an avoidance reflex in response to object contacts with far lateral objects is overridden by a resistance reflex.

6.1.2.2 Reflex types in crustaceans. Studies on rock lobsters and crayfish have contributed much insight into the sensory-motor link between antennal proprioception and reflex movements. In the rock lobster *Palinurus vulgaris*, mechanical stimulation of the MCF-chordotonal organ can elicit both a resistance- and an assistance reflex (Clarac and Vedel, 1975). Induced reflex activity is restricted to the four dedicated extensor/flexor motoneurons, to the accessory extensor motoneuron and to the common inhibitor. The common extensor and common flexor motoneuron as well as the phasic CF-extensor motoneuron do not contribute (Vedel, 1980). During a resistance reflex, the tonic motoneurons that would assist the imposed movement stop firing. Extensor reflex activity to imposed flexion is velocity-dependent, whereas flexor reflex activity to imposed extension is not. Passive movement of the CF-joint causes reflex activity also in the muscles of the neighbouring MC-joint, but this inter-joint effect is weaker than the intra-joint effect on the CF-musculature.

Assistance reflex activity occurs in extensor motoneurons when the animal becomes spontaneously active. In this case, extension of the flagellum elicits extensor activity and a corresponding decrease in the antagonistic flexor activity. The result is a positive feedback action upon extension. The assistance reflex habituates and its gain is strongly velocity-dependent. Imposed extension also causes velocity-dependent reflex activity in the common inhibitor. Furthermore, assistance reflex activity causes history dependence in that a subsequent resistance reflex to imposed flexion is no longer velocity-dependent. Vedel (1980) suggested that the behavioural role of the extension assistance reflex is to support voluntary extension of the flagellum during the threat and defence behaviour of *Palinurus*.

In the lobster *Homarus americanus*, passive flexion at the MC- or CF-joint causes a resistance reflex involving the two dedicated extensor motoneurons and one of the common extensor motoneurons (Sigvardt, 1977). Conversely, extension of these joints causes activation of the dedicated flexor motoneurons. Accordingly, passive flexion and extension of the CF-joint elicits a resistance reflex in the crayfish *Cherax destructor* (Sandeman, 1989). As in *Palinurus*, these reflexes are mediated by the bi-articular MCF chordotonal organ.

Tactile hairs on the flagellum of the crayfish *C. destructor* subserve an avoidance reflex. Brushing the medial surface of the flagellum elicits inhibitory potentials in the tonic extensor motoneuron, whereas brushing the lateral surface causes an increase of the spike rate in the same neuron (Sandeman, 1989). As extension of the flagellum is equivalent to movement towards the medial

direction, the resulting movement is directed away from the stimulus. Bending of the flagellum of *C. destructor* has a pronounced effect on tonic motoneurons whose activity leads to the relief of the imposed tension. With regard to movement direction of the flagellum, this is an assistance reflex, but the strain sensors are part of a negative feedback loop.

6.1.2.3 Reflex modulation in crustaceans. All studies on crustacean antennal reflexes have emphasised the presence of modulatory effects. In *P. vulgaris*, the motoneuron activity associated with the resistance reflex is dependent on ambient temperature and on ascending influences from the uropods and pereopods (Clarac and Vedel, 1975). In crayfish (*Euastacus armatus*), reflex-like extension of the antennae occurs in response to tactile stimulation of the leg bases; to passive rotation of the body; or as part of the escape behaviour (Sandeman and Wilkens, 1983). In *Euastacus*, flexion of the antenna occurs in response to tactile stimulation of the uropods or telson. Also, motoneuron activity associated with the resistance reflex can be modulated by electrical stimulation of almost all ipsilateral brain nerves and of the contralateral optic nerve and connective, indicating the complexity of the neural circuits involved. Tactile stimulation of the branchiostegite enhances the resistance reflex by facilitation of motoneuron activity. The reverse effect occurs during voluntary leg movements. These are accompanied by a reduction of tonic motoneuron activity that is typically associated with the resistance reflex, and by a reduction of co-activation of flexor and extensor motoneurons (Sandeman and Wilkens, 1983).

6.1.3 Steering of locomotion

Several insect orders employ their antennae in orientation and steering behaviours, ranging from wind-directed orientation (anemotaxis or anemomenotaxis) in airborne (Section 6.1.3.1) and terrestrial locomotion, to tactile course control during running (Section 6.1.3.2). Furthermore, tactile contacts and, possibly, air-current-induced deflections of the flagellum can trigger fast turning movements in escape or defence behaviour (Section 6.1.3.3).

6.1.3.1 Course control during flight. The use of antennae in control of flight speed is not restricted to insect orders with highly specialised antennae, such as flies, but is also well documented in locusts and bees. For example, tethered flying honeybees decrease their wing beat amplitude in response to a frontal air current. Antennectomy, or preventing the air current from reaching the antennae abolishes this response (Heran, 1957). At an air speed of 7 m s^{-1} , the antennae cause a decrease of wing beat amplitude by 35° , compared to 10° in the antennectomised animal (Heran, 1959). In free flying bees, antennectomy prevents light-dependent reduction of flight speed that can be observed in bees flying out of a dark room towards a bright window.

When an air current blows towards the head of a tethered flying bee, both antennae move inward, requiring an adduction and an outward roll in the HS-joint, but also a flexion of the SP-joint (Heran, 1959). Both of these components increase with increasing speed of the air current. Various immobilisation and ablation experiments show that deflection of the flagellum is the adequate stimulus, and that the movements of each antenna are elicited by receptors located in the ipsilateral pedicel. Moreover, information of one antenna does not cross the midline to affect the posture of the contralateral antenna, in spite of the presence of a neural connection via the dorsal SOG (Maronde, 1991). Heran (1959) further suggested that it is the flagellar vibration at the PF-junction that is decisive, rather than its static deflection. Finally, because the air flow experienced during a yaw turn of the flying bee causes an outward deflection of the outer antenna, which in turn leads to a decrease in the wing beat amplitude, Heran (1959) proposed that antennal proprioceptors close a negative feedback control loop that assists course stabilisation during flight.

In flying locusts (*Locusta migratoria*), antennae serve as air-flow sensors that complement a set of wind-sensitive hairs on the head (Gewecke, 1970). At normal flight speeds, active adjustments of the antennal posture in response to air current keep deflections at the PF-junction below 1° (Gewecke and Heinzel, 1980). Mechanoreceptors monitoring this deflection drive postural adjustment of the antenna itself, but also a change in the wing beat pattern. The crucial mechanoreceptors involved in this behaviour are a ring of campaniform sensilla around the distal pedicel, which codes deflection of the flagellum at the PF-junction up to 1.3° (Heinzel and Gewecke, 1979) (see Sections 2.2.2 and 3.2.2).

Reminiscent of the locust flight behaviour, flying crickets (*G. bimaculatus*) also hold their antennae in a forward-directed posture, although the distal flagellum is bent posteriorly, because it is not stiff enough to maintain a straight posture against an air current. To what extent cricket antennae are involved in steering is unknown, but lesion experiments show that ascending pathways from the suboesophageal and thoracic ganglia are necessary to maintain this posture during flight (Horseman *et al.*, 1997).

In flying locusts, antennal sensory cues are also involved in postural adjustments of the hind legs. Tethered flying *Schistocerca gregaria* attempt to fly an upwind curve, by abducting the leg on the side facing the air current, thus increasing the drag on the inner side of the curve (Arbas, 1986). This 'upwind response' is elicited by wind-sensitive hairs on the head. Following occlusion of these hairs, locusts show a 'downwind response', in which they abduct the hind leg on the downwind side. Immobilisation of the antennal joints or antennectomy abolishes leg abduction in response to changes in air-current direction. The behavioural significance of the antagonistic steering tendencies triggered by antennae on the one hand, and wind-sensitive hair fields on the other remains unknown. Because the directionality of the response may be reversed in some animals, a fact that Arbas (1986) attributed to "changes in mood", the

neural pathway from antennae to hind legs appears to be subject to considerable plasticity.

As in locusts, cricket behaviour reveals a relationship between antennal movements and steering movements of the legs. Flying crickets (*Teleogryllus oceanicus*) show an acoustic startle response to ultrasound stimuli that involves abduction and levation of the contralateral hind leg and bilateral abduction of the antennae (Miles *et al.*, 1992). Yet, whereas in locusts the antennal displacement induces the leg movement, it does not do so in *Teleogryllus* because activation of the scapal muscles occurs with similar or even longer latency than activation of leg muscles. Hence, the functional significance of antennal movements in flying crickets is less clear than in flying locusts. Nevertheless, the same neural pathway may be recruited as in Arbas' downwind response.

6.1.3.2 Course control during walking and running. Much as in flying, many insects perform wind-related steering actions during terrestrial locomotion. Linsenmair (1973) lists 38 insect species that show anemomenotactic behaviour, most of which are Coleoptera and Heteroptera. In walking beetles, amputation of a single antenna changes the preferred orientation angle with respect to wind direction (Linsenmair, 1970).

A comparative study on the cockroach species *Blatella germanica*, *Periplaneta americana* and *Blaberus craniifer* shows that spontaneous locomotor behaviour varies considerably in terms of looping and other aspects of path curvature. Yet, the walking paths of all species get increasingly straighter with increasing wind speed (Bell and Kramer, 1979). Nevertheless, the preferred angle with respect to wind direction is not the same for all species. Bilateral antennectomy abolishes anemotactic orientation in *Blaberus*, but unilateral antennectomy or removal of the cerci has no detrimental effect, indicating that the information of at least one intact antenna is sufficient and necessary to elicit this orientation behaviour (Fig. 16). Furthermore, immobilisation of the SP-joint, or of the antennal base beyond the PF-junction revealed that mechanoreceptors of the pedicel (either at the SP-joint or at the PF-junction) are necessary to detect wind direction. It is worth noting, however, that the measures of path straightness and preferred orientation angle, as applied by Bell and Kramer (1979), are insensitive to more subtle changes in locomotor activity. For instance, the trajectories of unilaterally antennectomised *B. craniifer* turned out to be as straight as in intact animals, but the size of the turns and the duration of straight periods in-between was rather different (Fig. 16B, C). Thus, antennal mechanoreceptive information is likely to be part of a course control circuit that involves both antennae.

Further evidence for such a control circuit stems from experiments on running cockroaches of the species *P. americana*. When running on the ground adjacent to vertical walls, cockroaches often keep one antenna in touch with the wall and use the contact information from their antennae for fast course corrections to regulate the distance to the wall (Camhi and Johnson, 1999).

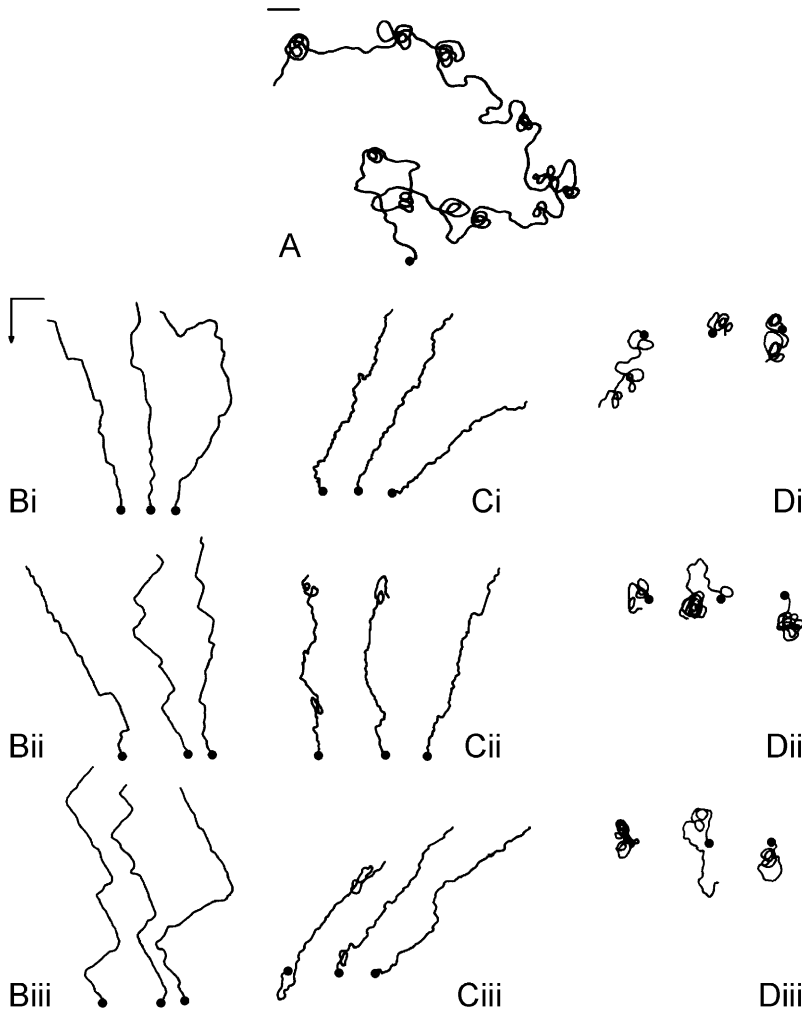


FIG. 16 Antennae are involved in wind orientation of the cockroach *Blaberus craniifer*. (A) Representative walking track of a cockroach (1000 s) in the absence of visual and wind stimuli. The track is curved and interspersed by repetitive loops. Scale bar is 1 m. (Bi–Biii) Representative walking tracks of intact cockroaches (200 s) at different wind speeds (Bi: 0.03 m s^{-1} ; Bii: 0.12 m s^{-1} ; Biii: 0.24 m s^{-1}). Upwind direction is indicated by the arrow. Scale bar is 1 m. The preferred upwind direction of locomotion is indicated by a dot at the end of the path. Walking tracks become straightened in the presence of wind. Conspicuous kinks occur regularly. (Ci–Ciii) Same details as in Bi–Biii, but after unilateral antennectomy. Kinks are less pronounced than in B, and the path is more irregular. (Di–Diii) Same details as in Bi–Biii, but after removal of both antennae. The looping behaviour is similar to the no-wind condition shown in A. Adapted from Bell and Kramer (1979) with permission from Elsevier Publishing.

This effect is dependent on the speed of locomotion, and decreases as the average distance relative to the wall decreases with increasing speed (Fig. 17). When a cockroach runs along a pleated wall, orientation of the body axis can change by 20° (peak to peak) or more at frequencies around 10 cycles s^{-1} (Fig. 17D–G), which is close to one turn per step cycle.

Camhi and Johnson (1999) also found that bend position and contact point along the flagellum may be a good indicator for the distance to the wall, partly because the antennal angle near the base is held rather stable by the cockroach. Accordingly, immobilisation of the antennal base and of the neck does not cause significant changes in contact-mediated turns, whereas loss of flagellar information abolishes turning altogether. Although it is still unknown how a cockroach manages to make the appropriate course corrections on a step-to-step basis, the data of Camhi and Johnson (1999) indicate that the tripod leg coordination pattern is not disturbed during rapid turning. Apparently, antennal contact information causes changes in leg kinematics and/or dynamics, rather than in timing of entire step cycles.

In a similar wall-following paradigm, Cowan *et al.* (2004) measured the time course of steering actions of running cockroaches in response to a step change in sensed heading angle with respect to the wall. Cockroaches alter their heading angle faster in response to a large step in sensed angle than in response to a small step. Because a change in heading angle is always accompanied by a change in lateral distance to the wall, Cowan *et al.* (2004) suggest that cockroaches control a combination of heading angle and lateral distance, a variable they termed ‘tactile flow’ (see also Section 7.1). Apart from the stunningly fast steering abilities of a running cockroach, the question about the biological significance of a wall-following behaviour in insects remains. Possibly, vertical surfaces in a typical cockroach environment tend to lead to shelters or holes.

Other than the mentioned fast course corrections in response to an outward bend of the flagellum, walking cockroaches (*P. americana*) turn towards an object that they touch with their antenna. If the object is presented to a tethered walking animal, it may continue to turn for several seconds. In this case, angular velocity of turning increases with abduction angle of the antenna (Okada and Toh, 2000). Because the turning tendency is strongly reduced after ablation of the hair plates of the HS-joint, these proprioceptors appear to serve as angular sensors in tactile orientation behaviour.

A further role of the antennae in steering, the ability to follow conspecifics, becomes apparent from descriptions of queuing aggregations in spiny lobsters, but also in the so-called tandem runs of ants, a form of recruiting behaviour. For example, Hölldobler (1985) shows a picture of a tandem run by an Australian *Hypoponera* species, where the antennae of the follower touch the sides of the leader’s abdomen. Tandem runs are interpreted as a primitive form of communication among nestmates (see Section 6.2.4.2). The antennal sensory information necessary for establishing a tandem run have been studied in some detail for *Bothroponera tesserinoda*, where it involves both mechanical and

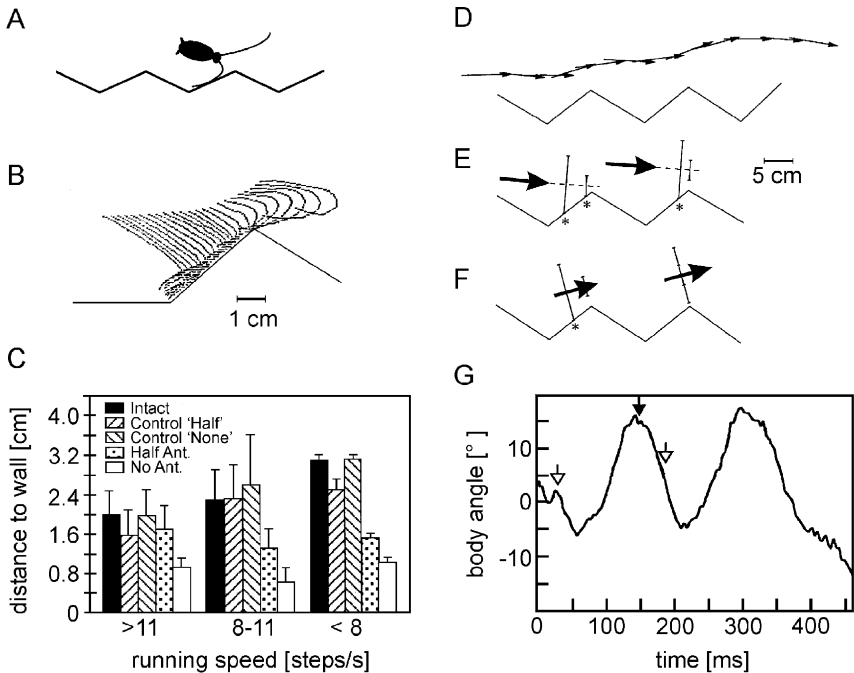


FIG. 17 Tactile wall-following in the cockroach: (A) Cockroaches of the species *P. americana* tend to maintain antennal contact when running along a wall. Observation of body orientation and distance to a pleated wall reveals fast steering responses. (B) Superimposed image sequence of a flagellum as a cockroach runs past a kink of the wall, beginning with first contact with the outward wall projection until 12 ms after it lost contact. Exposures are separated by 4 ms. (C) The distance of the head to the wall depends on the running speed (compare 'Intact') and on the length of the flagellum. Unilateral shortening or complete removal of an antenna results in significant differences between runs, in which the shortened antenna faces the wall, and control runs, in which the intact antenna faces the wall (compare 'Half Ant' with 'Control Half' and 'No Ant' with 'Control None'). All pairs significant except Half Ant/Control for speeds >11 steps/s). Distance is significantly correlated with flagellum length. (D–G) Example run along a pleated wall. (D) Arrows show body axis every 40 ms. (E) If body orientation had been unaltered after the two indicated instances, the cockroach would have touched the wall with a leg (long line segment with *) and its body (short line segment with *). (F) Arrows show body axis 80 ms after the corresponding arrow in E. Owing to fast course corrections, the cockroach only touches the wall once, and only with a leg (*). (G) The time course of body axis angle reveals fast turning after both antennal contacts with the outward projecting wall (white arrows). The black arrow marks the time of leg contact. Adapted from Camhi and Johnson (1999), with permission from the Company of Biologists Ltd.

contact-chemosensory cues (Maschwitz *et al.*, 1974), and in *Camponotus sericeus*, where chemosensory signals are obligatory (Hölldobler *et al.*, 1974). Whereas it seems to be clear that the actual steering, i.e. the close following of the leader's course, involves tactile cues sensed by the follower's antennae, the required sensory structures are still unknown.

In a peculiar form of migration behaviour involving up to thousands of individuals, the spiny lobster *Panulirus argus* forms long queues that rely on tactile contact. As the queue moves, each animal keeps physical contact with the animal in front (Herrnkind, 1969). Contact occurs most often with the inner branch of the antennule (which is homologous to the insect antenna) and with the first pair of pereopods, but also with the antennae. In blinded animals, or in the dark, queue formation is most often initiated by antennal contact. Experiments by Bill and Herrnkind (1976) show that spiny lobsters can greatly reduce the drag opposing their locomotion effort by joining queues (see Section 2.2.2.1).

6.1.3.3 Tactile elicited emergency behaviour. Escape behaviour involves steering, provided that the direction of the escape is correlated to the direction of the triggering stimulus. A prominent example of a steered escape is the reaction of a cockroach to a wind puff, where the air current is sensed by the cerci, and a short latency reaction is mediated by the ascending giant interneuron (GI) pathway. The escape reaction consists of a turn away from the wind stimulus (contraversive turn) and an escape run.

First experimental evidence that a descending neural pathway, mediating information from the antennae, i.e. one that does not involve GI, is part of the escape system was presented by Comer *et al.* (1988) and Stierle *et al.* (1994). Ablation of the GI in the cockroach *P. americana* causes a decrease in responsiveness to wind stimuli (97–30%) and an increase in response latency (53–104 ms by Comer *et al.*, 1988; 34–74 ms by Stierle *et al.*, 1994), but does not affect the direction of escape. Bilateral antennectomy further reduces the responsiveness to wind stimuli to 4%, without impairing the motor ability to turn or to escape from tactile stimuli to the body (Comer *et al.*, 1988). However, unilateral sectioning of the neck connective, i.e. introducing an asymmetry in the descending pathway of otherwise intact animals, alters the direction of wind-evoked escape responses without changing responsiveness or latency of escape (Keegan and Comer, 1993). Tactile stimulation of the antenna triggers escape behaviour more reliably when touching the lesioned side, though at longer latency (Comer *et al.*, 1994). Wind-evoked escape of lesioned animals are shorter in distance and duration. Also, lesioned animals show a bias to turn away from the lesioned side, during spontaneous walking as well as during wind- or touch-evoked escape. Nevertheless, lesions do not preclude turning in either direction, indicating that they affect the steering command rather than the neural circuitry underlying the execution of the turn.

Indeed, nerve recordings suggest that the descending information modulates the GI-mediated escape responses: whereas, the lesion does not affect the typical asymmetry of wind-evoked activity in the GI (more activity in one connective than in the other), it does so in flexor motoneurons of the middle leg (Keegan and Comer, 1993) which are known to be important for the initiation of an escape turn (Nye and Ritzmann, 1992). Chronic recordings from descending neurons in the neck connective of stationarily running cockroaches show stronger excitation in the connective contralateral to the touched antenna (Ye and Comer, 1996), i.e. the direction of the escape. This neuronal activity is at least partially due to the identified descending neurons DM1a-1 and DM1b-1 (see Section 5.2).

Selective wind stimulation of antennae or cerci corroborates the findings of ablation studies: ascending cercal information elicits escape more reliably and at shorter latency than does descending antennal information (Stierle *et al.*, 1994). Consistent with this result, removal of the cerci strongly reduces responsiveness to strong wind stimuli and nearly abolishes escape responses to low wind stimuli. Also, response latencies below 40 ms are hardly ever observed after GI ablation. Bilateral antennectomy in animals with intact GI and cerci has no effect on responsiveness or latency.

In a realistic prey–predator encounter, antennal tactile cues prove to be of great behavioural relevance for the appropriate escape of the cockroach (Comer *et al.*, 1994). In a behavioural assay with four different predators (toad, mouse, preying mantis and wolf spider), 96% of escapes from spiders and 18–55% of escapes from other predators commenced only after tactile contact. In spider strikes, 93% of escapes were directed away from the side of first contact, even though this meant that many were directed towards the attacking spider (33% were directed towards the lunge of the spider). Response latency to touch of the predator is about 20 ms, and 25 ms in response to the touch of a mechanical probe. After antennectomy, significantly fewer escape reactions are elicited in response to spider strikes, but not to toad strikes. The reverse is true after cercectomy, indicating that the behavioural relevance of the two ‘emergency warning systems’ – antennae and cerci – differs according to predator-specific stimulus profiles.

Touching mechanoreceptors on the pronotum or on the hind leg with an artificial probe is more effective in eliciting escape than is touch of the antennae. On the other hand, stimulation of the antennae but not of the pronotum or hind leg, with probes having different surfaces (e.g. a glass rod and a pipe cleaner), leads to a differential responses (Comer *et al.*, 1994; see Section 6.2.3 on related work by Comer *et al.*, 2003). This suggests that antennal contact triggers escape only if a certain stimulus quality is detected. This is also supported by the observation that a glass rod may as well elicit a simple antennal avoidance reflex (Frings and Frings, 1949; Section 6.1.2.1). In contrast to surface properties, antennal posture does not affect responsiveness, but escape direction. Since the direction of the tactile stimulus, i.e. imposing a

small medial or lateral deflection of the flagellum, has no effect on escape direction (Ye *et al.*, 2003), triggering and steering of escape must involve different neural mechanisms. Immobilisation of the HS- and SP-joint greatly reduces the likelihood of a tactile stimulus to elicit escape, whereas replacement of the flagellum by a prosthesis causes much less reduction of responsiveness (Comer *et al.*, 2003). This result is paralleled by electrophysiological findings, which show that the typical escape-related activity of descending interneurons is altered significantly by immobilisation of the basal joints but not after replacement of the flagellum with a prosthesis. The fact that escape behaviour can be impaired significantly in some individuals with flagellar prosthesis, and that neither treatment completely abolishes escapes, indicates that deflection of the basal joints is neither sufficient nor necessary to trigger escape. In any case, some information processing of both proprioceptive and exteroceptive information must be postulated. Comer *et al.* (2003) suggested that the escape-triggering cue is deflection of the antennal joints, whereas the likelihood of eliciting escape depends on flagellar information.

In another kind of emergency reaction, crickets use their antennae in detection of predators and early initiation of defensive behaviour. When attacked by a digger wasp, crickets (*Acheta domesticus*) often react by turning their abdomen to the offender, and by raising their body axis to assume postures called head-stand and stilt-stand. The latter is the posture that a cricket assumes for a defensive kick with its hind legs. Although the main difference between head- and stilt-stands is only the degree of abdominal elevation, Gnatzy and Heußlein (1986) found that head stands are often elicited in crickets without antennae but rarely in animals without cerci, whereas stilt-stands are often elicited in crickets without cerci but rarely in animals without antennae. Thus, similar to the escape system of the cockroach, the predator defence system of the cricket uses synergistic action of descending antennal and ascending cercal information. Also, both 'emergency behaviours' are initiated by a fast-turning reaction. According to Gnatzy and Hustert (1989), the stilt-stand of the cricket is probably elicited by campaniform and scolopidial sensilla of the pedicel. They report "weak stimulation of this area [in the pedicel] triggered stilt-stand, whereas stronger stimuli often led to a 180° turn followed by stilt-stand with the cerci and hind legs toward the predator" (p. 225).

6.2 ACTIVE SENSING

Some of the behaviours discussed previously involve active antennal movements as part of reflex reactions (e.g. assistance reflexes), or are a facultative result of preceding active exploration (e.g. escape). In contrast, the active antennal movements discussed in the following sections are integral parts of active sensing behaviour, i.e. the animals act to increase the likelihood of tactile stimulation, or to change the quality of stimulus reception. Accordingly, the behavioural functions of active antennal movements and mechanoreception

include active exploration of the animal's environment (Section 6.2.1); aiding complex locomotor actions such as climbing over obstacles (Section 6.2.2); aiding recognition and learning of surface structures or movement sequences (Section 6.2.3); or acting as obligatory components in complex behavioural sequences such as fighting or mating (Section 6.2.4).

6.2.1 *Exploratory behaviour and tracking of objects*

When regarding an antenna as an articulated, elongated, multi-sensory probe, it is trivial to observe that the likelihood with which any of its sensors will be stimulated depends on both the distribution of stimulants within the probe's workspace and on its posture. Whereas the antennae of some insects can assume very complicated postures, for example with multiple kinks along the flagellum of *Collembola* (Simon, 1979), the flagellum of many other insects does not change shape during active movement, simplifying the analysis of active movement strategies. In the case of solid objects and, to some extent, of volatiles too, active movement of the antenna increases the likelihood of contacting a stimulant. Hence, active movement patterns of the antennae must be interpreted as explorative strategies that serve to sample the workspace of the flagellum for contact surfaces, odorants, concentration gradients, and the like. Moreover, not only the arrangement of the antennal joint axes and segment lengths, but also the distribution of the exteroceptors on the flagellum are morphological parameters that determine the limits of spatial resolution (see Sections 2.2.1 and 3.2.4). These parameters set the frame for sampling efficiency and, therefore, the behavioural significance of any movement strategy.

For example, the length of the antenna of the stick insect *Carausius morosus* is virtually the same as that of an outstretched leg of the same animal (Dürr *et al.*, 2001) and a large fraction of the antennal workspace overlaps with the workspace of the leg. Movements of the antenna are therefore likely to be relevant to locomotion. If the stick insect moves its antenna without touching anything, the ipsilateral front leg 'knows' a trajectory where obstacles are unlikely to be met. If the antenna does touch something, the front leg must 'beware of' an obstacle. Accordingly, the fact that the cockroach *P. americana* preferentially points its antenna some 30° above the horizon and within a sector ranging from 60° to 90° lateral (Ye *et al.*, 2003), means that this species 'pays most attention' to this part of its antennal workspace. Antennal movements that are part of the searching behaviour of walking cockroaches (cf. Okada and Toh, 2000) explore the space ahead for obstacles.

Stick insects (*C. morosus*) not only continuously move their antennae during walking (Fig. 18A), but also coordinate the cyclic movement pattern of each antenna with the step cycle of the ipsilateral front leg (Dürr *et al.*, 2001). Each antenna typically moves through one abduction/adduction cycle and two levation/depression cycles per step. The timing of largest abduction and subsequent adduction follows the start of stance movement in the ipsilateral front

leg (Fig. 18B). Thus, the pattern of fast antennal abduction and slower adduction becomes part of the gait of the legs. Moreover, the antennal movement pattern is also spatially coordinated with the leg movements. If the antennal searching efficiency was considered the likelihood of the antenna to touch an obstacle prior to leg contact, the typical antennal movement pattern of walking stick insects predict the animal's behaviour: searching efficiency increases sigmoidally with obstacle height. The strongest increase occurs at obstacle heights of about half the clearance of the body (Dürr *et al.*, 2001), which is equivalent to the height range in which model simulations have shown that it becomes necessary to change the normal locomotor pattern in order to climb the obstacle (Cruse *et al.*, 1998; Kindermann, 2002). Furthermore, antennectomised stick insects detect obstacles later and fail to raise their body axis as smoothly as intact animals (Fig. 18C; Dürr *et al.*, 2003), showing that stick insects indeed exploit antennal tactile information during walking.

Like stick insects, crickets (*G. bimaculatus*, Horseman *et al.*, 1997) and locusts (*Schistocerca gregaria*, Saager and Gewecke, 1989) move their antennae during walking. Although no obvious, simple cycle-to-cycle coupling between antennal and leg movements has been documented in either species, ascending neural pathways are crucially involved in this behaviour in crickets (Horseman *et al.*, 1997): transection of a circumesophageal connective completely abolishes ipsilateral antennal movements during walking. Transection of the neck connective can reduce the angular velocity in both antennae during walking.

When walking, the crayfish *C. destructor* typically moves its antennae independently of each other (Sandeman, 1985). During voluntary movements of the lobster antenna (*H. americanus*), the MC- and CF-joint either move in synchrony, or the CF-joint moves alone (Sigvardt, 1977). This is similar in the crayfish. Sandeman and Wilkens (1983) describe antennal movements of the freshwater crayfish *E. armatus* that occur in a number of behavioural situations. This species typically points its antennae backward during forward walking and forward during backward walking, possibly to reduce drag. Coupling between antennal and leg movements during walking occurs, for instance, after obstruction of the stance movement. The crayfish responds to such disturbance by moving its antennae in the opposite direction: flexion in response to a forward pull, extension in response to a rearward pull of the walking substrate. Although such neural coupling between leg movements and antennal movements is likely to influence active searching behaviour, it is reminiscent of the equilibrium responses of *Palinurus* sp. to substrate shift or tilt (Schöne *et al.*, 1976; Barnes and Neil, 1982; Neil *et al.*, 1982, 1984). Occasional rhythmic antennal whipping movements of the walking *Euastacus* show no coupling to the stepping movement made by the legs (Sandeman and Wilkens, 1983). Thus, a clear relation between leg movements and antennal exploration movements remains to be demonstrated in crayfish.

There is evidence that both insects and crayfish adapt their antennal posture and/or movement pattern to the locomotor state of the animal. For example, in

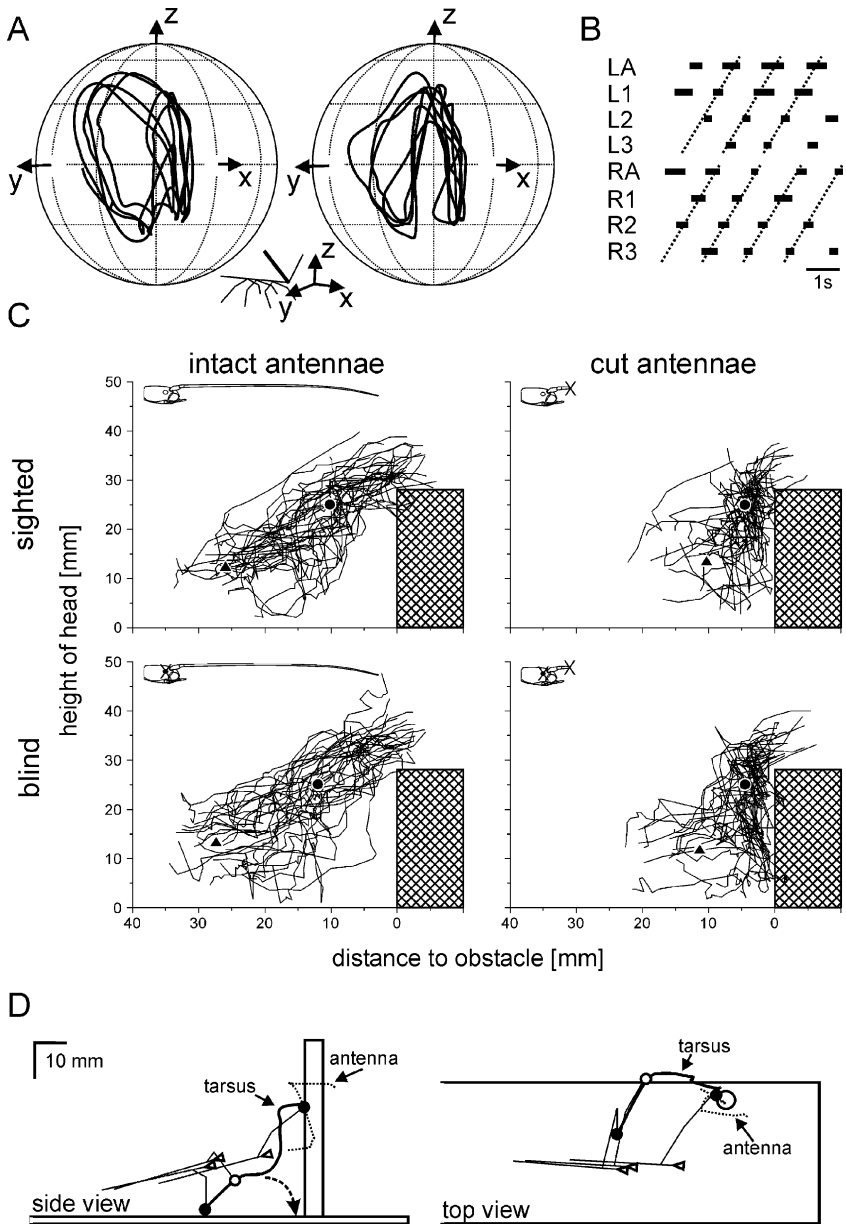


FIG. 18 Active exploration and use of antennae in stick insect locomotor behaviour: (A) Antennal movements of the stick insect *C. morosus* during walking are rhythmical and of fairly regular appearance. Trajectories of antennal tips are drawn to a sphere that is centred on the head and oriented as shown by the inset. The trajectory of the left antenna (right sphere) is drawn as a mirror image, to match the orientation of the right antenna. (B) Abduction phases of the antennae (first and fourth row of black bars) are

stick insects, in case of lack of foothold, antennal movements back-up the searching effort of the front leg (Dürr, 2001). In stick insects walking a curve, the average abduction angle for the antennae increases for the antenna on the inside of the curve, while it decreases for that on the outside of the curve (Dürr and Ebeling, 2005). Similarly, the crayfish *C. destructor* moves both of its antennae into the walking direction before initiation of walking and before turning (Zeil *et al.*, 1985).

Exploratory movements of antennae also occur in response to external stimuli. For example, honeybees move their antennae in response to an air puff containing an odorant (Suzuki, 1975). Both antennal joints contribute to this orienting movement, and fixation of one antennal joint does not preclude movement in the other. Similar stimulus-induced searching movements of the antennae have been described for several insect species, not only in response to odours but also to thermal gradients (e.g. in the bug *Triatoma infestans*; Flores and Lazzari, 1996).

Spatial mapping of external stimulus direction and antennal posture occurs in two forms: first, as directed, open-loop responses, the magnitude of which correlates with the stimulus direction, and second, as antennal tracking of the object, probably in a closed loop. An example of an open-loop response is the reflex-like posture adjustment of the antennae of the spiny lobster *P. argus*. During rest, the spiny lobster holds its antennae laterally, approximately perpendicular to the body long axis. Upon hydrodynamic stimulation with water jets, or tactile stimuli to the body, the ipsilateral antenna is moved anteriorly or posteriorly, always appropriately directed to reduce the angle between stimulus direction and flagellum (Wilkins *et al.*, 1996). Although response amplitudes

continued

coupled to the pattern of swing movements of the ipsilateral walking legs. Diagonal dotted lines indicate the back-to-front sequence (a metachronal wave) of leg swing movements and antennal abduction. LA and RA: left and right antenna, respectively; L1–L3: left front, middle and hind leg; R1–R3: right legs. (C) The tactile sense is exploited for efficient locomotion. Head trajectories of stick insects walking towards a rectangular obstacle (hatched area, side view) show how the insect raises to climb the edge (walking direction: left to right). Stick insects with intact antennae (left) detect the obstacle earlier (triangles = mean position at first contact) and climb the obstacle with more clearance (circles = position when reaching twice the average walking height) than antennectomised animals (right). Sighted (top) and blind (bottom) animals behave the same. (D) Tactile antennal cues can cause rapid re-targeting of an on-going front leg swing movement. Side view (left) and top view (right) of a stick insect walking sequence towards a vertical pole. Three stick figures show body axis, right front leg and tarsus position (circles) at times of lift-off, first antennal contact with the pole (open circle) and leg contact with the pole (solid line: tarsus trajectory; dotted line: trajectory of the antennal tip). A normal swing movement would have been continued as indicated by the dashed arrow. Instead, no more than 60 ms after antennal contact, the swing movement is re-directed to grasp the pole. Data taken from Dürr *et al.* (2001) (A, B); Dürr *et al.* (2003) (C) and Dürr and Krause (2001) (D).

are generally low and rarely exceed 30° , it shows that *Panulirus* has a sense of spatial direction, allowing it to transform stimulus angle into a proportional antennal angle. Similarly, the crayfish *Procambarus clarkii* responds to hydrodynamic stimuli to mechanoreceptors on the telson by backward sweeps of both antennae. As in *Panulirus*, the angle of the sweep is small ($30\text{--}60^\circ$) but correlated with the angle of stimulation (Schmitz, 1992).

Evidence for spatial mapping of exteroceptive information to proprioceptive cues from the antenna stems from observations on antennal tracking behaviour. For example, the crayfish *Procambarus alleni* points its antennae at moving objects and tracks them (Bovbjerg, 1956). *C. destructor* points its antennae towards novel visual stimuli (Sandeman, 1989), and *Orconectes limosus* directs its antennae towards the source of water vibration (Tautz, 1987).

To date, the best-studied antennal tracking behaviour is that of crickets (*Gryllus* sp.). In experimental trials with visual objects moving horizontally through an arc of 180° , these insects track the object with their ipsilateral antenna through a range of 110° lateral towards the midline (Honegger, 1981). When the object crosses the midline, typically the antenna stops moving and the contralateral antenna continues tracking. Occasionally, both antennae move in parallel until the adducted antenna points 30° contralaterally. Various visual cues influence antennal tracking in crickets, including the direction of movement (determining the tracked edge), object size and orientation (summarised by Honegger and Campan, 1989). The antennal tracking response of crickets is strongly habituating (Honegger, 1981). Habituation depends on the size of the tracked object, being stronger for small objects (e.g. 13° diameter) than for large objects (e.g. 25° diameter). When tracking small disks the gain of the response is small, i.e. the antenna lags behind, and the response may even subside completely (Honegger *et al.*, 1985). However, a 30 s interval between trials is sufficient for dishabituation of the tracking response. Vibration of a tethered cricket reduces habituation significantly, increasing the correlation between the antennal angle and the target angle. Vibrating the stimulus has no dishabituating effect, indicating that dishabituation is caused mechanically, not visually. There is no frequency dependency of dishabituation by vibration within the tested range of 25–450 Hz.

Walking crickets do not track visual objects, indicating that the visual-antennal connection is gated by signals related to thoracic motor activity (Honegger, 1981). Much like active antennal movements during walking, antennal tracking also requires functioning ascending pathways, as transection of a circumoesophageal connective abolishes tracking by the ipsilateral antenna (Horseman *et al.*, 1997). So far, all of the published data refer to horizontal tracking that relies on movement of the SP-joint. To what extent the HS-joint allows tracking of vertically moving objects is unknown.

The cockroach *P. americana* points its antennae towards passing conspecifics and tracks them (Ye *et al.*, 2003), but tracking performance has not been quantified. A visual cue presented at an azimuth beyond the preferred antennal

orientation angle causes abduction of the ipsilateral antenna, a movement that often leads to object contact.

Whether or not the visually mediated pointing and tracking behaviours of crustaceans and insects are genuine closed-loop behaviours that reduce the visual angle between the image of the target and the image of the antenna is unknown. In fact, it is unclear whether crickets or crayfish can see their own antenna well enough to measure its posture, as visual spatial resolution is fairly low. It is unlikely that these animals can see their antennal tips, but possibly very good lighting conditions could provide sufficient contrast to crudely judge the orientation of the antennal base. An alternative mechanism would be that the animal matches the visual angle of the target with the output of a neural forward model of its own motor action, or by mapping of proprioceptive cues to external angles (see Section 7.2).

6.2.2 Guidance of locomotion and tactile localisation

As has been indicated in the discussion of stick insect exploratory behaviour (Section 6.2.1), active antennal movements during walking effectively detect obstacles and permit the insect to adjust their gait or trajectories of single leg movements appropriately. Like the stick insect, the potato beetle *Leptinotarsa decemlineata* raises its body in response to contact with obstacles (Pelletier and McLeod, 1994). While antennectomised beetles bump into obstacles that are higher than 2 mm, blind beetles with intact antennae increase the angle between the body long axis and the substrate with increasing obstacle height. In response to touching very high obstacles (7 mm), some beetles lift both front legs off the ground, elevating the prothorax by a distance that exceeds the length of the front legs.

Levitation of the prothorax also occurs in stick insect climbing (Dürr, 2000; Fig. 18C). Similarly, Watson *et al.* (2002) show that the cockroach *Blaberus discoidalis* often stops walking to begin a rearing movement, i.e. body axis inclination, before climbing an obstacle. The kinematic basis underlying this rearing movement is a tibial rotation of the middle legs, caused by the coxa-trochanter joint and the trochanter-femur joint (which is not fused in this species). Because this rotation can occur prior to leg contact with the obstacle, Watson *et al.* (2002) conclude that either visual or antennal tactile cues trigger rearing and subsequent climbing.

Antennal movements are part of the searching behaviour of walking cockroaches (*P. americana*). Upon contact with a vertical rod, unrestrained cockroaches decrease walking speed, orient towards the rod, repeatedly contact the rod with the antennae and climb up it (Okada and Toh, 2000), much like the stick insect *C. morosus* does in similar situations (Dürr, 2000; Dürr and Krause, 2001). In stick insects, antennal contact with a vertical rod may lead to short-latency re-targeting of an ongoing swing movement of the ipsilateral front leg (Dürr and Krause, 2001; Fig. 18D). This suggests that antennal

mechanosensory information is passed rapidly to leg motoneurons and that swing movements are modulated by descending neural input. In the cockroach, shaving off the scapal hair plates affects the antennal contact interval of the repeated contacts with the rod, but not the overall cockroach behaviour, i.e. orienting and climbing (Okada and Toh, 2000).

In free walking stick insects of the species *Aretaon asperimus*, antennal contact with the far edge of a gap is accompanied by a rearward shift of the lift-off position of the front leg (Bläsing and Cruse, 2004). This suggests that the antennae signal the presence of a 'support area within reach', leading to an increased forward shift of the thorax which precedes the step across the gap.

Tethered walking cockroaches persistently attempt to orient themselves towards a touched object, a behaviour that requires the hair plates of the HS-joint (Okada and Toh, 2000). The ability to orient towards a touched object indicates the ability to code the contact angle. Provided that the contact distance is also coded, a neural polar coordinate system could subserve tactile localisation. However, the fact that cockroaches without HS-joint hair plates are not impaired in their ability to locate and climb obstacles (Okada and Toh, 2000) suggests that these proprioceptors are not necessary for tactile localisation. To date, not a single study unambiguously demonstrates that insects can indeed code the spatial location of a touched object, i.e. both distance and angle, using solely antennal tactile cues. This will require careful evaluation of a targeted behaviour that is directed by tactile information.

In Crustacea, Zeil *et al.* (1985) were the first to present quantitative evidence that crayfish use their antennae for tactile localisation of external objects. Studying active antennal movements of hungry freshwater crayfish *C. destructor*, they observed how the animals attacked objects which they had previously touched with their antenna. Vision is of no importance to this behaviour, as blinded crayfish perform as well as sighted ones. Touching a crayfish flagellum with a brush causes retraction of the antenna, turning of the animal towards the contact point, and an attack (Fig. 19A, B). Zeil *et al.* (1985) show that the turning angle correlates with the angle of the antenna at contact time, and that attack distance correlates with the distance of the contact point on the flagellum. Furthermore, turning angle varies systematically with contact distance (Fig. 19C, D), accounting for the spatial separation of the antennal joints and the centre of rotation of the body. Because the centre of rotation in *C. destructor* is located some 5 cm behind the antennal base, a distant object, that is contacted at the same antennal angle as a proximal object, requires a larger turning angle than the latter. For the same reason, objects that are touched by the same point on the flagellum but at a different antennal angle, require different attack distances following an appropriate turn. Indeed, crayfish take their own body geometry into account, suggesting that they make appropriate use of both polar coordinates, i.e. contact distance and angle.

In a follow-on study, Sandeman and Varju (1988) recorded the average movement trajectory of the crayfish with normal and with manipulated

antennae. During an attack, the crayfish rotates so that the touched object is aligned with its long axis, in the region of the mouthparts (Fig. 19E). The turning direction of the crayfish is always correct, and turning angles correlate with the angle that is expected if the crayfish was to turn first and then move towards the object. For distant contact points, however, the turning angle is closer to the antennal angle than to the expected angle. Furthermore, Sandeman and Varju (1988) show that the crayfish turns during the forward movement, i.e. that rotation and translation of the body are not sequential but simultaneous (Fig. 19F). Thus, deviation between the turning angle and the expected angle may be explained by the ability of the crayfish to take the time course of its own movement into account. After immobilisation of the basal antennal joints with a splint, i.e. without impairing proprioception, attack accuracy remains unaffected. However, bending of the flagellum by a splint biases the attack towards the location of the phantom contact point, i.e. the point at which antennal contact would be expected if the bend of the flagellum had been ignored (Fig. 19E). This bias is stronger with a medial bend than with a lateral bend, suggesting that some proprioceptive information from the flagellum is integrated to localise the contact point.

6.2.3 *Pattern recognition and learning*

The ability to discriminate mechanical surface properties on the basis of tactile information is difficult to show unequivocally, at least in animals that cannot be conditioned to associate a tactile cue with a reward. It is probably for this reason that most knowledge about tactile pattern recognition in arthropods has been gained from experiments with honeybees. Moreover, most experimental results on tactile recognition are linked to learning performance. This must be kept in mind because in some learning paradigms it is not evident whether the animal's ability to succeed is of particular relevance to its natural behaviour. We are aware of only two studies that demonstrate the use of the antennal tactile sense in recognition or evaluation of surface properties in a behavioural experiment without rewarding appropriate performance. One of them concerns the differential assessment of surface structure in situations that are potentially dangerous for a cockroach (Comer *et al.*, 2003), and the other concerns the fine-tuning of honeycomb building activity of the honeybee (Martin and Lindauer, 1966).

In the cockroach *P. americana*, flagellar information qualifies the contact with information about the surface properties, thus modulating the likelihood of an escape. For example, touching a wolf spider is more effective in triggering escape than touching a conspecific, provided there is sufficient time for palpation, i.e. repeated brief contacts (3–5 s used by Comer *et al.*, 2003). Extraction of surface chemicals does not alter the differential response to predator and conspecific, and spider- and cockroach-scented cardboard models do not cause different responsiveness. Therefore, contact-chemoreception is not

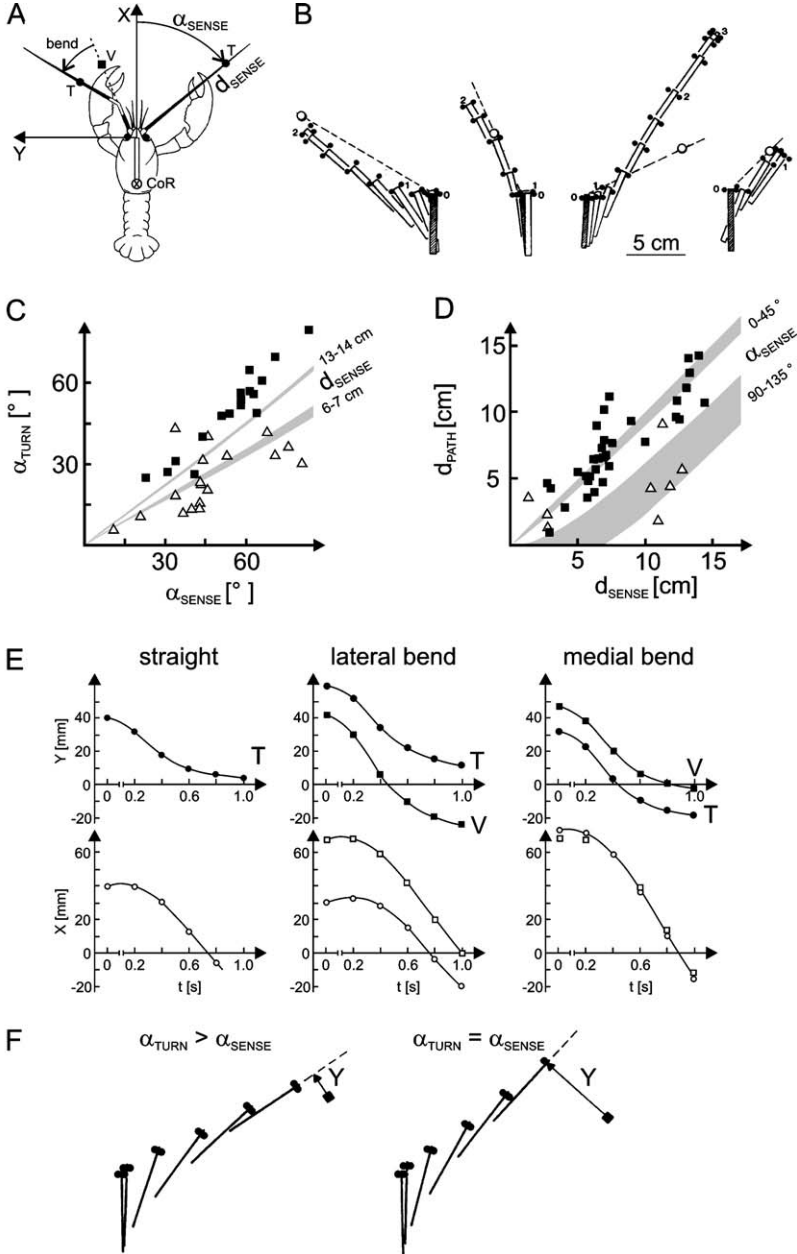


FIG. 19 Tactile localisation in the crayfish *C. destructor*: (A) Hungry crayfish attack an object (solid circles) by moving towards the location touched by the antenna. α_{SENSE} : contact angle of flagellum; d_{SENSE} : contact distance along flagellum; CoR: centre of rotation; T: Target position at contact time; V: virtual target position when disregarding

necessary to raise or lower the likelihood of an escape response. Comer *et al.* (2003) took a lot of care to standardise the stimulus presentation in that palpation time, stimulus movement and 'unscentedness' were well controlled. Nevertheless, the different body shapes of a spider and cockroach provide many more cues than just cuticular surface structure, and further studies will be necessary to conclude whether it is genuine surface structure information or a mix of structural and positional cues, e.g. the distance or posture of the spider's legs rather than their hairiness.

Honeybees are well known for their ability to build highly ordered and evenly spaced honeycombs. Apart from the geometrical regularity, honeybees keep the thickness of the wax walls within a narrow range, and appropriately manufacture different wall thicknesses for worker cells and drone cells. The thickness of the wall is set by a pressing and scraping the mandibles against the wall, while continuously probing the deformation around the pressure point with the antennal tips. Various impairments of the antennae of entire honeybee colonies, such as unilateral antennectomy or bilateral clipping the distal annuli of the flagellum, have no detrimental effect on average cell diameters or angular arrangement of a newly built honeycomb (Martin and Lindauer, 1966). However, loss or numbing of the antennal tips (by clipping or by dipping into phosphoric acid, respectively) does impair the bees' fine-tuning ability in controlling the thickness and evenness of the wax walls. On average, walls become much thicker, and occasional holes can occur. Martin and Lindauer (1966) attribute the mechanosensory significance of the antennal tip to a special

continued

the bend; X , Y : Axes of the body coordinate system. (B) Schematic illustrations of four example attacks towards objects located in different directions and distances (dashed lines with open circles. Numbers indicate time in s after first contact). Attacks consist of a turn and forward translation. (C) Turning angle depends on contact angle and contact distance. Attacks to near objects (triangles; $d_{\text{SENSE}} = 6-7$ cm) have smaller turning angles than attacks to distant objects (squares; $d_{\text{SENSE}} > 13$ cm). Grey areas mark predicted angles for exact localisation, if turning preceded translation. (D) Attack distance depends on contact distance and contact angle. Attack paths to lateral objects (triangles; $d_{\text{SENSE}} > 90^\circ$) are shorter than paths to rostro-lateral objects (squares; $d_{\text{SENSE}} < 45^\circ$). Grey areas mark predicted path lengths for exact localisation, if turning preceded translation. (E) Average relative movement of target, T , in body coordinates. Both X - (open symbols) and Y -coordinates (solid symbols) decrease to a value near zero, corresponding to an ideal position near the mouth. If the flagellum is bent laterally, average attack paths are intermediate to what would be the ideal localisation of the real (T , circles) or the virtual target (V , squares). If the flagellum is bent medially, attacks aim at the virtual target. (F) In a real attack, the animal rotates and translates simultaneously. In attacks to a distant target, the turn angle is larger than the sensed angle (left). If the turn angle was equal to the sensed angle, the Y -error would be larger. Adapted from Zeil *et al.* (1985) (B–D) and from Sandeman and Varju (1988) (A, E, F) with permission from Springer Verlag.

arrangement of sensilla on the last annulus, which they call the “sensory plate” (Sinnesplatte of Martin and Lindauer, 1966). Following measurements of the required force to deform the wax wall and the time constant of the damped return of the wall upon release of the pressure point, Martin and Lindauer (1966) suggest that the bees sense these parameters by the ‘sensory plate’.

Compared to the experimental evidence on tactile pattern recognition in insects, there is not much known about this ability in Crustacea. Although there are many studies that discuss the role of antennal movements in crayfish behaviour, we are unaware of any study that reports active tactile exploration of an object similar to that seen in insects. Rather, Sandeman (1989) explicitly states “crayfish [*Cherax destructor*] have not been seen to explore the contours of a novel object with their flagella in a way to provide them with information about its shape or size” (p. 216).

Honeybees can be trained to tactually discriminate surface structures, both in unrestrained (Martin, 1965; Kevan and Lane, 1985; Erber *et al.*, 1998) and restrained training situations (e.g. Kevan and Lane, 1985; Erber *et al.*, 1998; Scheiner *et al.*, 1999). Martin (1965) used a system of four horizontal, odourless, narrow corridors for simultaneous presentation of four alternative patterns, only one of which was rewarded with food. Each pattern had a different surface structure (ca. 1-mm wide longitudinal or transverse furrows, 1-mm wide holes, and a plane surface), and lined all four surfaces of a corridor, immediately adjacent to the entrance. Bees were able to associate any of the four surface structures with a reward, though much longer training sessions were necessary than for odour learning (1.5–3 days, compared to 5–6 h). Martin (1965) further showed that clipping the distal one to three annuli of the flagellum abolishes tactile pattern recognition. So does partial fixation of the scape that does not preclude antennal contact with the test stimuli, but prevents the normal contact pattern with the antennal tip. This led Martin (1965) to suggest that appropriate contacts of the ‘sensory plate’ at the tip of the flagellum are crucial for tactile pattern recognition in honeybees.

Honeybees are able to discriminate microtextural patterns of flower petals, where the components of the epidermal structures of the petal are of the same size range (ca. 10 μm) as the spacing of the sensilla at the antennal tip (Kevan and Lane, 1985). In a two-alternative learning paradigm, unrestrained bee workers can be trained to associate the petal of the flower *Helianthus annuus* with a sucrose reward. Thereafter, bees successfully discriminate the trained surface structure against the petals of other flower species, against the reverse petal surface of the same flower, and even against the same petal surface with reversed orientation. In an operant conditioning paradigm, restrained bees learn to extend their proboscis upon presentation of a tactile stimulus (Kevan and Lane, 1985). When conditioned to the petal surface of *H. annuus*, bees show proboscis extension in 88–100% of presentations of *H. annuus* (median: 100% correct responses), but only in 0–22% of presentations of petals of another plant species (*Xylorhiza wrightii*, median: 83% correct responses).

When presented with feeder holes with adjacent tactile stimulus plates (on which they can land on) free-flying bees can learn to associate grain size of abrasive paper or the orientation of a coarse grating (1.2 mm spatial wavelength) with a sucrose reward (Erber *et al.*, 1998). However, in the paradigm used in this study, it is difficult to exclude the possibility that bees use other sensory structures than those of their antennae. For instance, free-flying bees fail to learn the same task with finer gratings, despite the fact that they do learn very similar stimuli in a more controlled operant conditioning paradigm that only allows for antennal input. Restrained bees can be conditioned to extend their proboscis in response to tactile contact with an object (Erber *et al.*, 1997), and conditioning can be refined to particular aspects of the touched surface. For example, bees conditioned to a vertical grating of 450 μm spatial wavelength, can differentially respond to various surface properties, such as orientation angle (e.g. detect a 22.5° difference), or spatial wavelength (e.g. detect a 170 μm difference; Erber *et al.*, 1998). Applying the same training paradigm, Erber *et al.* (1998) also show that bees can learn to distinguish shape, orientation and size of a smooth metal plate.

Nevertheless, it should be noted that bees always also respond to the unrewarded alternative in at least 30% of trials. Compared to other sensory modalities, texture learning in free-flying bees is slower than olfactory or colour learning, but faster than visual form learning. Applying the same experimental paradigm as Erber *et al.* (1998), but only testing vertical and horizontal gratings, Scheiner *et al.* (1999) show that reversal learning is possible at a similar acquisition rate as initial learning (Fig. 20). Learning performance is different between pollen foragers and nectar foragers, but this difference can be attributed to differences in response threshold to reward concentration, and the resulting 'perceived value' of the reward.³ Furthermore, antennal tactile sensing behaviour can be sensitised by a sucrose stimulus and modulated experimentally by injection of the biogenic amines serotonin and octopamine (Pribbenow and Erber, 1996). Although these aspects of neuromodulation are unlikely to be specific to tactile learning (e.g. visually induced antennal movements can be modulated by the same substances; Erber and Kloppenburg, 1995), they are noteworthy when considering response variability within the same and between different behavioural experiments.

Apart from discriminating the surface structure, honeybees can also be conditioned to touch a metal plate with their antenna or, in the presence of two plates, to prefer touching one plate over touching the other (Kisch and Erber, 1999). If an object is placed within the workspace of the antennae of a restrained bee, the bee repeatedly antennates the object. This tendency can be enhanced by operant conditioning to a sucrose reward (Fig. 21A, B). Following differential

³According to Scheiner *et al.* (1999), the 'perceived value' is a function of the true value, i.e. the sucrose concentration, but also of a subjective value that depends on the internal, motivational state of the bee.

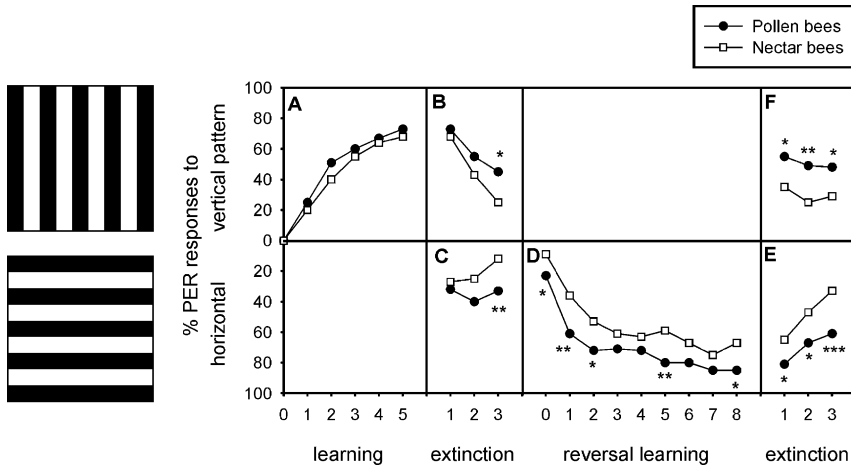


FIG. 20 Tactile antennal learning and discrimination in pollen and nectar foragers of the honey bee (*Apis mellifera*). Restrained bees, whose eyes were occluded with paint, were trained to associate the proboscis extension response (PER) with a tactile pattern. The conditioned stimulus consisted of a 3×4 mm copper plate with grooves which were $150\text{--}190\text{ }\mu\text{m}$ wide and $30\text{--}40\text{ }\mu\text{m}$ deep. In each training trial, bees were free to make antennal contacts with the training stimulus (vertical grating) for 3 s. Subsequently, the PER was elicited by applying a droplet of sucrose to the antennae. When the bee extended its proboscis, it was rewarded with a droplet of sucrose. If the bee responded with a conditioned PER, a drop of sucrose was also offered to the proboscis. The inter-trial interval was 5 min. Learning performance is presented as percentage of bees showing conditioned PER. (A) Learning curve of 75 pollen foragers (circles) and 75 nectar foragers (squares) over a sequence of 5 trials. (B, C) In the subsequent extinction series, bees were presented with either a vertical or a horizontal grating, but no reward was offered. In the first trial, performance is much better in response to the vertical grating compared to the horizontal grating, revealing orientation specificity of learning. Extinction after three unrewarded trials differs between types of foragers (*). (D) Reversal learning over a series of eight trials with the same procedure as in A, but with horizontal grating. Both forager types show successful reversal learning, but differ in performance (*). (E, F) Extinction sequence as in B, C. In the first trial, orientation specificity is reversed compared to B, and differences between forager types are more pronounced. Adapted from Scheiner *et al.* (1999) permission from Springer Verlag.

conditioning to one out of two objects, the learnt preference can be reversed by a second training period (Kisch and Erber, 1999). Very similar antennal movements can be conditioned by offering a reward only if the muscle activity of the SP-joint flexor muscle reaches a certain threshold (Erber *et al.*, 2000), indicating that tactile contact is not necessary for operant conditioning of antennal movements (Fig. 21A, C).

Assuming that the conditioned movement pattern is identical to the movement pattern that occurs in tactile scanning of an object, associative learning of some object properties, e.g. the shape, could be done by learning a movement pattern rather than a contact pattern. The latter interpretation gains support

by the fact that the training procedure of Erber *et al.* (2000) rewarded the activity of a single fast flexor motoneuron. Moreover, this motoneuron is known to govern antennal tactile scanning behaviour (Pribbenow and Erber, 1996; Erber and Pribbenow, 2000). Since the honeybee can recognise object properties by antennal movements, and important features of the movement pattern are dominated by a single motoneuron, the activity of which can be conditioned experimentally, one might speculate that the activity pattern of this motoneuron is indeed sufficient for the honeybee to learn a tactile stimulus. However, because fixation of the antennal joints abolishes associative plasticity of flexor activity, and fixation of only the HS-joint reduces the performance (Erber *et al.*, 2000; Fig. 21C), feedback about the resulting antennal movement appears to be crucial. Nevertheless, it is likely that at least some aspects of tactile learning involve learning of the antennal movement pattern rather than learning the actual contact information. Erber *et al.* (1997) demonstrate that non-associative, i.e. non-rewarded, learning of antennal movement patterns occurs in restrained honeybees that antennate a presented object for periods of at least 10 min or more. When a metal plate is placed into the workspace of the antennae, bees spontaneously and repetitively touch the object at contact frequencies between 3 and 300 contacts min^{-1} , and a median contact duration below 10 ms (Erber *et al.*, 1993). This antennation behaviour is called 'antennal scanning' by Erber *et al.* (1997). The pattern of antennal scanning after removal of the object significantly differs from the spontaneous movement pattern before object presentation: the antennal tip more frequently moves through the area where the object was located and, after 30-min presentation, also avoids areas where the edge of the object was.

Total contact time is not the decisive parameter governing this form of motor plasticity, because the required number of subsequent presentation periods depends on the duration of the single presentations (e.g. 1×10 min is enough, but not 2×5 min; Erber *et al.*, 1997). Mechanosensory information about contacts with the object is not necessary either, because cutting the antennal nerve in the flagellum or replacing the flagellum by a prosthesis does not affect it (cf. Erber and Pribbenow, 2000). However, non-associative plasticity cannot be induced when presenting a wire rather than a plate, although the wire is repeatedly touched by the antenna (Erber *et al.*, 1997). Also, fixing the SP-joint precludes the object-related change. This suggests that the observed motor plasticity is not simply a matter of entraining a movement sequence, but also involves some information related to the touched object, e.g. its size or shape. Although Erber and Pribbenow (2000) demonstrate that sensory cues have no impact on the 'scanning parameters', i.e. contact duration and frequency, the possibility that antennal mechanoreception plays a role in motor plasticity should not be excluded completely. Significant sensory impact on kinematic parameters of the antennal movements may well prove to be important to the natural honeybee behaviour, without showing up in the mere contact sequence.

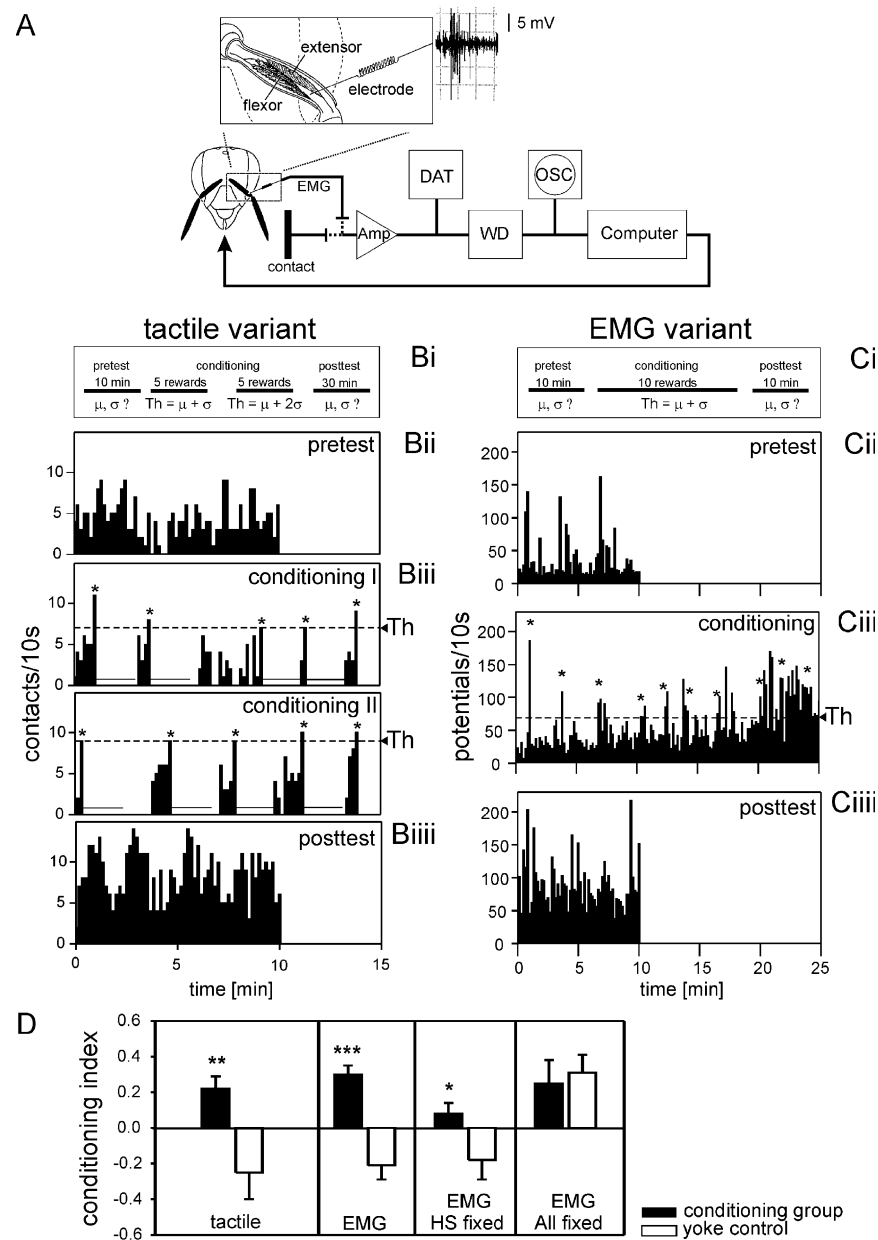


FIG. 21 Operant conditioning of antennal movements in the honeybee works both with and without tactile contact. (A) Restrained honeybees, *A. mellifera*, were rewarded either if they touched a small metal plate with their antenna with sufficient contact frequency (tactile variant), or if the EMG of the flexor muscle of the SP-joint contained sufficient spikes of the fast flexor motoneuron (EMG variant). In both variants,

6.2.4 Communication

The list of reports about the use of antennal movements or tactile contact in intraspecific encounters is long, and by no means can a review of these reports be complete. Many of the published observations are of anecdotal character in that they do not investigate the kind of information that is emitted or received, nor which cues are the decisive ones. The following sections provide an overview of the range of behaviours in which intraspecific communication is known to involve antennal mechanoreception. Emphasis is given where ablation or training procedures have specified the significance of antennal contact.

In intraspecific communication, tactile and vibration cues that are sensed by antennal mechanoreceptors can be of great importance. For example, [Martin and Lindauer \(1966\)](#) immobilised the antennae of each insect in an entire honeybee colony. The effect on the colony's social behaviour was summarised in a footnote by [Martin \(1965, p. 284\)](#):⁴ “Bees of an experimental colony (consisting of some

continued

electronic signals were amplified (Amp, connected to either of the signal sources), stored on a DAT recorder, digitised by a window discriminator (WD) and monitored on an oscilloscope (OSC). Spontaneous occurrence of the measured signal was determined by its mean frequency, μ , and standard deviation σ . A computer showed a signal if a sucrose reward was to be offered. (Bi–Biiii) Tactile variant, with experimental protocol shown in Bi. The reward threshold $Th \geq \mu + \sigma$ was calculated from the contact frequency during a 10 min pretest period (Bii). After a first conditioning session with five rewards (* in Biii), Th was increased to $\mu + 2\sigma$ and another five rewards were given in a second conditioning sequence (Biii). Each reward was followed by a 2 min period without evaluation (horizontal lines). Finally, training success was evaluated during a 10 min post-test interval (Biiii). Biiii shows twice the average contact frequency of Bii (8.6 ± 3.2 compared to 4.3 ± 2.3 per 10 s). (Ci–Ciiii) EMG variant. Experimental protocol (Ci), pre-test (Cii), conditioning with 10 rewards (Ciii) and post-test (Ciiii) equivalent to the tactile variant, except that only one conditioning threshold Th was used. Graph details as in Bi–Biiii. Conditioning caused an increase of flexor spike frequency from 36.0 ± 32.7 in Cii to 99.4 ± 38.5 potentials per 10 s in Ciiii. (D) Comparison of tactile variant and three versions of the EMG variant by means of a conditioning index, CI, that quantifies the difference of pre- and post-test signal frequencies, divided by the sum of pre- and post-test signal frequencies. In both, the tactile variant (tactile, $n_{Exp} = 25$, $n_{Control} = 10$) and the standard EMG variant (EMG, $n_{Exp} = 10$, $n_{Control} = 10$), differences in CI are statistically different between the conditioned group and a yoke control group (bees that receive the same reward sequences as conditioned group, but independent of their behaviour). If the scape is fixed (EMG HS fixed, $n_{Exp} = 10$, $n_{Control} = 9$), the conditioning of the EMG variant is still successful. A non-associative increase in CI occurs in the EMG variant, if the antenna is completely fixed (EMG, all fixed, $n_{Exp} = 5$, $n_{Control} = 5$). Adapted from [Kisch and Erber \(1999\) \(B\)](#) and [Erber et al. \(2000\) \(A, C, D\)](#) with permission from Springer Verlag and Elsevier Publishing.

⁴Translated from German. The ‘operation done’ was, as one of the experimenters recalls, immobilisation of both antennal joints ([Martin Lindauer](#), personal communication).

500 bees) were refrained from mutual antennation by an operation done to their antennae. Bees no longer aggregated to form a building cohort. Social behaviour ceased completely. The animals dispersed uniformly in the small observation hive. Even the queen was disregarded! Moreover, mutual food exchange no longer took place. Control experiments showed that the sense of smell was completely intact". Apart from qualitative descriptions, at least three behavioural contexts in arthropods have been shown experimentally to involve communication via antennal contact: agonistic behaviours that serve to establish dominance order or territoriality (Section 6.2.4.1), transfer of information about resources (Section 6.2.4.2), and courtship and mating (Section 6.2.4.3).

6.2.4.1 Agonistic behaviour. Some species of crustaceans and insects establish dominance orders and/or conquer and defend territories, and in the corresponding agonistic behaviours, antennae can be used as both emitters and receivers of information.

Many insect species fight over territories and resources, and aggressive encounters in social species may involve whole colonies. For example, colonies of the honeypot ant *Myrmecocystus mimicus* carry out ritualised tournaments to settle territorial fights. During these tournaments, pairs of ants engage in a peculiar lateral display during which they raise their abdomen and employ their antennae to drum on each other's behind with great intensity (Hölldobler, 1976). The kind of information transmitted by antennal drumming is yet unknown. It is likely to signal the opponent's strength, but may also involve additional contact-chemosensory information about the opponent's chemistry.

Agonistic encounters among male crickets (*G. bimaculatus*) are an ideal behavioural model to study the role of sensory cues in a complex behavioural chain. This is because males express different levels of aggression in an obligatory sequence of behaviours, beginning with antennal fencing, continuing with unilateral and bilateral mandible spreading, escalating to mandible engagement and, finally, wrestling (Fig. 22A). Body weight, or externally visible parameters that are correlated to weight (e.g. size, head width), are not good predictors of the outcome of a fight, or even of the aggression level reached. Even in pairs with a large size difference, larger males tend to win only some 70% of encounters (Hofmann and Schildberger, 2001). Antennal fencing appears to serve the assessment of the opponent's strength, possibly by sensing the vibrations of the flagella. Males without antennae do not fight, even if two previously dominant males are paired (Fig. 22B). During fencing, the two antennae are moved independently. If the opponent's flagella hit each other, they often vibrate for periods of 1 s or longer. Hofmann and Schildberger (2001) show that the frequency spectrum of the induced flagellar vibration contains information that correlates with the level of aggression that will be reached subsequently. In particular, the ratio of the main frequency modes of the opponent's spectra appears to be a good predictor (main mode for animal 1: main mode for animal 2; see Fig. 22D, E): fights between opponents with similar vibration frequencies

tend to escalate, whereas opponents with large differences in vibration frequency settle the fight after fencing or mandible spreading. Thus, crickets appear to assess the relative stiffness and/or hit strength, both of which would affect the vibration frequency of the flagellum. This is interesting because it is obvious that a sensor such as Johnston's organ could sense the animal's own flagellar vibrations, but sensing the opponent's flagellar vibration must be rather complicated. To date, it is unclear how this could be done.

Antennal fencing also occurs in agonistic encounters of cockroaches. When excised male antennae are used to fence artificially with a conspecific male, *P. americana* typically responds by backing away from the stimulus, or by elevating its body axis to assume a stilt posture (Bell, 1978). The same experimental situation causes rather different responses in the more aggressive species *Nauphotea cinerea*. Depending on the social status, dominant males tend to lunge towards the stimulus, whereas the sub-ordinate males rather assume a submissive posture and may even retreat. The stilt posture can be assumed by both groups, showing that the transferred information is sufficiently graded to allow for a range of responses.

American lobsters (*H. americanus*) use their antennae to whip the opponent on the claws, thus emitting information about their aggressive state. The whipped lobster senses the offensive signal with mechanosensory hairs on its claws (Solon and Cobb, 1980). The range of possible responses to the received signal (e.g. escape, attack, etc.) suggests that lobsters transmit graded information about their size or strength, a cue that is likely to be relevant in the establishment of a dominance order among conspecifics. Huber and Kravitz (1995) analysed components of the behavioural sequence of agonistic encounters among juvenile *H. americanus*, including two kinds of whipping and lashing movements, and a distinct antennal posture where the flagella are pointed upward. The latter "antenna up" posture is part of a threat pattern, and is frequently accompanied by raised claws and an inclined body axis. Single antennal whips occur in a behavioural pattern termed "do-si-do", in which the animal tends to approach its opponent with its claws held downward. Repeated antennal whipping and lashing characterises a further behavioural pattern. The statistical analysis of Huber and Kravitz (1995) indicates that these patterns occur in a fixed sequence that, like the levels of aggression described for the cricket (see above), indicates the intensity of aggression. As encounters reach different intensity, the components of the behavioural patterns appear to serve an exchange of information that determines the course of the encounter.

In the crayfish species *Procambarus alleni*, antennal cues provide sufficient sensory cues to establish a dominance order (Bovbjerg, 1956). Ablation experiments show that blind crayfish still know which end of their opponent to attack, an ability that is abolished by additional antennectomy. Lack of antennae and eyesight precludes the establishment of a dominance order. Whether or not antennal tactile cues are augmented by contact-chemosensory cues in this behaviour is unknown. Because either intact vision or intact antennal sensing are

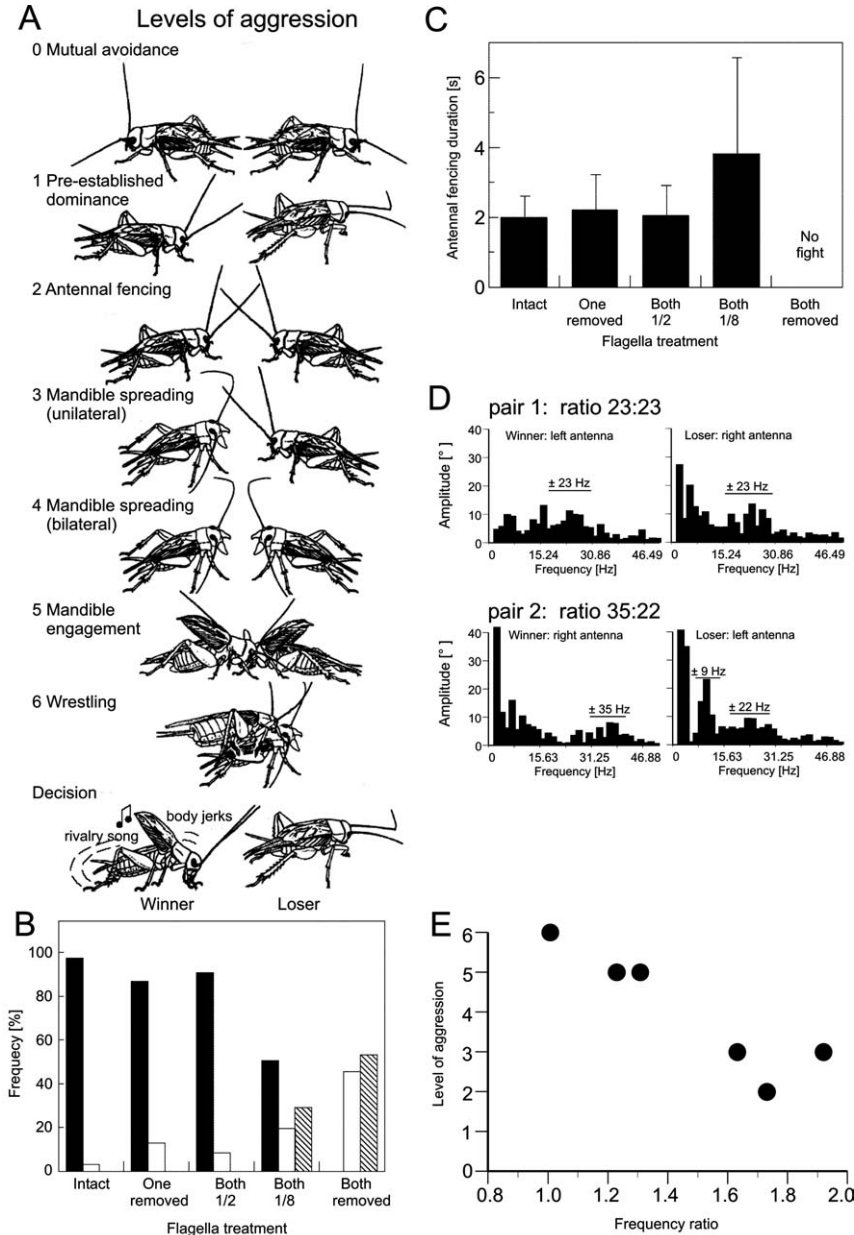


FIG. 22 Antennal fencing behaviour is an indicator for intensity of cricket fights: (A) Agonistic encounters in the cricket *G. bimaculatus* follow a behavioural sequence that can be described by six distinct levels of aggression. Antennal fencing (level 2) is the first fighting stage. (B) Antennae are crucial for the initiation of fighting. Pairs of crickets either fight (black bars), show pre-established dominant or sub-ordinate behaviour

sufficient but not necessary to establish a dominance order, it is likely that normal crayfish use cues received by both of these senses in aggressive encounters. Crayfish of the related species *Orconectes virilis* respond strongly to the antennal posture of an approaching conspecific, and antennal posture is a good predictor for the intensity of the encounter in this species (Heckenlively, 1970). *Orconectes* also performs antennal waving displays during agonistic encounters, a behaviour that is also executed in the dark (Bruski and Dunham, 1990) and even with antennal stumps (Ameyaw-Akumfi, 1979). In the latter situation, the display lasts longer and the performing animal is held longer by its opponent, leading Ameyaw-Akumfi (1979) to interpret antennal waving as an 'appeasement display' that transmits mechanical cues.⁵ In support of this view, Tierney *et al.* (2000) describe antennal waving to be performed by the subordinate animal, and in one species of *Orconectes*, it is sex-specific to males. The display results in release of chela contact by the dominant animal. Visual cues affect various aspects of antennal waving displays (Bruski and Dunham, 1990), but whether visual cues transmit information to the opponent is still a matter of debate.

6.2.4.2 Transferring information about resources. Possibly the most famous form of insect communication is the waggle dance of the honeybee (*Apis mellifera*), a locomotor display by which a worker bee transmits information about the location of a food source to its colony companions. Although it has been known for many decades that the directional information is coded by the dance direction relative to gravity, the mode by which this information was transmitted was unknown for a long time. Owing to the detailed analyses of

continued

(white bars), or court each other (hatched bars). Shortening of antennae to half of the flagellum length does not change a pair's behaviour. If both flagella are shortened to 1/8th, encounters without fights are more frequent compared to intact animals, and courtship may occur. Bilateral antennectomy completely abolishes fighting behaviour and further increases the frequency of courtship behaviour. (C) Duration of antennal fencing in cricket encounters with same treatments as in (B). The duration increases significantly in crickets with antennae shortened to 1/8th of the normal length. (D) Frequency spectra of flagellar vibrations after mutual antennal contact during fencing. In pair 1 the modes with highest frequency components were similar (ratio 23:23). This fight escalated to level 5. In pair 2 the frequency spectra differ more strongly (ratio 35:22). This fight stopped at level 3. (E) Among six fights where flagellar frequency spectra were obtained, the final level of aggression correlated significantly with the frequency ratio of the opponents' high-frequency modes ($r_s = -0.88, p < 0.02$). Adapted from Hofmann and Schildberger (2001) with permission from Elsevier Publishing.

⁵Although Ameyaw-Akumfi (1979) uses the word 'antennule' in one table and in a few places of the text, the remaining text and images suggest that the appendage referred to really is the 2nd antenna, not the antennule (1st antenna).

Michelsen *et al.* (reviewed in Michelsen, 1993), it is now established that the vibrations produced by the waggle dancer are airborne sounds, and cannot be detected via substrate vibrations. Rather, the cues produced by the dancer are characterised by an acoustic near field with 94 dB sound pressure near the wings and a steep pressure gradient that declines within a few millimetres. Moreover, peak to peak particle velocities are large (ca. 1 m s^{-1}), and the sound spectrum varies at different locations around the dancer's abdomen, potentially allowing a follower bee to detect the location behind the dancer as it moves from the side of the dancer towards her rear end (which is the course a follower normally takes). All of these cues could be picked up by an appropriate vibration sensor on the antenna.

Indeed, honeybees can be trained to associate a sound of a given frequency with an electric shock, and to avoid the sound thereafter in a frequency-specific manner (Towne and Kirchner, 1989). The conditioned sound is also avoided during flight behaviour, even if the conditioned frequency (265 Hz) is close to the bee's own wing beat frequency. When placed in a standing-wave tube, bees perform characteristic antennal lifting movements at locations where the air particle velocity is high (1 m s^{-1} , at 18 Pa rms sound pressure), but not if air pressure oscillations are strong and air particle velocity is low (0.07 m s^{-1} , 60 Pa rms sound pressure). This indicates that bees perceive air particle movements rather than sound pressure.

Unilateral antennectomy reduces the likelihood that a bee gets recruited to a new food source, and increases the time needed to find the feeder (Dreller and Kirchner, 1993b). In contrast, clipping the antennal tips or shaving all hairs on the head is not detrimental to recruitment success. Because controls showed that bees with only one antenna can fly sufficiently well, attend waggle displays and are motivated to leave the hive for foraging, Dreller and Kirchner (1993b) conclude that tactile cues are not necessary for information transmission in waggle displays, but a 'binaural', or rather bi-antennal mechanism must be postulated.

The sensory organ involved was shown to be located in the pedicel (Dreller and Kirchner, 1993a). In an operant conditioning paradigm developed by Kirchner *et al.* (1991), honeybees can learn to associate a sound with sucrose reward in a two-alternative paradigm (Dreller and Kirchner, 1993a). Subsequent testing of bees with a variety of ablations reveals that a single antenna is sufficient to maintain discrimination, albeit with reduced performance. Removal of both antennae further reduces the performance to chance level. Immobilisation of the SP-joint and the PF-junction also causes significant impairments, but fixing only the SP-joint has no effect. Also, shaving the hairs off the bee's head is ineffective. In conclusion, the bee's flagellum is well suited to transmit the sound frequencies emitted by the dancer (see Section 2.2.2.2) to the PF-junction, where Johnston's organ can transduce the information into a neural activity. There is little or no contralateral transfer of auditory information from the antennae, since unilateral antennectomy impairs choice performance only if the sound is

presented on the manipulated side, but not on the intact side (Dreller and Kirchner, 1993a).

Within the honeybee genus *Apis*, not only the Western species *A. mellifera* can be operantly conditioned to sound stimuli, but also the Asian species *A. dorsata* and *A. florea* (Dreller and Kirchner, 1994). Because all three species recruit other foragers by a waggle dance, but *A. florea* does not emit any sound while dancing, it is likely that the ability to detect sound preceded the use of sound production in the evolution of the waggle dance. According to this scenario, the sensory ability to detect near-field sound using antennal mechanoreceptors allowed the introduction of an additional cue into intraspecific communication.

Apart from receiving the information of the waggle dancer, antennae appear to play a role in attracting bees to follow the waggle dance, i.e. in initiating the behaviour that will lead to information transmission about food sources. Tautz and Rohrseitz (1998) traced back the paths of follower bees to determine when and where surrounding bees became interested in the waggle dance. Their results show that the fraction of bees that become follower bees only after they make antennal contact with the dancer or another follower bee, is dependent on the lighting conditions and on the properties of the dance floor (e.g. open vs. sealed cells). The worse the transmission of vibration of the floor, and the darker the 'room', the more bees need physical contact with others before following the waggle dance. A closer look at the antennal contact pattern of the follower bees with the dancer reveals that follower bees make regular brief contact with the body of the dancer, during which the flagellum is bent laterally (Rohrseitz and Tautz, 1999). The duration of coincident antennal contact of both antennae decreases as the follower bee gradually moves from a position beside the dancer to one behind the dancer. Given that a follower bee knows its own orientation relative to gravity, the correlation between bilateral contact coincidence and orientation relative to the dancer may allow the follower to either 'calculate' the orientation of the dancer with respect to gravity, or help the follower to align herself with the dancer, i.e. find the location where it is easiest for her to determine the orientation of the waggle dance.

Antennation is frequently observed in other hymenopterans too, and often has been particularly emphasised in reports on social behaviour in ants (e.g. see reviews by Lenoir and Jaisson, 1982; Hölldobler, 1999). For example, in a study on liquid food exchange in the ponerine ant *Pachycondyla villosa*, Hölldobler (1985) describes how antennation precedes food transmission between the forager and the solicitor. Food transmission in *Pachycondyla* is entirely external, i.e. food is not carried in the crop and then regurgitated. As regurgitation of food from the crop occurs in species that are considered to be more 'advanced', this behaviour may represent an evolutionarily intermediate stage between social food transmission with and without regurgitation. Because other ponerine ant species that do not transmit food, use antennation cues to 'invite' nest mates to follow them in a tandem run back to the nest,

Hölldobler (1985) supposes that antennation, being a widely used behavioural component in ant communication, became a ritualised signal for food solicitation, allowing ant colonies to form an external 'social bucket' at first, and later even a distributed internal 'social stomach'. Similar evolutionary scenarios are discussed by Lenoir and Jaisson (1982).

Solicitation behaviours in different ant species may also serve as a tool to determine the relative significance and benefit of the two sensory modalities that are involved: mechano- and chemoreception. For example, the ant species *Bothroponera tesserinoda* and *Camponotus sericeus* both recruit conspecifics by tandem runs, a behaviour during which both participating ants expect regular physical contact with the other. Whereas both of these species use contact-chemosensory cues to identify a leader as 'being worth-while following', *B. tesserinoda* can be convinced to follow an unscented dummy, provided she has searched for a lost leader for long enough to raise her motivation (Maschwitz *et al.*, 1974). *C. sericeus* on the other hand, does not follow unscented dummies (Hölldobler *et al.*, 1974). Thus, *C. sericeus* is more restrained and more specific in its requirements to complete successful recruitment. If a leader is lost, only an appropriately scented substitute will do. *B. tesserinoda* appears to weigh the importance of mechano- and chemosensory cues, depending on its behavioural state. Whether or not such differences are linked to the likelihood of an individual of a given species to lose its leader before reaching the nest remains to be shown.

To our knowledge, antennation between ant species or between ants and myrmecophile insects is to date the only use of antennae in interspecific communication. For example, the staphylinid beetle *Atemeles pubicollis* imitates antennation signals used by colonies of the Formicidae for food solicitation, allowing the beetle to parasitise the intraspecific food flow (Hölldobler, 1970).

6.2.4.3 Courtship and mating. In a study on the behavioural sequence in the mating cycle of the cricket *Teleogryllus commodus*, Loher and Rence (1978) repeatedly emphasise the significance of antennal movements at various times of the mating cycle. Pair formation begins with the male calling-song that attracts the female. Upon the first encounter, the male seeks antennal contact with the approaching cricket. In a male-female encounter, the pair engages either in agitated antennal contact for several seconds, or the male antennates the female's elytra, abdomen and hind legs, depending on whether the female is approached from the front or from the rear, respectively. Following this antennation, the male withdraws and vibrates its antennae. Blinded and deafened crickets begin courtship upon contact with a female antenna, indicating that mutual antennal contact is important for sex recognition. Antennectomised females respond appropriately to the male calling sound but are "reluctant to mount the male and experienced difficulties in assuming the right position on the back of the male" (Loher and Rence, 1978, p. 231). During courtship, female antennation of the male elytra and head releases a range of courtship-specific

behaviours in the male, such as searching movements of the cerci. Once the female has mounted the male, the male maintains antennal contact by reaching back and flagellating her antennae, keeping her in place until copula. After copulation, the male stays close to the female for some 50–60 min and lays his antennae on her back, as if to guard her. If, during this ‘guarding period’, antennal contact is lost due to restlessness of the female, the male reacts with antennal vibration and flagellation, apparently trying to calm her down. Guarding is important, because unguarded females get rid of the spermatophore within a few minutes. Loher and Rence (1978) claim that lack of antennation is “the single most disruptive influence on spermatophore formation and mating frequency” (p. 240).

The specific role of antennal tactile cues in parts of the cricket mating cycle, particularly in sex recognition, has been studied to some detail. Whereas Huber (1955) suggested that male *Gryllus campestris* use tactile cues about the strength of antennal contacts to tell the opponent’s sex, species of the genus *Teleogryllus* appear to use contact-chemosensory cues. In *T. commodus*, there are no structural differences between male and female flagella, nevertheless male crickets respond to contact with the cut-off antenna of a conspecific with sex-specific behaviours (courtship song vs. aggressive song; Rence and Loher, 1977). This differential response is abolished if the cuticular wax layer on the antenna is removed with chloroform. Also, olfactory cues can be ruled out, because neither behavioural choice of appropriately scented containers, nor electroantennograms reveal any sex-specificity. In the closely related *Teleogryllus oceanicus*, selective blockage of antennal chemoreceptors by zinc sulphate abolishes initiation of courtship in males, without impairing tactile avoidance responses (Balakrishnan and Pollack, 1997). Males of *G. bimaculatus* frequently respond to encounters with dead females with courtship song, but not if cuticular hydrocarbons are removed by a solvent. However, re-painting the dead female’s flagellum with the hydrocarbon extract is sufficient to re-establish courtship response in some males (Tregenza and Wedell, 1997). The latter study also shows that male and female hydrocarbon extracts differ strongly in concentrations of individual components, but not in the presence of a single pheromone. Note, however, that the extracts were not obtained from antennae only, and that pronounced sex differences in hydrocarbon composition could be restricted to particular parts of the cuticle, for instance to the flagellum.

Apart from sex recognition, a second component of the mating cycle in *T. oceanicus*, the female mounting the male, is significantly impaired by antennectomy of the females (Balakrishnan and Pollack, 1997). Because immobilisation of the female’s pedicel has no effect on mounting frequency, it is not crucial for the female to sense the wing vibration of the courting male with her Johnston’s organ, but other cues seem to aid mounting. Possibly, these are cues related to those involved in rearing or climbing obstacles (see Section 6.2.2).

Similar to crickets, antennal contact of the male cockroach *Blatella* with a female is necessary for sex recognition, and a non-volatile compound produced by the female's integument is the decisive cue (Roth and Willis, 1952; cf. Cornwell, 1968, p. 179ff). As soon as the contacted cockroach is recognised as a female, the couple repeatedly stroke each other's antenna, whereby a sexually stimulating substance is transferred to the male antenna. The male then twitches his abdomen and turns it towards the female, lifting his fore and hind wings, thereby exposing a gland on the dorsal side of its eighth abdominal segment. 'Sparring' the antennae between the couple, and active movement by the female, are both described as important factors in getting the female to take the secretion of the dorsal gland. Sparring is not essential though, as males will also court antennectomised females. Antennectomy of the male strongly reduces his response.

Antennal movements also play a conspicuous role in various components of the mating cycle in cockroaches of the genus *Blaberus* (Wendelken and Barth, 1987). A male *B. discoidalis* approaches the female upon reception of a volatile sex pheromone, waves his antennae and, upon contact with her, strokes her dorsal surface with the antennae. The female indicates her receptivity by turning towards the male, holding her antennae in front of her, sometimes followed by a brief period of antennal fencing. Following female antennation, the male almost always begins his courtship behaviour by wing pumping. The direction of the subsequent 180° turn of the male appears to be away from the side that receives more tactile contact from the female.

B. giganteus males maintain antennal contact with the female as they perform a wing pumping display with clockwise circling movements. The females of *B. giganteus* can assume an unreceptive posture, in which the antennae are held posteriorly, below the body and between the legs. Thus, females actively avoid antennal contact with the male. If receptive, she antennates the male.

Unreceptive females of *B. craniifer* also hide their antennae, but beneath the tegmina. *B. craniifer* males typically maintain antennal contact during wing pumping behaviour, and even during intermittent turns. In conclusion, antennal waving and antennation of the male are fixed components in the courtship behaviour of all *Blaberus* species observed by Wendelken and Barth (1987). So is female antennation to indicate her receptivity: a prerequisite for the behavioural sequence that leads to copulation. Very likely, the described behaviours in *Blaberus* include both contact-chemosensory and tactile cues.

7 Biomimetics and 'antennal engineering'

In recent years, more and more experiments have been performed to steer insects by implanting electrodes into their nervous systems, adjacent structures or muscles, or to build robots that, in one way or another, mimic insect function (e.g. Espenschied *et al.*, 1993; Frik and Amendt, 1995; Pfeiffer *et al.*, 1995;

Ritzmann *et al.*, 2000). Naturally, such attempts have stirred public interest beyond the scientific community (Suplee, 1997; Yam, 1998; Trachtman, 2000). The following sections will review some attempts to use insect antennae to steer robots, or to remote control an insect via its antennal system (Section 7.1), as well as various engineering issues concerning the design and fabrication of artificial tactile antennae (Section 7.2).

7.1 STEERING INSECTS AND ROBOTS

In two cases, attempts to steer a cockroach by electrical stimulation of antennal afferents and the connecting descending pathways have been successful. In the first study, electrical stimulation of $\sim 10\text{--}100\ \mu\text{A}$ via electrodes, inserted into antennal stumps, was used to steer cockroaches (*Periplaneta*), which were held stationary on top of an air-supported ball (Holzer and Shimoyama, 1997). The animals responded by turning away from the stimulated side, but the variance of the response angles was rather large. In the second study, electrodes were inserted at the base of the antennae of *Gromphadorhina portentosa* (Moore *et al.*, 1998), by penetrating the soft cuticle of the SP-joint. The indifferent electrode was located in the dorsal abdomen. The effects of the electrical stimulation mimic escape responses described for *Periplaneta americana* (e.g. Comer *et al.*, 1994; Stierle *et al.*, 1994; see Section 6.1.3.3), but it is unknown whether the underlying neural escape circuit (Burdohan and Comer, 1990; Ye and Comer, 1996) was actually recruited. The rather large variations of the behavioural responses suggest that the electrical stimulation recruited a multitude of circuits.

Passive tactile sensors have been used on mobile robots, including legged robots, for detecting contact with walls or with large obstacles, but typically they send only an unspecific binary stop signal to the controller (e.g. hexapod robot ‘Ghengis’; Brooks, 1989). A mobile robot that uses tactile antennae for steering in a more sophisticated way is the small hexapod robot ‘Sprawlette’, which is equipped with a flexible probe with several strain gauges (Cowan *et al.*, 2004). The sensor reading is used to estimate the lateral distance to the contacted wall, and simple proportional–differential (PD) control of the distance is sufficient to achieve reasonably stable course control. Intuitively, the PD control loosely matches the idea of keeping stable both the lateral distance to the wall and the angle between wall and heading direction. As a mismatch in the angle relative to the wall soon leads to a change in lateral distance to the wall, controlling both lateral distance and its derivative stabilises a variable that Cowan *et al.* (2004) call “tactile flow”. This was inspired by experimental results on the cockroach *P. americana* (Johnson and Camhi, 1999; see Fig. 17 and relating text).

Instead of building an artificial antenna, Kuwana *et al.* (1999) use the antennae of the silk moth *Bombyx mori* as a chemosensory device to steer a small wheel-driven robot. Pheromone-induced electrical activity in the antennal

nerve can be picked up by electrodes, amplified, drift-cancelled, band-pass filtered and compared to an offset voltage. After a period of about 25 min, the baseline drift has decreased sufficiently, and the sensor produces reliable signal-to-noise ratios for a period of half an hour. During this test interval, the robot can track a realistic odour plume to proceed towards the pheromone source, although its track is considerably different to that of a moth.

7.2 ENGINEERING OF ACTIVE TACTILE SENSORS

Similar to the approach of Kuwana *et al.* (1999), insect antennae have been coupled repeatedly to stationary electronic devices, e.g. via field-effect transistors, to detect volatile chemicals at very low concentrations (e.g. Schöning *et al.*, 1998, 2000; Schütz *et al.*, 1999, 2000; Schroth *et al.*, 2001; Park *et al.*, 2002; Park and Baker, 2002). In principle, similar arrangements could be employed to record mechanoreceptor activity for tactile contact sensing, although the mechanical interaction with the contact surface would probably make a stable recording even more difficult than it is already for chemical sensors. To our knowledge, no attempts have been made to use an insect antenna as a natural contact sensor.

The mechatronics of tactile sensors is a strongly growing field of engineering. Research areas concerning transduction methods, incorporation of sensors into the contact material, integration of information about tactile contacts with information about posture and movement (haptics), and data processing schemes have witnessed considerable improvement over the past decade (see Lee and Nicholls, 1999, for review).

Artificial fabrication of an antenna-like tactile sensor still poses many problems, however, some of which concern the mechanics of the sensor-carrying probe and others, which concern the actual sensing process. In case of the probe, problems mainly concern conflicting requirements. For instance, an ideal probe would be reasonably stiff to simplify determination of the contact point, but also to avoid strong movement-induced vibrations. On the other hand, the probe would need to be sufficiently light to allow for fast active sampling strategies, and fairly compliant for passive bending upon object contact. Moreover, it has to be durable.

In case sensors are to be mounted to the probe, mechanical robustness is required, but also very basic issues of transduction and coding are still far from the so-called standard solutions. Lee and Nicholls (1999) highlight the three main engineering difficulties of tactile sensing as being (1) the lack of a localised sensory organ, i.e. transduction involves signals from a much wider area than the sensor area alone, (2) the signal complexity, i.e. the mixed and differing impact of shape, texture, friction, force, etc. on the transduction process and (3) the “difficulty to imitate” the decisive sensory quality, i.e. to define the physical properties that are most appropriate to be quantified.

The importance of the right choice of probe is emphasised by studies on bending and resonance properties. Recently, Barnes *et al.* (2001) showed that the tapered shape of a mechanical probe allows distinction of drag-induced bending, caused by flow of the surrounding medium, from contact-induced bending, caused by a touched obstacle. In a biological context this is interesting, as the antennae of crayfish and lobsters, but also those of locusts, are tapered (see Section 2.3.2). In a series of papers, Ueno and co-workers analysed the possibility of using contact-induced oscillations of an insensitive beam to determine the contact distance (Ueno and Kaneko, 1995; Ueno *et al.*, 1996, 1998). Their experiments show that the fundamental and second-order natural frequencies of a beam are sufficient to give a good estimate of the contact point. The major difficulty in this application, however, is to measure the frequencies, because repeated object contacts preclude sufficiently long oscillations to apply standard routines such as a Fourier transformation. Ueno *et al.* (1996) also demonstrate that an appropriate mass distribution along the beam is a prerequisite for a unique mapping of oscillation frequencies to contact point. In a biological context, these results imply that the afferent activity of Johnston's organ, a vibration sensor common to all higher insects, could in principle contain information about the contact point during tactile sensing.

Exploiting force and position measurements at the base of an insensitive compliant probe, Tsujimura and Yabuta (1992) were able to reconstruct the shape of a touched object. Additionally to such measurements, Kaneko *et al.* (1995, 1998) detect lateral slip on the contact surface and use an active movement strategy to determine the touch direction that avoids lateral slip. Provided that the contacted object is sufficiently static, an insensitive probe can, thus, determine the contour of a touched object.

Nevertheless, equipping a probe with a set of mechanosensors would certainly boost the applicability of an artificial tactile antenna. Recently, micro-electric mechanical systems (MEMS) of various types have been fabricated and mounted to a biomimetic lobster antenna (McGruer *et al.*, 2002), the aim being to simultaneously measure water flow, bending of the probe and contact point with an external object. The resulting MEMS antenna is part of an underwater ambulatory robot that is inspired by an eight-legged lobster (Ayers, 2002). The antennae can be positioned at one out of four postures, but active movement is not used in the sensing process.

In the control of legged locomotion, Dürr and Krause (2001) argue that it can be particularly efficient to use a leg-like sensor to guide a leg, much as insects use a pair of highly specialised sensor legs – their antennae – to keep locomotion efficient, even on rough terrain. A great potential in active tactile sensing certainly lies in the context-dependent choice of sampling strategy and its adaptive control. It will pay off to analyse further the tactile sampling strategies of insect antennae, that appear well-adapted to tactile detection of obstacles (see Sections 6.2.1 and 6.2.2). For adaptive control, Dürr *et al.* (2003) have advocated the use of forward models for the control of a tactile sensor. In

a simulation study, they used a parallel set of extended self-organising maps to track an external target with an antenna, a behaviour known from insects and Crustacea (see Section 6.2.1). Unless it is performed under closed-loop control, tracking requires the solution of the inverse kinematics problem. Whereas technical systems typically calculate the solution to this problem, animals cannot do so, because they do not know how to mathematically invert the analytical description of their forward kinematics. Yet, animals can learn their forward kinematics by comparing the motor signal to the corresponding sensory information. Moreover, the obtained forward model of a limb has to be adaptive, because the forward kinematics may change during growth or due to injury. Dürri and Krause (2002) implement a forward model by means of extended self-organising maps, i.e. extended Kohonen maps. The adaptive forward model is then applied to iteratively solve the inverse kinematics problem during object tracking.

Both in technical and biological systems, forward models are a very powerful concept for prediction of the system's state, including its proprioceptive information. In an active sensor, forward models are an appropriate means for comparing the expected input to the real input, thus permitting the separation of active and passive movement. Biologists have repeatedly claimed the necessity of a special case of forward model, the efference copy or corollary discharge (Holst and Mittelstaedt, 1950; Sperry, 1950), where a scaled copy of the motor command is used to cancel the sensory input that is induced by active movement. In the nervous system, many variants of such forward models are likely to be implemented. In technical systems, they are likely to be indispensable for the separation of actively and passively induced tactile inputs.

8 Conclusions

This review focuses on the biology of the antennal tactile sense of insects and crustaceans. It covers the biomechanics, kinematics and behavioural biology of antennal movements, the peripheral and central neurobiology of antennal mechanoreception and interdisciplinary aspects of biology and engineering of tactile sensing. The discussion of these topics is centred around a set of insect model organisms, all of which employ their antennae in active tactile sensing. Work on other species, particularly on decapod crustaceans, is used to complete and contrast the insights from the insect model organisms.

In conclusion, we will discuss three aspects of the antennal tactile sense that draw on information from various sections and, therefore, require separate treatment. Firstly, we will address the behavioural similarities between insect and crustacean antennal tactile sensing (Section 8.1), which are particularly remarkable when bearing in mind the morphological differences between these taxa. Secondly, we will contrast in hindsight the categories of passive and active sensing, as used in Section 6, by relating them to a behaviour-based

classification of tactile sensing (Section 8.2). Finally, we will discuss this classification in an evolutionary context (Section 8.3).

8.1 THE ANTENNAL TACTILE SENSE OF INSECTS AND CRUSTACEA

The antennae of insects are sensory organs that carry sensilla of at least three modalities: mechano-, chemo- and thermoreception. Accordingly, different insect taxa have evolved specialised antennae that are more suitable for one of these modalities than for another. For example, the pectinate antennae of some moths (e.g. *Manduca sexta*), are well-adapted for olfaction, but are less suitable for tactile sensing. This is because an increase of surface area also increases the delicateness of the structure, which can be detrimental where physical contacts are necessary for sensing. Generally, highly specialised antennae, i.e. those that strongly deviate from a linear morphology, are abundant in holometabolous insect orders (particularly in Coleoptera, Lepidoptera and Diptera; e.g. the Hymenoptera being an exception) but less common in hemimetabolous insect orders (e.g. Orthoptera, Phasmatodea, Blattodea; e.g. the Odonata and Hemiptera being exceptions).

Crickets, stick insects and cockroaches have long, straight flagella and move their long antennae constantly during walking, thus raising the likelihood of tactile contacts. In species of these orders, objects are rarely touched with the antennal tip. Rather, contacts typically occur along the side of the flagellum. Sensory hairs along the cricket and cockroach antennae, are arranged in rings around the distal ends of the annuli, and their number is highest at the tip of their antenna.

Honeybees, on the other hand, use their antennae in a very different way (e.g. Section 6.2.3). The geniculate morphology of the Hymenopteran antenna can be considered as a kinematic specialisation to targeted probing with the antennal tip, requiring a highly dexterous antenna. In the honeybee, the highest number of sensilla is also found at the tip of the antenna, but a peculiarity to the honeybee antenna is a specialised arrangement of sensory hairs, the sensory plate, on the terminal annulus (cf. Section 3.2.4.1).

The differences in kinematics, sensing strategies and the arrangement of receptors on the antenna are paralleled by neuroanatomical differences between the model systems of cricket and cockroach on the one side, and of the honeybee on the other. In crickets and cockroaches, antennal afferents project to three distinct neuropils, the AL, the DL and the VFA (Section 4.1.1). In the highly structured VFA, afferents connect to interneurons, some of which show conspicuously regular arborisation patterns. This indicates a topological organisation of this neuropil (Section 4.2.3). In comparison, although honeybees do have ALs and DLs, they appear to lack the VFA. Quite possibly, the mentioned differences in movement strategy and receptor arrangement on the flagellum imply differences in neural processing of antennal tactile information, reflected by a different layout of the deutocerebral neuroanatomy.

From a behavioural perspective, decapod Crustacea appear to use their antennae in similar ways as insects with long linear antennae. At the level of antennal reflexes, there seems to be little difference between the palinurid rock lobsters on the one hand, and locusts on the other (Section 6.1.2). Similarly, speed-dependent abduction, antennal-tracking behaviour and the use of antennae in agonistic encounters, bear many similarities in crustacea and insects. Yet, the underlying physiological mechanisms may be different. These differences may have their basis in differences in antennal form, size, 'construction' of the joints and the viscosity of the surrounding medium, but also in the involvement of different brain regions.

Differences in antennal movement strategies must be related to the kinematic peculiarity of the crustacean antenna, where two redundant joints (MC- and CF-joint) move the flagellum within a plane that is rotated by the complex BIM-joint. In comparison, the basic construction of an insect antenna is that of a Cardan joint (Section 2.2). The presence of redundant joints in crustaceans is mirrored in the existence of common excitatory motoneurons, whereas the antennal musculature in insects is innervated by 'dedicated' excitatory and common inhibitory motoneurons (Section 2.3). Also, the crayfish antenna has a bi-articular chordotonal organ, whereas functionally related proprioceptors in insects only span a single joint.

Similarly, environmental and biomechanical aspects probably had a bearing on the differences between some insect and crustacean mechanosensory structures and, possibly, their transduction mechanisms. For example, transduction may be affected by differences in cuticular characteristics, but also by the external medium.

Finally, unlike insects, crayfish and lobsters have two pairs of specialised antennae, the first pair of which are used primarily for olfaction, while the second antennae serve as tactile probes. Afferents from first and second antennae project to separate neuropils in the deuto- and tritocerebrum, respectively. The degree of structure of the tritocerebral antenna II neuropil correlates with the length of the flagellum of the different species. The length of the flagellum can be considered indicative for the tactile usage of the antenna. Thus, it is intriguing to hypothesise that the necessity for topological coding of contact location along the flagellum is related to the finding of a structured neuropil in crustaceans and insects with long tactile antennae (cf. Section 4.1.2).

Certainly, the behavioural significance of the antennal tactile sense in insects and Crustacea suggests that these taxa have indeed evolved serially homologous structures into an analogous set of active tactile sensors.

8.2 ACTIVE AND PASSIVE MECHANICAL SENSING

In the present review, behavioural functions have been loosely grouped into passive and active sensing. This distinction may appear somewhat artificial,

particularly as certain behavioural contexts presented in Section 6 may involve both passive and active aspects.

The reason for this choice of classification is the attempt to arrange behavioural contexts into a sequence of increasing complexity. For example, assistance reflexes (Section 6.1.2) are treated within the passive sensing context, although they occur during voluntary, hence ‘active’ behaviour. Similarly, tactile wall-following in cockroaches is classified as passive sensing (Section 6.1.3.2), although cockroaches must be expected to actively control their antennal posture.

Accordingly, the passive sensing section (Section 6.1) comprises behavioural contexts in which any active component solely concerns postural reflexes. Active components, such as posture stabilisation in tactile wall-following or during flight speed control, may be necessary to keep passive sensing simple, e.g. by keeping the SP-joint angle constant or the average torque acting on the PF-junction in a linear range. Thus, local postural reflexes may be prerequisites for simple control schemes, which, in turn, underlie orientation behaviours.

Exploration movements are the most basic form of active sensing, in that they serve to expand the workspace of an otherwise passive sensor. Similarly, tracking of visual targets adjusts the workspace to a region with likely sensory input. All further behavioural contexts discussed in Section 6 are variations of this scheme: antennal movements act either to raise the likelihood of gathering potentially important information (as during locomotion), or of seeking physical contact at a given location (as in pattern recognition) and possibly in a given sequence (as in communication).

Honeybee hearing does not fit this scheme, because it does not involve active antennal movement. Nevertheless, the small sensing range of honeybee hearing, e.g. a few millimetres in waggle dance communication, requires active movements of the follower bee and, therefore, active positioning of the antennal sensory structures. Moreover, the vibration cues heard by a honeybee bear much similarity to vibration cues associated with other examples of insect communication, such as antennal fencing in crickets and, possibly, temporal sequences of antennation in ant communication systems. Therefore, honeybee hearing may be considered as active sensing – although it is a special case.

Active movement contributes to the efficiency of tactile sensing by increasing the likelihood of the antennal contacts with an external object. On the other hand, active movement of a movement-sensitive sensor requires the ability to discriminate self-induced stimulation from external stimulation. Hence, active sensing produces ambiguities with passive sensing. The use of forward models can effectively resolve this ambiguity as discussed in Section 7.2 from an engineering point of view. The motor-dependent suppression of antennal mechanosensory inputs to a cricket descending interneuron probably represents an example where a feed-forward motor signal modifies a sensory signal (Section 5.2). Thus, this may be indicative for the biological implementation of a forward model, for instance an ‘efference copy’. Moreover, biomechanical

properties may also help to discriminate active from passive sensory cues (Section 2.2.2.1). One could even hypothesise that, depending on the specific quality of the antenna-object contact differential sets of antennal mechanoreceptors are activated by self-induced and external contacts, thus enabling an active organism to discriminate between these sources of stimulation.

Another aspect of active sensing, e.g. tactile scanning of external objects or tracking of a visual stimulus is dependent on the spatial and temporal alignment of the sensor and its target.

Firstly, spatial information from different sensors, e.g. the compound eye and the antenna, must be brought into congruence, either by transforming one of the coordinate frames into the other, or by mapping afferent inputs of both sensors to a common frame of reference. As a consequence on the neuronal level, interneurons must exist, which integrate the information from different sensory modalities to generate appropriate motor commands to adjust the position of the antenna. Accordingly, interneurons have been found which are the likely components of a multimodal integration stage (Sections 4.2.2 and 5.1). Assessing their behavioural significance, e.g. for a co-ordinate transformation between the eye and the antennal base, however, requires profound knowledge of their sensory physiology. This knowledge, however, is currently not available.

Secondly, spatial acuity of the tactile sense depends crucially on the acuity of the elements of the underlying neuronal circuitries. Passive and active sensing may be very similar in terms of acuity, assuming that acuity depends on sensor spacing and its regularity alone. This, however, and how sensor spacing is represented in the brain remains so far unknown. The data that are available for antennal sensory neurons (Sections 3.2 and 3.3) provide a reference that is necessary to judge the acuity of the coding of stationary and dynamic parameters of antennal posture and movement. Moreover, local interneurons in the deutocerebrum appear to process the primary sensory information to extract specific features of movement parameters (Section 5.1).

From a comparative point of view, the great extent to which passive and active sensing behaviours resemble each other in insects and crustaceans, emphasises the similarities rather than differences between both arthropod taxa. Except in the context of pattern recognition (Section 6.2.3), insect sensing movements find their counterparts in crustaceans, hinting at the evolutionary benefit that a sensor gains from adopting active properties.

8.3 LEVELS OF BEHAVIOURAL COMPLEXITY

Following the common assumption that antennae have evolved from serially homologous locomotor appendages, it follows that they have probably never passed an immobile stage in their evolution. Rather, they are likely to represent a body appendage that has always been mobile and has always been equipped with the sensory infrastructure that is common to arthropod legs. Therefore, it

may be hypothesised that antennae have always been actively moveable, complex sensors, and that their sensory potential initially subserved the behavioural function of posture stabilisation and propulsion. Consequently, antennal searching behaviour during locomotion, or antennal tracking (Sections 6.2.1 and 6.2.2) may be fairly close to the origins of antennal motor programs, i.e. rhythmic stepping movements, whereas apparently 'simple' behaviours like the 'static' antennal flight posture, may well represent a rather derived condition.

On the basis of the distinction of passive and active sensing, however, behavioural contexts may be arranged in a sequence of increasing complexity of motor action and of neural processing of sensory information.

Basically, orientation behaviours like geotaxis (Section 6.1.1) and anemotaxis (Section 6.1.3.2) are crude forms of course control, as it occurs in the control of flight speed (Section 6.1.3.1) and wall-following (Section 6.1.3.2). Moreover, the ability to sense small deflections is the prerequisite for appropriate 'turn-away' reactions, as is the case in antenna-mediated escape in cockroaches and the defensive stilt posture and corresponding kick reaction of the cricket (Section 6.1.3.3). It is reasonable to assume that the sensory abilities required for these behaviours were already present in an articulated leg.

Furthermore, assuming the gradual conversion of a front leg into a dedicated sensor, it is plausible that antennae served as sensors to signal the presence of an obstacle within the action range of the legs, thus aiding efficient locomotion on unpredictable terrain (Section 6.2.2). The involvement of antennae in climbing of obstacles and gaps suggests that lack of antennal contact signals 'space to move' to the legs, allowing the animal to stick with the current locomotor behaviour, whereas antennal contact elicits a fast change in locomotor strategy, e.g. by recruiting the thoracic joints to incline the body axis. The presence of dedicated, large calibre descending antennal mechanosensory interneurons that project to the thoracic ganglia, possibly mediating fast reactions to detected obstacles ahead of the animal would support this behavioural consideration (Sections 4.2.3 and 5.2).

Pattern recognition and communication are yet more complex behaviours that require the coding of spatial and/or of temporal information. In contrast to mere spatial coding of the location of a single contact point, pattern recognition requires sequential antennal contacts, and, therefore, the ability to either map each contact information into a common 3D representation, or to code a four-dimensional pattern by including time as a variable. Only in cases where the recognised microtextures are of the spatial range as the mechanoreceptor spacing (e.g. Kevan and Lane, 1985), may sequential contacts be unnecessary. The ability of honeybees to learn movement patterns (e.g. Erber *et al.*, 1997, 2000), which is reflected in the activity of a single motoneuron; and the similarity of tactile learning with movement learning (see Fig. 21), hints at the possibility that tactile pattern recognition in insects may in part be associated with the serial coding of a movement pattern and, thus,

temporal coding of a sequence, in addition to spatial coding of location. So far, the literature suggests that both spatial and temporal information can be learned. As a consequence, further studies will have to reveal to what extent these two aspects are linked.

Finally, antenna-mediated communication in insects requires the ability to control a 'voluntary' action sequence by the emitter, as well as the ability to decode the transmitted temporal information and/or the stimulus intensity. Most reported forms of communication that involve the antennae rely either on vibration cues (e.g. antennal fencing in crickets, waggle dance in honeybees) or repetitive antenation (e.g. the tournaments of *Myrmecocystus*). A special case may be courtship behaviour, as contact-chemoreception appears to be more important than in other forms of communication.

In conclusion, the utilisation of active properties appears to be a major theme in the design of tactile antennal behaviours. This may be an '*a priori*' condition of arthropod antennae, which is expressed in both insects and crustaceans. The various organisational levels of antennal movements range from simple reflexes to complex sequences. Some of the aspects of the biology of arthropod antennae relate to core interests of behavioural biology, neuroethology and neurophysiology. Accordingly, we are confident that continued comparative research on the antennal tactile sense and its underlying neural substrates will provide insight into the evolution and organisation of sensorimotor aspects of arthropod behaviour in general.

Acronyms and abbreviations

ACH	acetylcholine
ACHE	acetylcholine esterase
AL	antennal lobe of the insect deutocerebrum
AMMC	antennal mechanosensory and motor centre
BE-joint	basipodite-exopodite joint
BIM-joint	basipodite-ischiopodite-meropodite (3rd) joint complex of the crustacean 2nd antenna
CB-joint	coxopodite-basipodite (2nd) joint of the crustacean 2nd antenna
CF-joint	carpopodite-flagellum (5th) joint of the crustacean 2nd antenna
CI	common inhibitor neuron
CNS	central nervous system
DBN	descending brain neuron
DC	deutocerebrum
DL	dorsal lobe of the insect deutocerebrum

DUM	dorsal unpaired median neuron
EJP	excitatory junction potential
EMG	Electromyogram
GABA	γ -amino butyric acid
GI	giant interneurons and corresponding ascending neural pathway of the cockroach escape system
HC-joint	'head'-coxopodite (1st) joint of the crustacean 2nd antenna
HS-joint	head-scape (or 1st) joint of the insect antenna
IJP	inhibitory junction potential
IM-joint	ischiopodite-meropodite joint
IR	immunoreactive
LAN	lateral antennular neuropil
MAN	medial antennular neuropil
MC-joint	meropodite-carpopodite (4th) joint of the crustacean 2nd antenna
MCF-CO	biarticular chordotonal organ of the crustacean antenna, spanning the MC- and CF-joint.
MEMS	micro-electric-mechanical systems
N1	main antennal nerve, Nervus antennalis, that branches inside the antenna into a lateral N1l and a medial N1m.
NO	nitric oxide
OSC	oscilloscope
PF-junction	not actively moveable joint between the pedicell and the flagellum of the insect antenna
sCO	scapal chordotonal organ
SOG	suboesophageal ganglion
SP-joint	scape-pedicel (or 2nd) joint of the insect antenna
T1-7	tract number one to seven
TC	tritocerebrum
TCG	tritocerebral commissure giant interneuron
VFA	ventral area of flagellar afferents, a neuropil of the insect deutocerebrum
VUM	ventral unpaired median neuron
WD	window discriminator

Acknowledgements

V.D. is supported by the Deutsche Forschungsgemeinschaft grant DU-380/1. V.D. thanks J. Camhi (Jerusalem), J. Erber (Berlin), H. Hofmann (Cambridge/MA), D. Sandeman (Wellesley) and J. Zeil (Canberra) for consenting and/or commenting the presentation of their original figures and corresponding legends, J. Tautz and M. Lindauer (Würzburg) for information related to the translated citation in Section 6.2.4, as well as S. Kühn and H. Cruse (Bielefeld) for

constructive feedback on large portions of the manuscript. M.G. thanks P. Bräunig (Aachen), for providing access to his literature database and for commenting on Fig. 14. M.G. is also grateful to H. W. Honegger (Nashville), for granting permission to re-use Fig. 3. E.M.S. thanks G. A. Jacobs (Bozeman), and P. Bräunig (Aachen) for helpful comments on the manuscript and P. Bräunig (Aachen), for some discussions. The authors thank T. Matheson for helpful suggestions on the manuscript.

Note added in proof

- Dürr, V. (2005). Biomechanics of an active tactile sensor: the flagellum of *Carausius morosus*. *Proc. German Zool. Soc.*, in press.
- Dürr, V. and Ebeling, W. (2005). The behavioural transition from in straight to curve walking: kinetics of leg movement parameters and the initiation of turning. *J. Exp. Biol.* **208**, 2237–2252.
- Fields, D. M. and Weissburg, M. J. (2004). Rapid firing rates from mechanosensory neurons in copepod antennules. *J. Comp. Physiol. A* **190**, 877–882.
- Han, Q., Hansson, B. S., and Anton, S. (2005). Interactions of mechanical stimuli and sex pheromone information in antennal lobe neurons of a male moth, *Spodoptera littoralis*. *J. Comp. Physiol. A* **191**, 521–528.
- Krause, A.F. and Dürr, V. (2005). Proprioceptive influence of sensory hair fields on antennal movements of *Carausius morosus*. *Proc. German Zool. Soc.*, in press.
- Kung, C. (2005). A possible unifying principle for mechanosensation. *Nature* **436**, 647–654.
- Kwon, H.-W., Lent, D. D., and Strausfeld, N. J. (2004). Spatial learning in the restrained American cockroach *Periplaneta americana*. *J. Exp. Biol.* **207**, 377–383.
- Lambin, M., Deglise, P., and Gauthier, M. (2005). Antennal movements as indicators of odor detection by worker honeybees. *Apidologie* **36**, 119–126.
- Lange, O., Reimann, B., Saenz, J., Dürr, V., and Elkmann, N. (2005). Insectoid obstacle detection based on an active tactile approach. *Proc. Int. Symp. Adapt. Motion Anim. Mach. 5* (AMAM5), in press.
- Lewinger, W. A., Harley, C. M., Ritzmann, R. E., Branicky, M. S., and Quinn, R. D. (2005). Insect-like antennal sensing for climbing and tunneling behavior in a biologically-inspired mobile robot. In: *Proc. IEEE Int. Conf. Robotics Automation (ICRA'05)*, Barcelona, April 18–22.
- Lent, D. D. and Kwon, H.-W. (2004). Antennal movements reveal associative learning in the American cockroach *Periplaneta americana*. *J. Exp. Biol.* **207**, 369–375.
- Okada, J. and Toh, Y. (2004). Spatio-temporal patterns of antennal movements in the searching cockroach. *J. Exp. Biol.* **207**, 3693–3706.

References

- Abzhanov, A. and Kaufman, T. C. (1999). Homeotic genes and the arthropod head: expression patterns of the *labial*, *proboscipedia*, and *deformed* genes in crustaceans and insects. *Proc. Nat. Acad. Sci. USA* **96**, 10224–10229.
- Allgäuer, C. (1989). Analyse der Antennenbewegungen bei Grillen (*Gryllus campestris* und *Gryllus bimaculatus*): motorische Kontrolle, sensorische Bewegungskodierung, integrative Interneurone. Doctoral dissertation, TU München.

- Allgäuer, C. and Honegger, H.-W. (1993). The antennal motor system of crickets: modulation of muscle contractions by a common inhibitor, DUM neurons, and proctolin. *J. Comp. Physiol. A* **173**, 485–494.
- Altner, H. (1977a). Insect sensilla: principles of structure and function. *Verh. Dtsch. Zool. Ges.* **1977**, 139–153.
- Altner, H. (1977b). Insect sensillum specificity and structure: an approach to a new typology. In: *6th International Symposium on Olfaction and Taste*, (eds Magnen, J. and MacLeod, P.), pp. 295–303. Paris, London: Gif-sur-Yvette Information Retrieval Ltd.
- Altner, H. and Prillinger, L. (1980). Ultrastructure of invertebrate chemo-, thermo-, and hygroreceptors and its functional significance. *Int. Rev. Cytol.* **67**, 69–139.
- Altner, H., Routil, C. and Loftus, R. (1981). The structure of bimodal chemoreceptive, thermoreceptive, and hygroreceptive sensilla on the antenna of *Locusta migratoria*. *Cell Tiss. Res.* **215**, 289–308.
- Altner, H., Sass, H. and Altner, I. (1977). Relationship between structure and function of antennal chemo-, hygro- and thermoreceptive sensilla in *Periplaneta americana*. *Cell Tiss. Res.* **176**, 389–405.
- Altner, H., Schaller-Selzer, L., Stetter, H. and Wohlrab, I. (1983). Poreless sensilla with inflexible sockets. A comparative study of a fundamental type of insect sensilla probably comprising thermo- and hygroreceptors. *Cell Tiss. Res.* **234**, 279–307.
- Ameyaw-Akumfi, C. (1979). Appeasement displays in cambarid crayfish (Decapoda, Astacoidea). *Crustaceana Suppl.* **5**, 135–141.
- Arbas, E. A. (1986). Control of hindlimb posture by wind-sensitive hairs and antennae during locust flight. *J. Comp. Physiol. A* **159**, 849–857.
- Arnold, G., Masson, C. and Budharugsa, S. (1985). Comparative study of the antennal lobes and their afferent pathway in the worker bee and the drone (*Apis mellifera*). *Cell Tiss. Res.* **242**, 593–605.
- Ayers, J. (2002). A conservative biomimetic control architecture for autonomous underwater robots. In: *Neurotechnology for biomimetic robots*, (eds Ayers, J., Davis, J. L. and Rudolph, A.), pp., 241–259. Cambridge/MA, London: MIT Press.
- Baba, Y. (2000). New methods of dye application for staining motor neurons in an insect. *J. Neurosci. Meth.* **98**, 165–169.
- Baba, Y. N. and Comer, C. M. (2003). Antennal mechanosensory neurons in the cockroach their location and projection sites. *Abstract Viewer/Itinerary Planner*. Washington, DC: Society for Neuroscience. CD-ROM. **2003**, 269.12.
- Bacon, J. (1980). A homologous interneurone in a locust, cricket and a mantid. *Verh. Dtsch. Zool. Ges.* **85**, 300.
- Bacon, J. and Möhl, B. (1983). The tritocerebral commissure giant (TCG) wind-sensitive interneuron in the locust. I. Its activity in straight flight. *J. Comp. Physiol.* **150**, 439–452.
- Bacon, J. and Tyrer, M. (1978). The tritocerebral commissure giant (TCG): a bimodal interneurone in the locust *Schistocerca gregaria*. *J. Comp. Physiol.* **126**, 317–325.
- Balakrishnan, R. and Pollack, G. S. (1997). The role of antennal sensory cues in females responses to courting males in the cricket *Teleogryllus oceanicus*. *J. Exp. Biol.* **200**, 511–522.
- Ball, E. and Young, D. (1974). Structure and development of the auditory system in the prothoracic leg of the cricket *Teleogryllus commodus* (Walker) II. Postembryonic development. *Z. Zellforsch.* **147**, 313–324.
- Balss, H. (1940). Decapoda (Zehnfüßer). In: *Bronn's Klassen und Ordnungen des Tierreichs* **5**(1), 116–123.

- Barnes, T. G., Truong, T. Q., Adams, G. G. and McGruer, N. E. (2001). Large deflection analysis of a biomimetic lobster antenna due to contact and flow. *ASME J. Appl. Mech.* **68**, 948–951.
- Barnes, W. J. P. and Neil, D. M. (1982). Reflex antennal movements in the spiny lobster, *Palinurus elephas*. II. Feedback and motor control. *J. Comp. Physiol.* **147**, 269–280.
- Barth, F. G. (2002). *A Spider's World: Senses and Behavior*. Heidelberg, Berlin: Springer.
- Bartos, M., Allgäuer, C., Eckert, M. and Honegger, H.-W. (1994). The antennal motor system of crickets: proctolin in slow and fast motoneurons as revealed by double labelling. *Eur. J. Neurosci.* **6**, 825–836.
- Bartos, M. and Honegger, H.-W. (1997). Impact of motor activity and antennal mechanosensory input on the intensity of proctolin-like immunoreactivity in antennal motoneurons of crickets (*Gryllus bimaculatus*). *J. Comp. Physiol. A* **181**, 59–70.
- Bartos, M., Lauer, B. and Honegger, H. W. (1995). Colocalisation of glutamate and proctolin in antennal motoneurons: an immunocytochemical double-labelling study in crickets. *Proc. Göttingen. Neurobiol. Conf.* **23**, 600.
- Bauer, C. (1991). Modulatory action of proctolin in the locust (*Locusta migratoria*) antennal motor system. *J. Insect Physiol.* **37**, 519–532.
- Bauer, C. and Gewecke, M. (1991). Motoneuronal control of antennal muscles in *Locusta migratoria*. *J. Insect Physiol.* **37**, 551–562.
- Bässler, U. (1971). Zur Bedeutung der Antennen für die Wahrnehmung der Schwerkraftfrichtung bei der Stabheuschrecke *Carausius morosus*. *Kybernetik* **9**, 31–34.
- Bässler, U. (1993). The walking- (and searching-) pattern generator of stick insects, a modular system composed of reflex chains and endogenous oscillators. *Biol. Cybern.* **69**, 305–317.
- Bässler, U. and Büschges, A. (1998). Pattern generation for stick insect walking movements – multisensory control of a locomotor program. *Brain Res. Rev.* **27**, 65–88.
- Bell, W. J. (1978). Directional cues in tactile stimuli involved in agonistic encounters in cockroaches. *Physiol. Entomol.* **3**, 1–6.
- Bell, W. J. and Kramer, E. (1979). Search and anemotactic orientation of cockroaches. *J. Insect Physiol.* **25**, 631–640.
- Bender, M., Gnatzy, W. and Tautz, J. (1984). The antennal feathered hairs in the crayfish: a non-innervated stimulus-transmitting system. *J. Comp. Physiol. A* **154**, 45–47.
- Bentley, D. (1977). Control of cricket song patterns by descending interneurons. *J. Comp. Physiol.* **116**, 19–38.
- Bicker, G. (1992). Taurine in the insect nervous system. *Comp. Biochem. Physiol. C* **103**, 423–428.
- Bicker, G., Schaefer, S., Ottersen, O. P. and Storm-Mathisen, J. (1988). Glutamate-like immunoreactivity in identified neuronal populations of insect nervous systems. *J. Neurosci.* **8**, 2108–2122.
- Bicker, G., Schafer, S. and Kingan, T. G. (1985). Mushroom body feedback interneurons in the honeybee show GABA-like immunoreactivity. *Brain Res.* **360**, 394–397.
- Bill, R. G. and Herrnkind, W. F. (1976). Drag reduction by formation movement in spiny lobsters. *Science* **193**, 1146–1148.
- Bläsing, B. and Cruse, H. (2004). Mechanisms of stick insect locomotion in a gap-crossing paradigm. *J. Comp. Physiol. A* **190**, 173–183.
- Blochlinger, K., Jan, L. Y. and Jan, Y. N. (1991). Transformation of sensory organ identity by ectopic expression of cut in *Drosophila*. *Genes Dev.* **5**, 1124–1135.
- Bloechl, R. and Selzer, R. (1988). Embryogenesis of the connective chordotonal organ in the pedicel of the American cockroach: cell lineage and morphological differentiation. *Cell Tiss. Res.* **252**, 669–678.

- Bodmer, R., Barbel, S., Sheperd, S., Jack, J. W., Jan, L. Y. and Jan, Y. N. (1987). Transformation of sensory organs by mutations of the cut locus of *Drosophila melanogaster*. *Cell* **51**, 293–307.
- Boeckh, J., Ernst, K. D. and Polz, H. (1983). Convergence of inputs from different sense organs upon neurons of the central antennal pathway in insects. *Fortschr. Zool.* **28**, 143–148.
- Boeckh, J., Ernst, K. D., Sass, H. and Waldow, U. (1984). Anatomical and physiological characteristics of individual neurones in the central antennal pathway of insects. *J. Insect Physiol.* **30**, 15–26.
- Boeckh, J. and Tolbert, L. P. (1993). Synaptic organization and development of the antennal lobe in insects. *Micr. Res. Tech.* **24**, 260–280.
- Bornhauser, B. C. and Meyer, E. P. (1997). Histamine-like immunoreactivity in the visual system and brain of an orthopteran and a hymenopteran insect. *Cell Tiss. Res.* **287**, 211–221.
- Bovbjerg, R. V. (1956). Some factors affecting aggressive behavior in crayfish. *Physiol. Zool.* **29**, 127–136.
- Boyan, G., Williams, L. and Meier, T. (1993). Organization of the commissural fibers in the adult brain of the locust. *J. Comp. Neurol.* **332**, 358–377.
- Boyan, G. S. and Williams, J. L. D. (1981). Descending interneurons in the brain of the cricket. *Naturwissenschaften* **68**, 486–487.
- Boyan, G. S., Williams, J. L. D., Posser, S. and Bräunig, P. (2002). Morphological and molecular data argue for the labrum being non-apical, articulated, and the appendage of the intercalary segment in the locust. *Arthropod Struct. Dev.* **31**, 65–76.
- Boyan, G. S. and Williams, L. (2002). A single cell analysis of *engrailed* expression in the early embryonic brain of the grasshopper *Schistocerca gregaria*: ontogeny and identity of the secondary headspot cells. *Arthropod Struct. Dev.* **30**, 207–218.
- Böhm, H. and Schildberger, K. (1992). Brain neurons involved in the control of walking in the cricket *Gryllus bimaculatus*. *J. Exp. Biol.* **166**, 113–130.
- Böhm, L. (1911). Die antennalen Sinnesorgane der Lepidopteren. *Arbeiten aus dem Zoologischen Institut der Universität Wien und der Zoologischen Station in Triest* **19**, 219–246.
- Bräunig, P. (1985). Mechanoreceptive neurons in an insect brain. *J. Comp. Neurol.* **236**, 234–240.
- Bräunig, P. (1990). The morphology of suboesophageal ganglion cells innervating the nervus corporis cardiaci III of the locust. *Cell Tiss. Res.* **260**, 95–108.
- Bräunig, P. (1991). Subesophageal DUM neurons innervate the principal neuropils of the locust brain. *Philos. Trans. R. Soc. Lond. B* **332**, 221–240.
- Bräunig, P., Allgäuer, C. and Honegger, H.-W. (1990). Suboesophageal DUM neurons are part of the antennal motor system of locusts and crickets. *Experientia* **46**, 259–261.
- Bräunig, P. and Hustert, R. (1980). Proprioceptors with central cell bodies in insects. *Nature* **283**, 768–770.
- Bräunig, P., Pflüger, H.-J. and Hustert, R. (1983). The specificity of central nervous projections of locust mechanoreceptors. *J. Comp. Neurol.* **218**, 197–207.
- Bräunig, P. and Pflüger, H. J. (2001). The unpaired median neurons of insects. *Adv. Insect Physiol.* **28**, 185–266.
- Breer, H. (1981). Comparative studies on cholinergic activities in the central nervous system of *Locusta migratoria*. *J. Comp. Physiol.* **141**, 271–275.
- Breer, H. and Sattelle, D. B. (1987). Molecular properties and functions of insect acetylcholine receptors. *J. Insect Physiol.* **33**, 771–790.
- Brewster, R. and Bodmer, R. (1995). Origin and specification of type-II sensory neurons in drosophila. *Development* **121**, 2923–2936.

- Brewster, R., Hardiman, K., Deo, M., Khan, S. and Bodmer, R. (2001). The selector gene cut represses a neural cell fate that is specified independently of the Achaete-Scute-Complex and atonal. *Mech. Dev.* **105**, 57–68.
- Brodfehner, P. D. and Hoy, R. R. (1990). Ultrasound sensitive neurons in the cricket brain. *J. Comp. Physiol.* **166**, 651–662.
- Brooks, R. A. (1989). A robot that walks; emergent behaviors from a carefully evolved network. *Neural Computat.* **1**, 253–262.
- Brusca, R. C. and Brusca, G. J. (1990). *Invertebrates*. Sunderland, MA: Sinauer Associates.
- Bruski, C. A. and Dunham, D. W. (1990). Antennal waving in the crayfish *Orconectes rusticus* (Girard, 1852) (Decapoda, Astacidea). *Crustaceana* **58**, 83–87.
- Burdohan, J. A. and Comer, C. M. (1990). An antennal-derived mechanosensory pathway in the cockroach: descending interneurons as a substrate of evasive behaviour. *Brain Res.* **535**, 347–352.
- Burdohan, J. A. and Comer, C. M. (1996). Cellular organization of an antennal mechanosensory pathway in the cockroach, *Periplaneta americana*. *J. Neurosci.* **16**, 5830–5843.
- Burrows, M. (1992). Local circuits for the control of leg movements in an insect. *TINS* **15**, 226–232.
- Burrows, M. (1996). *The Neurobiology of an Insect Brain*. Oxford: Oxford University Press.
- Burrows, M. and Newland, P. L. (1993). Correlation between the receptive fields of locust interneurons, their dendritic morphology, and the central projections of mechanosensory neurons. *J. Comp. Neurol.* **329**, 412–426.
- Büschges, A. and El Manira, A. (1998). Sensory pathways and their modulation in the control of locomotion. *Curr. Opin. Neurobiol.* **8**, 733–739.
- Camhi, J. M. and Johnson, E. N. (1999). High-frequency steering maneuvers mediated by tactile cues: antennal wall-following in the cockroach. *J. Exp. Biol.* **202**, 631–643.
- Campbell, F. L. (1972). A new antennal sensillum of *Blattella germanica* (Dictyoptera: Blattellidae) and its presence in other Blattaria. *Ann. Entomol. Soc. Am.* **65**, 888–892.
- Chapman, K. M., Duckrow, R. B. and Moran, D. T. (1973). Form and role of deformation in excitation of an insect mechanoreceptor. *Nature* **244**, 453–454.
- Chapman, R. F. (2003). Contact chemoreception in feeding by phytophagous insects. *Ann. Rev. Entomol.* **48**, 455–484.
- Chapman, R. F. and Lee, J. C. (1991). Environmental effects on numbers of peripheral chemoreceptors on the antennae of a grasshopper. *Chem. Senses* **16**, 607–616.
- Clarac, F., Neil, D. M. and Vedel, J.-P. (1976). The control of antennal movements by leg proprioceptors in the rock lobster, *Palinurus vulgaris*. *J. Comp. Physiol.* **107**, 275–292.
- Clarac, F. and Vedel, J.-P. (1971). Etude des relations fonctionnelles entre le muscle fléchisseur accessoire et les organes sensoriels chordotonaux des appendices locomoteurs de la langouste *Palinurus vulgaris*. *Z. Vergl. Physiol.* **72**, 386–410.
- Clarac, F. and Vedel, J. P. (1975). Neurophysiological study of the antennal motor patterns in the rock lobster. I. Reflex modulation of extensor and flexor motoneurone activity. *J. Comp. Physiol.* **102**, 201–221.
- Comer, C. M., Dowd, J. P. and Stubblefield, T. G. (1988). Escape responses following elimination of the giant interneuron pathway in the cockroach, *Periplaneta americana*. *Brain Res.* **445**, 370–375.
- Comer, C. M., Mara, E., Murphy, K. A., Getman, M. and Mungy, M. C. (1994). Multisensory control of escape in the cockroach *Periplaneta americana*. II. Patterns of touch-evoked behavior. *J. Comp. Physiol. A* **174**, 13–26.

- Comer, C. M., Parks, L., Halvorsen, M. B. and Breese-Terteling, A. (2003). The antennal system and cockroach evasive behavior. II. Stimulus identification and localization are separable antennal functions. *J. Comp. Physiol. A* **189**, 97–103.
- Cornwell, P. B. (1968). *The Cockroach. I. A laboratory Insect and an Industrial Pest*. London: Hutchinson.
- Cowan, N. J., Ma, E. J., Cutkosky, M. and Full, R. J. (2004). A biologically inspired passive antenna for steering control of a running robot. *Proceedings of 11th International Symposium on Robotics Research (ISRR 2004)*, in press.
- Crouau, Y. (1997). Comparison of crustacean and insect mechanoreceptive setae. *Int. J. Ins. Morphol. Embryol.* **26**, 181–190.
- Cruse, H., Kindermann, T., Schumm, M., Dean, J. and Schmitz, J. (1998). Walknet – a biologically inspired network to control six-legged walking. *Neural Networks* **11**, 1435–1447.
- Dambly-Chaudiere, C., Jamet, E., Burri, M., Bopp, D., Basler, K., Hafen, E., Dumont, N., Spielmann, P., Ghysen, A. and Noll, M. (1992). The paired box gene *pox-neuro* – a determinant of poly innervated sense-organs in drosophila. *Cell* **69**, 159–172.
- Damen, W. G. M., Hausdorf, M., Seyfarth, E. A. and Tautz, D. (1998). A conserved mode of head segmentation in arthropods revealed by the expression pattern of *hox* genes in a spider. *Proc. Nat. Acad. Sci. USA* **95**, 10665–10670.
- Dietz, A. and Humphreys, W. J. (1971). Scanning electron microscopic studies of antennal receptors of the worker honeybee, including sensilla campaniformia. *Ann. Entomol. Soc. Am.* **64**, 919–925.
- Distler, P. G. and Boeckh, J. (1996). Synaptic connection between olfactory receptor cells and uniglomerular projection neurons in the antennal lobe of the American cockroach, *Periplaneta americana*. *J. Comp. Neurol.* **370**, 35–46.
- Dostal, B. (1958). Riechfähigkeit und Zahl der Riechsinneselemente bei der Honigbiene. *Z. Vergl. Physiol.* **41**, 179–203.
- Dreller, C. and Kirchner, W. H. (1993a). Hearing in honeybees: localization of the auditory sense organ. *J. Comp. Physiol. A* **173**, 275–279.
- Dreller, C. and Kirchner, W. H. (1993b). How bees perceive the information of the dance language. *Naturwissenschaften* **80**, 319–321.
- Dreller, C. and Kirchner, W. H. (1994). Hearing in the Asian honeybees *Apis dorsata* and *Apis florea*. *Insectes Sociaux* **41**, 291–299.
- Dürr, V. (2000). Grasping movements in an insect: effects of tactile cues and step cycle phase. *Eur. J. Neurosci.* **12** (Suppl. 11), 150.
- Dürr, V. (2001). Stereotypic leg searching-movements in the stick insect: kinematic analysis, behavioural context and simulation. *J. Exp. Biol.* **204**, 1589–1604.
- Dürr, V., König, Y. and Kittmann, R. (2001). The antennal motor system of the stick insect *Carausius morosus*: anatomy and antennal movement pattern during walking. *J. Comp. Physiol. A* **187**, 131–144.
- Dürr, V. and Krause, A. (2001). The stick insect antenna as a biological paragon for an actively moved tactile probe for obstacle detection. In: *Climbing and Walking Robots – From Biology to Industrial Applications. Proc. 4th Int. Conf. Climbing and Walking Robots (CLAWAR 2001, Karlsruhe)*, (eds Berns, K. and Dillmann, R.), pp. 87–96. Bury St. Edmunds, London: Professional Engineering Publishing.
- Dürr, V. and Krause, A. (2002). Design of a biomimetic active tactile sensor for legged locomotion. In: *Climbing and Walking Robots, and the Support Technologies for Mobile Machines. Proceedings of 5th International Conference on Climbing and Walking Robots, CLAWAR 2002, Paris*, (eds Bidaud, P. and Ben Amar, F.), pp. 255–262. Bury St. Edmunds, London: Professional Engineering Publishing.
- Dürr, V., Krause, A., Schmitz, J. and Cruse, H. (2003). Neuroethological concepts and their transfer to walking machines. *Int. J. Robot. Res.* **22**, 151–167.

- Dürr, V. and Ebeling, W. (2005). The behavioural transition from in straight to curve walking: kinetics of leg movement parameters and the initiation of turning. *J. Exp. Biol.* **208**, 2237–2252.
- Ehmer, B. and Gronenberg, W. (1997a). Antennal muscles and fast antennal movements in ants. *J. Comp. Physiol. B* **167**, 187–296.
- Ehmer, B. and Gronenberg, W. (1997b). Proprioceptors and fast antennal reflexes in the ant *Odontomachus* (Formicidae, Ponerinae). *Cell Tiss. Res.* **290**, 153–165.
- Eichmüller, S. and Schäfer, S. (1995). Sensory neuron development revealed by taurine immunocytochemistry in the honeybee. *J. Comp. Neurol.* **352**, 297–307.
- Ellerkmann, E. (1994). Neuronale Kontrolle und Modulation der Antennenbewegung bei der Honigbiene. Doctoral dissertation, Technische Universität Berlin.
- Erber, J., Kierzek, S., Sander, E. and Grandy, K. (1998). Tactile learning in the honeybee. *J. Comp. Physiol. A* **183**, 737–744.
- Erber, J. and Kloppenburg, P. (1995). The modulatory effects of serotonin and octopamine in the visual system of the honeybee (*Apis mellifera* L.). Behavioral analysis of the motion-sensitive antennal reflex. *J. Comp. Physiol. A* **176**, 111–118.
- Erber, J. and Pribbenow, B. (2000). Antennal movements in the honeybee: how complex tasks are solved by a simple neuronal system. In: *Prerational Intelligence. Adaptive Behaviour and Intelligent Systems without Symbols and Logic*, (eds Cruse, H., Dean, J. and Ritter, H.), pp. 109–121. Dordrecht: Kluwer.
- Erber, J., Pribbenow, B., Bauer, A. and Kloppenburg, P. (1993). Antennal reflexes in the honeybee – tools for studying the nervous system. *Apidologie* **24**, 283–296.
- Erber, J., Pribbenow, B., Grandy, K. and Kierzek, S. (1997). Tactile motor learning in the antennal system of the honeybee (*Apis mellifera* L.). *J. Comp. Physiol. A* **181**, 355–365.
- Erber, J., Pribbenow, B., Kisch, J. and Faensen, D. (2000). Operant conditioning of antennal muscle activity in the honeybee (*Apis mellifera* L.). *J. Comp. Physiol. A* **186**, 557–565.
- Ernst, K. D. and Boeckh, J. (1983). A neuroanatomical study on the organisation of the central antennal pathways in insects. III. Neuroanatomical characterization of physiologically defined response types of deutocerebral neurons in *Periplaneta americana*. *Cell Tiss. Res.* **229**, 1–22.
- Espenschied, K. S., Quinn, R. D., Chiel, H. J. and Beer, R. D. (1993). Leg coordination mechanisms in stick insect applied to hexapod robot locomotion. *Adapt. Behav.* **1**, 455–468.
- Esslen, J. and Kaissling, K. E. (1976). Number and distribution of sensilla on antennal flagellum of honeybee (*Apis mellifera* L.). *Zoomorphology* **83**, 227–251.
- Field, L. H. and Matheson, T. (1998). Chordotonal organs of insects. *Adv. Insect Physiol.* **27**, 1–228.
- Flanagan, D. and Mercer, A. R. (1989). Morphology and response characteristics of neurons in the deutocerebrum of the brain in the honeybee *Apis mellifera*. *J. Comp. Physiol. A* **164**, 483–494.
- Fleming, R. C. (1968). Head musculature of sphinx moths (Lepidoptera: Sphingidae). *Contrib. Am. Entomol. Inst.* **3**, 1–32.
- Flores, G. B. and Lazzari, C. R. (1996). The role of the antennae in *Triatoma infestans*: orientation towards thermal sources. *J. Insect Physiol.* **42**, 433–440.
- Foelix, R. F. (1996). *Biology of Spiders*. 2nd ed. Oxford, New York: Oxford University Press.
- French, A. S. and Sanders, E. J. (1979). The mechanism of sensory transduction in the sensilla of the trochanteral hair plate of the cockroach, *Periplaneta americana*. *Cell Tiss. Res.* **198**, 159–174.
- French, A. S. and Sanders, E. J. (1981). The mechanosensory apparatus of the femoral tactile spine of the cockroach, *Periplaneta americana*. *Cell Tiss. Res.* **219**, 53–68.

- Frik, M. and Amendt, O. (1995). Neural control of a walking robot in variable terrain. IFTOMM 1995, Mailand, pp. 2297–2301.
- Frings, H. and Frings, M. (1949). The loci of contact chemoreceptors in insects. *Am. Mid. Nat.* **41**, 602–658.
- Fudalewicz-Niemczyk, W. and Rosciszewska, M. (1973). The peripheric nervous system of the larva of *Gryllus domesticus* L. (Orthoptera). Part I. Antenna. *Acta Biol. Cracov. Ser. Zool.* **XVI**, 209–217.
- Gaffal, K. P., Tichy, H., Theiss, J. and Seelinger, G. (1975). Structural polarities in mechanosensitive sensilla and their influence on stimulus transmission (Arthropoda). *Zoomorphology* **82**, 79–103.
- Gebhardt, M. (1996). Intersegmentale und lokale Informationsverarbeitung im neuronalen Netzwerk zur Kontrolle von Antennenbewegungen bei Grillen (*Gryllus campestris* (L.), *Gryllus bimaculatus* De Geer). Doctoral dissertation, Technische Universität München.
- Gebhardt, M. (2004). In vivo-application of anti-proctolin-antiserum affects antennal flight posture in crickets. *J. Comp. Physiol. A* **190**, 359–364.
- Gebhardt, M. and Honegger, H.-W. (2001). Physiological characterisation of antennal mechanosensory descending interneurons in an insect (*Gryllus bimaculatus*, *Gryllus campestris*) brain. *J. Exp. Biol.* **204**, 2265–2275.
- Gewecke, M. (1970). Antennae: another wind sensitive receptor in locusts. *Nature* **225**, 1263–1264.
- Gewecke, M. (1972a). Antennen und Stirn-Scheitelhaare von *Locusta migratoria* L. als Luftströmungs-Sinnesorgane bei der Flugsteuerung. *J. Comp. Physiol.* **80**, 57–94.
- Gewecke, M. (1972b). Bewegungsmechanismus und Gelenkrezeptoren der Antennen von *Locusta migratoria*. *Z. Morphol. Tiere* **71**, 128–149.
- Gewecke, M. (1975). The influence of air current sense organs on the flight behaviour of *Locusta migratoria*. *J. Comp. Physiol. A* **103**, 79–95.
- Gewecke, M. (1979). Central projection of antennal afferents for the flight motor in *Locusta migratoria* (Orthoptera: Acrididae). *Entomol. Gen.* **5**, 317–320.
- Gewecke, M. and Heinzel, H.-G. (1980). Aerodynamic and mechanical properties of the antennae as air-current sense-organs in *Locusta migratoria*. I. Static characteristics. *J. Comp. Physiol. A* **139**, 357–366.
- Ghysen, A. and Dambly-Chaudiere, C. (1993). The specification of sensory neuron identity in drosophila. *Bioessays* **15**, 293–298.
- Gillespie, P. G. and Walker, R. G. (2001). Molecular basis of mechanosensory transduction. *Nature* **413**, 194–202.
- Gnatzy, W. (1984). Campaniform structures on lobster antennae are dermal glands. *Cell Tiss. Res.* **236**, 729–731.
- Gnatzy, W. and Heußlein, R. (1986). Digger wasp against crickets. I. Receptors involved in the antipredator strategies of the prey. *Naturwissenschaften* **73**, 212–215.
- Gnatzy, W. and Hustert, R. (1989). Mechanoreceptors in behavior. In: *Crickets Behavior and Neurobiology*, (eds Huber, F., Moore, T. E. and Loher, W.), pp. 198–226. Ithaca, London: Cornell University Press.
- Gnatzy, W., Schmidt, M. and Rombke, J. (1984). Are the funnel-canal organs the campaniform sensilla of the shore crab *Carcinus maenas* (Crustacea, Decapoda). 1. Topography, external structure and basic organization. *Zoomorphology* **104**, 11–20.
- Goodman, L. J. (1981). Organisation and physiology of the insect dorsal ocellar system. In: *Comparative Physiology and Evolution of Vision in Invertebrates. C: Invertebrate Visual Centers and Behavior II* C, (eds Autrum, H., Goodman, L. J., Messenger, J. B. and Wehner, R.), pp. 201–286. Springer: Berlin.
- Goulding, S. E., zur Lage, P. and Jarman, A. P. (2000). Amos, a proneural gene for *Drosophila* olfactory sense organs that is regulated by lozenge. *Neuron* **25**, 69–78.

- Göpfert, M. C., Briegel, H. and Robert, D. (1999). Mosquito hearing: sound-induced antennal vibrations in male and female *Aedes aegypti*. *J. Exp. Biol.* **202**, 2727–2738.
- Göpfert, M. C. and Robert, D. (2001a). Active auditory mechanics in mosquitoes. *Proc. R. Soc. Lond. B* **268**, 333–339.
- Göpfert, M. C. and Robert, D. (2001b). Turning the key on *Drosophila* audition. *Nature* **411**, 908.
- Gras, H. and Hörner, M. (1992). Wind-evoked escape running of the cricket *Gryllus bimaculatus*. *J. Exp. Biol.* **171**, 189–214.
- Greenwood, M. and Chapman, R. F. (1984). Differences in numbers of sensilla on the antennae of solitary and gregarious *Locusta migratoria* L. (Orthoptera: Acrididae). *Int. J. Insect Morphol. Embryol.* **13**, 295–301.
- Griss, C. and Rowell, H. F. (1986). Three descending interneurons reporting deviation from course in the locust. I. Anatomy. *J. Comp. Physiol. A* **158**, 765–774.
- Gronenberg, W. and Strausfeld, N. J. (1990). Descending neurons supplying the neck and flight motor of diptera: physiological and anatomical characteristics. *J. Comp. Neurol.* **302**, 973–991.
- Guharay, F. and Sachs, F. (1984). Stretch-activated single ion channel currents in tissue-cultured embryonic chick skeletal muscle. *J. Physiol. Lond.* **352**, 685–701.
- Gundel, M. and Penzlin, H. (1995). Anatomy of neurons crossing the tritocerebral commissures of the cockroach *Periplaneta americana* (Blattaria). *J. Morphol.* **223**, 225–242.
- Gupta, B. P. and Rodrigues, V. (1997). Atonal is a proneural gene for a subset of olfactory sense organs in *Drosophila*. *Genes Cells* **2**, 225–233.
- Haas, M. S., Brown, S. J. and Beeman, R. W. (2001). Homeotic evidence for the appendicular origin of the labrum in *Tribolium castaneum*. *Dev. Genes Evol.* **211**, 96–102.
- Habig, C. and Taylor, R. C. (1982a). The crayfish second antennae. I. Muscles, innervation patterns and movements of antennal segments. *Comp. Biochem. Physiol.* **72A**, 339–347.
- Habig, C. and Taylor, R. C. (1982b). The crayfish second antennae. II. Motoneuron structure as revealed by cobalt chloride backfilling. *Comp. Biochem. Physiol. A* **72**, 349–358.
- Hallberg, E. and Hansson, B. (1999). Arthropod sensilla: morphology and phylogenetic considerations. *Microsc. Res. Tech.* **47**, 428–439.
- Hansson, B. S. (ed.). (1999). *Insect Olfaction*. Berlin: Springer.
- Heckenlively, D. B. (1970). Intensity of aggression in the crayfish *Orconectes virilis* (Hagen). *Nature* **225**, 180–181.
- Hedwig, B. (1994). A cephalothoracic command system controls stridulation in the acridid grasshopper *Omocestus viridulus* L. *J. Neurophysiol.* **72**, 2015–2025.
- Hedwig, B. (1996). A descending brain neuron elicits stridulation in the cricket *Gryllus bimaculatus* (de Geer). *Naturwissenschaften* **83**, 428–429.
- Heinrich, R. (2002). Impact of descending brain neurons on the control of stridulation, walking, and flight in Orthoptera. *Microsc. Res. Tech.* **56**, 301.
- Heinzel, H.-G. and Gewecke, M. (1979). Directional sensitivity of the antennal campaniform sensilla in locusts. *Naturwissenschaften* **66**, 212–213.
- Heinzel, H.-G. and Gewecke, M. (1987). Aerodynamic and mechanical properties of the antennae as air-current sense-organs in *Locusta migratoria*. II. Dynamic characteristics. *J. Comp. Physiol. A* **161**, 671–680.
- Hennig, W. (1986). *Taschenbuch der speziellen Zoologie. Wirbellose II. Gliedertiere*. 4th ed. Frankfurt/M., Thun: Verlag Harry Deutsch.
- Hensler, K. (1988). The pars intercerebralis neurone PI(2)5 of locusts: convergent processing of inputs reporting head movements and deviations from straight flight. *J. Exp. Biol.* **140**, 511–533.

- Hensler, K. (1992). Neuronal co-processing of course deviation and head movements in locusts. I. Descending deviation detectors. *J. Comp. Physiol. A* **171**, 257–271.
- Hensler, K. and Rowell, C. H. F. (1990). Control of opromotor responses by descending deviation detector neurons in intact flying locusts. *J. Exp. Biol.* **149**, 191–205.
- Heran, H. (1957). Die Bienenantenne als Meßorgan der Flugeigengeschwindigkeit. *Naturwissenschaften* **44**, 475.
- Heran, H. (1959). Wahrnehmung der Flugeigengeschwindigkeit bei *Apis mellifica* L. *Z. Vergl. Physiol.* **42**, 103–163.
- Herrnkind, W. (1969). Queuing behavior of spiny lobsters. *Science* **164**, 1425–1427.
- Hertel, H. and Maronde, U. (1987). The physiology and morphology of centrally projecting visual interneurons in the honeybee brain. *J. Exp. Biol.* **133**, 301–315.
- Hicks, J. B. (1857). On a new organ in insects. *J. Proc. Linn. Soc. (London)* **1**, 136–140.
- Higashijima, S., Michiue, T., Emori, Y. and Saigo, K. (1992). Subtype determination of *Drosophila* embryonic external sensory organs by redundant homeo box genes BarH1 and BarH2. *Genes Dev.* **6**, 1005–1018.
- Hildebrand, J. G. (1996). Olfactory control of behavior in moths: central processing of odor information and the functional significance of olfactory glomeruli. *J. Comp. Phys. A* **178**, 5–19.
- Hill, E. S., Iwano, M., Gatellier, L. and Kanzaki, R. (2002). Morphology and physiology of the serotonin-immunoreactive putative antennal lobe feedback neuron in the male silkworm *Bombyx mori*. *Chem. Senses* **27**, 475–483.
- Hofmann, H. A. and Schildberger, K. (2001). Assessment of strength and willingness to fight during aggressive encounters in crickets. *Anim. Behav.* **62**, 337–348.
- Hofmann, T., Koch, U. T. and Bässler, U. (1985). Physiology of the femoral chordotonal organ in the stick insect, *Cuniculina impigra*. *J. Exp. Biol.* **114**, 207–223.
- Holst, E. V. and Mittelstaedt, H. (1950). Das Reafferenzprinzip. Wechselwirkungen zwischen Zentralnervensystem und Peripherie. *Naturwissenschaften* **37**, 464–476.
- Holzer, R. and Shimoyama, I. (1997). Locomotion control of a bio-robotic system via electric stimulation. *Proc. Int. Conf. Intell. Robots Systems* 1514–1519.
- Homberg, U. (1984). Processing of antennal information in extrinsic mushroom body neurons of the bee brain. *J. Comp. Physiol.* **154**, 825–836.
- Homberg, U. (1994). Flight-correlated activity changes in neurons of the lateral accessory lobes in the brain of the locust *Schistocerca gregaria*. *J. Comp. Physiol. A* **175**, 597–610.
- Homberg, U. (2002). Neurotransmitters and neuropeptides in the brain of the locust. *Microsc. Res. Tech.* **56**, 189–209.
- Homberg, U., Christensen, T. A. and Hildebrand, J. G. (1989). Structure and function of the deutocerebrum in insects. *Ann. Rev. Entomol.* **34**, 477–501.
- Homberg, U. and Hildebrand, J. G. (1989). Serotonin-immunoreactive neurons in the median protocerebrum and suboesophageal ganglion of the sphinx moth *Manduca sexta*. *Cell Tiss. Res.* **258**, 1–24.
- Homberg, U., Kingan, T. G. and Hildebrand, J. G. (1987). Immunocytochemistry of GABA in the brain and suboesophageal ganglion of *Manduca sexta*. *Cell Tiss. Res.* **248**, 1–24.
- Homberg, U., Kingan, T. G. and Hildebrand, J. G. (1990). Distribution of FMRF-amide-like immunoreactivity in the brain and suboesophageal ganglion of the sphinx moth *Manduca sexta* and colocalization with SCPB-, BPP-, and GABA-like immunoreactivity. *Cell Tiss. Res.* **259**, 401–419.
- Homberg, U., Wurden, S., Dirksen, H. and Rao, K. R. (1991). Comparative anatomy of pigment-dispersing hormone-immunoreactive neurons in the brain of orthoptoid insects. *Cell Tiss. Res.* **266**, 343–357.

- Honegger, H.-W. (1981). A preliminary note on a new optomotor response in crickets: antennal tracking of moving targets. *J. Comp. Physiol. A* **142**, 419–421.
- Honegger, H.-W., Allgäuer, C., Klepsch, U. and Welker, J. (1990a). Morphology of antennal motoneurons in the brains of two crickets, *Gryllus bimaculatus* and *Gryllus campestris*. *J. Comp. Neurol.* **291**, 256–268.
- Honegger, H.-W., Bartos, M., Gramm, T. and Gebhardt, M. J. (1995). Peripheral modulation and plasticity of antennal movements in crickets. *Verh. Dt. Zool. Ges.* **88**, 129–137.
- Honegger, H.-W., Brunner, B., Bräunig, P. and Elekes, K. (1990b). GABA-like immunoreactivity in a common inhibitory neuron of the antennal motor system of crickets. *Cell Tiss. Res.* **260**, 349–354.
- Honegger, H.-W. and Campan, R. (1989). Vision and visually guided behavior. In: *Crick Behavior and Neurobiology*, (eds Huber, F., Moore, R. and Loher, W.), pp. 147–177. Ithaca, London: Cornell University Press.
- Honegger, H. W., Garcia-Scheible, I., Kammerer, R. and Laurent, G. (1985). Vibration blocks habituation of an optical released antennal response in crickets. In: *Acoustic and Vibrational Communication in Insects*, (eds Kalmring, K. and Elsner, N.), pp. 223–230. Berlin, Hamburg: Paul Parey.
- Horn, E. (1983). Gravity. In: *Comprehensive Insect Physiology, Biochemistry and Pharmacology*, Vol. 6 (eds Kerkut, G. A. and Gilbert, L. I.), pp. 223–230. Oxford, NY: Pergamon Press.
- Horn, E. and Bischof, H.-J. (1983). Gravity reception in crickets: the influence of cercal and antennal afferents on the head position. *J. Comp. Physiol. A* **150**, 93–98.
- Horn, E. and Föller, W. (1985). Tonic and modulatory subsystems of the complex gravity receptor system in crickets, *Gryllus bimaculatus*. *J. Insect Physiol.* **31**, 937–946.
- Horseman, B. G., Gebhardt, M. J. and Honegger, H.-W. (1997). Involvement of the suboesophageal and thoracic ganglia in the control of antennal movements in crickets. *J. Comp. Physiol. A* **181**, 195–204.
- Hoskins, S. G., Homberg, U., Kingan, T. G., Christensen, T. A. and Hildebrand, J. G. (1986). Immunocytochemistry of GABA in the antennal lobes of the sphinx moth *Manduca sexta*. *Cell Tiss. Res.* **244**, 243–252.
- Howse, P. E. (1965). The structure of the subgenual organ and certain other mechanoreceptors of the termite *Zootermopsis angusticollis* (Hagen). *Proc. R. Ent. Soc. Lond. A* **40**, 137–146.
- Howse, P. E. (1968). The fine structure and functional organization of chordotonal organs. *Symp. Zool. Soc. Lond.* **23**, 167–198.
- Howse, P. E. and Claridge, M. F. (1970). Fine structure of Johnston's organ of leaf hopper, *Oncopsis flavicollis*. *J. Insect Physiol.* **16**, 1665.
- Hoyle, G. (1978). Distribution of nerve and muscle fibre types in locust jumping muscle. *J. Exp. Biol.* **73**, 203–233.
- Hölldobler, B. (1970). Zur Physiologie der Gast-Wirt-Beziehungen (Myrmecophilie) bei Ameisen: II. Das Gastverhältnis des imaginalen *Atemeles pubicollis* Bris. (Col. Staphylinidae) zu *Myrmica* und *Formica* (Hym. Formicidae). *Z. Vergl. Physiol.* **66**, 215–250.
- Hölldobler, B. (1976). Tournaments and slavery in a desert ant. *Science* **192**, 912–914.
- Hölldobler, B. (1985). Liquid food transmission and antennation signals in ponerine ants. *Israel J. Entomol.* **XIX**, 89–99.
- Hölldobler, B. (1999). Multimodal signals in ant communication. *J. Comp. Physiol. A* **184**, 129–141.
- Hölldobler, B., Möglich, M. and Maschwitz, M. (1974). Communication by tandem running in the ant *Camponotus sericeus*. *J. Comp. Physiol.* **90**, 105–127.

- Hörner, M. (1992). Wind-evoked escape running of the cricket *Gryllus bimaculatus*. II. Neurophysiological analysis. *J. Exp. Biol.* **171**, 215–245.
- Hörner, M. and Gras, H. (1985). Physiological properties of some descending neurons in the cricket brain. *Naturwissenschaften* **72**, 603–604.
- Huber, F. (1955). Sitz und Bedeutung nervöser Zentren für Instinkthandlungen beim Männchen von *Gryllus campestris* L. *Z. Tierpsychol.* **12**, 12–48.
- Huber, R. and Kravitz, E. A. (1995). A quantitative analysis of agonistic behavior in juvenile American lobsters (*Homarus americanus* L.). *Brain Behav. Evol.* **46**, 72–83.
- Hughes, G. M. and Wiersma, A. G. (1960). Neuronal pathways and synaptic connexions in the abdominal cord of the crayfish. *J. Exp. Biol.* **37**, 291–307.
- Imms, A. D. (1939). On the antennal musculature in insects and other arthropods. *Quart. J. Microsc. Sci.* **81**, 273–320.
- Jacobs, G. A. and Theunissen, F. E. (1996). Functional organization of a neural map in the cricket cercal sensory system. *J. Neurosci.* **16**, 769–784.
- Janet, C. (1911). Sur l'existence d'un organe chordotonal et d'une vesicle pulsatile antennaires chez l'abeille et sur la morphologie de la tete de cet. *Comptes Rendus hebdomadaires Seances l'Academie Sci.* **152**, 110–113.
- Jhaveri, D., Sen, A., Reddy, G. V. and Rodrigues, V. (2000). Sense organ identity in the *Drosophila* antenna is specified by the expression of the proneural gene atonal. *Mech. Dev.* **99**, 101–111.
- Kämper, G. (1992). Development of cricket sensory hairs changes of dynamic mechanical properties. *J. Comp. Physiol. A* **170**, 49–55.
- Kammerer, R. and Honegger, H. W. (1988). The role of mechanoreceptors in the control of antennal tracking movements of crickets. *Proc. Göttingen. Neurobiol. Conf.* **201**.
- Kaneko, M., Kanayma, N. and Tsuji, T. (1995). 3-D active antenna for contact sensing. *Proc. IEEE Int. Conf. Robot. Automat.* **1995**, 1113–1119.
- Kaneko, M., Kanayma, N. and Tsuji, T. (1998). Active antenna for contact sensing. *IEEE Trans. Robot. Automat.* **14**, 278–291.
- Kanzaki, R., Arbas, E. A. and Hildebrand, J. G. (1991). Physiology and morphology of descending neurons in the pheromone-processing olfactory pathways in the male moth *Manduca sexta*. *J. Comp. Physiol. A* **169**, 1–14.
- Kanzaki, R., Arbas, E. A., Strausfeld, N. J. and Hildebrand, J. G. (1989). Physiology and morphology of projection neurons in the antennal lobe of the male moth *Manduca sexta*. *J. Comp. Physiol. A* **165**, 427–453.
- Keegan, A. P. and Comer, C. M. (1993). The wind-elicited escape response of cockroaches (*Periplaneta americana*) is influenced by lesions rostral to the escape circuit. *Brain Res.* **620**, 310–316.
- Keil, T. A. (1982). Contacts of pore tubules and sensory dendrites in antennal chemosensilla of a silkworm – demonstration of a possible pathway for olfactory molecules. *Tissue Cell* **14**, 451–462.
- Keil, T. A. (1989). Fine structure of the pheromone-sensitive sensilla on the antenna of the hawkmoth, *Manduca sexta*. *Tissue Cell* **21**, 139–151.
- Kevan, P. G. and Lane, M. A. (1985). Flower petal microtexture is a tactile cue for bees. *Proc. Natl. Acad. Sci. USA* **82**, 4750–4752.
- Kim, J. L. and Yamasaki, T. (1996). Sensilla of *Carabus (Isiocarabus) fiduciaris saishutoicus* Csiki (Coleoptera: Carabidae). *Int. J. Insect Morphol. Embryol.* **25**, 153–172.
- Kim, Y., Lee, H. and Kim, K. (1990). Projection of antennal receptor cells into the dorsal lobe of brain of *Pieris rapae* (Insecta, Lepidoptera). *Korean J. Zool.* **33**, 28–34.
- Kindermann, T. (2002). Behavior and adaptability of a six-legged walking system with highly distributed control. *Adapt. Behav.* **9**, 16–41.

- Kirchner, W. H. (1994). Hearing in honeybees: The mechanical response of the bee's antenna to near-field sound. *J. Comp. Physiol. A* **175**, 261–265.
- Kirchner, W. H., Dreller, C. and Towne, W. F. (1991). Hearing in honeybees: operant conditioning and spontaneous reactions to airborne sound. *J. Comp. Physiol. A* **168**, 85–89.
- Kisch, J. and Erber, J. (1999). Operant conditioning of antennal movements in the honeybee. *Behav. Brain Res.* **99**, 93–102.
- Klein, U., Bock, C., Kafka, W. A. and Moore, T. E. (1988). Antennal sensilla of *Magicicada cassini* Fisher (Homoptera, Cicadidae). Fine structure and electrophysiological evidence for olfaction. *Int. J. Insect Morphol. Embryol.* **17**, 153–167.
- Kloppenburger, P. (1995). Anatomy of the antennal motoneurons in the brain of the honeybee (*Apis mellifera*). *J. Comp. Neurol.* **363**, 333–343.
- Kloppenburger, P., Camazine, S. M., Sun, X. J., Randolph, P. and Hildebrand, J. G. (1997). Organization of the antennal motor system in the sphinx moth *Manduca sexta*. *Cell Tiss. Res.* **287**, 425–433.
- Kloppenburger, P. and Erber, J. (1995). The modulatory effects of serotonin and octopamine in the visual system of the honeybee (*Apis mellifera* L.): II. Electrophysiological analysis of motion-sensitive neurons in the lobula. *J. Comp. Physiol. A* **176**, 119–129.
- Kloppenburger, P., Kirchhof, B. S. and Mercer, A. R. (1999). Voltage-activated currents from adult honeybee (*Apis mellifera*) antennal motor neurons recorded *in vitro* and *in situ*. *J. Neurophysiol.* **81**, 39–48.
- Klose, M. (1996). Development of leg chordotonal sensory organs in normal and heat shocked embryos of the cricket *Teleogryllus* (Walker). *Roux's. Arch. Dev. Biol.* **205**, 344–355.
- Knyazeva, N. I. (1974). Campaniform sensilla of *Locusta migratoria* L. (Orthoptera, Acrididae). *Entom. Rev.* **53**, 62–66.
- Knyazeva, N. I., Fudalewicz-Niemczyk, W. and Rosciszewska, M. (1975). Proprioceptors of the house-cricket (*Gryllus domesticus* L.) (Orthoptera). *Acta Biol. Cracov. Ser. Zool.* **18**, 33–44.
- Knyazeva, N. I., Fudalewicz-Niemczyk, W. and Rosciszewska, M. (1978). Proprioceptors in the nymph of the house cricket *Gryllus domesticus* (Orthoptera, Gryllidae). *Entom. Rev.* **58**, 16–19.
- Koch, M. (2000). The cuticular cephalic endoskeleton of primarily wingless hexapods: ancestral state and evolutionary changes. *Pedobiologia.* **44**, 374–385.
- Kondoh, Y., Okuma, J. and Newland, P. L. (1995). Dynamics of neurons controlling movements of a locust hind leg – wiener kernel analysis of the responses of proprioceptive afferents. *J. Neurophysiol.* **73**, 1829–1842.
- Konings, N. M., Vullings, G. B., Geffard, M., Buijs, R. M., Diederer, H. B. and Jansen, W. F. (1988). Immunocytochemical demonstration of octopamine-immunoreactive cells in the nervous system of *Locusta migratoria* and *Schistocerca gregaria*. *Cell Tiss. Res.* **251**, 371–379.
- Krause, A. F. and Dürre, V. (2004). Tactile efficiency of insect antennae with two hinge joints. *Biol. Cybern.* **91**, 168–181.
- Kreissl, S. and Bicker, G. (1989). Histochemistry of acetylcholinesterase and immunocytochemistry of an acetylcholine receptor-like antigen in the brain of the honeybee. *J. Comp. Neurol.* **286**, 71–84.
- Kreissl, S., Eichmüller, S., Bicker, G., Rapus, J. and Eckert, M. (1994). Octopamine-like immunoreactivity in the brain and suboesophageal ganglion of the honeybee. *J. Comp. Neurol.* **348**, 583–595.
- Kros, C. J., Rusch, A. and Richardson, G. P. (1992). Mechano-electrical transducer currents in hair cells of the cultured neonatal mouse cochlea. *Proc. R. Soc. Lond. B* **249**, 185–193.

- Kuster, J. E., French, A. S. and Sanders, E. J. (1983). The effects of microtubule dissociating agents on the physiology and cytology of the sensory neuron in the femoral tactile spine of the cockroach, *Periplaneta americana* L. *Proc. R. Soc. Lond. B* **219**, 397–412.
- Kuwana, Y., Nagasawa, S., Shimoyama, I. and Kanzaki, R. (1999). Synthesis of the pheromone-oriented behaviour of silkworm moths by a mobile robot with moth antennae as pheromone sensors. *Biosensors Bioelectron.* **14**, 195–202.
- Lambin, M. (1973). Le sensilles de l'antenne chez quelques blattes et en particulier chez *Blaberus cranifer* (Burm.). *Z. Zellforsch.* **143**, 183–206.
- Laurent, G. (1996). Dynamical representation of odors by oscillating and evolving neural assemblies. *TINS* **19**, 489–496.
- Lee, J. K. and Strausfeld, N. J. (1990). Structure, distribution and number of surface sensilla and their receptor cells on the olfactory appendage of the male moth *Manduca sexta*. *J. Neurocytol.* **19**, 519–538.
- Lee, M. H. and Nicholls, H. R. (1999). Tactile sensing for mechatronics – a state of the art survey. *Mechatronics* **9**, 1–31.
- Lenoir, A. and Jaisson, P. (1982). Evolution et rôle des communications antennaires chez les insectes sociaux. In: *Social Insects in the Tropics*, (ed. Jaisson, P.), pp. 157–180. Paris: Presses de l'Université de Paris XIII.
- Linsenmair, K. E. (1970). Die Interaktion der paarigen antennalen Sinnesorgane bei der Windorientierung laufender Käfer (Insecta, Coleoptera). *Z. Vergl. Physiol.* **70**, 247–277.
- Linsenmair, K. E. (1973). Die Windorientierung laufender Insekten. *Fortschr. Zool.* **21**, 59–79.
- Loftus, R. (1966). Cold receptor on the antenna of *Periplaneta americana*. *Z. Vergl. Physiol.* **52**, 380–385.
- Loftus, R. (1968). The response of the antennal cold receptor of *Periplaneta americana* to rapid temperature changes and to steady temperature. *Z. Vergl. Physiol.* **59**, 413–455.
- Loftus, R. (1969). Differential thermal components in the response of the antennal cold receptor of *Periplaneta americana* to slowly changing temperature. *Z. Vergl. Physiol.* **63**, 415–433.
- Loftus, R. (1976). Temperature-dependent dry receptor on antenna of *Periplaneta* – Tonic response. *J. Comp. Physiol.* **111**, 153–170.
- Loher, W. and Rence, B. (1978). The mating behavior of *Teleogryllus commodus* (Walker) and its central and peripheral control. *Z. Tierpsychol.* **46**, 225–259.
- Lutz, E. M. and Tyrer, N. M. (1988). Immunohistochemical localization of serotonin and choline-acetyltransferase in sensory neurons of the locust. *J. Comp. Neurol.* **267**, 335–342.
- Malun, D. (1991). Inventory and distribution of synapses of identified uniglomerular projection neurons in the antennal lobe of *Periplaneta americana*. *J. Comp. Neurol.* **305**, 348–360.
- Malun, D., Waldow, U., Kraus, D. and Boeckh, J. (1993). Connections between the deutocerebrum and the protocerebrum, and neuroanatomy of several classes of deutocerebral projection neurons in the brain of male *Periplaneta americana*. *J. Comp. Neurol.* **329**, 143–162.
- Manning, A. (1967). Antennae and sexual receptivity in *Drosophila melanogaster* females. *Science* **158**, 136–137.
- Markl, H. (1962). Borstenfelder an den Gelenken als Schweresinnesorgane bei Ameisen und anderen Hymenopteren. *Z. Vergl. Physiol.* **45**, 475–569.
- Maronde, U. (1991). Common projection areas of antennal and visual pathways in the honeybee brain *Apis mellifera*. *J. Comp. Neurol.* **309**, 328–340.

- Martin, H. (1965). Leistungen des topochemischen Sinnes bei der Honigbiene. *Z. Vergl. Physiol.* **50**, 254–292.
- Martin, H. and Lindauer, M. (1966). Sinnesphysiologische Leistungen beim Wabenbau der Honigbiene. *Z. Vergl. Physiol.* **53**, 372–404.
- Maschwitz, U., Hölldobler, B. and Möglich, M. (1974). Tandemlaufen als Rekrutierungsverhalten bei *Bothroponera tesserinoda* Forel (Formicidae: Ponerinae). *Z. Tierpsychol.* **35**, 113–123.
- Mathenia, N. R., Leung, V., Baba, Y. and Comer, C. M. (2003). Comparison of a specific descending mechanosensory interneuron in cockroaches and crickets. *Abstract Viewer/Itinerary Planner. Washington, DC: Society for Neuroscience. CD-ROM.* **2003**, 269.9.
- Matheson, T. (1990). Responses and locations of neurons in the locust metathoracic femoral chordotonal organ. *J. Comp. Physiol. A* **166**, 915–927.
- Matheson, T. (1992). Range fractionation in the locust metathoracic femoral chordotonal organ. *J. Comp. Physiol. A* **170**, 509–520.
- Matsuda, R. (1965). *Morphology and Evolution of the Insect Head*. Ann Arbor, MI: American Entomological Institute.
- McFarlane, J. E. (1953). The morphology of the chordotonal organs of the antenna, mouthparts and legs of the lesser migratory grasshopper, *Melanoplus mexicanus mexicanus* (Saussure). *Can. Entomol.* **LXXXV**, 81–103.
- McGruer, N. E., Adams, G. G., Truong, T. Q., Barnes, T. G., Lu, X. and Aceros, J. C. (2002). Biomimetic flow and contact/bending MEMs sensors. In: *Neurotechnology for Biomimetic Robots*, (eds Ayers, J., Davis, J. L. and Rudolph, A.), pp. 11–30. Cambridge/MA, London: MIT Press.
- McIndoo, N. E. (1914). The olfactory sense of the honey-bee. *J. Exp. Biol.* **16**, 265–346.
- McIndoo, N. E. (1922). The auditory sense of the honey-bee. *J. Comp. Neurol.* **34**, 173–199.
- McIver, S. B. (1975). Structure of cuticular mechanoreceptors of arthropods. *Ann. Rev. Entomol.* **20**, 381–397.
- McIver, S. B. (1985). Mechanoreception. In: *Comprehensive Insect Physiology, Biochemistry and Pharmacology*, (eds Kerkut, G. A. and Gilbert, L. I.), pp. 71–132. Oxford, NY: Pergamon Press.
- Menzel, R. and Müller, U. (1996). Learning and memory in honeybees: from behavior to neural substrates. *Ann. Rev. Neurosci.* **19**, 379–404.
- Mercer, A., Mobbs, P. G., Evans, P. D. and Davenport, A. (1983). Biogenic amines in the brain of the honeybee, *Apis mellifera*. *Cell Tiss. Res.* **234**, 655–677.
- Merritt, D. J. (1997). Transformation of external sensilla to chordotonal sensilla in the cut mutant of *Drosophila* assessed by single-cell marking in the embryo and larva. *Microsc. Res. Tech.* **39**, 492–505.
- Michelsen, A. (1993). The transfer of information in the dance language of honeybees: progress and problems. *J. Comp. Physiol. A* **173**, 135–141.
- Milde, J. J. and Strausfeld, N. J. (1990). Cluster organization and response characteristics of the giant fiber pathway of the blowfly *Calliphora erythrocephala*. *J. Comp. Neurol.* **294**, 59–75.
- Miles, C. I., May, M. L., Holbrook, E. H. and Hoy, R. R. (1992). Multisegmental analysis of acoustic startle in the flying cricket (*Teleogryllus oceanicus*): kinematics and electromyography. *J. Exp. Biol.* **169**, 19–36.
- Mill, P. J. (1976). Chordotonal organs of crustacean appendages. In: *Structure and Function of Proprioceptors in the Invertebrates*, (ed. Mill, P. J.), London: Chapman & Hall.
- Monteforti, G., Angeli, S., Petacchi, R. and Minnocci, A. (2002). Ultrastructural characterization of antennal sensilla and immunocytochemical localization of a

- chemosensory protein in *Carausius morosus* Brunner (Phasmida : Phasmatidae). *Arthropod Struct. Dev.* **30**, 195–205.
- Moore, T. E., Crary, S. B., Koditschek, D. E. and Conklin, T. A. (1998). Directed locomotion in cockroaches: “Biobots”. *Acta Entomol. Slovenica* **6**, 71–78.
- Moran, D. T. and Varela, F. G. (1971). Microtubules and sensory transduction. *Proc. Natl. Acad. Sci. USA* **68**, 757–760.
- Moran, D. T., Varela, F. J. and Rowley, J. C. (1977). Evidence for active role of cilia in sensory transduction. *Proc. Natl. Acad. Sci. USA* **74**, 793–797.
- Möhl, B. and Bacon, J. (1983). The tritocerebral commissure giant (TCG) wind-sensitive interneuron in the locust. II. Directional sensitivity and role in flight stabilisation. *J. Comp. Physiol.* **150**, 453–465.
- Murphey, R. K., Possidente, D., Pollack, P. and Ghysen, A. (1989). Compartments and topography of leg afferent projections in *Drosophila*. *J. Neurosci.* **9**, 3209–3217.
- Mücke, A. and Lakes-Harlan, R. (1995). Central projections of sensory cells of the midleg of the locust, *Schistocerca gregaria*. *Cell Tiss. Res.* **280**, 391–400.
- Myers, C. M. and Evans, P. D. (1987). An FMRFamide antiserum differentiates between populations of antigens in the brain and retrocerebral complex of the locust, *Schistocerca gregaria*. *Cell Tiss. Res.* **250**, 93–99.
- Naessel, D. R., Cantera, R. and Karlsson, A. (1992). Neurons in the cockroach nervous system reacting with antisera to the neuropeptide Leukokinin I. *J. Comp. Neurol.* **322**, 45–67.
- Neil, D. M., Barnes, W. J. P. and Burns, M. D. (1982). Reflex antennal movements in the spiny lobster *Palinurus elephas*. I. Properties of reflexes and their interaction. *J. Comp. Physiol.* **147**, 259–268.
- Neil, D. M., Priest, T. D., Miyan, J. A., Wotherspoon, R. M. and Schoene, H. (1984). Co-ordinated equilibrium responses at two joints in the spiny lobster antenna in relation to the pattern of movements imposed upon the legs. *J. Comp. Physiol.* **155**, 351–363.
- Neuhäuser, T., Sorge, D., Stay, B. and Hoffmann, K. H. (1994). Responsiveness of the adult cricket (*Gryllus bimaculatus* and *Acheta domestica*) retrocerebral complex to allatostatin-1 from a cockroach, *Diploptera punctata*. *J. Comp. Physiol. B* **164**, 23–31.
- Newland, P. L. (1990). Morphology of a population of mechanosensory ascending interneurons in the metathoracic ganglion of the locust. *J. Comp. Neurol.* **299**, 242–260.
- Newland, P. L., Rogers, S. M., Gaaboub, I. and Matheson, T. (2000). Parallel somatotopic maps of gustatory and mechanosensory neurons in the central nervous system of an insect. *J. Comp. Neurol.* **425**, 82–96.
- Nottebohm, E., Dambly-Chaudiere, C. and Ghysen, A. (1992). Connectivity of chemosensory neurons is controlled by the gene *poxn* in *Drosophila*. *Nature* **359**, 829–832.
- Nye, S. W. and Ritzmann, R. E. (1992). Motion analysis with leg joints associated with escape turns of the cockroach, *Periplaneta americana*. *J. Comp. Physiol.* **171**, 183–194.
- Ochieng, S., Hallberg, E. and Hansson, B. (1998). Fine structure and distribution of antennal sensilla of the desert locust, *Schistocerca gregaria* (Orthoptera: Acrididae). *Cell Tiss. Res.* **291**, 525–536.
- Okada, J., Kanamaru, Y. and Toh, Y. (2002). Mechanosensory control of antennal movement by the scapal hair plate in the American cockroach. *Zool. Sci.* **19**, 1201–1210.
- Okada, J. and Toh, Y. (2000). The role of antennal hair plates in object-guided tactile orientation of the cockroach (*Periplaneta americana*). *J. Comp. Physiol. A* **186**, 849–857.

- Okada, J. and Toh, Y. (2001). Peripheral representation of antennal orientation by the scapal hair plate of the cockroach *Periplaneta americana*. *J. Exp. Biol.* **204**, 4301–4309.
- Okada, R., Ikeda, J. and Mizunami, M. (1999). Sensory responses and movement-related activities in extrinsic neurons of the cockroach mushroom bodies. *J. Comp. Physiol. A* **185**, 115–129.
- Okada, R., Sakura, M. and Mizunami, M. (2003). Distribution of dendrites of descending neurons and its implications for the basic organisation of the cockroach brain. *J. Comp. Neurol.* **458**, 158–174.
- Ott, S. R. and Elphick, M. R. (2002). Nitric oxide synthase histochemistry in insect nervous systems: methanol/formalin fixation reveals the neuroarchitecture of formaldehyde-sensitive NADPH diaphorase in the cockroach *Periplaneta americana*. *J. Comp. Neurol.* **448**, 165–185.
- Ott, S. R., Jones, I. W., Burrows, M. and Elphick, M. R. (2000). Sensory afferents and motor neurons as targets for nitric oxide in the locust. *J. Comp. Neurol.* **422**, 521–532.
- Pareto, A. (1972). Die zentrale Verteilung der Fühlerafferenz bei Arbeiterinnen der Honigbiene, *Apis mellifera* L. *Z. Zellforsch.* **131**, 109–140.
- Park, K. C. and Baker, T. C. (2002). Improvement of signal-to-noise ratio in electroantennogram responses using multiple insect antennae. *J. Insect Physiol.* **48**, 1139–1145.
- Park, K. C., Ochieng, S. A., Zhu, J. W. and Baker, T. C. (2002). Odor discrimination using insect electroantennogram responses from an insect antennal array. *Chem. Senses* **27**, 343–352.
- Paydar, S., Doan, C. A. and Jacobs, G. A. (1999). Neural mapping of direction and frequency in the cricket cercal sensory system. *J. Neurosci.* **19**, 1771–1781.
- Pelletier, Y. and McLeod, C. D. (1994). Obstacle perception by insect antennae during terrestrial locomotion. *Physiol. Entomol.* **19**, 360–362.
- Petit, L. (1958). Le déterminisme génétique et psychophysiologique de la compétition sexuelle chez *Drosophila melanogaster*. *Bull. Biol. France Belg.* **92**, 248–327.
- Petryszak, A. (1975). Sensory peripheral nervous system of *Periplaneta americana* L. (Blattoidea). II. Antenna. *Acta Biol. Cracov. Ser. Zool.* **18**, 257–263.
- Pfaffmann, C. (1955). Gustatory nerve impulses in rat, cat and rabbit. *J. Neurophysiol.* **18**, 429–440.
- Pfeiffer, F., Eltze, J. and Weidemann, H. J. (1995). Six-legged technical walking considering biological principles. *Robot. Auton. Systems* **14**, 223–232.
- Poulet, J. F. A. and Hedwig, B. (2002). A corollary discharge maintains auditory sensitivity during sound production. *Nature* **418**, 872–876.
- Poulet, J. F. A. and Hedwig, B. (2003). A corollary discharge mechanism modulates central auditory processing in singing crickets. *J. Neurophysiol.* **89**, 1528–1540.
- Pribbenow, B. and Erber, J. (1996). Modulation of antennal scanning in the honeybee by sucrose stimuli, serotonin, and octopamine: behavior and electrophysiology. *Neurobiol. Learn. Memory* **66**, 109–120.
- Pringle, W. S. (1938a). Proprioception in insects. I. A new type of mechanical receptor from the palps of the cockroach. *J. Exp. Biol.* **15**, 100–113.
- Pringle, W. S. (1938b). Proprioception in insects. II. The action of the campaniform sensilla on the legs. *J. Exp. Biol.* **15**, 114–131.
- Pringle, W. S. (1948). The gyroscopic mechanism of the halteres of diptera. *Philos. Trans. R. Soc. Lond. B* **233**, 347–384.
- Rehder, V., Bicker, G. and Hammer, M. (1987). Serotonin-immunoreactive neurons in the antennal lobes and suboesophageal ganglion of the honeybee. *Cell Tiss. Res.* **247**, 59–66.

- Reichert, H. (1993). Sensory inputs and flight orientation in locusts. *Comp. Biochem. Physiol. A* **104**, 647–657.
- Reichert, H. and Rowell, C. H. F. (1986). Neural circuits controlling flight in the locust: how sensory information is processed for motor control. *TINS* **9**, 281–283.
- Remy, C., Guy, J., Pelletier, G. and Boer, H. H. (1988). Immunohistological demonstration of a substance related to neuropeptide-Y and FMRFamide in the cephalic and thoracic nervous systems of the locust *Locusta migratoria*. *Cell Tiss. Res.* **254**, 189–195.
- Rence, B. and Loher, W. (1977). Contact chemoreceptive sex recognition in the male cricket, *Teleogryllus commodus*. *Physiol. Entomol.* **2**, 225–236.
- Richard, D., Preteur, V., Campan, R., Beugnon, G. and Williams, L. (1985). Visual interneurons of the neck connectives in *Gryllus bimaculatus*. *J. Insect Physiol.* **31**, 407–417.
- Rind, F. C. and Leitter, G. (2000). Immunocytochemical evidence that collision sensing neurons in the locust visual system contain acetylcholine. *J. Comp. Neurol.* **423**, 389–401.
- Risler, H. (1977). The construction of the auditory organ in male mosquitoes. *Fortschr. Zool.* **24**, 143–147.
- Ritzmann, R. E., Quinn, R. D., Watson, J. T. and Zill, S. N. (2000). Insect walking and biorobotics: a relationship with mutual benefits. *Bioscience* **50**, 23–33.
- Rogers, S. M. and Simpson, S. J. (1997). Experience-dependent changes in the number of chemosensory sensilla on the mouthparts and antennae of *Locusta migratoria*. *J. Exp. Biol.* **200**, 2313–2321.
- Rogers, S., Matheson, T., Despland, E., Dodgson, T., Burrows, M. and Simpson, S. (2003). Mechanosensory-induced behavioural gregarization in the desert locust *Schistocerca gregaria*. *J. Exp. Biol.* **206**, 3991–4002.
- Rohrseitz, K. and Tautz, J. (1999). Honeybee dance communication: waggle run direction coded in antennal contacts?. *J. Comp. Physiol. A* **184**, 463–470.
- Rosciszewska, M. and Petryszak, A. (1986). The sense organs of the antennae of *Periplaneta americana* L. (Blattoidea), *Carausius morosus* B. (Cheleutoptera), *Gryllus domesticus* L. (Orthoptera). *Folia histochemica et cytobiologica* **24**, 318–319.
- Rospars, J. P. (1988). Structure and development of the insect antennodeutocerebral system. *Int. J. Insect Morphol. Embryol.* **17**, 243–294.
- Rospars, J. P. and Hildebrand, J. G. (1992). Anatomical identification of glomeruli in the antennal lobes of the male sphinx moth *Manduca sexta*. *Cell Tiss. Res.* **270**, 205–227.
- Rossi-Durand, C. and Vedel, J. P. (1981). Antennal proprioception in the rock lobster *Palinurus vulgaris*: anatomy and physiology of a bi-articular chordotonal organ. *J. Comp. Physiol.* **145**, 505–516.
- Rossi, E. L. and Vedel, J. P. (1980). Encoding of the angular position of the two distal joints of the spiny lobster antennae by a single mechanoreceptor. *Neurosci. Lett.* **18**, 301–308.
- Roth, L. M. and Willis, E. R. (1952). A study of cockroach behaviour. *Amer. Mid. Nat.* **47**, 65–129.
- Rowell, C. H. F. and O'Shea, M. (1976a). Neuronal basis of a sensory analyser, acridid movement detector system. 1. Effects of simple incremental and decremental stimuli in light and dark adapted animals. *J. Exp. Biol.* **65**, 273–288.
- Rowell, C. H. F. and O'Shea, M. (1976b). Neuronal basis of a sensory analyser, acridid movement detector system. 3. Control of response amplitude by tonic lateral inhibition. *J. Exp. Biol.* **65**, 617–625.
- Rowell, H. F. and Reichert, H. (1986). Three descending interneurons reporting deviation from course in the locust. II. Physiology. *J. Comp. Physiol. A* **158**, 775–794.

- Römer, H. (1983). Tonotopic organization of the auditory neuropile in the bushcricket *Tettigonia viridissima*. *Nature* **306**, 60–62.
- Römer, H., Marquart, V. and Hardt, M. (1988). Organization of a sensory neuropile in the auditory pathway of two groups of Orthoptera. *J. Comp. Neurol.* **275**, 201–215.
- Ruppert, E. E. and Barnes, R. D. (1994). *Invertebrate Zoology*. 6th ed. New York: Saunders College Publishing.
- Rüth, E. (1976). Electrophysiology of sensilla chaetica on antennae of *Periplaneta americana*. *J. Comp. Physiol.* **105**, 55–64.
- Saager, F. and Gewecke, M. (1989). Antennal reflexes in the desert locust *Schistocerca gregaria*. *J. Exp. Biol.* **147**, 519–532.
- Salecker, I. and Distler, P. (1990). Serotonin-immunoreactive neurons in the antennal lobes of the American cockroach *Periplaneta americana* – light microscopic and electron-microscopic observations. *Histochemistry* **94**, 463–473.
- Sandeman, D., Sandeman, R., Derby, C. and Schmidt, M. (1992). Morphology of the brain of crayfish, crabs, and spiny lobsters: a common nomenclature for homologous structures. *Biol. Bull.* **183**, 304–326.
- Sandeman, D. C. (1985). Crayfish antennae as tactile organs: their mobility and the responses of their proprioceptors to displacement. *J. Comp. Physiol. A* **157**, 363–374.
- Sandeman, D. C. (1989). Physical properties sensory receptors and tactile reflexes of the antenna of the Australian freshwater crayfish *Cherax destructor*. *J. Exp. Biol.* **141**, 197–218.
- Sandeman, D. C., Scholtz, G. and Sandeman, R. E. (1993). Brain evolution in decapod crustacea. *J. Exp. Zool.* **265**, 112–133.
- Sandeman, D. C. and Varju, D. (1988). A behavioral study of tactile localization in the crayfish *Cherax destructor*. *J. Comp. Physiol. A* **163**, 525–536.
- Sandeman, D. C. and Wilkens, L. (1983). Motor control of movements of the antennal flagellum in the Australian crayfish, *Euastacus armatus*. *J. Exp. Biol.* **105**, 253–273.
- Schafer, R. and Reagan, P. D. (1981). Colchicine reversibly inhibits electrical activity in arthropod mechanoreceptors. *J. Neurobiol.* **12**, 155–166.
- Schafer, R. and Sanchez, T. V. (1973). Antennal sensory system of cockroach, *Periplaneta americana*: postembryonic development and morphology of sense organs. *J. Comp. Neurol.* **149**, 335–353.
- Schafer, R. and Sanchez, T. V. (1976). Nature and development of sex attractant specificity in cockroaches of genus *Periplaneta*. I. Sexual dimorphism in distribution of antennal sense organs in 5 species. *J. Morphol.* **149**, 139–157.
- Schaller, D. (1978). Antennal sensory system of *Periplaneta americana* L. *Cell Tiss. Res.* **191**, 121–139.
- Schäfer, S. and Bicker, G. (1986a). Common projection areas of 5-HT-like and GABA-like immunoreactive fibers in the visual system of the honeybee. *Brain Res.* **380**, 368–370.
- Schäfer, S. and Bicker, G. (1986b). Distribution of GABA-like immunoreactivity in the brain of the honeybee. *J. Comp. Neurol.* **246**, 287–300.
- Schäfer, S., Bicker, G., Ottersen, O. P. and Storm-Mathisen, J. (1988). Taurine-like immunoreactivity in the brain of the honeybee. *J. Comp. Neurol.* **268**, 60–70.
- Schäfer, S. and Rehder, V. (1989). Dopamine-like immunoreactivity in the brain and suboesophageal ganglion of the honeybee. *J. Comp. Neurol.* **280**, 43–58.
- Scheidler, A., Kaulen, P., Bruning, G. and Erber, J. (1990). Quantitative autoradiographic localization of I-125 alpha-bungarotoxin binding sites in the honeybee brain. *Brain Res.* **534**, 332–335.
- Scheiner, R., Erber, J. and Page, R. E. (1999). Tactile learning and the individual evaluation of the reward in honeybees (*Apis mellifera* L.). *J. Comp. Physiol. A* **185**, 1–10.

- Schildberger, K. (1984). Multimodal interneurons in the cricket brain – properties of identified extrinsic mushroom body cells. *J. Comp. Physiol.* **154**, 71–79.
- Schildberger, K. and Agricola, H. (1992). Allatostatin-like immunoreactivity in the brains of crickets and cockroaches. *Proc. Göttingen. Neurobiol. Conf.* **20**, 494.
- Schmidt-Ott, U., Gonzales-Gaitan, M., Jaeckle, H. and Technau, G. M. (1994). Number, identity, and sequence of the *Drosophila* head segments as revealed by neural elements and their deletion patterns in mutants. *Proc. Natl. Acad. Sci. USA* **91**, 8363–8367.
- Schmidt-Ott, U. and Technau, G. M. (1992). Expression of *en* and *wg* in the embryonic head and brain of *Drosophila* indicates a refolded band of seven segment remnants. *Development* **116**, 111–125.
- Schmidt, K. (1973). Comparative morphology of insect mechanoreceptors. *Verh. Dtsch. Zool. Ges.* **1972**, 15–25.
- Schmidt, M. (1990). Ultrastructure of a possible new type of crustacean cuticular strain receptor in *Carcinus maenas* (Crustacea, Decapoda). *J. Morphol.* **204**, 335–344.
- Schmidt, M. and Ache, B. W. (1992). Antennular projections to the midbrain of the spiny lobster. II. Sensory innervation of the olfactory lobe. *J. Comp. Neurol.* **318**, 291–303.
- Schmidt, M. and Ache, B. W. (1993). Antennular projections to the midbrain of the spiny lobster. III. Central arborisations of motoneurons. *J. Comp. Neurol.* **336**, 583–594.
- Schmidt, M. and Ache, B. W. (1996). Processing of antennular input in the brain of the spiny lobster, *Panulirus argus*: I. Non-olfactory chemosensory and mechanosensory pathway of the lateral and median antennular neuropils. *J. Comp. Physiol. A* **178**, 579–604.
- Schmidt, M., Ekeris, L. V. and Ache, B. W. (1992). Antennular projections to the midbrain of the spiny lobster. I. Sensory innervation of the lateral and medial antennular neuropils. *J. Comp. Neurol.* **318**, 277–290.
- Schmidt, M. and Gnatzy, W. (1984). Are the funnel-canal organs the campaniform sensilla of the shore crab *Carcinus maenas* (Crustacea, Decapoda). 2. Ultrastructure. *Cell and Tiss. Res.* **237**, 81–93.
- Schmitz, B. (1992). Directionality of antennal sweeps elicited by water jet stimulation of the tailfan in the crayfish *Procambarus clarkii*. *J. Comp. Physiol. A* **171**, 617–627.
- Schneider, D. (1964). Insect antennae. *Ann. Rev. Entomol.* **9**, 103–122.
- Schneider, D. and Kaissling, E. (1957). Der Bau der Antenne des Seidenspinners *Bombyx mori* L. II. Sensillen, cuticuläre Bildungen und innerer Bau. *Zool. Jahrb. Abt. Anat. Ontol. Tiere.* **76**, 223–250.
- Schoch, K., Rillich, J., Stevenson, P. A. and Schildberger, K. (2002). Aggression in crickets IV: sensory and motor systems – functional anatomy. *Proc. Int. Congr. Neuroethol.* **6**, 248.
- Schoch, K., Stevenson, P. A. and Schildberger, K. (2001). Neurochemical architecture of a putative mechanosensory neuropil in the deutocerebrum of the cricket *Gryllus bimaculatus*. *Proc. Göttingen. Neurobiol. Conf.* **28**, 324.
- Schöne, H., Neil, D. M., Stein, A. and Carlstead, M. K. (1976). Reactions of the spiny lobster, *Palinurus vulgaris*, to substrate tilt (I.). *J. Comp. Physiol.* **107**, 113–128.
- Schöning, M. J., Schroth, P. and Schütz, S. (2000). The use of insect chemoreceptors for the assembly of biosensors on semiconductor field-effect transistors. *Electroanal.* **12**, 645–652.
- Schöning, M. J., Schütz, S., Schroth, P., Weißbecker, B., Steffen, A., Kordos, P., Hummel, H. E. and Lüth, H. (1998). A BioFET on the basis of intact insect antennae. *Sensors Actuat. B* **47**, 235–238.
- Schram, F. R. (1986). *Crustacea*. New York: Oxford University Press.

- Schroth, P., Schöning, M. J., Lüth, H., Weißbecker, B., Hummel, H. E. and Schütz, S. (2001). Extending the capabilities of an antenna/chip biosensor by employing various insect species. *Sensors Actuat. B* **78**, 1–5.
- Schürmann, F. W., Ottersen, O. P. and Honegger, H. W. (2000). Glutamate-like immunoreactivity marks compartments of the mushroom bodies in the brain of the cricket. *J. Comp. Neurol.* **418**, 227–239.
- Schütz, S., Schöning, M. J., Schroth, P., Weißbecker, B., Schmitz, H. and Bleckmann, H. (2000). An insect-based BioFET as a bioelectronic nose. *Sensors Actuat. B* **65**, 291–295.
- Schütz, S., Weissbecker, B., Hummel, H. E., Apel, K. H. *et al* (1999). Insect antenna as a smoke detector. *Nature* **398**, 298–299.
- Sen, A., Reddy, G. V. and Rodrigues, V. (2003). Combinatorial expression of prospero, seven-up, and elav identifies progenitor cell types during sense-organ differentiation in the *Drosophila* antenna. *Dev. Biol.* **254**, 79–92.
- Shepherd, D. and Murphey, R. K. (1986). Competition regulates the efficacy of an identified synapse in crickets. *J. Neurosci.* **6**, 3152–3160.
- Shields, D. C. and Hildebrand, J. G. (1999a). Fine structure of antennal sensilla of the female sphinx moth, *Manduca sexta* (Lepidoptera: Sphingidae). I. Trichoid and basiconic sensilla. *Can. J. Zool.* **77**, 290–301.
- Shields, D. C. and Hildebrand, J. G. (1999b). Fine structure of antennal sensilla of the female sphinx moth, *Manduca sexta* (Lepidoptera: Sphingidae). II. Auriculate, coeloconic, styliform complex sensilla. *Can. J. Zool.* **77**, 302–313.
- Shields, V. D. C. and Hildebrand, J. G. (2001). Recent advances in insect olfaction, specifically regarding the morphology and sensory physiology of antennal sensilla of the female sphinx moth *Manduca sexta*. *Micros. Res. Tech.* **55**, 307–329.
- Siegler, M. V. S. and Pousman, C. A. (1990a). Distribution of motor neurons into anatomical groups in the grasshopper metathoracic ganglion. *J. Comp. Neurol.* **297**, 313–327.
- Siegler, M. V. S. and Pousman, C. A. (1990b). Motor neurons of grasshopper metathoracic ganglion occur in stereotypic anatomical groups. *J. Comp. Neurol.* **297**, 298–312.
- Sigvardt, K. A. (1977). Sensory-motor interactions in antennal reflexes of the American lobster. *J. Comp. Physiol. A* **118**, 195–214.
- Simon, H. R. (1979). Räumliches Orientierungsverhalten durch Antennenbewegungen von epigeisch-hemiedaphischen Collembolen. *Dtsch. Ent. Z.* **26**, 89–100.
- Slifer, E. H. (1966). Sense organs on the antennal flagellum of a walkingstick *Carausius morosus* Brünner (Phasmida). *J. Morphol.* **120**, 189–202.
- Smith, D. (1969). The fine structure of haltere sensilla in the blowfly, *Calliphora erythrocephala* (Meig.), with scanning electron microscopic observations on the haltere surface. *Tissue and Cell* **1**, 443–484.
- Snodgrass, R. E. (1935). *Principles of Insect Morphology*. New York, London: McGraw-Hill Book Company.
- Snodgrass, R. E. (1956). *Anatomy of the Honeybee*. Ithaca, London: Cornell University Press.
- Solon, M. H. and Cobb, J. S. (1980). Antennae-whipping behavior in the American lobster, *Homarus americanus*. *J. Exp. Mar. Biol. Ecol.* **48**, 217–224.
- Sperry, R. W. (1950). Neural basis of the spontaneous optokinetic response produced by visual inversion. *J. Comp. Physiol. Psychol.* **43**, 482–489.
- Spörhase-Eichmann, U., Vullings, H. G. B., Buijs, R. M., Hörner, M. and Schürmann, F. W. (1992). Octopamine-immunoreactive neurons in the central nervous system of the cricket, *Gryllus bimaculatus*. *Cell Tiss. Res.* **268**, 287–304.

- Staudacher, E. (1998a). Confocal imaging of antennal afferents and their overlap with arborisations of descending neurons in the cricket. *Proc. Göttingen Neurobiol. Conf.* **26**, 551.
- Staudacher, E. (1998b). Distribution and morphology of descending brain neurons in the cricket *Gryllus bimaculatus*. *Cell Tiss. Res.* **294**, 187–202.
- Staudacher, E. M. (2001). Sensory responses of descending brain neurons in the walking cricket, *Gryllus bimaculatus*. *J. Comp. Physiol. A* **187**, 1–17.
- Staudacher, E. and Schildberger, K. (1998). Gating of sensory responses of descending brain neurons during walking in crickets. *J. Exp. Biol.* **201**, 559–572.
- Staudacher, E. and Schildberger, K. (1999). A newly described neuropile in the deutocerebrum of the cricket: antennal afferents and descending interneurons. *Zoology: Anal. Complex Systems*, **102**, 212–226.
- Stay, B., Tobe, S. S. and Bendena, W. G. (1994). Allatostatins – identification, primary structures, functions and distribution. *Adv. Ins. Physiol.* **25**, 267–337.
- Steinbrecht, R. A. (1997). Pore structures in insect olfactory sensilla: a review of data and concepts. *Int. J. Insect Morphol. Embryol.* **26**, 229–245.
- Stengl, M., Homberg, U. and Hildebrand, J. G. (1990). Acetylcholinesterase activity in antennal receptor neurons of the sphinx moth *Manduca sexta*. *Cell Tiss. Res.* **262**, 245–252.
- Stern, M. (1999). Octopamine in the locust brain: cellular distribution and functional significance in an arousal mechanism. *Microsc. Res. Tech.* **45**, 135–141.
- Stern, M., Thompson, K. S. J., Zhou, P., Watson, D. G., Midgley, J. M., Gewecke, M. and Bacon, J. P. (1995). Octopaminergic neurons in the locust brain – morphological, biochemical and electrophysiological characterization of potential modulators of the visual-system. *J. Comp. Physiol. A* **177**, 611–625.
- Stevenson, P. A. (1999). Colocalisation of taurine with transmitter-immunoreactivities in the nervous system of the migratory locust. *J. Comp. Neurol.* **404**, 86–96.
- Stevenson, P. A. and Spöhrhase-Eichmann, U. (1995). Localization of octopaminergic neurones in insects. *Comp. Biochem. Physiol. B* **110**, 203–215.
- Stierle, I. E., Getman, M. and Comer, C. M. (1994). Multisensory control of escape in the cockroach *Periplaneta americana*. I. Initial evidence from patterns of wind-evoked behavior. *J. Comp. Physiol. A* **174**, 1–11.
- Stocker, R. F. (2001). *Drosophila* as a focus in olfactory research: mapping of olfactory sensilla by fine structure, odor specificity, odorant receptor expression, and central connectivity. *Microsc. Res. Tech.* **55**, 284–296.
- Strausfeld, N. J. (1976). *Atlas of an Insect Brain*. Berlin: Springer.
- Strausfeld, N. J. (1998). Crustacean-insect relationships: the use of brain characters to derive phylogeny amongst segmented invertebrates. *Brain Behav. Evol.* **52**, 186–206.
- Strausfeld, N. J., Bassemir, U., Singh, R. N. and Bacon, J. P. (1984). Organizational principles of outputs from dipteran brains. *J. Insect. Physiol.* **30**, 73–93.
- Strausfeld, N. J., Hansen, L., Li, Y., Robert, S. G. and Ito, K. (1998). Evolution, discovery, and interpretations of arthropod mushroom bodies. *Learning and Memory* **5**, 11–37.
- Sun, X. J., Tolbert, L. P. and Hildebrand, J. G. (1993). Ramification pattern and ultrastructural characteristics of the serotonin-immunoreactive neuron in the antennal lobe of the moth *Manduca sexta*. A laser-scanning confocal and electron-microscopic study. *J. Comp. Neurol.* **338**, 5–16.
- Supplee, C. (1997). Robot revolution. *Natl. Geogr.* **192**, 81–85.
- Suzuki, H. (1975). Antennal movements induced by odour and central projection of the antennal neurones in the honey-bee. *J. Insect Physiol.* **21**, 831–847.

- Tamarelle, M., Belles, X., Piulachs, D. and Girardie, J. (1990). Gastrin-cholecystokinin-like and neuroparsin-like immunoreactivities in the brain and retrocerebral neuroendocrine complex of the cockroach *Blatella germanica*. *Histochemistry* **93**, 433–438.
- Tamarelle, M., Romeuf, M. and Vanderhaeghen, J. J. (1988). Immunocytochemical localization of gasterin-cholecystokinin-like immunoreactivity in the central nervous system of the migratory locust. *Histochemistry* **89**, 201–207.
- Tautz, J. (1987). Water vibration elicits active antennal movements in the crayfish, *Orconectes limosus*. *Anim. Behav.* **35**, 748–754.
- Tautz, J., Masters, W. M., Aicher, B. and Markl, H. (1981). A new type of water vibration receptor on the crayfish antenna. I. Sensory physiology. *J. Comp. Physiol.* **144**, 533–541.
- Tautz, J. and Müller-Tautz, R. (1983). Antennal neuropile in the brain of the crayfish: morphology of neurons. *J. Comp. Neurol.* **218**, 415–425.
- Tautz, J. and Rohrseitz, K. (1998). What attracts honeybees to a waggle dancer?. *J. Comp. Physiol. A* **183**, 661–667.
- Taylor, R. C. (1975). Physical and physiological properties of the crayfish antennal flagellum. *J. Neurobiol.* **6**, 501–519.
- Telford, M. J. and Thomas, R. H. (1998). Expression of homeobox genes shows chelicerate arthropods retain their deutocerebral segment. *Proc. Natl. Acad. Sci. USA* **95**, 10671–10675.
- Thany, S. H., Lenaers, G., Crozatier, M., Armengaud, C. and Gauthier, M. (2003). Identification and localization of the nicotinic acetylcholine receptor alpha3 mRNA in the brain of the honeybee, *Apis mellifera*. *Insect Mol. Biol.* **12**, 255–262.
- Thompson, K. S. and Bacon, J. P. (1991). The vasopressin-like immunoreactive (VPLI) neurons of the locust, *Locusta migratoria*. II. Physiology. *J. Comp. Physiol. A* **168**, 619–630.
- Thompson, K. S., Tyrer, N. M., May, S. T. and Bacon, J. P. (1991). The vasopressin-like immunoreactive (VPLI) neurons of the locust, *Locusta migratoria*. I. Anatomy. *J. Comp. Physiol. A* **168**, 605–617.
- Thurm, U. (1963). Die Beziehungen zwischen mechanischen Reizgrößen und stationären Erregungszuständen bei Borstenfeld-Sensillen von Bienen. *Z. Vergl. Physiol.* **46**, 351–382.
- Thurm, U. (1964). Mechanoreceptors in the cuticle of the honeybee: fine structure and stimulus mechanism. *Science* **145**, 1063–1065.
- Thurm, U. (1965). An insect mechanoreceptor. Part I: fine structure and adequate stimulus. *Cold Spring Harbor Symp. Quant. Biol.* **30**, 75–82.
- Thurm, U. (1983). Biophysics of Mechanoreception. In: *Biophysics*, (eds Hoppe, W. B. and Bauer, R. D.), pp. 666–671. Berlin, N. Y.: Springer.
- Thurm, U., Stedtler, A. and Foelix, R. (1975). Stimulus-effective deformations of the terminal structures of a mechanoreceptor. *Verh. Dtsch. Zool. Ges.* **67**, 37–41.
- Tichy, H. (1979). Hygro- and thermoreceptive triad in the antennal sensillum of the stick insect, *Carausius morosus*. *J. Comp. Physiol. A* **132**, 149–152.
- Tichy, H. and Loftus, R. (1983). Relative excitability of antennal olfactory receptors in the stick insect, *Carausius morosus* L.: in search of a simple, concentration-independent odor-coding parameter. *J. Comp. Physiol. A* **152**, 459–473.
- Tichy, H. and Loftus, R. (1987). Response characteristics of a cold receptor in the stick insect *Carausius morosus*. *J. Comp. Physiol. A* **160**, 33–42.
- Tichy, H. and Loftus, R. (1990). Response of moist-air receptor on antenna of the stick insect *Carausius morosus* to step changes in temperature. *J. Comp. Physiol. A* **166**, 507–516.
- Tichy, H. and Loftus, R. (1996). Hygroreceptors in insects and a spider: humidity transduction models. *Naturwissenschaften* **83**, 255–263.

- Tierney, A. J., Godleski, M. S. and Massanari, J. R. (2000). Comparative analysis of agonistic behavior in four crayfish species. *J. Crust. Biol.* **20**, 54–66.
- Toh, Y. (1977). Fine structure of antennal sense organs of the male cockroach, *Periplaneta americana*. *J. Ultrastruct. Res.* **60**, 373–394.
- Toh, Y. (1981). Fine-structure of sense organs on the antennal pedicel and scape of the male cockroach, *Periplaneta americana*. *J. Ultrastruct. Res.* **77**, 119–132.
- Tominaga, Y. and Yokohari, F. (1981). External structure of the sensillum capitulum, a hygro- and thermoreceptive sensillum of the cockroach, *Periplaneta americana*. *Cell Tiss. Res.* **226**, 309–318.
- Torkkeli, P. H. and French, A. S. (1999a). Primary culture of antennal mechanoreceptor neurons of *Manduca sexta*. *Cell Tiss. Res.* **297**, 301–309.
- Torkkeli, P. H. and French, A. S. (1999b). Stretch activated ion channels in cultured mechanosensory neurons of *Manduca sexta*. *Biophys. J.* **76**, A203.
- Towne, W. F. and Kirchner, W. H. (1989). Hearing in honeybees: detection of air-particle oscillations. *Science* **244**, 686–688.
- Trachtman, P. (2000). Redefining robots. *Smithsonian* **30**, 97–112.
- Trezenza, T. and Wedell, N. (1997). Definitive evidence for cuticular pheromones in a cricket. *Anim. Behav.* **54**, 979–984.
- Tsujimura, T. and Yabuta, T. (1992). A tactile sensing method employing force/torque information through insensitive probes. *Proc. IEEE Int. Conf. Robot. Autom.* **1992**, 1315–1320.
- Tyler, N. M., Pozza, M. F., Humbel, U., Peters, B. H. and Bacon, J. P. (1988). The tritocerebral commissure “dwarf” (TCD): a major GABA-immunoreactive descending interneuron in the locust. *J. Comp. Physiol. A* **164**, 141–150.
- Tyler, N. M., Turner, J. D. and Altman, J. S. (1984). Identifiable neurons in the locust central nervous system that react with antibodies to serotonin. *J. Comp. Neurol.* **227**, 313–330.
- Ueno, N. and Kaneko, M. (1995). On a new contact sensing strategy for dynamic active antenna. *Proc. IEEE Int. Conf. Robot. Autom.* **1995**, 1120–1125.
- Ueno, N., Kaneko, M. and Svinin, M. (1996). Theoretical and experimental investigation on dynamic active antenna. *Proc. IEEE Int. Conf. Robot. Autom.* **1996**, 3557–3563.
- Ueno, N., Svinin, M. M. and Kaneko, M. (1998). Dynamic contact sensing by flexible beam. *IEEE-ASME Trans. Mechatron.* **3**, 254–264.
- Urvoy, J., Fudalewicz-Niemczyk, W., Petryszak, A. and Rosciszewska, M. (1984). Etude des organes sensoriels externes de l’antenne de Phasme *Carausius morosus* B. (Cheleutoptera). *Acta Biol. Cracoviensia Ser.: Zool.* **XXVI**, 57–68.
- vande Berg, J. S. (1971). Fine structural studies of Johnston’s organ in the Tobacco Hornworm moth, *Manduca sexta* (Johannson). *J. Morphol.* **133**, 439–456.
- Vedel, J. P. (1980). The antennal motor system of the rock lobster: competitive occurrence of resistance and assistance reflex patterns originating from the same proprioceptor. *J. Exp. Biol.* **87**, 1–22.
- Vedel, J. P. (1985). Cuticular mechanoreception in the antennal flagellum of the rock lobster *Palinurus vulgaris*. *Comp. Biochem. Physiol. A* **80**, 151–158.
- Vedel, J. P. and Monnier, S. (1983). A new muscle receptor organ in the antenna of the rock lobster *Palinurus vulgaris*: mechanical, muscular and proprioceptive organization of the two proximal joints JO and J1. *Proc. R. Soc. Lond. B* **218**, 95–110.
- Verhaert, P., Grimmlikhuijzen, J. P. and Deloof, A. (1985). Distinct localization of FMRFamide-like and bovine pancreatic polypeptide-like material in the brain, retrocerebral complex and suboesophageal ganglion of the cockroach *Periplaneta americana* L. *Brain Res.* **348**, 331–338.

- Vitzthum, H., Homberg, U. and Agricola, H. (1996). Distribution of dip-allatostatin I-like immunoreactivity in the brain of the locust *Schistocerca gregaria* with detailed analysis of immunostaining in the central complex. *J. Comp. Neurol.* **369**, 419–437.
- Vitzthum, H., Homberg, U., Würden, S. and Agricola, H. (1995). Allatostatin-like immunoreactivity in the central complex of the locust *Schistocerca gregaria*. *Proc. Göttingen. Neurobiol. Conf.* **23**, 598.
- Waldow, U. (1975). Multimodal neurons in deutocerebrum of *Periplaneta americana*. *J. Comp. Physiol.* **101**, 329–341.
- Waldow, U. (1977). CNS units in cockroach (*Periplaneta americana*) – specificity of response to pheromones and other odor stimuli. *J. Comp. Physiol.* **116**, 1–17.
- Watson, A. H. and Burrows, M. (1983). The morphology, ultrastructure, and distribution of synapses on an intersegmental interneuron of the locust. *J. Comp. Neurol.* **214**, 154–169.
- Watson, A. H. and Pflüger, H. J. (1989). Regional specialisation for synaptic input and output on a locust intersegmental interneurone with multiple spike-initiating zones. *J. Comp. Neurol.* **279**, 515–527.
- Watson, J. T., Ritzmann, R. E., Zill, S. N. and Pollack, A. J. (2002). Control of obstacle climbing in the cockroach, *Blaberus discoidalis*. I. Kinematics. *J. Comp. Physiol. A* **188**, 39–53.
- Weber, H. and Weidner, H. (1974). *Grundriss der Insektenkunde*. Stuttgart: Gustav Fischer.
- Weide, W. (1960). Einige Bemerkungen über die antennalen Sensillen sowie über das Fühlerwachstum der Stabheuschrecke *Carausius* (*Dixippus*) *morosus* Br. (Insecta: Phasmida). *Wiss. Z. Univ. Halle Math.-Nat.* **IX/2**, 247–250.
- Wendelken, P. W. and Barth, R. H. (1987). The evolution of courtship phenomena in neotropical cockroaches of the genus *Blaberus* and related genera. *Adv. Ethol.* **27**, 1–98.
- Wendler, G. (1965). Über den Anteil der Antennen an der Schwererezeption der Stabheuschrecke *Carausius morosus* Br. *Z. Vergl. Physiol.* **51**, 60–66.
- Wendt, B. and Homberg, U. (1992). Immunocytochemistry of dopamine in the brain of the locust *Schistocerca gregaria*. *J. Comp. Neurol.* **321**, 387–403.
- Wilkens, L. A., Schmitz, B. and Herrnkind, W. F. (1996). Antennal responses to hydrodynamic and tactile stimuli in the spiny lobster *Panulirus argus*. *Biol. Bull.* **191**, 187–198.
- Williams, L. D. (1975). Anatomical studies of the insect central nervous system: a ground-plan of the midbrain and an introduction to the central complex in the locust *Schistocerca gregaria* (Orthoptera). *J. Zool. Lond.* **176**, 67–86.
- Wolf, H. and Burrows, M. (1995). Proprioceptive sensory neurons of a locust leg receive rhythmic presynaptic inhibition during walking. *J. Neurosci.* **15**, 5623–5636.
- Wolf, H. and Laurent, G. (1994). Rhythmic modulation of the responsiveness of locust sensory local interneurons by walking patten generating networks. *J. Neurophysiol.* **71**, 110–118.
- Yagodin, S. V. (1980). Role of antenna in flight maintenance of the cockroach *Periplaneta americana*. *J. Evol. Biochem. Physiol.* **16**, 30–36.
- Yam, P. (1998). Roaches at the wheel. *Sci. Am.* **278**, 45.
- Ye, S. and Comer, C. M. (1996). Correspondence of escape-turning behavior with activity of descending mechanosensory interneurons in the cockroach, *Periplaneta americana*. *J. Neurosci.* **16**, 5844–5853.
- Ye, S., Leung, V., Khan, A., Baba, Y. and Comer, C. M. (2003). The antennal system and cockroach evasive behavior. I. Roles for visual and mechanosensory cues in the response. *J. Comp. Physiol. A* **189**, 89–96.

- Yildiz, F. and Gebhardt, M. (2003). Complex innervation of the antennal base of crickets. *Proc. Göttingen. Neurobiol. Conf.* **29**, 238.
- Yokohari, F. (1981). The sensillum capitulum, an antennal hygro- and thermoreceptive sensillum of the cockroach, *Periplaneta americana* L. *Cell Tiss. Res.* **216**, 525–543.
- Young, D. (1970). The structure and function of a connective chordotonal organ in the cockroach leg. *Philos. Trans. R. Soc. Lond. B*, **256**, 401–426.
- Younossi-Hartenstein, A. and Hartenstein, V. (1997). Pattern, time of birth, and morphogenesis of sensillum progenitors in *Drosophila*. *Microsc. Res. Tech.* **39**, 479–491.
- Zacharuk, R. Y. (1985). Antennae and Sensilla. In: *Comprehensive insect physiology, biochemistry and pharmacology*, (eds Kerkut, G. A. and Gilbert, L. I.), pp. 1–69. Oxford, NY: Pergamon Press.
- Zeil, J., Sandeman, R. and Sandeman, D. C. (1985). Tactile localisation: the function of active antennal movements in the crayfish *Cherax destructor*. *J. Comp. Physiol. A* **157**, 607–617.
- Zeiner, R. and Tichy, H. (1998). Combined effects of olfactory and mechanical inputs in antennal lobe neurons of the cockroach. *J. Comp. Physiol. A* **182**, 467–476.